

TEXTBOOK FOR B.SC. 5TH SEMESTER · VAC: AI

# Patterns, Decisions & Consequences

**Artificial Intelligence** for Indian Undergraduate Students



Alok Patel, Ph.D.

KUCHINDA COLLEGE · SAMBALPUR UNIVERSITY

# **PATTERNS, DECISIONS, AND CONSEQUENCES**

*Artificial Intelligence for Indian Undergraduate Students*

**Dr. Alok Patel, Ph.D.**

Head, Department of Zoology  
Kuchinda College (Sambalpur University)  
Kuchinda, Odisha

*Value-Added Course — Artificial Intelligence · NEP 2020*

# Patterns, Decisions, and Consequences: Artificial Intelligence for Indian Undergraduate Students

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This book is written for the Value-Added Course on Artificial Intelligence under the National Education Policy 2020, for undergraduate students of Sambalpur University and other Indian universities.

*To my parents,*

*who taught me to observe before I could read, and to keep looking for  
the pattern beneath the surface long after everyone else had stopped  
asking.*

# Preface

This book began with a question that I suspect every teacher in a new course eventually asks: what would I want my students to understand, not just know? The AI literacy courses being introduced across Indian universities under the National Education Policy 2020 give us an unusual opportunity. We are not retrofitting a curriculum to accommodate a new subject. We are building something from the beginning, and that means we get to decide what it should be for.

The answer I arrived at is this: a student who finishes this course should be able to think clearly about AI. Not operate AI tools, though that matters too. Not pass examinations about AI terminology, though that has its place. But think: to observe an AI system carefully, to ask the right questions about who built it and from what data and for whose benefit, to identify when something is going wrong and articulate why, and to contribute, in whatever role their life places them in, to making AI work better for the communities they are part of.

The Pattern Hunters philosophy that organises this book is my attempt to operationalise that answer. Pattern recognition is not a metaphor for what AI does: it is literally what AI does, and it is also, as I hope the Prologue makes clear, something that living creatures of every kind have done for as long as there has been life in this world. The fisherwoman reading the water surface, the doctor reading a face, the farmer reading the sky: they are all doing what the algorithms in Chapter 3 do, with the difference that they are doing it with embodied experience, with moral stakes, and with accountability to the communities their judgments affect. Understanding that continuity, and understanding what is genuinely new about the AI era, is the beginning of wisdom about it.

The book is deliberately Indian in its examples and unapologetically particular in its regional grounding. Western Odisha, where I have spent my life, appears throughout: the earthworms of Kuchinda, the Kalajeera rice of Koraput, the Mahanadi's floods, the Sambalpuri weaver's ikat, the gauntia — village head, or pradhan — mediating a village dispute. This is not closed-mindedness but it is a methodological commitment. AI is not an abstract universal: it is always deployed in a specific place, affecting specific people, trained on data from specific histories. The abstract universal disappears when you ask: which Indian? Which language? Which community? Which farmer? Grounding the analysis in concrete Indian contexts is the only honest way to do it.

The book covers technical material across Units 1 and 2 without requiring mathematical background. It covers applications across Unit 3 with genuine attention to Indian deployment contexts. It covers ethical, social, and governance questions across Unit 4 with the seriousness those questions deserve: not as add-ons to the technical material, but as the reason the technical material matters. And it ends with Unit 5's mini-project, because thinking about AI and doing something with AI are not the same activity, and the course should require both.

This book owes debts to many people. To my departmental colleagues — Pooja, Susama, and Ritu — for their steady collegiality and support. To the students of Kuchinda College, who over several semesters asked the kind of questions only someone encountering an idea for the first time can ask. To my research collaborators in earthworm genomics and soil health science, who keep reminding me that the most important questions in science begin in a specific field, in a specific season, with a specific soil type. To the members of the Pattern Hunters community, who tested early versions of these materials. And to my family — my wife, Suchismita, and my daughter, Arya — whose patience with a writer is a form of love I do not sufficiently acknowledge.

A note on what this book is not. It is not a technical manual: readers who need to build production AI systems should look elsewhere. It is not a policy brief: the governance frameworks described in Chapter 14 are starting points for analysis, not finished recommendations. It is not a polemic against AI or an advertisement for it: both reactions miss the point. AI is a tool, which means its value is entirely determined by the purposes it is put to, the communities it is built to serve, and the governance frameworks that shape its operation. This book is an attempt to help students understand enough to participate in those determinations. The rest is up to them.

**Alok Patel**

*Head, Department of Zoology*

*Kuchinda College (Sambalpur University)*

*Kuchinda, Odisha*

*June 2026*



# Table of Contents

Right-click and choose "Update Field" to generate the table of contents.

## UNIT ONE

# What Is Intelligence?

*Before this book explains a single algorithm, it asks an older question. A fisherwoman reading the sky, a doctor reading a rash, a farmer reading a leaf: none of them run code, yet all of them do what every intelligent system does. They notice a regularity, act on it, and watch what happens next.*

*That loop — observe, decide, act, learn — is not a metaphor for intelligence. It is intelligence, in its oldest and plainest form. The chapters ahead give it a name: the Pattern-Decision-Action loop. Everything that follows in this book, human or artificial, is a variation on it.*

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1. Patterns All the Way Down
  2. When Machines Learn to See Patterns
  3. The Anatomy of an Intelligent System

## Chapter 1

# Patterns All the Way Down

*Every act of understanding is an act of pattern recognition. Before there was mathematics, before there was language, before there was science, there was a living creature noticing that certain things happen together, that certain shapes mean danger, that certain colours mean food. Intelligence, in its most ancient and honest sense, is the ability to find structure in experience.*

*This chapter begins there. Not with computers. Not with algorithms. Not with code. It begins with the question that makes all of those things possible: what is a pattern, and why does finding one matter?*

## Learning Objectives

By the end of this chapter, you will be able to:

- Define a pattern and distinguish it from noise or coincidence
- Describe the Pattern-Decision-Action loop with examples from everyday Indian life
- Explain the difference between a useful pattern and a spurious one
- Trace the ladder of abstraction from raw data to actionable decision
- Recognise pattern recognition as the cognitive foundation of all intelligence, natural and artificial

## 1.1 What Is a Pattern?

Consider three moments from ordinary Indian life.

A woman in a fishing village on the Odisha coast steps out of her home before dawn. She looks at the colour of the sky, feels the direction of the breeze on her cheek, notices the behaviour of the birds along the shoreline. Without consulting any instrument, without performing any calculation, she makes a decision: today the fishing will be good, or today the sea is not safe. She has been making this judgement for thirty years. She is almost never wrong.

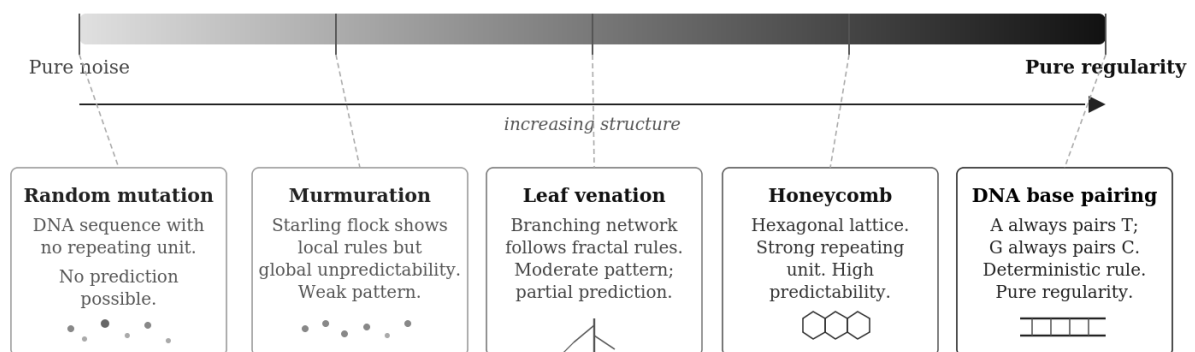
A doctor in a district hospital in Chhattisgarh examines a patient. She looks at the distribution of the rash on the skin, asks about the sequence in which symptoms appeared, notes the patient's age and village. Within minutes, before any laboratory result arrives, she has a provisional diagnosis. She has seen this pattern before, many times, in many variations.

A farmer in the Vidarbha region of Maharashtra has been watching his cotton crop for weeks. The leaves at the edges of certain rows have a particular yellowing pattern. The soil in those rows is slightly different. He remembers a similar pattern from twelve years ago. He knows what is coming and he knows what to do.

In each case, the person is doing the same fundamental thing. They are looking at a set of observations and recognising a structure within them. The colour of the sky, the shape of a rash, the yellowing of a leaf are not random pieces of information. They are clues, and the person reading them has learned, through experience, that certain clues appear together, that certain combinations mean certain things. That is pattern recognition.

A pattern, in the most general sense, is any regularity that appears in data. It is a relationship between observations that holds across multiple instances. The sky is red at morning more often than not before a storm. That particular distribution of a rash appears more often in patients with a specific bacterial infection. That yellowing at leaf edges, in that soil, with that timing, precedes a specific fungal disease. The regularity is the pattern.

**Fig. 1.1**  
**The Pattern Spectrum**



*The pattern spectrum from pure noise (left) to pure regularity (right). Most biological phenomena, and most AI training data, lie in the middle — partially structured, partially unpredictable.*

*Figure 1.1 — Pattern Spectrum*

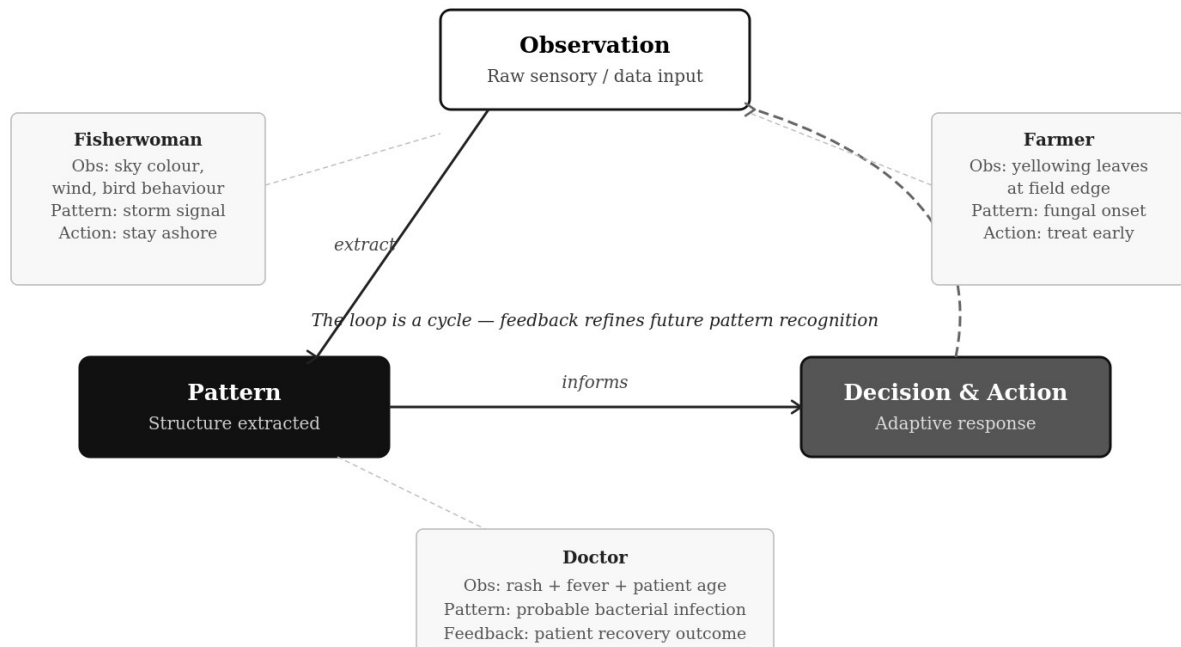
Figure 1.1 shows a spectrum. At one extreme is pure noise, where no structure exists and no prediction is possible. At the other extreme is pure regularity, where the pattern is so strong that every observation can be predicted perfectly. Most of what we encounter in the real world lies somewhere in between. A good intelligence, whether human or artificial, is one that can find the signal within the noise: the structure that is genuinely there, even when it is partially obscured.

## 1.2 From Pattern to Decision: The Loop That Drives Intelligence

Finding a pattern is not, in itself, useful. What makes pattern recognition valuable is what happens next. The fisherwoman does not simply observe the red sky and file it away as an interesting fact. She acts on it. The pattern is the input to a decision, and the decision is the

input to an action. This three-part sequence, Pattern-Decision-Action, is the basic loop of all intelligent behaviour.

**Fig. 1.2**  
**The Pattern-Decision-Action Loop**



*The Pattern-Decision-Action loop is the basic structure of all intelligent behaviour. The dashed feedback arrow distinguishes true intelligence (which learns from consequences) from mere reactive response (which does not).*

*Figure 1.2 — Pattern Decision Action Loop*

The loop is more than a sequence. It is a cycle. The action taken in response to a pattern changes the world, and that changed world generates new observations, which feed back into the pattern-recognition process. The farmer who treats his cotton crop based on a recognised pattern observes whether the treatment worked. That observation becomes new data. Next year, his pattern recognition is slightly more refined.

This feedback loop is what allows intelligence to improve over time. An intelligence that acts but never learns from the consequences of its actions is not truly intelligent. It is merely reactive. True intelligence, in any form, includes a mechanism for updating its patterns based on new evidence.

Consider how this loop plays out across different domains:

In healthcare, a doctor recognises a pattern of symptoms (observation), forms a diagnosis (decision), prescribes treatment (action), and observes the patient's response (feedback). A doctor who has treated thousands of patients has refined this loop to an extraordinary degree.

In language, a child hearing sentences repeatedly (observation) begins to infer grammatical rules (pattern), applies those rules to form new sentences (action), and receives correction or confirmation from listeners (feedback). Language acquisition is pattern recognition learning in action.

In ecology, a predator hunting in the forests of the Satpura hills observes animal tracks and sounds (observation), infers the location and direction of prey (pattern and decision), moves accordingly (action), and either succeeds or fails (feedback). Every hunt updates the predator's internal model of its prey's behaviour.

The Pattern-Decision-Action loop is not a human invention. It is a biological universal. Every organism that survives does so because it is capable, at some level, of recognising patterns in its environment and responding adaptively. What artificial intelligence does is take this loop and implement it in a machine, using data and computation in place of sensory experience and neural learning.

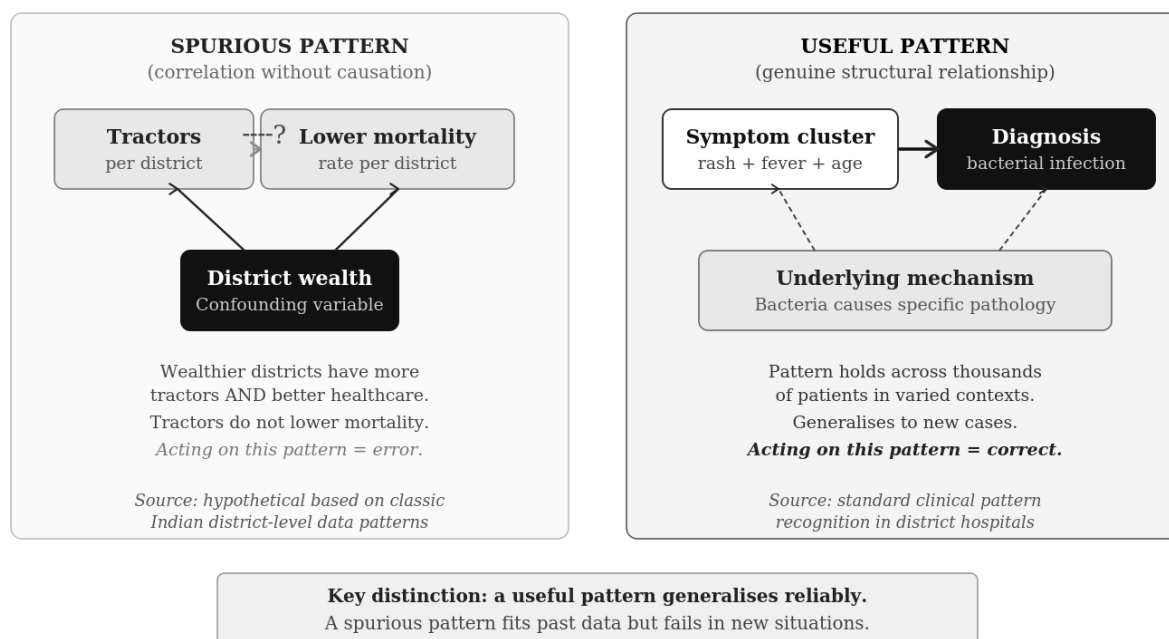
### **1.3 The Difference Between a Useful Pattern and a Spurious One**

Not all patterns are created equal. The history of human error, in medicine, in economics, in daily life, is substantially the history of mistaking spurious patterns for meaningful ones. A spurious pattern is one that appears in data but does not reflect any genuine underlying relationship. It is a coincidence elevated to the status of a rule.

Consider a well-known example from Indian public health statistics. For several decades, mortality rates in certain districts were observed to be lower in districts with higher numbers of registered tractors. There is a genuine, measurable pattern here. But the tractors are not protecting people from disease. Both the tractors and the lower mortality are consequences of a third factor: wealth. Wealthier districts have more agricultural machinery and also have better healthcare access. The pattern between tractors and mortality is real but spurious. Acting on it, by purchasing tractors to improve public health, would be absurd.

This is the problem of correlation versus causation. Two things can vary together, can show a genuine statistical pattern, without either causing the other. A useful pattern is one where the relationship reflects something real about the structure of the world, something that will remain true in new situations and that can be acted upon reliably. A spurious pattern is one that exists in the data examined but does not generalise.

**Fig. 1.3**  
**Useful vs. Spurious Patterns**



Both examples show genuine statistical correlations in Indian data. Only the right-hand pattern reflects a real underlying mechanism and can safely guide action. AI systems can learn either type of pattern indiscriminately.

Figure 1.3 — Useful Vs Spurious Patterns

Distinguishing useful patterns from spurious ones is one of the central challenges of both human reasoning and artificial intelligence. Humans are, unfortunately, rather bad at this. We are wired to find patterns, and we find them whether they are there or not. This tendency, which psychologists call apophenia, served our ancestors well on the savannah, where it was better to mistake a shadow for a predator than to mistake a predator for a shadow. But it makes us unreliable pattern-recognisers in complex statistical domains.

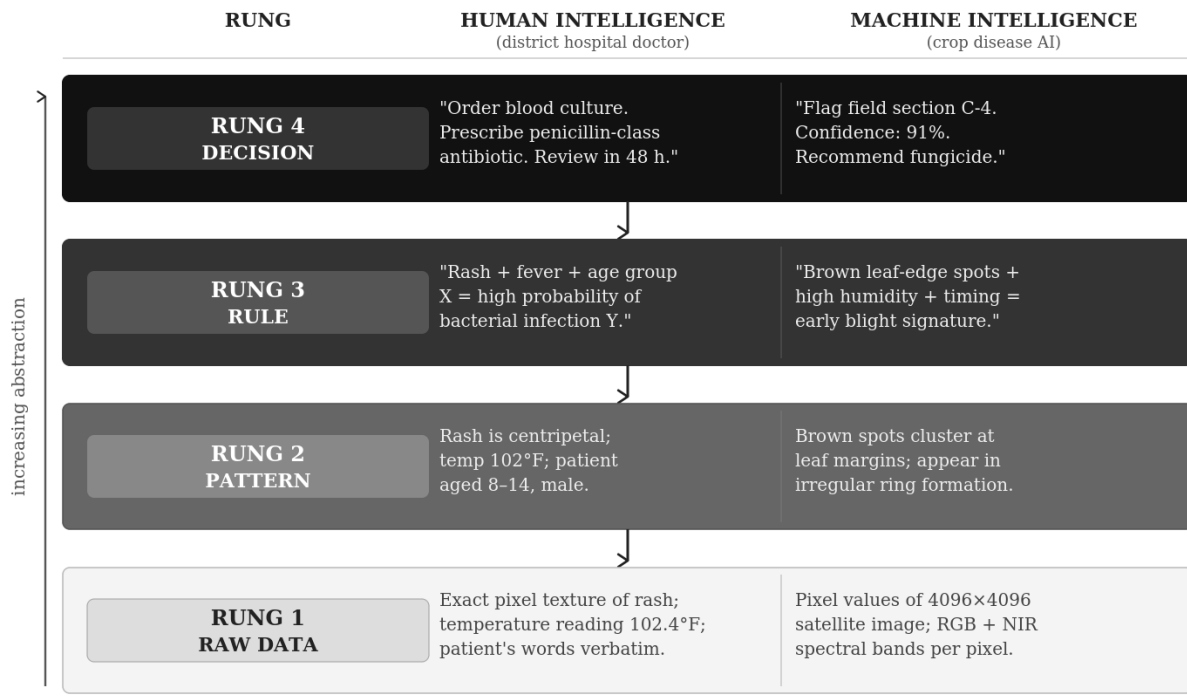
Artificial intelligence systems face the same problem, and they can fail in the same way. An AI trained on data that contains a spurious pattern will learn that pattern just as readily as a real one. This is not a flaw of AI specifically. It is a flaw of any learning system that cannot distinguish between real structure and statistical coincidence. Understanding this limitation is the first step toward using AI responsibly.

## 1.4 The Ladder of Abstraction

Raw experience is not directly useful as the basis for decision-making. The fisherwoman does not make her decision based on every photon that enters her eyes, every air molecule that moves past her skin, every sound wave that reaches her ears. Her nervous system performs an enormous amount of processing before any of that raw sensory data becomes usable. It filters, compresses, categorises, and abstracts. What she consciously perceives is not raw data but a structured interpretation of raw data.

This process of moving from raw data to structured interpretation is what we call abstraction. It happens in multiple stages, and each stage moves further away from the raw sensory input and closer to a form that can guide decision-making.

**Fig. 1.4**  
**The Ladder of Abstraction**



*Both human and machine intelligence move up this ladder automatically. Raw data is too voluminous to act on directly; abstraction compresses it into forms that can guide decisions efficiently.*

**Figure 1.4 — Ladder Of Abstraction**

Consider the ladder in Figure 1.4. At the bottom rung is raw data: the actual measurements and observations, unprocessed and uninterpreted. For the doctor examining the patient, this is the precise pixel-level appearance of the rash, the exact temperature reading, the specific words the patient uses to describe the pain.

At the second rung is pattern: the regularities extracted from the raw data. The doctor notices that the rash has a particular geographical distribution, that the temperature is in a specific range, that the pain description matches a known presentation. These are patterns within the raw data.

At the third rung is rule: the generalisation extracted from many patterns. The doctor has seen this combination of patterns in many previous patients and has distilled from that experience a rule: when symptoms A, B, and C appear together in a patient of age group X, the probable diagnosis is condition Y.

At the fourth rung is decision: the specific action dictated by the rule applied to the current situation. Given this patient's presentation, which matches the rule, the appropriate action is to order test Z and prescribe treatment W.

This ladder, from raw data to pattern to rule to decision, is not unique to medicine. It describes the structure of all intelligent decision-making. A machine learning system does exactly the same thing. It takes raw data, extracts patterns, generalises those patterns into rules (which it calls a model), and applies the model to new data to generate decisions. The machinery is different, but the logical structure is identical.

What artificial intelligence offers is the ability to climb this ladder automatically, from raw data to decision, using statistical computation rather than years of embodied human experience. The question of when that automatic ascent produces reliable decisions, and when it produces dangerous ones, will occupy us throughout this book.

## **1.5 Intelligence as the Universal Pattern-Finder**

We are now in a position to offer a working definition of intelligence. Intelligence is the capacity to recognise patterns in experience and to use those patterns to guide adaptive behaviour. This definition is deliberately broad. It is intended to cover the intelligence of the crow that drops nuts on pedestrian crossings so that cars will crack them, the intelligence of the octopus that learns to open jars, the intelligence of the chess grandmaster reading board positions, and the intelligence of the language model predicting the next word in a sentence.

All of these are, at the most fundamental level, examples of the same process: a system that takes in information, extracts structure from it, and uses that structure to do something useful. The difference between them is in the nature of the information, the sophistication of the pattern-extraction, and the complexity of the action taken.

This is why the title of this chapter is what it is. Patterns all the way down. Below the behaviour of the crow is a nervous system learning through reinforcement. Below that learning is the structure of neurons and synapses. Below that is the molecular biology of memory formation. At every level, the same fundamental process: structure extracted from noise, signal separated from background, regularity found in apparent chaos. Intelligence is not a special property of human brains. It is a property of any sufficiently complex system that processes information in structured ways. What makes human intelligence remarkable is not that it finds patterns, but the extraordinary richness and flexibility of the patterns it can find.

What makes artificial intelligence possible is the discovery, over the last seventy years, that the formal structure of this pattern-finding process can be implemented in silicon and software. Machines cannot yet match the breadth of human intelligence. But they can, in

specific domains, and with specific kinds of data, find patterns faster, more consistently, and at larger scale than any human being ever could.

That is the premise on which this entire book rests. And it begins, as all intelligence begins, with a pattern.

## **1.6 Patterns in My Field: Stream Perspectives**

The concepts developed in this chapter appear across every domain of human knowledge. The following examples show how pattern recognition as a foundation for intelligence manifests in the three streams.

### **Arts Stream: How Spell-Checkers Learned Odia and Hindi**

A spell-checker is a pattern-recognition system. It has been trained on millions of correctly spelled words and has learned the patterns of which letter combinations are valid in a given language. When it encounters an unfamiliar string of letters, it compares it against those learned patterns and either accepts it or flags it as anomalous. Building a spell-checker for Odia or Hindi requires first collecting the patterns of those languages, their morphological rules, their common word formations, their script-specific regularities. The challenge of building such tools for Indian languages is a direct application of the Pattern-Decision-Action loop to the domain of language technology.

### **Science Stream: How Satellites Detect Deforestation in the Sundarbans**

Satellite images of the Sundarbans mangrove forest in West Bengal and Bangladesh are updated regularly. A deforestation detection system compares each new image against historical ones, looking for changes in the spectral signature of pixels, changes that indicate the replacement of dense vegetation with open ground or water. The pattern being sought is the signature of forest loss: specific changes in specific wavelengths of reflected light, appearing in clusters that are too large and too rapid to be natural variation. The system has learned, from labelled examples of known deforestation events, what those patterns look like. It now scans new images for the same signatures.

### **Commerce Stream: How UPI Fraud Detection Works**

Every UPI transaction generates data: the amount, the time, the location, the device used, the history of the account, the pattern of the recipient. A fraud detection system has been trained on millions of past transactions, both legitimate and fraudulent, and has learned the patterns that distinguish one from the other. An unusually large transaction at an unusual time, to an account with no transaction history, from a device that has never been used before, matches a pattern associated with fraud. The system flags it, not because any single feature is definitive, but because the combination of features matches a learned pattern of risk.

## When the Pattern Is a Prejudice

The concepts in this chapter have a shadow side that must be acknowledged here and will be developed more fully in Unit 4. If intelligence is pattern recognition, and if artificial intelligence learns its patterns from human-generated data, then what happens when the patterns in that data encode human prejudice?

Consider a hiring algorithm trained on ten years of employment data from an Indian technology company. The data shows that the employees who were promoted most rapidly share certain characteristics: they attended certain institutions, they lived in certain cities, they had certain surnames. The algorithm learns these patterns faithfully. It then applies them to new candidates, screening out applicants who do not match the historical patterns. The pattern is real. The promoted employees do, statistically, share those characteristics. But the pattern reflects historical hiring bias, not genuine predictive power. The algorithm has learned a prejudice and is now implementing it at scale, faster and more consistently than any human recruiter could.

This is not a hypothetical risk. It is a documented phenomenon in AI systems deployed in hiring, credit scoring, healthcare triage, and judicial sentencing in multiple countries. The pattern is real. The pattern is also unjust. Understanding the difference between a pattern that reflects genuine structure and a pattern that encodes historical inequity is one of the central ethical challenges of our moment in history. We will return to it repeatedly.

## — CHAPTER SUMMARY —

### Chapter 1 — Key Points

1. A pattern is any regularity that appears across multiple observations; it is structure within data.
2. Intelligence, in all its forms, begins with pattern recognition.
3. The Pattern-Decision-Action loop describes the basic structure of intelligent behaviour.
4. Feedback from action refines future pattern recognition; this is how intelligence improves.
5. Not all patterns are equally valid. Spurious patterns (coincidences) must be distinguished from genuine structural relationships.
6. The ladder of abstraction describes the movement from raw data to pattern to rule to decision.
7. Artificial intelligence automates the same pattern-finding process that underlies human and animal cognition.
8. Patterns learned from biased data will themselves be biased; this is a central ethical challenge of AI.

## Section A — MCQ (5 Questions)

**Q1. Which of the following best defines a pattern in the context of intelligence?**

- A) A single observation that stands out from others
- B) A regularity that appears across multiple observations
- C) A mathematical formula derived from laboratory experiments
- D) A prediction made by a computer algorithm

**Answer: B**

*Explanation: A pattern is defined by its regularity across multiple observations, not by any single striking observation or computational process.*

**Q2. The Pattern-Decision-Action loop is best described as:**

- A) A one-time sequence that ends when an action is taken
- B) A cycle in which action feeds back into future pattern recognition
- C) A process unique to human cognition and not found in animals
- D) A computational algorithm developed by AI researchers

**Answer: B**

*Explanation: The loop is a cycle because the consequences of action generate new observations that refine future pattern recognition, making it a feedback mechanism rather than a linear sequence.*

**Q3. In the example of tractors and mortality rates in Indian districts, the pattern observed is called spurious because:**

- A) The data was incorrectly collected
- B) The correlation does not reflect a causal relationship
- C) The pattern only appears in one district
- D) Tractors are not related to human health at all

**Answer: B**

*Explanation: A spurious pattern is one where a genuine statistical relationship exists but does not reflect direct causation. Both tractors and lower mortality are effects of a third variable, wealth, not causes of each other.*

**Q4. Which rung of the ladder of abstraction involves generalising from many observed patterns to a rule that can be applied to new situations?**

- A) Raw data
- B) Pattern
- C) Rule
- D) Decision

**Answer: C**

*Explanation: The rule rung represents the generalisation step, where patterns observed across many instances are distilled into a principle that can be applied to new, previously unseen situations.*

**Q5. Why is apophenia a relevant concept for understanding both human cognition and AI systems?**

- A) It describes the tendency to find patterns even where none genuinely exist
- B) It is the process by which AI systems improve through feedback
- C) It refers to the ability to distinguish correlation from causation
- D) It is a statistical technique for validating machine learning models

**Answer: A**

*Explanation: Apophenia is the tendency to perceive meaningful patterns in random data. Both humans and AI systems can exhibit this tendency, making pattern validation a critical part of responsible intelligence.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Using one example from Indian daily life, explain the three stages of the Pattern-Decision-Action loop.**

**Model Answer:**

The Pattern-Decision-Action loop can be illustrated through the example of a farmer in Odisha observing that the sky has turned dark green before previous hailstorms. Recognising this pattern, he decides to cover his vegetable plot with tarpaulin. The action protects his crop. If it succeeds, this outcome becomes new data that reinforces the pattern; if it fails, the pattern is refined with additional information.

*Key terms: Pattern-Decision-Action loop, observation, feedback, adaptive behaviour*

**Q7. Explain the difference between a useful pattern and a spurious pattern. Why does this distinction matter for artificial intelligence?**

**Model Answer:**

A useful pattern is a genuine regularity in data that reflects an underlying causal or structural relationship and generalises reliably to new situations. A spurious pattern is a statistical coincidence that appears in one dataset but does not reflect genuine structure. This distinction matters critically for AI because machine learning systems learn patterns indiscriminately from training data. If they learn spurious patterns, they will make systematically wrong predictions when applied to new situations.

*Key terms: Spurious pattern, useful pattern, generalisation, correlation, causation, training data*

**Q8. What is the ladder of abstraction, and why is it necessary for intelligent decision-making?**

**Model Answer:**

The ladder of abstraction describes the process of moving from raw sensory or measurement data through pattern recognition and rule formation to actionable decisions. It is necessary because raw data is too voluminous, too noisy, and too context-specific to be directly acted upon. The abstraction process filters and compresses information into forms that can guide behaviour efficiently. Without this ladder, an intelligence would be overwhelmed by the sheer volume of unprocessed sensory input and unable to respond in a timely and appropriate way.

*Key terms: Abstraction, raw data, pattern, rule, decision, compression, generalisation*

**Q9. How does the concept of feedback distinguish true intelligence from mere reactive behaviour?**

**Model Answer:**

A merely reactive system responds to patterns but does not update its internal model based on the outcomes of its responses. True intelligence, by contrast, includes a feedback mechanism whereby the results of actions are observed, compared to expectations, and used to refine future pattern recognition. This is what allows intelligence to improve over time and to adapt to changing environments. A thermostat responds to temperature but does not learn from its responses. A doctor, by contrast, continuously updates their clinical knowledge based on patient outcomes.

*Key terms: Feedback, learning, adaptation, update, model refinement, reactive*

**Q10. Why do the chapter authors argue that intelligence is not a uniquely human property? What implications does this have for our understanding of artificial intelligence?**

**Model Answer:**

The chapter argues that intelligence, defined as the capacity to recognise patterns and use them to guide adaptive behaviour, is found across many forms of life, from bacteria responding to chemical gradients to crows using traffic to crack nuts. What distinguishes different forms of intelligence is not the fundamental process but the richness and flexibility of the patterns involved. This understanding implies that artificial intelligence is not attempting to replicate something mystical or uniquely human, but rather to implement a process that is already widespread in nature, using computational rather than biological machinery.

*Key terms: Adaptive behaviour, biological intelligence, pattern recognition, universality, implementation*

## **Section C — Reflective Questions (3 Questions)**

**Q11. Think of a skill you have learned through practice, whether it is cooking, playing a sport, reading maps, or any other activity. Describe how the Pattern-Decision-Action loop operated in your learning of that skill.**

**Thinking prompts:**

- What observations were you making when you were a beginner?
- What patterns did you gradually learn to recognise?
- How did your actions change as your pattern recognition improved?
- Can you identify a moment when feedback from a failed action changed your pattern?
- How is your current skilled performance different from your early reactive responses?

*Rubric hint: Strong responses will connect all three stages of the loop explicitly and will include at least one example of how feedback changed a pattern. Responses that only describe the skill without analysing its cognitive structure will score lower.*

**Q12. The chapter describes how a hiring algorithm can learn prejudice from biased training data. Do you think humans who make hiring decisions are subject to the same problem? Is there a meaningful difference between a human prejudice and an algorithmic one?**

**Thinking prompts:**

- What kinds of patterns might a human recruiter learn from experience that could be biased?
- In what ways are humans and algorithms similar as pattern-learning systems?
- Are there ways in which human bias and algorithmic bias differ in their effects?
- What does scale mean in this context, and why does it matter?
- What would a fair pattern-learning system look like?

*Rubric hint: Strong responses will engage with both the similarity and the difference between human and algorithmic bias. Responses that simply say one is worse than the other without analysis will score lower. Credit for responses that introduce the concept of scale as a distinguishing factor.*

**Q13. The chapter claims that AI finds patterns faster and at larger scale than any human being, but also that it can find the wrong patterns just as readily as the right ones. Does this make AI more useful or more dangerous than human intelligence? Or both?**

**Thinking prompts:**

- In what domains might the speed and scale of AI pattern-finding be clearly beneficial?
- In what domains might the same speed and scale be clearly dangerous?
- How might the concept of the spurious pattern help answer this question?
- What role does human oversight play in making AI useful rather than dangerous?
- Can you think of a current example from India where AI pattern recognition is being used, and evaluate it using these criteria?

*Rubric hint: Strong responses will resist the binary and argue for a contextual answer. Credit for identifying specific Indian applications and for connecting the speed-and-scale advantage to both the benefit and the risk.*

## **Section D — Stream Project Prompts (2 Questions)**

**Q14. [Arts / Science] Choose any one traditional knowledge practice from your region, whether in agriculture, weather prediction, medicine, or craft, and analyse it using the Pattern-Decision-Action framework. Document at least three specific patterns used in this practice, the decisions they inform, and the actions they guide. Then compare this traditional pattern-recognition system to a modern AI application in the same domain. What has the modern system gained? What, if anything, has it lost?**

This project requires both fieldwork (interviewing a practitioner) and research (finding an equivalent AI application). The comparison should be analytical and specific. Students are encouraged to document the traditional knowledge in the local language as well as English.

**Q15. [Commerce] Select any one digital financial service widely used in your area, such as UPI, a mobile banking application, or a microfinance platform, and investigate what patterns it uses to make decisions. Interview at least two users about their experience. Then analyse one decision the system makes (approving or rejecting a transaction, assigning a credit score, recommending a financial product) using the ladder of abstraction. What raw data feeds into the decision? What patterns are extracted? What rule is applied? Present your findings as a short report with a diagram.**

This project combines primary research (user interviews) with conceptual analysis (ladder of abstraction). The diagram should be original, drawn or created by the student, not copied from any source.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                          |
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| <b>Core</b> | Hofstadter, D. R. (1979). Gödel, Escher, Bach: An Eternal Golden Braid. Basic Books. [Chapters 1 and 2 for the foundational discussion of pattern, meaning, and intelligence as emergent properties of structure.]                       |
| <b>Core</b> | Mitchell, M. (2019). Artificial Intelligence: A Guide for Thinking Humans. Farrar, Straus and Giroux. [Chapter 1 for an accessible, non-technical introduction to what AI is and what it does, written for a general educated audience.] |
| <b>Core</b> | Srinivasan, U., & Umamaheswaran, S. (2021). State of AI in India: Opportunities and Challenges. NITI Aayog Discussion Paper. [For an Indian policy and application context relevant to the stream fork examples.]                        |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson. [Chapter 1 for a more formal treatment of the definition of intelligence and the historical development of AI.]                                                                                                          |
| <b>Extended</b> | Kahneman, D. (2011). Thinking, Fast and Slow. Farrar, Straus and Giroux. [Parts I and II for the psychological foundation of pattern recognition, heuristics, and the distinction between System 1 intuitive pattern recognition and System 2 deliberate reasoning.]                                                      |
| <b>Extended</b> | Ganguli, K., & Sharma, P. (2022). Traditional Ecological Knowledge and Machine Learning: Convergences and Tensions. Economic and Political Weekly, 57(18), 34-41. [For the comparison between traditional pattern knowledge and AI-based pattern detection, directly relevant to the Arts/Science stream project prompt.] |

### Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Google AI Education. (2023). Machine Learning Crash Course: Introduction to ML. <a href="https://developers.google.com/machine-learning/crash-course">developers.google.com/machine-learning/crash-course</a> . [Free, accessible introduction to pattern recognition and machine learning fundamentals, no programming background required.] |
| <b>Online</b> | NITI Aayog. (2022). Responsible AI for All: Adopting the Framework. <a href="https://niti.gov.in">niti.gov.in</a> . [India's national AI strategy document, freely available, relevant to the What Could Go Wrong? section and Unit 4 themes.]                                                                                                |
| <b>Online</b> | 3Blue1Brown. (2017). But What Is a Neural Network? [Video]. <a href="https://youtube.com/3blue1brown">youtube.com/3blue1brown</a> . [A visually clear explanation of how pattern recognition scales from simple to deep networks, directly relevant to the ladder of abstraction concept.]                                                    |

### *End of Chapter 1*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 2

# When Machines Learn to See Patterns

*The previous chapter established that all intelligence begins with pattern recognition. This chapter asks a harder question: how did human beings manage to make machines do it? The answer is not obvious, and the path to it is not straight. It winds through decades of failed experiments, unexpected insights, and a fundamental change in how programmers thought about what a computer was for.*

*The story matters. Not because it is interesting history, though it is, but because understanding how AI went from rule-following to pattern-learning is the best way to understand what current AI systems actually do and why they sometimes fail.*

## Learning Objectives

By the end of this chapter, you will be able to:

- Distinguish between rule-based and learning-based AI systems, with examples of each
- Trace the key moments in AI history through the lens of Indian technological development
- Explain how an expert system works, using a crop advisory application as the model
- Describe the recommendation engine loop and identify it in services used daily in India
- Articulate why learning-based systems eventually outperformed rule-based ones for complex tasks

## 2.1 Two Ways to Build an Intelligent Machine

Imagine you are trying to teach a computer to diagnose whether a mango is ripe. You have two approaches available.

The first approach is to write down rules. You consult an expert, perhaps a fruit vendor who has been selecting mangoes for thirty years, and you ask her to articulate what she knows. She tells you: a ripe Alphonso mango should be yellow with a slight blush at the top, should give slightly under gentle thumb pressure, should smell distinctly sweet at the stem end, and should feel heavy for its size. You translate these into code. Your program checks each condition in sequence and outputs a verdict.

This is a rule-based system. It works by encoding expert knowledge as explicit conditional logic. If condition A is true and condition B is true and condition C is true, then conclude D.

The intelligence is entirely in the rules, which were written by a human being and then executed by a machine.

The second approach is to show the computer examples. You bring it ten thousand mangoes, each labelled ripe or unripe by experts. The computer does not receive any rules. Instead, it examines the examples, finds the patterns that distinguish the two categories, and builds its own internal model of the difference. When you show it a new mango, it compares it against that internal model and makes a prediction.

This is a learning-based system. The intelligence is not in explicit rules written by a human but in patterns extracted automatically from data. The machine has learned, in a limited but real sense, rather than been programmed.

**Fig. 2.1**  
**Rule-Based vs. Learning-Based Systems**

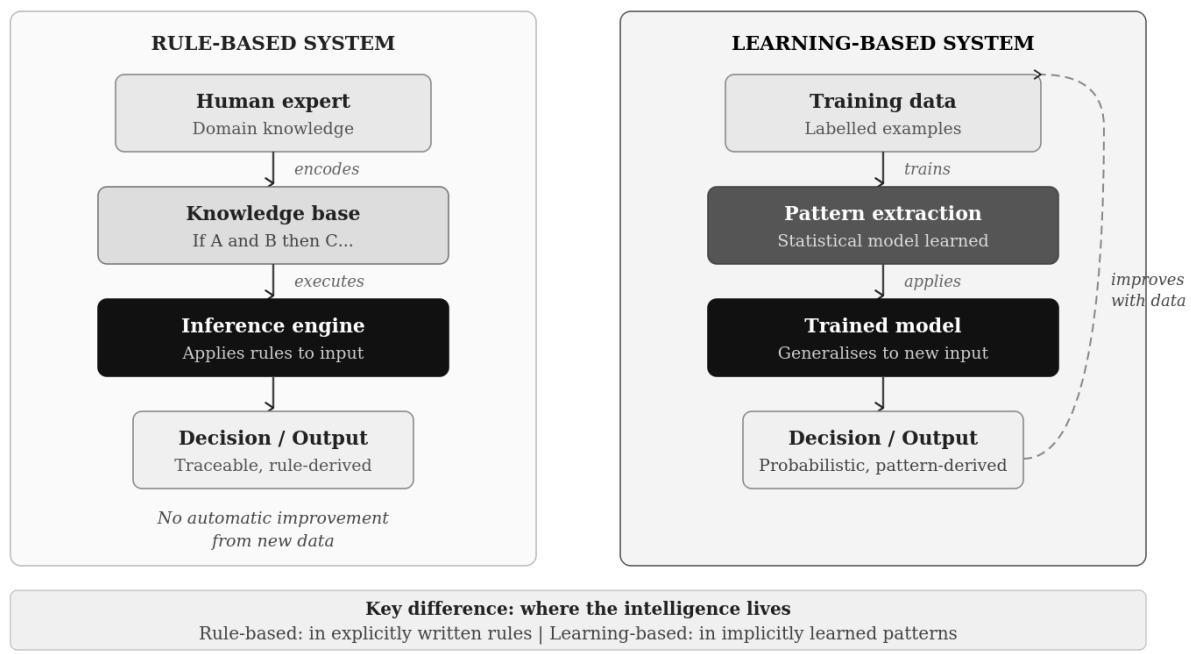


Figure 2.1 — Rule Vs Learning

Figure 2.1 shows the structural difference between these two approaches. In a rule-based system, human expertise flows in at the design stage, encoded as rules, and the system applies those rules rigidly to each new input. In a learning-based system, data flows in at the training stage, patterns are extracted automatically, and the resulting model is applied to new inputs. The key difference is where the intelligence lives: in explicitly written rules, or in implicitly learned patterns.

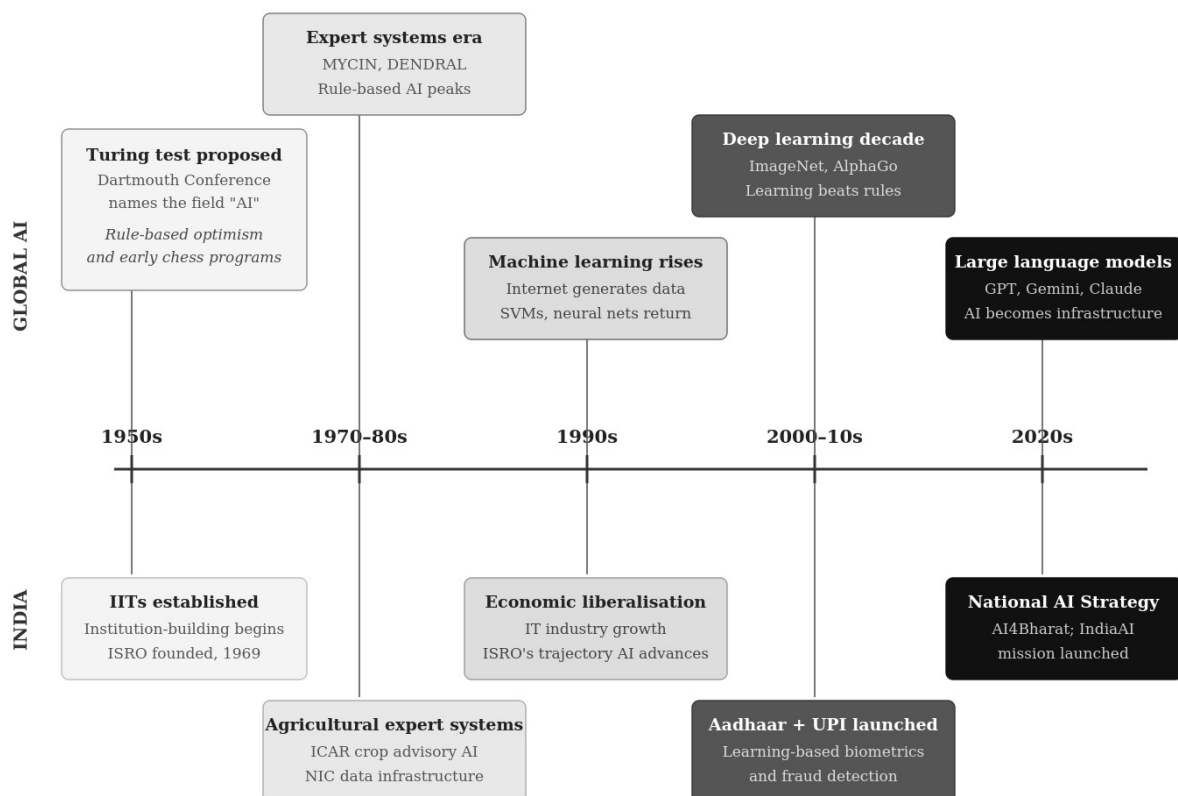
For most of AI's history, the first approach dominated. And for good reason: it was the only one that worked reliably. But rule-based systems had a fundamental limitation that became increasingly apparent as people tried to apply them to more complex problems.

That limitation is what eventually drove the shift to learning-based systems, and understanding it is essential to understanding why modern AI looks the way it does.

## 2.2 A Brief History of AI, Told Through Indian Eyes

The standard history of artificial intelligence is told from the perspective of American and British universities. ENIAC and Turing, Dartmouth and MIT, Stanford and Carnegie Mellon. This history is accurate but incomplete. It misses the parallel story of how the ideas developed in those universities arrived in India, were adapted to Indian problems, and eventually fed back into the global development of the field. Telling the history through Indian examples is not a concession to local pride. It is a correction that makes the history more complete and more useful to Indian students.

**Fig. 2.2**  
**AI History and Indian Technological Milestones**



*Global AI development and Indian technological growth have been parallel since the 1950s and increasingly intertwined since the 1990s. Understanding this parallel history contextualises how AI is deployed in India today.*

**Figure 2.2 — AI History and Indian Technological Milestones**

The 1950s and 1960s produced the foundational ideas. Alan Turing asked whether machines could think. The Dartmouth Conference in 1956 coined the term artificial intelligence and set an agenda of optimism that would take decades to fulfil and repeatedly disappoint. Early AI programs could play chess, prove theorems, and solve algebra problems. They worked by exhaustive search through possibility spaces, rules all the way

down. In India, this was the decade of the IITs, established from 1951 onwards, and of the beginning of serious scientific and technological institution-building. The foundations were being laid in both places simultaneously, though the connection between them would come later.

The 1970s and 1980s brought expert systems to prominence. An expert system encodes the knowledge of a human specialist as a set of if-then rules, managed by an inference engine that applies those rules to new problems. The most famous early examples were medical systems developed in the United States. But India was quick to see the agricultural potential. By the late 1980s, Indian agricultural universities and research stations were experimenting with expert systems for crop disease diagnosis, pest management, and irrigation scheduling. These were rule-based systems built on the knowledge of experienced agronomists, and they represented a serious attempt to scale expert knowledge to farmers who had no access to an agronomist in person.

The 1990s brought two developments that would transform the field. The first was the internet, which began to generate data at a scale that had never existed before. The second was a quiet revolution in machine learning, driven by researchers who had been working for years on learning-based approaches that had been repeatedly dismissed as impractical. In India, this was also the decade of economic liberalisation, the growth of the software industry, and the beginning of what would become the world's largest pool of software engineering talent. ISRO was refining its trajectory prediction algorithms. The National Informatics Centre was building data infrastructure for government. The connection between Indian technical capacity and global AI development was tightening.

The 2000s saw machine learning displace rule-based systems in domain after domain. Search engines, spam filters, recommendation systems, fraud detection: all of these shifted from explicit rules to learned models. In India, the mobile revolution was beginning, and with it the generation of data at a scale that would eventually make India one of the world's most important AI markets. The Aadhaar programme, the world's largest biometric identification system, was conceived in this decade. Its face recognition and fingerprint matching systems are learning-based, not rule-based.

The 2010s brought deep learning, and with it a qualitative shift in what AI could do. Image recognition, natural language processing, speech synthesis, game playing: all of these reached or exceeded human-level performance in specific benchmarks during this decade. In India, UPI was launched in 2016 and grew to process billions of transactions. The fraud detection systems that protect it use deep learning models trained on vast transaction datasets. By the end of the decade, AI had ceased to be a research topic and had become infrastructure.

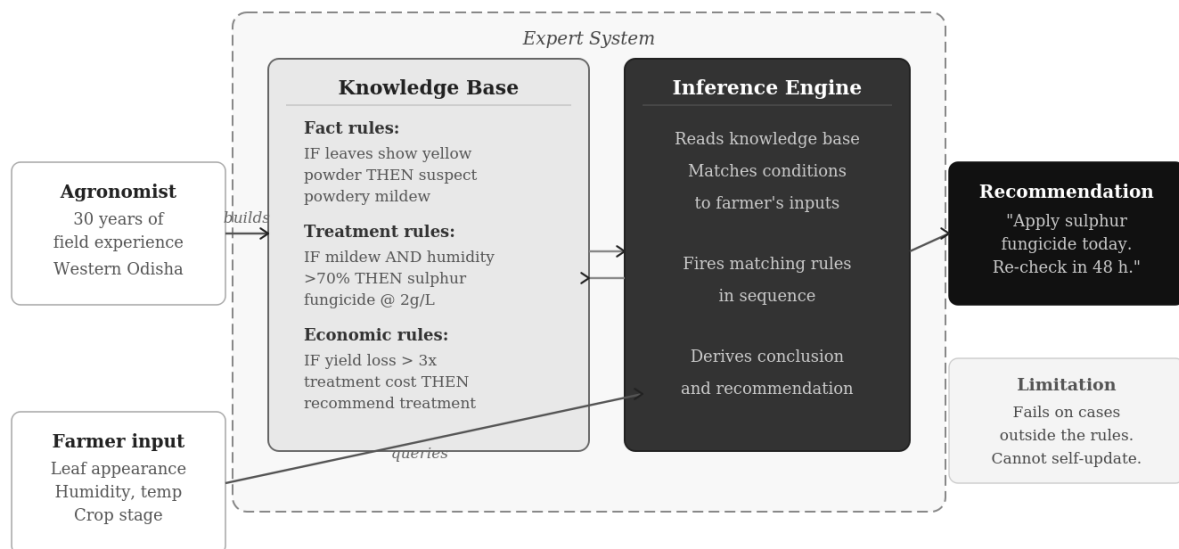
The 2020s are the decade in which this book is being written. Large language models can converse, write, translate, and reason at a level that was considered impossible five years ago. India has launched a National AI Strategy and is producing AI researchers and engineers at scale. The question is no longer whether AI will be part of Indian life. It

already is. The question is whether Indian citizens, including the students reading this book, will engage with it critically and shape it, or simply receive it passively.

## 2.3 The Expert System: Intelligence in a Box

To understand the shift from rule-based to learning-based AI, it helps to understand rule-based AI at its most sophisticated: the expert system. An expert system has two components. The knowledge base is a structured collection of facts and rules about a specific domain, built by interviewing human experts and encoding their knowledge. The inference engine is the computational mechanism that applies those rules to a new problem and derives conclusions.

**Fig. 2.3**  
**Expert System Anatomy: Crop Advisory for Western Odisha**



*An expert system for cotton disease advisory in Western Odisha. The knowledge base encodes rules from experienced agronomists; the inference engine applies them to farmer inputs. Transparent and reliable within its rule set, but brittle beyond it.*

*Figure 2.3 — Expert System Anatomy*

Consider a crop advisory expert system designed for cotton farmers in Western Odisha. The knowledge base contains rules derived from agronomists who have worked in the region for decades. Some rules concern pest identification: if the leaves show circular yellow spots that are powdery to the touch, and if the underside of the leaf shows a white coating, then the crop is affected by powdery mildew. Other rules concern treatment: if the diagnosis is powdery mildew and the humidity has been above seventy percent for three consecutive days, then apply a sulphur-based fungicide at a dosage of two grams per litre of water. Still other rules concern timing and economics: if the expected yield loss from the infection exceeds the cost of treatment by a factor of three or more, then treatment is recommended; otherwise it is optional.

The inference engine applies these rules in sequence. A farmer inputs observations: leaf appearance, humidity, temperature, crop stage. The engine works through the rules, firing each one whose conditions are satisfied, until it reaches a conclusion and a

recommendation. The farmer receives specific, expert-quality advice without having direct access to an expert.

This is a genuinely powerful idea, and it worked well within its limits. Expert systems drove real value in agriculture, medicine, financial planning, and industrial diagnosis throughout the 1980s and 1990s. But the limits were real. Building a knowledge base required intensive effort from both domain experts and knowledge engineers who could translate expert knowledge into formal rules. Maintaining the knowledge base as understanding of the domain evolved was expensive. And most critically, expert systems were brittle: they performed well within the situations their rules covered but failed completely outside those situations. They could not handle the novel, the ambiguous, or the case that fell between the cracks of the rules.

The real world, as every farmer knows, does not stay within the cracks of any rule system.

## **2.4 The Recommendation Engine: Intelligence from Behaviour**

While expert systems were being built and deployed in professional domains, a different kind of pattern-learning system was quietly becoming one of the most influential AI applications in the world. The recommendation engine does not encode expert knowledge. It learns from behaviour.

The principle is straightforward. A music streaming service observes what millions of users listen to, skip, replay, and add to playlists. It finds patterns: users who like one thing also tend to like certain other things. A user who has not yet heard a particular song is shown it, because users with similar listening patterns tended to like it. The recommendation is not based on any expert opinion about the quality of the music. It is based entirely on pattern similarity in observed behaviour.

**Fig. 2.4**  
**The Recommendation Engine Loop**

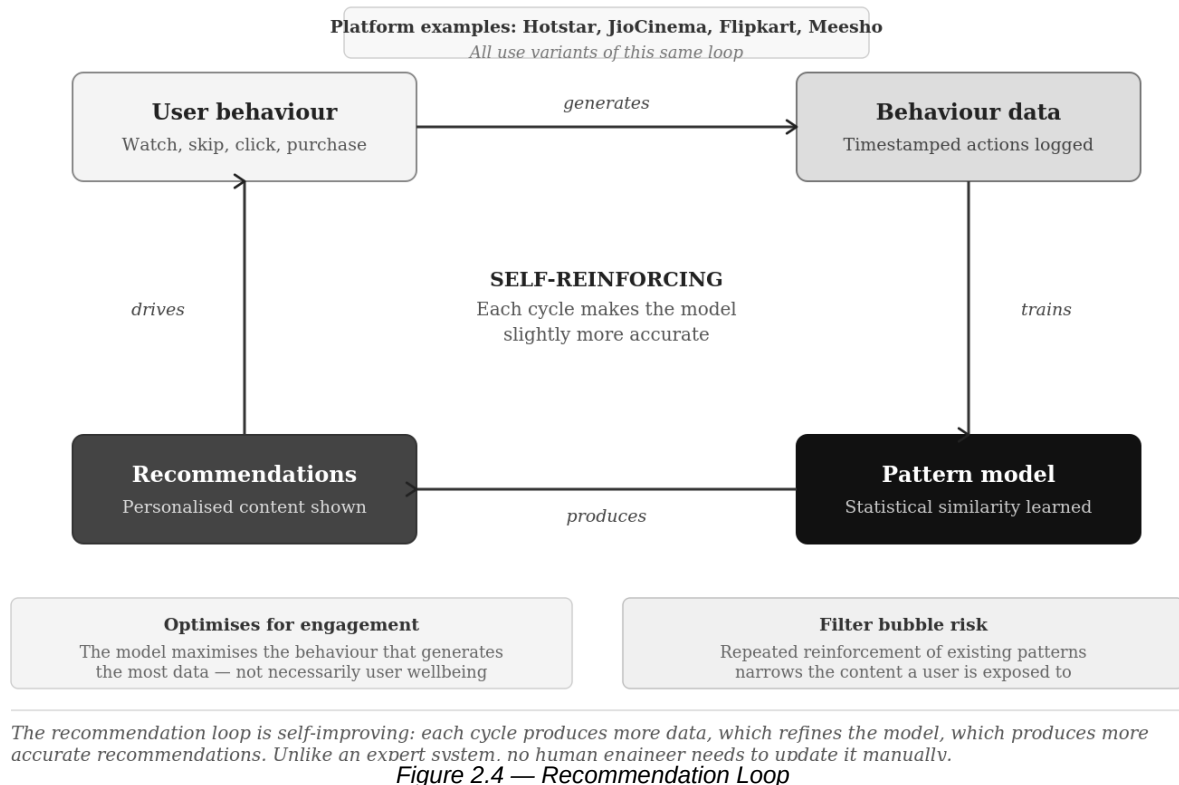


Figure 2.4 shows the recommendation loop. The key feature to notice is that the loop is self-reinforcing. A recommendation leads to behaviour, which becomes data, which refines the model, which generates better recommendations. This is a learning system in a stronger sense than an expert system: it improves continuously as it accumulates more data. An expert system does not improve unless a human engineer updates its rules. A recommendation engine improves automatically, every time a user takes an action.

In India, this loop is everywhere. Consider what happens when a user opens Hotstar or JioCinema. The platform knows what the user has watched before, how long they watched each programme, what they paused, what they abandoned, and what they watched twice. It uses this data to predict what the user will want to watch next. The recommendations shown are not chosen by an editor or a critic. They are generated by a model trained on the behaviour of millions of viewers, finding the patterns that connect this user's history to content they have not yet seen.

The same principle operates in e-commerce. When a user browses Flipkart or Meesho, the platform is continuously learning: which items were looked at and not bought, which were bought immediately, which were compared against each other. Every search query, every click, every abandoned cart is a data point that feeds the pattern-learning system and makes the next recommendation slightly more accurate.

Recommendation engines are consequential far beyond their commercial applications. The same loop that recommends films and products also operates in social media feeds, news aggregators, and educational platforms. In each case, the system is optimising for engagement, for the behaviour that generates the most data and the most revenue. Understanding that this loop exists, and that it is shaping what information reaches you, is a form of literacy that this book considers essential. The recommendation engine is not neutral. It is a pattern-learning system trained on behaviour, and it will find whatever patterns in your behaviour it can exploit to keep you engaged, whether those patterns are good for you or not.

## 2.5 Why Learning Beat Rules

The shift from rule-based to learning-based AI did not happen because researchers decided that learning was philosophically more interesting than rules. It happened because learning-based systems turned out to be dramatically better at specific, practically important tasks. Understanding why illuminates something deep about the nature of intelligence itself.

Consider the task of filtering spam email. In the early days of email, spam filters were rule-based. A filter might block any message containing the words free money, or any message from a domain that had been manually flagged. Spammers responded by modifying their language, using fr33 m0ney, or rotating through new domains faster than filters could be updated. The rule-based filter was in a permanent arms race with the spammers, and it was losing, because rules can only catch what they anticipate and spammers are inventive.

A learning-based spam filter takes a different approach. It examines millions of emails that users have labelled as spam or not spam, and it learns the statistical patterns that distinguish them. These patterns are complex, multidimensional, and not easily expressible as simple rules. The structure of the email headers, the ratio of images to text, the presence of certain combinations of words, the behaviour of the sending domain over time: a learning-based filter can integrate thousands of such signals simultaneously. More importantly, as spammers adapt their tactics, user labelling behaviour updates the model. The filter learns continuously from the evolving spam landscape.

This is the fundamental advantage of learning over rules: the capacity to find patterns that are too complex, too numerous, and too rapidly changing for any human expert to anticipate and encode. For tasks where the relevant patterns are simple and stable, rule-based systems can be excellent. For tasks where the patterns are complex, high-dimensional, and changing over time, learning-based systems have no realistic alternative.

Language is perhaps the clearest example. The rules of natural language, the full grammar, pragmatics, and contextual understanding that would be needed to process it reliably, are so complex that no rule-based system has ever successfully captured them.

Machine translation systems that were rule-based for decades were replaced overnight by learning-based systems that were trained on millions of translated documents and learned the patterns of equivalence between languages from examples rather than from grammar books.

In India, this shift had particular implications. India has twenty-two scheduled languages and hundreds of dialects. Building rule-based systems for even a fraction of this linguistic diversity was practically impossible. Learning-based systems, trained on the growing digital corpus of Indian language content, have made possible language tools, translation services, and voice interfaces that no rule-based approach could have achieved. The learning paradigm was not just technically superior in general. It was the only paradigm that could scale to India's linguistic reality.

## **2.6 Patterns in My Field: Stream Perspectives**

### **Arts Stream: How AI Learned to Write in the Style of Premchand**

Munshi Premchand, the great Hindi and Urdu author of the early twentieth century, wrote in a distinctive style: direct, compassionate, focused on the lives of ordinary rural people, with a particular gift for dialogue that captured the speech of specific communities without caricature. Could a rule-based system capture his style? In principle, a literary scholar could try to encode rules: use short sentences, prefer Hindi words over Urdu-derived ones in this context but not that, include specific social class markers. In practice, the rules would be impossibly numerous and subtly interdependent. A learning-based system, trained on a large corpus of Premchand's writing, can learn the statistical patterns of his style without any of those rules being explicitly articulated. It does not understand what it is doing, but it produces output that shares measurable stylistic properties with his writing. This raises profound questions about authorship, creativity, and the nature of style, questions that the Arts stream project in Chapter 15 will invite you to explore.

### **Science Stream: ISRO's Trajectory Prediction**

When ISRO launches a spacecraft, the mission involves solving an extraordinarily complex trajectory optimisation problem. The spacecraft must navigate gravitational fields from multiple bodies, account for atmospheric drag during launch and orbital insertion, and manage fuel consumption across a mission that may last years. Early ISRO trajectory calculations were done by hand, and then by simple rule-based computational methods. Modern trajectory planning uses learning-based methods alongside physics-based models, with machine learning components trained on historical mission data to predict anomalies and optimise fuel use. The Chandrayaan and Mangalyaan missions, and the more recent Aditya-L1 solar mission, all used AI-assisted trajectory planning. The pattern-learning systems involved are not replacing the physics but augmenting the physicists, finding patterns in historical mission data that improve future planning.

## Commerce Stream: How Aadhaar's Fraud Detection Works

The Aadhaar system currently holds biometric data for over 1.3 billion Indian residents. Authentication attempts, both legitimate and fraudulent, generate an enormous volume of transaction data. Fraud detection in a system of this scale is impossible with rule-based approaches alone: fraudsters adapt, rules become stale, and the sheer volume of transactions exceeds human review capacity. The fraud detection layer uses learning-based models trained on historical authentication data, looking for patterns that distinguish legitimate authentication attempts from suspicious ones. Unusual time of day, unusual location, unusual frequency, mismatches between claimed identity and biometric confidence score: a learning system integrates these signals and flags anomalies without any single rule explicitly encoding each combination. The system improves continuously as new fraud patterns emerge and are detected.

## 2.7 AI Problem Solving and Search Techniques

Before machine learning became the dominant paradigm, and even alongside it today, a large part of what “intelligent” software does is best described not as learning but as searching. Consider a very different kind of problem from anything discussed so far in this chapter: not “is this ripe or unripe” but “what is the best route from here to there.”

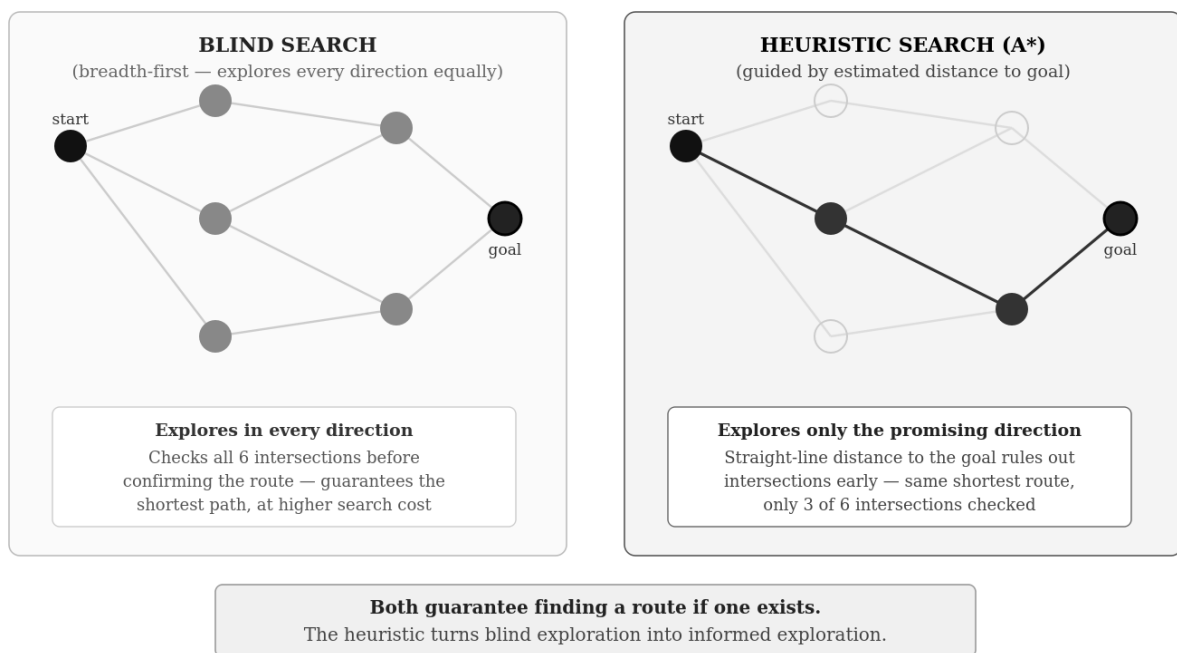
When a ride-hailing app in Bengaluru calculates the route from a passenger's pickup point to their destination, it is not consulting a learned model of typical journeys. It is searching. The city's road network can be represented as a graph: intersections are nodes, roads connecting them are edges, and each edge has a cost, typically expected travel time given current traffic. The app's task is to find the sequence of edges from the start node to the destination node that minimises total cost. This is a search problem, and it has been studied since the earliest days of artificial intelligence, well before anyone thought to make machines learn from data.

A search problem has a precise structure: a starting state, a set of possible actions that move from one state to another, a goal test that recognises when the destination has been reached, and, usually, a cost associated with each action. Solving the problem means finding a sequence of actions, a path through the space of possible states, that reaches a goal state at the lowest possible cost. This state-space framing turns out to describe an enormous range of problems that have nothing to do with roads: rearranging tiles in a sliding puzzle, planning a delivery route across a city, scheduling a set of exams so that no student has two exams at once, or finding a sequence of chess moves that leads to checkmate.

The simplest search strategies explore blindly. Breadth-first search examines every state one step away from the start, then every state two steps away, and so on, guaranteeing it finds the shortest path but at real computational cost as the space grows. Depth-first search plunges down one path as far as it can before backing up, using far less memory but with no guarantee of finding the shortest route. Both strategies share a limitation: they know nothing about which direction is more promising. They search the way a person would search a dark room, feeling along every wall in turn.

Heuristic search adds a crucial ingredient: an estimate of how far a given state is from the goal, even before that distance is known exactly. In route planning, the straight-line distance to the destination is a natural heuristic: it can be computed instantly from map coordinates and is always an honest underestimate of the true road distance, since roads rarely travel in a perfectly straight line. The A\* algorithm, one of the most widely used search methods in practical software, combines the actual cost travelled so far with this heuristic estimate of the cost remaining, and always expands the most promising state first. The effect is that A\* behaves like a person navigating with a rough sense of direction rather than one wandering blindly: it still checks its surroundings, but it checks the surroundings that are more likely to lead somewhere useful first.

**Fig. 2.5**  
**Blind Search vs. Heuristic Search**



*This is exactly what ride-hailing and food-delivery apps run continuously over Indian city road networks.*

*Figure 2.5 — Blind Search vs. Heuristic Search*

The Indian Railways route and fare enquiry systems, and every major food delivery and ride-hailing platform operating in Indian cities, rely on variants of this kind of heuristic search running continuously, millions of times a day, over road and rail networks that change their costs by the minute as traffic and demand shift. None of this requires the system to have seen a single labelled example of a “good route.” The intelligence is not learned from data; it is the product of a well-specified problem and an efficient way of exploring the space of possible solutions.

This distinguishes search from the learning-based systems discussed elsewhere in this chapter, but the two are not rivals. Modern AI systems routinely combine them: a learned model might predict how long a given road segment will actually take at six in the evening on a Friday, based on patterns in historical traffic data, and that learned prediction becomes the edge cost that a classical search algorithm then uses to find the best route. Rule-based systems, learning-based systems, and search-based systems are three different tools in the same toolbox, and knowing which one a given AI application is

actually using, rather than assuming every “smart” system must be a learning system, is part of the AI literacy this book is trying to build.

## — WHAT COULD GO WRONG? —

### The Arms Race and the Black Box

The shift from rule-based to learning-based AI introduced two new categories of problem that rule-based systems largely avoided.

The first is the adversarial arms race. Rule-based systems are transparent: their logic can be inspected, challenged, and gamed, but the gaming is bounded by the rules themselves. Learning-based systems, because they find complex statistical patterns rather than following explicit logic, can be gamed in more subtle and powerful ways. Adversarial examples in machine vision, carefully constructed images that fool a classifier while appearing completely normal to a human eye, are one manifestation. Adversarial manipulation of recommendation engines, where content creators learn to game the patterns that drive engagement, is another. The spam example from Section 2.5 cuts both ways: learning beats rules, but learners can also be fooled by learners on the other side.

The second problem is opacity. When an expert system reaches a conclusion, its reasoning is traceable: the inference engine fired specific rules in a specific sequence, and that sequence can be examined. When a deep learning model reaches a conclusion, its internal process is a chain of matrix multiplications across millions of parameters, and there is no simple way to say why it reached the conclusion it did. This matters enormously when the decision affects a person's life: a loan denial, a medical diagnosis, a hiring decision. The person affected has a reasonable claim to an explanation, and the learning-based system may be unable to provide one. In India, where AI is being deployed in government services affecting hundreds of millions of people, this opacity is not a technical curiosity. It is a governance challenge of the first order.

## — CHAPTER SUMMARY —

### Chapter 2 — Key Points

1. Rule-based AI encodes expert knowledge as explicit conditional logic; learning-based AI extracts patterns automatically from data.
2. Expert systems achieved real practical value in the 1980s and 1990s but were brittle: they failed outside the situations their rules anticipated.
3. AI history and Indian technological development have been parallel and increasingly intertwined since the 1950s.
4. The recommendation engine learns from behaviour rather than encoding expert knowledge, and improves continuously as data accumulates.
5. Learning-based systems outperformed rule-based systems when patterns were

complex, high-dimensional, and rapidly changing.

6. India's linguistic diversity made the shift to learning-based NLP especially consequential — rule-based approaches could not scale.

7. Learning-based systems introduced two new problems: adversarial vulnerability and opacity of reasoning.

8. Understanding the recommendation loop — and its non-neutrality — is a form of civic and digital literacy.

## — MODEL QUESTIONS —

### Section A — MCQ (5 Questions)

**Q1. In a rule-based AI system, where does the intelligence primarily reside?**

- A) In the data fed to the system at runtime
- B) In the explicit rules written by human experts during design
- C) In the statistical patterns extracted from training examples
- D) In the feedback loop that improves the system over time

**Answer: B**

*Explanation: In a rule-based system, the intelligence is encoded as explicit conditional logic by human experts during the design stage. The system executes these rules but does not learn or modify them from data.*

**Q2. Which of the following best explains why expert systems were described as 'brittle'?**

- A) They required expensive hardware to run
- B) They could only process numerical data, not text or images
- C) They failed completely when faced with situations their rules did not anticipate
- D) They could not connect to the internet for real-time updates

**Answer: C**

*Explanation: Expert systems were brittle because their performance was entirely bounded by their rule set. Any situation falling outside those rules produced either no output or incorrect output, with no mechanism for graceful degradation or learning.*

**Q3. A recommendation engine on a video streaming platform improves over time primarily because:**

- A) Engineers manually update its content catalogue weekly
- B) It learns from the continuous stream of user behaviour data
- C) It uses rules written by film critics and content experts
- D) It downloads updated models from a central AI server each night

**Answer: B**

*Explanation: The recommendation engine's self-improvement comes from the continuous feedback loop of user behaviour — watching, skipping, replaying — which updates the model's understanding of patterns without manual intervention.*

**Q4. The shift from rule-based to learning-based AI in natural language processing was particularly significant for India because:**

- A) India has a larger internet population than any other country
- B) Indian languages use different scripts that rule-based systems could not handle
- C) India has twenty-two scheduled languages, making rule-based coverage impractical
- D) Indian engineers were the primary developers of early learning-based NLP systems

**Answer: C**

*Explanation: India's extraordinary linguistic diversity meant that rule-based NLP systems, which require intensive manual construction per language, could never achieve broad coverage. Learning-based systems, trained on digital corpora, offered the only practical path to multilingual scale.*

**Q5. Which of the following is an example of the 'opacity problem' associated with learning-based AI systems?**

- A) A spam filter that can be tricked by replacing letters with numbers
- B) A loan approval model that cannot explain why a specific application was rejected
- C) A recommendation engine that recommends content the user has already seen
- D) An expert system that gives incorrect advice due to an outdated rule

**Answer: B**

*Explanation: The opacity problem refers to the inability of deep learning models to provide traceable, human-interpretable explanations for their decisions. A loan denial without explanation is a paradigmatic example with real ethical and legal implications.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Explain the difference between a rule-based and a learning-based AI system using the example of mango ripeness detection.**

**Model Answer:**

A rule-based mango ripeness detector encodes expert criteria as explicit conditional statements: if the colour is yellow with a certain blush, and the texture yields under specific pressure, and the aroma at the stem is sweet, then the mango is ripe. A learning-based system instead examines thousands of labelled examples of ripe and unripe mangoes and derives its own internal model of the distinguishing patterns. The rule-based system is transparent and predictable but can fail for mangoes that fall outside its rules, such as rare varieties or unusual growing conditions. The learning-based system is more flexible and can generalise from examples but its internal criteria are not directly interpretable.

*Key terms: rule-based, learning-based, explicit rules, training examples, generalisation, brittleness*

**Q7. What were the two major limitations of expert systems that eventually drove the shift toward learning-based AI?**

**Model Answer:**

The first limitation was the cost and difficulty of knowledge acquisition. Building a knowledge base required intensive collaboration between domain experts and knowledge engineers, a process that was slow, expensive, and difficult to scale across many domains or to maintain as knowledge evolved. The second and more fundamental limitation was brittleness: expert systems performed reliably only within the situations their rules explicitly anticipated. When confronted with novel cases, edge cases, or situations that fell between the rules, they could not respond gracefully. Learning-based systems, which generalise from examples rather than executing fixed rules, are inherently more robust to novelty.

*Key terms: knowledge acquisition, knowledge engineering, brittleness, generalisation, novelty, maintenance cost*

**Q8. Describe the recommendation engine loop and explain one way in which it could have unintended consequences for users.**

**Model Answer:**

The recommendation engine loop consists of four stages: a user takes an action such as watching, clicking, or purchasing; this action generates data; the data updates a pattern-learning model that identifies similarities between users and content; the model generates recommendations that influence the user's next action, generating more data and closing the loop. One significant unintended consequence is the filter bubble effect. Because the model recommends content similar to what the user has already engaged with, and because engagement-maximising recommendations tend toward emotionally stimulating content, users may gradually receive an increasingly narrow and extreme set of content that confirms existing beliefs without exposure to alternative perspectives.

*Key terms: recommendation loop, user behaviour, pattern learning, filter bubble, engagement optimisation, feedback*

**Q9. Why was India's linguistic diversity a factor in the transition from rule-based to learning-based approaches in natural language processing?**

**Model Answer:**

Building a rule-based NLP system for a single language requires years of work by computational linguists who must formally encode the grammar, morphology, and pragmatic conventions of that language. India has twenty-two scheduled languages and hundreds of dialects, each with distinct scripts, grammatical structures, and oral traditions. Replicating this effort for even a fraction of India's languages was practically and economically impossible with rule-based methods. Learning-based NLP systems, which can be trained on digital text corpora, require language-specific data rather than language-specific engineering. As Indian language digital content has grown, learning-based systems have been able to cover an increasing range of Indian languages at a fraction of the effort that rule-based coverage would require.

*Key terms: linguistic diversity, scheduled languages, computational linguistics, training corpus, scalability, morphology*

**Q10. What is the 'adversarial arms race' in learning-based AI, and how does the spam filtering example illustrate it?**

**Model Answer:**

The adversarial arms race describes the dynamic in which a learning-based system learns to detect a pattern, and adversarial actors adapt their behaviour to avoid detection, forcing the system to learn the new pattern, which the adversaries then adapt to again. In spam filtering, early rule-based filters blocked specific words and phrases. Spammers responded by substituting characters and rotating domains. When learning-based filters began to recognise statistical patterns in spam more generally, spammers responded by

studying the patterns the filters had learned and generating spam designed to fall outside them. Each adaptation by either side drives adaptation by the other, in a continuous escalating cycle. This dynamic appears wherever AI systems are deployed in adversarial contexts: fraud detection, content moderation, plagiarism detection, and cybersecurity.

*Key terms: adversarial examples, arms race, adaptation, robustness, adversarial context, spam filtering*

## Section C — Reflective Questions (3 Questions)

**Q11. Think about a time when you received a recommendation from a digital platform, whether for a film, a song, a product, or a news article, that you found either surprisingly accurate or surprisingly wrong. Using the recommendation loop described in this chapter, try to explain how the system arrived at that recommendation.**

**Thinking prompts:**

- What data did the platform likely have about your past behaviour?
- What pattern might it have detected in that behaviour?
- What similar users or items might it have found that led to the specific recommendation?
- If the recommendation was wrong, at what point in the loop did the system likely go wrong?
- Did you act on the recommendation? If so, how did your action feed back into the loop?

*Rubric hint: Strong responses will trace the loop explicitly and engage with the specific platform's likely data sources. Credit for responses that identify a specific failure mode rather than simply saying the recommendation was inaccurate.*

**Q12. The chapter argues that the recommendation engine is 'not neutral' because it optimises for engagement rather than for the user's wellbeing. Do you agree that this is a problem? Who, if anyone, should be responsible for addressing it?**

**Thinking prompts:**

- What is the difference between what a user wants in the moment and what is good for them in the long run?
- Is the platform responsible for this distinction, or is that paternalistic?
- What interests does the platform have that may conflict with the user's interests?
- Are there examples from India where recommendation engine effects have had visible social consequences?
- What would a genuinely user-wellbeing-oriented recommendation system look like, and would anyone pay for it?

*Rubric hint: Strong responses will distinguish between the platform's commercial interest and the user's wellbeing interest, and will avoid the binary of either full platform responsibility or full user responsibility. Credit for Indian examples.*

**Q13. This chapter describes the opacity problem: learning-based AI systems often cannot explain their decisions. Do you think this matters? Are there**

### **situations where opacity is acceptable and situations where it is not?**

#### **Thinking prompts:**

- What is the difference between a decision that affects only the individual and one that affects their life opportunities?
- What does the right to an explanation mean in practical terms?
- Are there domains where opacity might be acceptable? What makes those domains different?
- How does the opacity problem interact with existing inequalities in Indian society?
- If you were a policy-maker, what would you require of AI systems used in government decisions affecting citizens?

*Rubric hint: Strong responses will not simply say opacity is always bad, but will contextualise it according to stakes and power imbalance. Credit for responses that connect the opacity problem to existing Indian governance and rights frameworks.*

## **Section D — Stream Project Prompts (2 Questions)**

**Q14. [Arts / Science] Identify one AI-powered tool currently available in an Indian language other than English, such as a translation service, a voice assistant, a keyboard, or a content recommendation system. Investigate how it works, whether it is rule-based or learning-based, what data it was trained on, and how well it performs for speakers of that language. Interview at least one fluent speaker of the language about their experience using the tool. Write a report that evaluates the tool's performance, identifies its limitations, and discusses what would be needed to improve it.**

This project requires both technical investigation and user research. The report should clearly distinguish between what you know from research and what you know from the user interview. Credit for identifying specific limitations that reflect the tool's training data or architecture, not just general complaints about quality.

**Q15. [Commerce] Map the recommendation loop for one digital platform you use regularly, whether for shopping, streaming, social media, or news. Document your own behaviour on the platform for one week, recording what you looked at, what you engaged with, and what you ignored. Analyse how the recommendations you received changed over the week in response to your behaviour. Then assess the commercial and social consequences of the loop you have documented: who benefits, who might be harmed, and what, if anything, should be done about it.**

This project requires disciplined self-observation over one week. The commercial and social analysis should draw on concepts from this chapter, particularly the non-neutrality of engagement-optimising systems. Credit for connecting the observation to broader patterns documented in research on recommendation engine effects.

## Core References

|             |                                                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Nilsson, N. J. (2010). <i>The Quest for Artificial Intelligence: A History of Ideas and Achievements</i> . Cambridge University Press. [Chapters 1–6 for a comprehensive, technically grounded history of AI from its origins through the expert systems era to the learning-based revolution. Freely available online.]          |
| <b>Core</b> | Minsky, M. (1968). <i>Semantic Information Processing</i> . MIT Press. [Classic collection showing the ambitions and methods of early rule-based AI; valuable for understanding what learning-based systems replaced and why.]                                                                                                    |
| <b>Core</b> | Sundararajan, A., & Dey, S. (2019). <i>Artificial Intelligence in India: Hype or Reality?</i> Brookings India Research Paper. [A grounded assessment of AI deployment in India across sectors, including agriculture, healthcare, and financial services, with specific attention to learning-based vs. rule-based distinctions.] |

## Extended References

|                 |                                                                                                                                                                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Pariser, E. (2011). <i>The Filter Bubble: What the Internet Is Hiding from You</i> . Penguin Press. [The foundational text on recommendation engine effects on information exposure; directly relevant to the What Could Go Wrong? section.]                                                                |
| <b>Extended</b> | Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision-making and a right to explanation. <i>AI Magazine</i> , 38(3), 50–57. [For the opacity problem and the right to explanation in AI-driven decisions; relevant to the Indian governance context.]                       |
| <b>Extended</b> | Joshi, P., & Kishore, S. (2021). Natural Language Processing for Indian Languages: Progress and Challenges. <i>Journal of the Indian Institute of Science</i> , 101(3), 847–862. [Directly relevant to the discussion of India's linguistic diversity and the shift from rule-based to learning-based NLP.] |

## Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | AI4Bharat. (2023). Indic NLP Resources. <a href="http://ai4bharat.iitm.ac.in">ai4bharat.iitm.ac.in</a> . [Open-source NLP tools and datasets for Indian languages developed at IIT Madras; directly relevant to the Arts/Science stream project prompt.]                                             |
| <b>Online</b> | UIDAI. (2022). Annual Report 2021–22. <a href="http://uidai.gov.in">uidai.gov.in</a> . [Primary source on the Aadhaar system's scale, authentication volumes, and fraud detection mechanisms; relevant to the Commerce stream section.]                                                              |
| <b>Online</b> | 3Blue1Brown. (2019). But How Do Neural Networks Actually Work? [Video series]. <a href="https://youtube.com/3blue1brown">youtube.com/3blue1brown</a> . [A visually clear, mathematically honest introduction to how learning-based systems differ from rule-based ones in their internal structure.] |

## *End of Chapter 2*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 3

# The Anatomy of an Intelligent System

*The first two chapters established the foundation. Intelligence is pattern recognition. Pattern recognition began with living creatures finding regularities in their environments. Machines learned to do the same thing, first through explicit rules and then through learning from data. This chapter takes the next step: it opens the intelligent system like a watch, looks at its moving parts, and asks how they fit together.*

*The answer is the same regardless of whether the system is a hospital's disease screening tool, a court document classifier, or a music recommendation engine. Beneath the surface variation, every intelligent system has the same five-part structure. Understanding that structure is what allows you to look at any AI application, however unfamiliar, and ask the right questions about it.*

## Learning Objectives

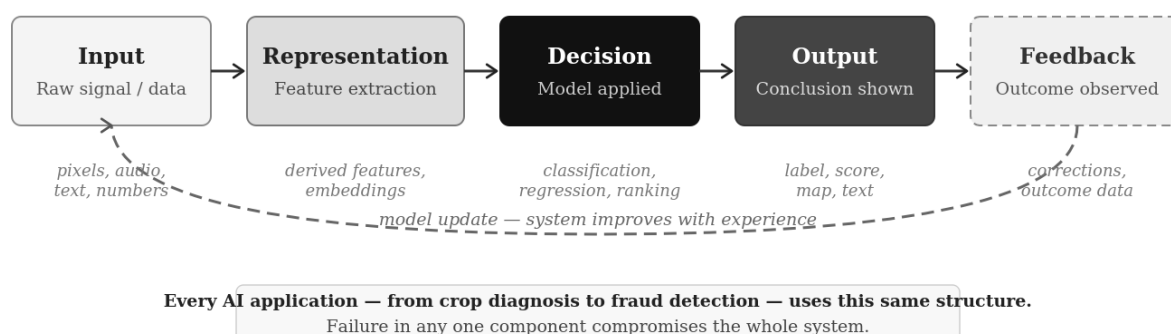
By the end of this chapter, you will be able to:

- Name and describe the five components of the universal AI system architecture
- Explain what a representation is, and why raw data must be transformed before it can be used
- Trace the feedback loop through a complete real-world AI application
- Apply the universal architecture to three distinct Indian AI systems in parallel
- Identify where in the architecture failures are most likely to occur, and why

## 3.1 The Five-Part Architecture

Every intelligent system, from the simplest spam filter to the most sophisticated language model, can be described in terms of five components arranged in a loop. The components are input, representation, decision, output, and feedback. Each component has a specific function, and the failure of any one of them compromises the whole system.

**Fig. 3.1**  
**The Universal AI System Architecture**



*The five-part universal architecture. The dashed feedback arc distinguishes a learning system (which improves with experience) from a static system (which does not). Without feedback, accuracy degrades silently as the world changes.*

*Figure 3.1 — Universal Architecture*

Figure 3.1 shows the loop. Let us examine each component in turn, using examples that will be familiar from daily Indian life before moving to the three extended case studies in Section 3.2.

Input is the raw information that enters the system. For a voice assistant responding to a spoken question, the input is the acoustic signal captured by the microphone: a sequence of pressure measurements sampled thousands of times per second. For a satellite image analysis system monitoring crop health in Madhya Pradesh, the input is a grid of pixels, each carrying spectral values in multiple wavelengths of light. For a fraud detection system monitoring UPI transactions, the input is a structured data record containing the transaction amount, timestamp, device identifier, location, and account history. In every case, the input is the system's only window onto the world. Whatever information is not captured in the input is invisible to the system, and no amount of sophisticated processing can recover it.

Representation is the transformation of raw input into a form that can be processed. This step is less visible than the others but arguably the most consequential. Raw input is almost never directly usable. The acoustic signal of a spoken sentence is a sequence of numbers that carries no obvious information about words or meaning. Satellite pixels are numbers representing reflectance values in specific wavelength bands that carry no obvious information about crop disease. A UPI transaction record is a set of fields that carry no obvious information about fraud risk. Representation transforms these raw signals into structured forms from which patterns can be extracted.

For the voice assistant, the representation step converts the acoustic signal into a sequence of phoneme probabilities, and then into a sequence of word probabilities, and then into a structured meaning representation. Each transformation loses some information but gains in tractability. For the satellite image, the representation step computes indices such as the Normalised Difference Vegetation Index, which combines

red and near-infrared reflectance values into a single number that is strongly correlated with vegetation health. For the fraud detection system, the representation step computes derived features: the ratio of this transaction amount to the account's typical transaction amount, the distance between the current location and the account's usual location, the time elapsed since the last transaction.

Decision is the step at which the system applies its learned model to the representation and produces a conclusion. This is the step that most people think of as the intelligence itself, but it is important to understand that the decision step is only as good as the representation feeding it. A sophisticated decision algorithm applied to a poor representation will produce poor decisions. The decision step for a fraud detection system might be a neural network that takes the derived feature vector as input and outputs a probability score for fraud. The decision step for a satellite crop health system might be a classification model that takes the vegetation index and other derived features and outputs a predicted crop stress level.

Output is the form in which the system's conclusion is communicated. The output of a fraud detection system might be a binary flag that triggers a transaction hold, or a probability score that is displayed to a human reviewer. The output of a satellite crop health system might be a colour-coded map that a state agricultural department can use to direct field visits. The output of a voice assistant is synthesised speech. The form of the output matters: a probability score communicates different information than a binary flag, and an interactive map communicates different information than a text report. Output design is a user experience problem as much as a technical one.

Feedback is the mechanism by which the system's performance is evaluated and its model updated. Without feedback, a system cannot improve. The feedback mechanism for a fraud detection system includes the investigation records of flagged transactions: did the flagged transactions turn out to be fraudulent? Did any unflagged transactions later prove to be fraudulent? These outcomes update the model's parameters, making future detections more accurate. The feedback mechanism for a voice assistant includes user corrections: when a user says no, I said Bhubaneswar, not Bhubani, that correction is a training signal. Feedback is what transforms a static system into a learning one.

### **3.2 Three Indian Systems, One Architecture**

The universal architecture is most clearly understood when you can see it operating across different domains simultaneously. The three applications examined in this section, a crop disease detection app, a court document classifier, and a raga recommendation system, are all real categories of AI deployment in India. They look completely different on the surface. Underneath, they share the same five-part structure.

**Fig. 3.2**  
**Three Indian AI Systems — One Architecture**

|            | Crop Disease App                                                                                   | Court Document Classifier                                                                               | Raga Recommender                                                                                            |
|------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| INPUT      | Leaf photograph from farmer's mobile phone<br><i>12 MP image, RGB pixels</i>                       | Scanned or digital court document text<br><i>Hindi, English, or regional script</i>                     | Listening history, time of day, season, device type<br><i>Multi-dimensional user context</i>                |
| REPRESENT. | Edge, texture, colour spot pattern feature vector<br><i>Discards background, orientation</i>       | Word and sentence embedding vectors from language model<br><i>Dense semantic representation</i>         | User profile vector + raga characteristic vector<br><i>Time and seasonal associations encoded</i>           |
| DECISION   | CNN classification model outputs disease probability<br><i>Trained on 500,000+ labelled images</i> | Transformer classifier predicts document type and urgency<br><i>Fine-tuned on court document corpus</i> | Similarity computed between user context and raga vectors<br><i>Collaborative + content-based filtering</i> |
| OUTPUT     | Diagnosis + confidence score + treatment recommendation<br><i>Displayed in farmer's language</i>   | Routing label + priority flag sent to case management system<br><i>Reduces manual sorting workload</i>  | Ranked raga recommendations shown in app interface<br><i>Time and mood contextualised</i>                   |
| FEEDBACK   | Farmer reports whether diagnosis was correct<br><i>Retrains model regionally</i>                   | Staff corrections when documents are misrouted<br><i>Logged and used for retraining</i>                 | Play duration, skips, playlist additions<br><i>Updates user and item vectors</i>                            |

Three completely different applications sharing one underlying architecture. Comparing them in parallel reveals that the same conceptual questions — what is the input? what is the representation? — apply to any AI system.

Figure 3.2 — Three Systems One Architecture

Crop Disease Detection

A farmer in Vidarbha notices unusual spots on the leaves of her cotton crop. She opens an app, photographs the affected leaves with her mobile phone, and submits the image. Within seconds, she receives a diagnosis and a treatment recommendation.

The input is the photograph: a grid of pixels captured by the phone camera, each pixel carrying red, green, and blue intensity values. The representation step processes this image through a series of transformations. Early transformations detect edges and textures. Later transformations identify colour patterns and spatial distributions of spots. The final representation is a compact numerical description of the visual features most relevant to disease identification, something like a fingerprint of the image's diagnostic content. The decision step applies a classification model trained on hundreds of thousands of labelled images of diseased and healthy cotton crops. It outputs a probability distribution across possible diagnoses. The output displayed to the farmer is the most probable diagnosis with a confidence score and a treatment recommendation linked to the local availability of agricultural inputs. The feedback mechanism collects outcomes: farmers who report whether the diagnosis was correct and whether the treatment worked. This data retrains the model periodically, improving its accuracy for the specific crop varieties and disease pressures common in that region.

Court Document Classification

India's court system generates an enormous volume of documents: petitions, judgements, orders, affidavits, and correspondence across twenty-five high courts and hundreds of

district courts. A court document classification system automatically categorises incoming documents, routes them to the correct department, and flags those requiring urgent attention.

The input is the text of the document, which may be in English, Hindi, or one of several regional languages. The representation step converts the text into a numerical format suitable for machine processing. Each word or word-piece is converted into a dense vector of numbers, its position in a high-dimensional meaning space learned from billions of words of training text. The entire document is then represented as a combination of these word vectors, capturing both the meanings of individual words and the relationships between them. The decision step applies a classification model to this representation, predicting the document type and urgency level. The output is a routing label and a priority flag, transmitted to the court's case management system. The feedback mechanism includes corrections from court staff: when a document is misrouted, the correction is logged and used to retrain the model.

### **Raga Recommendation**

A user opens a classical Indian music streaming application late on a winter evening. The application recommends a Bhairavi, a raga traditionally associated with late evening and the end of performances, and one that the user has not explicitly searched for.

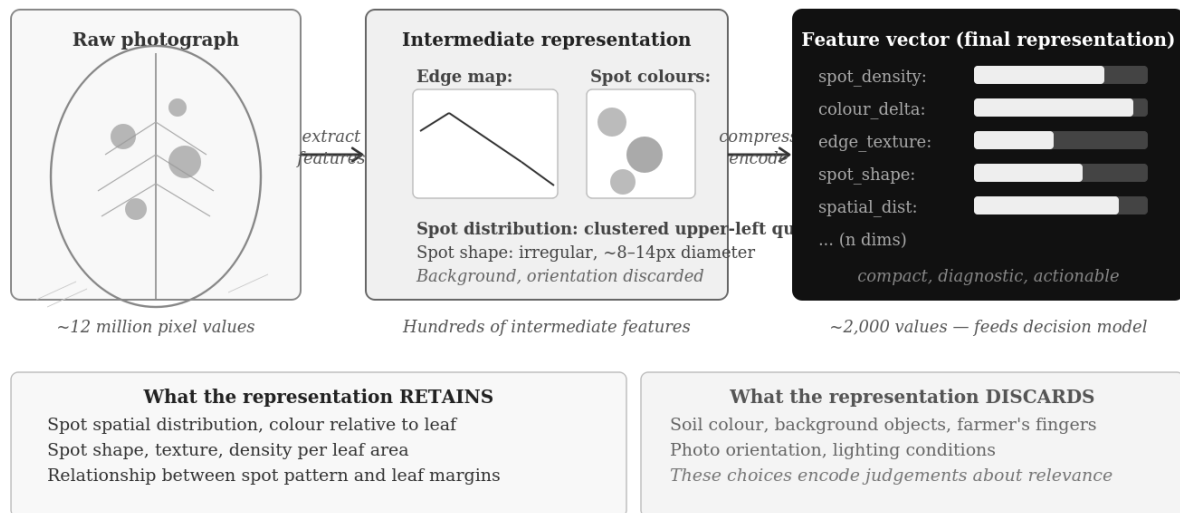
The input to this system is multi-dimensional: the user's listening history, the current time of day, the day of the week, the season, and the device being used. The representation step constructs a user profile vector that captures the statistical pattern of the user's listening behaviour across multiple dimensions: which ragas, which gharanas, which artists, what times, what session lengths. It also constructs an item vector for each raga in the catalogue, capturing its musical characteristics, its traditional associations with time of day and season, and its historical co-occurrence patterns with other ragas in listening sessions. The decision step computes the similarity between the user's current context vector and the catalogue of available ragas, weighted by the time and seasonal associations encoded during representation. The output is a ranked list of recommendations. The feedback mechanism records whether the user plays the recommended piece, how long they listen, and whether they add it to a playlist, all of which update both the user profile and the item vectors.

Three applications, three domains, three entirely different surface appearances. One architecture.

## **3.3 What Is a Representation?**

The representation step deserves more careful attention than it typically receives, because it is where most of the real intellectual work happens and where most of the consequential decisions are made.

**Fig. 3.3**  
**What Is a Representation? (Crop Disease Example)**



*The representation step compresses 12 million pixel values into a compact feature vector. The choices about what to retain and what to discard reflect domain knowledge and value judgements about relevance, not just technical preference.*

*Figure 3.3 — What Is A Representation*

Consider what happens when the crop disease app processes the farmer's photograph. The raw image contains approximately twelve million pixel values, one for each pixel's red, green, and blue intensity. This is the input. The representation process transforms this into something far smaller and more structured, perhaps a vector of a few thousand numbers that captures the essential diagnostic content while discarding everything else.

What is discarded? The representation discards the precise colour of the soil visible in the background. It discards the orientation of the photograph, because a leaf spotted with a disease looks the same whether photographed from above or from the side. It discards the presence of the farmer's fingers at the edge of the frame. It retains the spatial distribution of spots, their colour relative to the surrounding leaf tissue, their shape and texture, and their size relative to the leaf.

This selective retention is not arbitrary. It reflects a judgment about what information is diagnostically relevant. That judgment is encoded in the architecture of the representation system, which was designed by engineers and agronomists working together. If the judgment is wrong, if the representation discards information that is actually diagnostically relevant, or retains information that introduces bias, the decision system built on top of it cannot compensate. The representation is not a neutral step. It is a set of choices about what matters.

This point has consequences far beyond crop disease detection. Every AI system's representation step encodes assumptions about relevance. A loan assessment system's representation might include a feature derived from the applicant's postal code, which in India often correlates strongly with caste and socioeconomic background. If that feature is included, the model will learn to use it, and the system's outputs will encode those

correlations. The choice of what to include in a representation is a technical choice that has social and ethical consequences. It is not a purely engineering decision. It is a decision about what the world should look like.

### 3.4 The Feedback Loop as Learning

Every component of the architecture matters, but feedback is the component that separates a truly intelligent system from a sophisticated calculator. A calculator can perform complex operations, but it produces the same output for the same input every time, regardless of what happened the last time it ran. A learning system improves with experience. Feedback is the mechanism that makes improvement possible.

**Fig. 3.4**  
**The Feedback Loop as Learning**

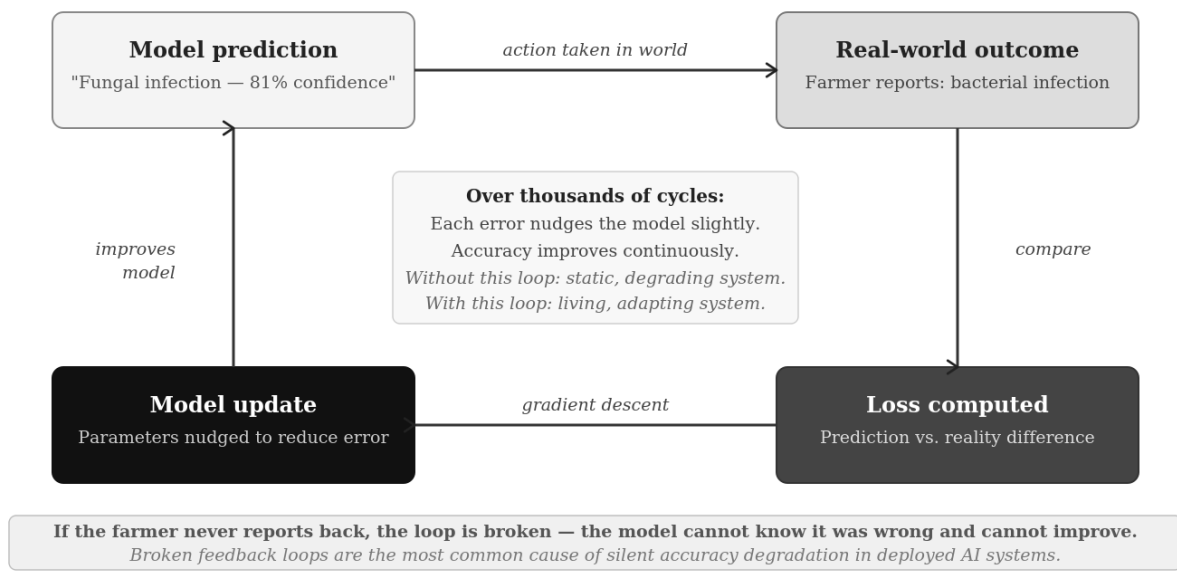


Figure 3.4 — Feedback Loop As Learning

The feedback loop works as follows. The system produces an output. That output has consequences in the world: a farmer treats or does not treat her crop, a transaction is approved or blocked, a user listens to a recommended raga or skips it. Some of those consequences are observable: the farmer reports back through the app, the blocked transaction is investigated, the user's play behaviour is logged. The observed consequences are compared against what the system predicted, and the difference between prediction and reality is used to update the system's model. If the crop disease app predicted fungal infection but the farmer reports it was bacterial, the model's parameters are nudged in a direction that makes it slightly less likely to make that mistake for similar images in the future. Over thousands and millions of such nudges, the model improves.

This process has a name in machine learning: it is called gradient descent, and the measure of difference between prediction and reality is called the loss function. The technical details will be covered in Unit 2. What matters here is the conceptual point: feedback is not an optional feature. It is what makes the system intelligent rather than merely computational. An AI system without a feedback mechanism is a snapshot of learning frozen at the moment of deployment. It will not improve as the world changes around it, and it will degrade as the gap between its training distribution and the current world grows wider.

In India, this degradation problem is particularly relevant. Many AI systems deployed in India were trained primarily on data from other countries or from urban Indian contexts, then applied to rural Indian contexts where the input distribution is substantially different. A crop disease detection model trained on images from American or European fields may perform poorly on Indian varieties grown in Indian conditions and photographed with lower-resolution cameras on slower network connections. The feedback loop is what allows such a system to adapt, but only if the feedback is actually collected, which requires that farmers report outcomes, that the company invests in retraining, and that the infrastructure for collecting rural feedback exists. Each of these conditions is more fragile than it looks.

### **3.5 Where Systems Fail**

Understanding the architecture allows us to locate failure precisely. Most AI failures can be attributed to a specific component of the five-part structure. Knowing which component has failed is the first step toward fixing it.

Input failures occur when the data entering the system does not accurately represent the phenomenon the system is trying to understand. A voice assistant trained primarily on standard Hindi spoken by educated urban speakers will perform poorly on Bhojpuri-inflected Hindi spoken in rural Bihar. Not because the decision step is wrong, but because the input distribution the system was trained on does not match the input distribution it is deployed on.

Representation failures occur when the transformation from raw input to structured features discards information that matters or retains information that introduces bias. A face recognition system trained primarily on lighter-skinned faces will develop a representation that captures fine-grained distinctions among lighter-skinned faces but blurs distinctions among darker-skinned faces. The representation has learned to encode the discriminative features present in its training data, and those features happen to work better for some populations than others.

Decision failures occur when the model is too simple to capture the relevant patterns, or too complex and has learned spurious patterns that do not generalise. A model that is too simple underfits, performing poorly even on familiar examples. A model that is too

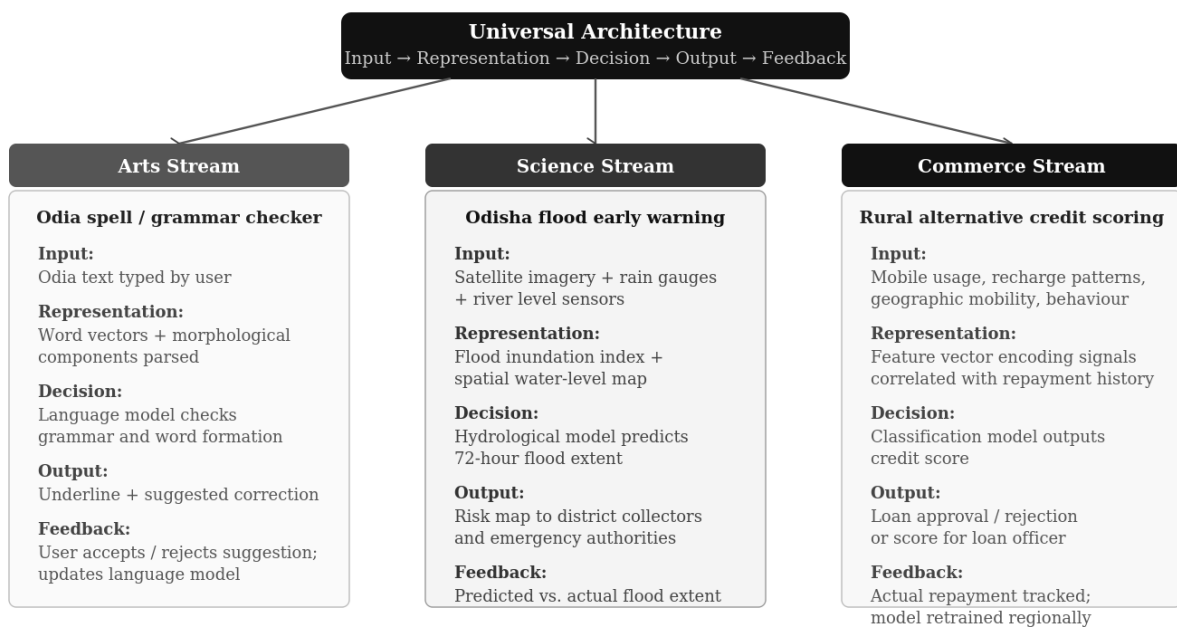
complex overfits, performing well on training examples but failing on new ones. Chapter 4 will examine these failure modes in detail.

Output failures occur when the system's conclusion is communicated in a form that leads to poor decisions by the humans receiving it. A probability score of 0.73 for fraud is meaningless to a bank officer who does not understand probability. If the output is presented as merely suspicious rather than as a specific score with a specific confidence interval, different officers will interpret and act on it differently, introducing inconsistency that the model itself does not contain.

Feedback failures occur when the loop is broken: when outcomes are not observed, not recorded, or not used to update the model. A medical AI deployed in a district hospital that has no mechanism for collecting whether its diagnoses were confirmed by laboratory tests is flying blind. It cannot know whether it is improving or degrading, and the gap between its training distribution and the current reality will widen silently until the system fails in a way that is impossible to ignore.

### 3.6 Patterns in My Field: Stream Perspectives

**Fig. 3.5**  
**Stream Fork: One Architecture, Three Contexts**



*The universal architecture applies unchanged across all three streams. What changes is the domain, the data, and the ethical stakes — not the underlying structure. Learning to read that structure is learning to understand any AI system.*

*Figure 3.5 — Stream Fork Architecture*

#### Arts Stream: The Odia Language Correction Tool

Spell and grammar checking tools for Odia, one of India's classical languages, are being actively developed by teams at institutions including IIT Bhubaneswar and the Odia

Wikimedia community. Tracing this system through the five-part architecture: the input is the raw text typed by the user in the Odia script. The representation step converts each word into a vector in a learned meaning space and identifies its morphological components, which in Odia are numerous and complex. The decision step applies a language model trained on a corpus of Odia text to determine whether the word is correctly formed and used in context. The output is an underline and a suggested correction. The feedback loop collects data on which suggestions users accept and which they reject, which updates the language model's understanding of correct Odia usage. The critical limitation is data: the corpus of digitised, high-quality Odia text is far smaller than the equivalent corpus for English, which means the representation and decision steps are less accurate. The feedback loop is the mechanism through which community participation can directly improve the tool's performance.

### **Science Stream: Satellite-Based Flood Early Warning in Odisha**

Odisha is among the most flood-prone states in India. The Mahanadi basin regularly floods during the monsoon season, affecting millions of people and thousands of square kilometres of agricultural land. A satellite-based flood early warning system uses the five-part architecture as follows: the input is multi-spectral satellite imagery updated every one to three days, combined with rainfall data from rain gauge networks and river level data from monitoring stations. The representation step computes flood inundation indices from spectral bands sensitive to water presence, integrates the satellite data with river level measurements, and produces a spatial representation of flood extent and water level. The decision step applies a hydrological model trained on historical flood events to predict flood extent and timing over the next seventy-two hours. The output is a flood risk map distributed to district collectors and emergency management authorities. The feedback loop compares predicted flood extent with actual observed extent after each event, updating the hydrological model's parameters. This system has been operationally deployed by the Odisha State Disaster Management Authority and has demonstrably reduced evacuation response time.

### **Commerce Stream: Credit Scoring for Microfinance in Rural India**

Tens of millions of rural Indians lack formal credit histories, making them invisible to traditional credit scoring systems. Microfinance institutions and fintech companies have developed alternative credit scoring systems that use non-traditional data to assess creditworthiness. Traced through the architecture: the input includes mobile phone usage patterns, such as call frequency, contact network size, and data consumption; mobile recharge patterns; geographic mobility; and in some systems, responses to short behavioural assessments administered through the app. The representation step constructs a feature vector from these diverse inputs, encoding them as signals that historical data has shown to correlate with loan repayment behaviour. The decision step applies a classification model that outputs a credit score. The output is presented to a loan officer or, in fully automated systems, directly to the applicant as an approval or rejection. The feedback loop tracks actual repayment behaviour, continuously updating the model's understanding of which input signals are genuinely predictive of repayment in specific regional contexts. The ethical questions raised by this system, particularly around consent, privacy, and the use of behavioural proxies for financial character, will be addressed in Unit 4.

## — WHAT COULD GO WRONG? —

### The Broken Feedback Loop

Of the five components of the architecture, feedback is the one most frequently neglected in real-world deployments. Building the input pipeline, the representation system, and the decision model requires substantial engineering investment, and there is understandable pressure to deploy and move on. The feedback infrastructure, which requires ongoing data collection, storage, analysis, and model retraining, is expensive and unglamorous. It is often deferred or omitted entirely.

The consequences of a broken feedback loop are insidious precisely because they develop slowly. At deployment, the model performs at its training-time accuracy. Over months and years, the world changes: new crop varieties are planted, new disease strains emerge, fraud patterns evolve, language usage shifts. Without feedback, the model cannot adapt. Its accuracy degrades silently, because there is no mechanism for the system to know that it is failing. Users who trusted the system's early performance continue to trust it, not knowing that its accuracy has declined. By the time the failure becomes visible, significant harm may have occurred.

In India, this problem is compounded by the scale of deployment. An AI system that makes credit decisions for a microfinance institution serving ten million rural borrowers, and whose feedback loop has been broken for two years, has made two years of systematically degrading decisions for ten million people. Restoring the feedback loop does not undo those decisions. This is why feedback is not an optional feature of responsible AI deployment. It is the infrastructure of accountability.

## — CHAPTER SUMMARY —

### Chapter 3 — Key Points

1. Every intelligent system has five components: Input, Representation, Decision, Output, and Feedback.
2. Input is the system's only window onto the world; information not captured in the input is invisible to the system.
3. Representation transforms raw input into structured features — this step encodes assumptions about what information matters.
4. The decision step is only as good as the representation feeding it; sophisticated algorithms cannot compensate for poor features.
5. Output design matters: the same conclusion communicated in different forms leads to different decisions by human recipients.
6. Feedback is the mechanism that makes a system intelligent rather than merely computational; without it, systems degrade silently.

7. Failures can be precisely located in one of the five components — diagnosis precedes repair.
8. The representation step encodes value judgements about relevance, not just technical choices — it has social and ethical consequences.

— MODEL QUESTIONS —

## Section A — MCQ (5 Questions)

**Q1. Which component of the five-part AI architecture is responsible for transforming raw input into a form that can be processed by the decision step?**

- A) Input
- B) Representation
- C) Output
- D) Feedback

**Answer: B**

*Explanation: The representation step transforms raw input — pixels, text, sensor readings — into structured numerical features from which patterns can be extracted. It is distinct from the input (which is the raw signal) and from the decision step (which applies the model to the features).*

**Q2. A crop disease detection app produces accurate results in Maharashtra but performs poorly on the same crops in Kerala. The most likely cause is:**

- A) The decision model is too simple
- B) The output format is unsuitable for Kerala farmers
- C) The input distribution differs between training and deployment contexts
- D) The feedback loop is collecting too much data

**Answer: C**

*Explanation: This is an input failure caused by distribution mismatch: the model was trained on images from Maharashtra conditions and performs poorly when the input distribution shifts to Kerala's different lighting, humidity, and crop variety conditions.*

**Q3. Why is the representation step described as encoding 'value judgements about relevance'?**

- A) Because it assigns monetary value to input data
- B) Because the choice of which features to retain reflects assumptions about what information matters
- C) Because it uses moral reasoning to filter inappropriate inputs
- D) Because it determines the user's values from their input behaviour

**Answer: B**

*Explanation: The representation step involves deliberate choices about which aspects of raw input to retain and which to discard. These choices reflect assumptions — made by engineers, data scientists, and domain experts — about what information is relevant to*

*the task, and those assumptions can encode social biases.*

**Q4. Which of the following best describes a 'feedback failure' in an AI system?**

- A) The system receives negative user reviews
- B) The model's decision step produces a wrong output
- C) Outcomes are not observed, recorded, or used to update the model
- D) The input pipeline stops receiving new data

**Answer: C**

*Explanation: A feedback failure occurs when the loop between output consequences and model update is broken. The system cannot know whether its predictions were correct, cannot learn from mistakes, and degrades silently as the gap between its training data and the current world grows.*

**Q5. The raga recommendation system recommends Bhairavi to a user late in the evening. This recommendation draws primarily on which component of the architecture?**

- A) Input only — it reads the current time from the device
- B) Representation — it constructs a context vector combining time, season, and user history
- C) Output — the display algorithm selects traditional ragas by time of day
- D) Feedback — previous evening users' ratings drive the recommendation

**Answer: B**

*Explanation: The recommendation reflects the representation step, which constructs a multi-dimensional context vector combining the user's listening history, current time, season, and raga characteristics including their traditional time associations. The decision step then computes similarity between this context vector and the raga catalogue.*

## **Section B — Short Answer Questions (5 Questions)**

**Q6. Explain the role of the representation step using the example of a UPI transaction fraud detection system.**

**Model Answer:**

In a UPI fraud detection system, the raw input is a transaction record containing fields such as amount, timestamp, device ID, location coordinates, and account history. The representation step transforms these fields into derived features that are more informative for fraud detection: the ratio of this transaction amount to the account's average transaction amount, the distance in kilometres between the current transaction location and the account's typical location, the number of transactions in the past hour, and the age of the device identifier. These derived features are not present in the raw record — they are computed from it, encoding domain knowledge about what makes a transaction suspicious. The resulting feature vector is what the decision model actually operates on. A poorly designed representation will produce a poor model even if the decision algorithm is sophisticated.

*Key terms: representation, feature engineering, derived features, domain knowledge, distribution, input transformation*

**Q7. What is meant by 'distribution mismatch' and why is it a particular challenge for AI systems deployed in India?**

**Model Answer:**

Distribution mismatch occurs when the statistical properties of the data a model is deployed on differ significantly from the properties of the data it was trained on. A model trained on data from one distribution will have learned patterns that are accurate for that distribution but may not generalise to a different one. In India, this problem is acute for several reasons. First, many AI systems deployed in India were trained primarily on Western or urban Indian data. Second, India has enormous internal variation across languages, dialects, crop varieties, climate zones, and socioeconomic contexts, meaning that a model trained even on Indian data from one region may mismatch when deployed in another. Third, the infrastructure for collecting feedback from rural and low-resource contexts is less developed, making it harder to detect and correct distribution mismatch after deployment.

*Key terms: distribution mismatch, generalisation, training distribution, deployment distribution, regional variation, transfer learning*

**Q8. Why is output design considered a user experience problem as much as a technical one? Use the fraud detection context as an example.**

**Model Answer:**

Output design determines how the system's conclusion is communicated to the human who will act on it, and different forms of the same conclusion can lead to different actions. A fraud detection model might compute a probability score of 0.78, meaning it believes there is a 78% chance the transaction is fraudulent. If this is communicated to a bank officer as the raw number 0.78, an officer unfamiliar with probability may not know whether to act on it. If it is communicated as Suspicious — Manual Review Required, the officer knows to investigate but does not know how confident the system is. If it is communicated as High Confidence Fraud — Recommend Block, the officer's action is heavily shaped by the system's presentation. Each output form creates a different human-machine collaboration pattern, with different error rates and different accountability structures. Output design is therefore a decision about who bears responsibility for the final action, which is fundamentally a user experience and governance question.

*Key terms: output design, human-machine collaboration, probability communication, accountability, uncertainty quantification*

**Q9. Describe one way in which the representation step of a credit scoring AI in rural India could encode social bias, even unintentionally.**

**Model Answer:**

Alternative credit scoring systems in rural India often include features derived from mobile phone usage patterns. One such feature might be the geographic mobility of the applicant, measured by the number of distinct cell tower locations the phone connects to over a month. Historical data might show that higher geographic mobility correlates with higher loan repayment rates in the training data, because mobile workers who travel to find employment tend to have more diverse income streams. However, women in rural India often have significantly lower geographic mobility than men due to social and safety constraints on travel, not due to any difference in creditworthiness. A model trained on this feature would encode this mobility disparity and systematically assign lower credit scores to women applicants. The bias is unintentional in the sense that it was not explicitly programmed, but it is predictable from the structure of the training data. The representation choice, including geographic mobility as a feature, is the point at which the bias enters the system.

*Key terms: proxy variable, social bias, representation choice, gender discrimination, training data, unintended discrimination*

**Q10. Explain why feedback is described as 'the infrastructure of accountability' in AI systems deployed at scale.**

**Model Answer:**

Accountability in AI deployment requires the ability to know whether a system is performing as intended, to detect degradation, and to correct errors. All three of these capabilities depend on feedback. Without a mechanism for collecting outcomes and comparing them against predictions, there is no way to know whether the system's decisions are accurate, whether its accuracy is changing over time, or whether specific categories of users are being systematically harmed. At scale, this matters enormously: a system making credit decisions for ten million people that has no feedback mechanism cannot be held accountable because there is no evidence of its performance. Feedback infrastructure, the processes and systems for collecting, storing, and acting on outcome data, is what makes accountability operationally possible. Calling it infrastructure, rather than a feature, signals that it is not optional and should be built before deployment, not added after a problem is discovered.

*Key terms: accountability, feedback infrastructure, performance monitoring, outcome data, degradation detection, deployment at scale*

## Section C — Reflective Questions (3 Questions)

**Q11. Think of an AI system you interact with regularly — perhaps a voice assistant, a social media feed, or a navigation app. Try to identify all five components of its architecture. Which component do you think is least visible to users, and does that invisibility matter?**

**Thinking prompts:**

- What is the raw input the system receives from you?
- What transformations might be applied before your input reaches the decision model?
- What does the system's output look like, and what information does it communicate?
- How does the system receive feedback on whether its outputs were useful or correct?
- Which component would you most want to be able to inspect, and why?

*Rubric hint: Strong responses will identify all five components specifically, not generically, and will engage with the question of visibility. Credit for responses that identify the representation step as particularly invisible and consequential, and that connect invisibility to accountability.*

**Q12. This chapter argues that the representation step is where most AI failures originate, because it encodes assumptions about what information matters. Do you think users should have the right to know what representation a system uses to process their data? Who would benefit from such transparency, and who might resist it?**

**Thinking prompts:**

- What would it mean in practice for a user to know their representation?
- Who currently has access to this information, and who does not?
- In which contexts would representation transparency be most valuable?
- What interests might lead companies to resist disclosing their representations?
- How does this connect to existing debates about data rights in India?

*Rubric hint: Strong responses will engage with the tension between transparency and commercial confidentiality, and will connect the question to Indian data governance frameworks. Credit for responses that identify specific contexts where representation transparency would change outcomes.*

**Q13. The chapter describes an Odisha flood early warning system as a successful deployment of the five-part architecture that has reduced evacuation response time. But what would a failure of this system look like, and what would its consequences be?**

**Thinking prompts:**

- Which component failure would be most catastrophic — input, representation, decision, output, or feedback?
- What would an input failure look like in this context?
- Who bears the consequences of a false negative, where the system fails to warn of a flood that occurs?
- Who bears the consequences of a false positive, where the system warns of a flood that does not occur?
- What safeguards would you put in place to reduce the risk of catastrophic failure?

*Rubric hint: Strong responses will engage with asymmetric consequences of different failure types. Credit for responses that recognise false negatives as potentially catastrophic and false positives as costly but recoverable, and that propose safeguards that address the specific failure mode most likely in an operational flood warning system.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Choose one AI-powered service that operates in your region — this could be a government service, a health application, an agricultural advisory tool, or a language technology product. Trace the service through the five-part architecture as completely as you can from publicly available information. Then identify which component you believe is most likely to fail, explain why, and propose one specific intervention that would reduce that risk.**

This project requires research into a real system. Use the organisation's website, annual reports, press coverage, and where possible direct contact with users of the system. The five-part trace should be as specific as possible — avoid generic descriptions. The failure analysis should be grounded in the specific context of the region and user population the system serves.

**Q15. [Commerce] Investigate one fintech or microfinance company operating in rural or semi-urban India that uses AI-based credit scoring. Map its credit**

**scoring system through the five-part architecture, paying particular attention to the representation step: what features does the system use, and where do those features come from? Then assess whether the feedback loop is adequately designed. Does the company collect outcome data? Does it use that data to retrain its model? What would you recommend to strengthen accountability?**

This project combines desk research with a critical analysis of the accountability structure of a real system. The representation step analysis is the most important part — it requires identifying the specific features used, which may require reading technical documentation, regulatory filings, or academic papers that describe the system. The accountability assessment should be specific and actionable.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Domingos, P. (2015). <i>The Master Algorithm: How the Quest for the Ultimate Learning Machine Will Remake Our World</i> . Basic Books. [Chapter 1 for an accessible account of what machine learning systems do, and Chapter 6 for a clear explanation of how representation and features relate to model performance.]                                                       |
| <b>Core</b> | Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. <i>Science</i> , 366(6464), 447–453. [A landmark paper demonstrating precisely how representation choices encode bias — a directly applicable case study for understanding the representation step's social consequences.] |
| <b>Core</b> | Odisha State Disaster Management Authority. (2022). <i>Integrated Flood Management and Early Warning System: Technical Documentation</i> . osdma.org. [Primary source on the Odisha flood early warning system described in the Science stream section; freely available and relevant to the Architecture analysis project.]                                                  |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Sculley, D., Holt, G., Golovin, D., et al. (2015). Hidden technical debt in machine learning systems. <i>Advances in Neural Information Processing Systems</i> , 28. [On the infrastructure costs of real-world ML deployment, including feedback loop maintenance; directly relevant to the 'infrastructure of accountability' argument.] |
| <b>Extended</b> | Barocas, S., Hardt, M., & Narayanan, A. (2023). <i>Fairness and Machine Learning: Limitations and Opportunities</i> . MIT Press. [Chapters 1 and 2 for a rigorous treatment of how representation choices introduce bias; freely available at <a href="https://fairmlbook.org">fairmlbook.org</a> .]                                       |
| <b>Extended</b> | Bhatt, U., et al. (2020). Explainability in AI for Medicine: Audit of COVID-19 Risk Prediction Models. <i>arXiv preprint</i> . [For the output design problem in high-stakes medical contexts, applicable to the output failure mode discussion.]                                                                                          |

## Accessible Online Resources

|        |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Online | Google Developers. (2023). Machine Learning Crash Course: Feature Engineering. <a href="https://developers.google.com/machine-learning/crash-course/representation">developers.google.com/machine-learning/crash-course/representation</a> . [A free, interactive introduction to the representation step — what features are, how they are constructed, and why they matter. No programming background required.] |
| Online | AI4Bharat. (2023). Odia NLP Tools and Resources. <a href="https://ai4bharat.iitm.ac.in">ai4bharat.iitm.ac.in</a> . [Primary source on the state of Odia language AI tools; relevant to the Arts stream section and to understanding the data limitations described.]                                                                                                                                               |
| Online | Partnership on AI. (2021). About ML: Annotation and Benchmarking. <a href="https://partnershiponai.org">partnershiponai.org</a> . [On the feedback and labelling infrastructure required for responsible AI deployment at scale; freely accessible and directly relevant to the feedback loop discussion.]                                                                                                         |

*End of Chapter 3*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## UNIT TWO

# How Machines Learn

*A machine does not understand a mango. It has never held one, smelled one, been disappointed by one. What it has is ten thousand examples of mangoes correctly labelled ripe or unripe, and a method for finding the pattern that separates them.*

*This unit asks how that finding happens — through labelled examples, through trial and reward, through layers of simple detectors stacked into something that can recognise a face or a failing crop. None of it resembles understanding as a person experiences it. Yet it works, often well enough to matter, and knowing exactly how it works is the difference between using a tool and being used by one.*

- 
4. Teaching by Example
  5. Learning Without Labels
  6. Learning by Doing
  7. The Shape of a Neural Network

## Chapter 4

# Teaching by Example

*Consider how a child learns to recognise a mango. Nobody gives the child a formal definition. Nobody explains that a mango is a drupe, that it belongs to the genus *Mangifera*, that its flesh should be a specific Munsell colour. Instead, the child is shown mangoes. Over dozens of encounters, handled in markets, eaten at the table, pointed out in orchards, the child builds an internal model of what a mango is. By the time the child can articulate anything about mangoes, the learning has already happened, accumulated from examples rather than constructed from rules.*

*Supervised machine learning works on the same principle. It is the most widely used form of machine learning, the foundation of most AI applications you encounter in daily life, and the subject of this chapter. Understanding it clearly requires no mathematics. It requires only the willingness to think carefully about what it means to learn from examples, and what can go wrong when the examples are imperfect.*

## Learning Objectives

By the end of this chapter, you will be able to:

- Explain supervised learning using the field guide metaphor and real Indian examples
- Distinguish between training data, validation data, and test data, and explain the purpose of each
- Describe overfitting and underfitting and identify them in practical contexts
- Explain how biased training data produces biased models, using Indian examples
- Interpret a simple decision boundary and understand what it means for a model to generalise

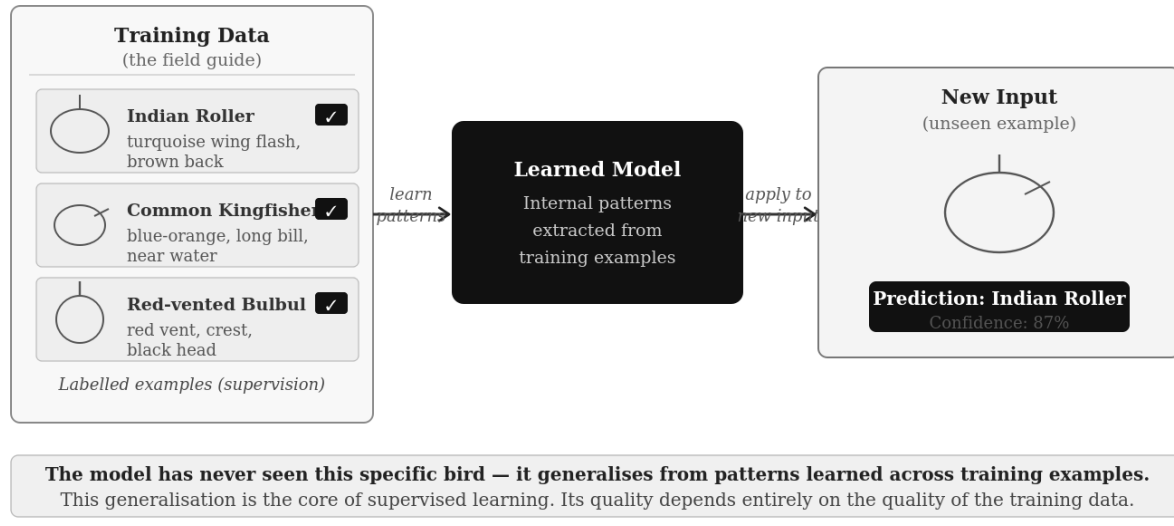
### 4.1 The Field Guide Metaphor

Imagine a naturalist learning to identify bird species from a field guide. The field guide contains hundreds of species, each illustrated with drawings or photographs and described with notes on distinctive features: the shape of the bill, the pattern of the plumage, the characteristic posture. The naturalist studies the guide and then goes into the field. They see a bird they have not encountered before, compare it mentally to the images and descriptions they have memorised, find the closest match, and make an identification.

This is exactly the structure of supervised machine learning. The field guide is the training data: a collection of labelled examples, each example consisting of an observation and a correct answer. The naturalist's mental model, built from studying the guide, is the learned

model. The bird encountered in the field is the new input. The identification is the model's prediction.

**Fig. 4.1**  
**The Field Guide Metaphor for Supervised Learning**



*Figure 4.1 — Field Guide Metaphor*

The critical word in that description is labelled. In supervised learning, every training example comes with a correct answer provided by a human annotator. A collection of ten thousand photographs of birds, each photograph tagged with the species name, is a supervised learning dataset. The tagging is the supervision. Without it, a machine learning algorithm has no way to know which features are relevant to the task of species identification and which are irrelevant.

The annotation process is expensive and consequential. Someone must look at each example and assign the correct label. For a bird identification dataset, this requires ornithological expertise. For a medical imaging dataset, this requires radiological expertise. For a dataset of legal documents, this requires legal expertise. The quality of the labels constrains the quality of the model: a model cannot be more accurate than the annotators who labelled its training data. And if the annotators hold systematic biases, those biases will appear in the labels and be learned by the model.

In India, this annotation problem has a particular character. Many AI applications deployed in India, in agriculture, healthcare, and language processing, rely on training data that was labelled by annotators who may not have been familiar with the specific regional context. A crop disease model trained on photographs labelled by experts in Pune may not correctly reflect the disease presentations common in the specific soil and climate conditions of Kalahandi. A medical AI trained on data labelled by urban hospital radiologists may not reflect the presentation patterns of diseases more common in rural populations. The field guide, in other words, may describe a different field.

## 4.2 Three Piles: Training, Validation, and Test

When a supervised learning system is built, the available labelled data is not used all at once. It is divided into three separate sets, each with a distinct purpose. Understanding this division is essential to understanding how machine learning models are evaluated honestly.

**Fig. 4.2**  
**Training, Validation, and Test Sets — The Three-Pile Division**



*Training accuracy is almost meaningless — models are optimised on training data. Only test set accuracy, from data that influenced no design decision, honestly estimates performance on genuinely new inputs during deployment.*

*Figure 4.2 — Three Data Sets*

The training set is the largest pile, typically comprising sixty to eighty percent of the available data. This is the set the model actually learns from. It is the field guide. The model examines every example in the training set, compares its predictions against the correct labels, and adjusts its parameters to reduce the error. By the end of training, the model has been exposed to every example in this set many times.

The validation set is the second pile, typically comprising ten to twenty percent of the data. It is kept separate from the training set and used during the development process to evaluate the model's performance on data it has not been trained on. The model designer uses the validation set to make decisions: which model architecture to use, which hyperparameters to set, when to stop training. Because these decisions are influenced by the validation set's performance, the validation set is not a truly independent evaluation. A model that performs well on the validation set may be doing so partly because the design decisions were tuned to it.

The test set is the third pile, typically comprising ten to twenty percent of the data. It is kept completely sealed throughout the development process and used only once, at the very end, to produce the model's final reported accuracy. This is the honest evaluation. Because no design decisions were influenced by the test set, the model has never, in any sense, been optimised for it. Its performance on the test set is the best available estimate of how it will perform on genuinely new data in deployment.

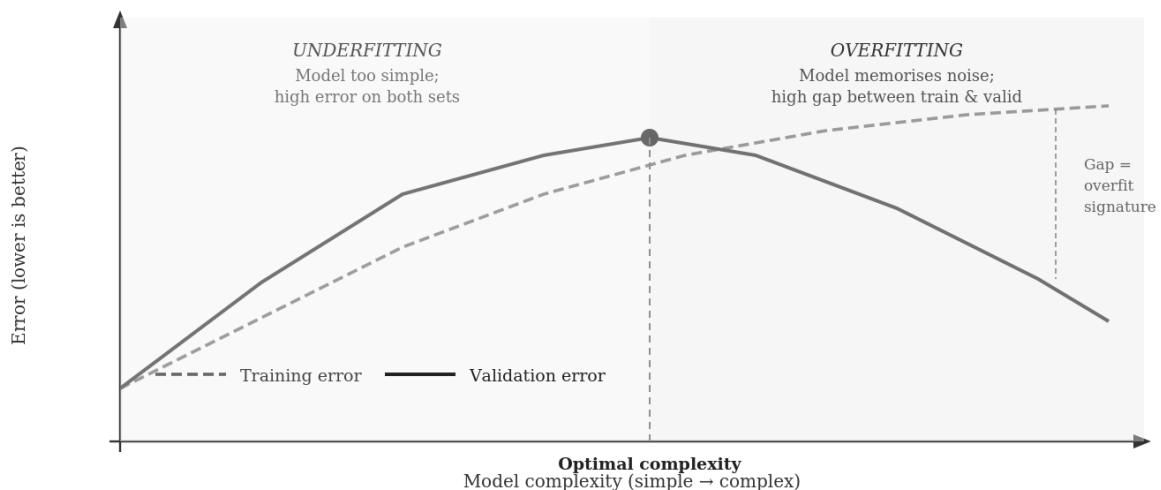
This three-way division matters enormously in practice. A model's accuracy on its training data is almost meaningless as an evaluation metric: the model has been specifically optimised to perform well on that data. Reporting training accuracy as if it were deployment accuracy is a common error, sometimes accidental and sometimes deliberate. Any credible evaluation of an AI system must report performance on held-out data that played no role in training or model selection.

In the Indian context, this standard is not always met. Government AI deployments and commercially marketed AI products sometimes report performance metrics without clearly distinguishing between training and test set accuracy, or without specifying whether the test set was drawn from the same population that the system will be deployed on. Asking these questions is not technical pedantry. It is basic due diligence for any institution considering adopting an AI system.

### 4.3 The U-Curve: Overfitting and Underfitting

Every machine learning model makes a tradeoff between memorisation and generalisation. Memorisation means performing well on the training data. Generalisation means performing well on new data. These two objectives are in tension, and the tension produces two characteristic failure modes.

**Fig. 4.3**  
**The Overfitting-Underfitting U-Curve**



*The optimal model is at the minimum of the validation error curve — not the training error curve. Reporting training accuracy as if it were deployment accuracy is a common error in AI evaluation reports. Always ask: what does the test set show?*

*Figure 4.3 — Overfitting U Curve*

Underfitting occurs when the model is too simple to capture the relevant patterns in the data. Consider a model for classifying bird photographs that uses only the colour of the brightest pixel in the image. This is an extremely simple model, and it will perform poorly even on the training data because colour alone is insufficient to distinguish species. It will also perform poorly on new data for the same reason. An underfitting model has high error

on both training and validation sets. The solution is a more expressive model that can represent more complex patterns.

Overfitting occurs when the model is too complex, capturing not only the genuine patterns in the training data but also the noise, the quirks, the accidental regularities that are specific to those particular examples and will not appear in new data. A model that has been trained on ten thousand bird photographs might learn not only the genuine visual features of each species but also the particular photographic style of the cameras used to collect the dataset, the particular lighting conditions common in the locations where photographs were taken, or the particular angles from which specific species were photographed. These learned patterns are real in the training data but do not generalise. The overfitting model performs well on the training set but poorly on the validation and test sets. The gap between training accuracy and validation accuracy is the diagnostic signature of overfitting.

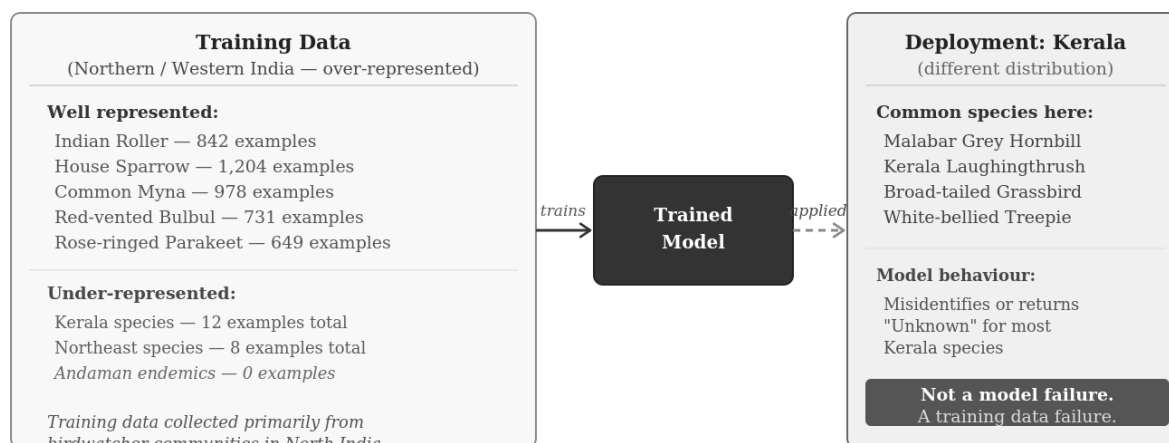
The U-curve in Figure 4.3 shows this dynamic. As model complexity increases from left to right, training error decreases monotonically: a more complex model can always fit the training data better. Validation error, by contrast, first decreases and then increases. The optimal model complexity is the minimum of the validation error curve, the point of best generalisation. To the left of that minimum is the underfitting regime; to the right is the overfitting regime.

In practice, the U-curve has a specific manifestation in Indian agricultural AI. A disease detection model trained on a large dataset of cotton disease photographs from Maharashtra may overfit to the specific visual characteristics of Maharashtra cotton varieties. When deployed on cotton from Telangana or Odisha, where the varieties differ, the overfitting means the model fails to generalise. This is not a hypothetical concern. It is a documented pattern in agricultural AI deployments in India, where models trained in one agroclimate are deployed in another without validation on local data.

## **4.4 The Problem of Biased Training Data**

The field guide metaphor, pursued further, reveals the deepest problem in supervised learning. If the field guide you are learning from is incomplete, or systematically biased toward certain species, your mental model of the avian world will be correspondingly incomplete or biased. You will be confident in identifications you have no right to be confident about, and uncertain about identifications that should be easy, for the simple reason that your learning was shaped by the particular field guide available to you rather than by birds in general.

**Fig. 4.4**  
**Biased Training Data: North Indian Field Guide Applied to Kerala**



*The model works exactly as designed — it has learned the patterns in its training data. The problem is that the training data was not representative of the deployment context. The fix is data collection, not algorithm improvement.*

*Figure 4.4 — Biased Training Data*

Consider the specific case of a bird identification app trained primarily on photographs contributed by birdwatchers in northern and western India, where the birdwatching community is larger and more digitally active. The training data contains many examples of species common in the Indo-Gangetic plains and the Western Ghats but far fewer examples of species endemic to northeastern India or the Andaman Islands. A naturalist using this app in Arunachal Pradesh will find that it performs unreliably on species common there, not because the app is technically flawed in its architecture but because its training data was geographically biased.

This geographic bias is a specific case of a general principle: a supervised learning model can only be as good as the training data it learns from. The training data is a sample of the world, and if the sample is unrepresentative, the model will be correspondingly wrong about the parts of the world that were undersampled. This is not a failure of the algorithm. It is a failure of the data.

The implications extend far beyond wildlife identification apps. A medical diagnosis AI trained on data from urban tertiary hospitals will be biased toward the disease presentations common in urban populations, which differ from rural populations in multiple relevant ways: nutrition, occupation, exposure history, co-morbidities. A hiring AI trained on the historical data of a company that has historically underrepresented women in senior roles will learn that women are less likely to succeed in senior roles, because that is what the historical data shows. The historical data accurately reflects the past. The question is whether we want the future to replicate the past.

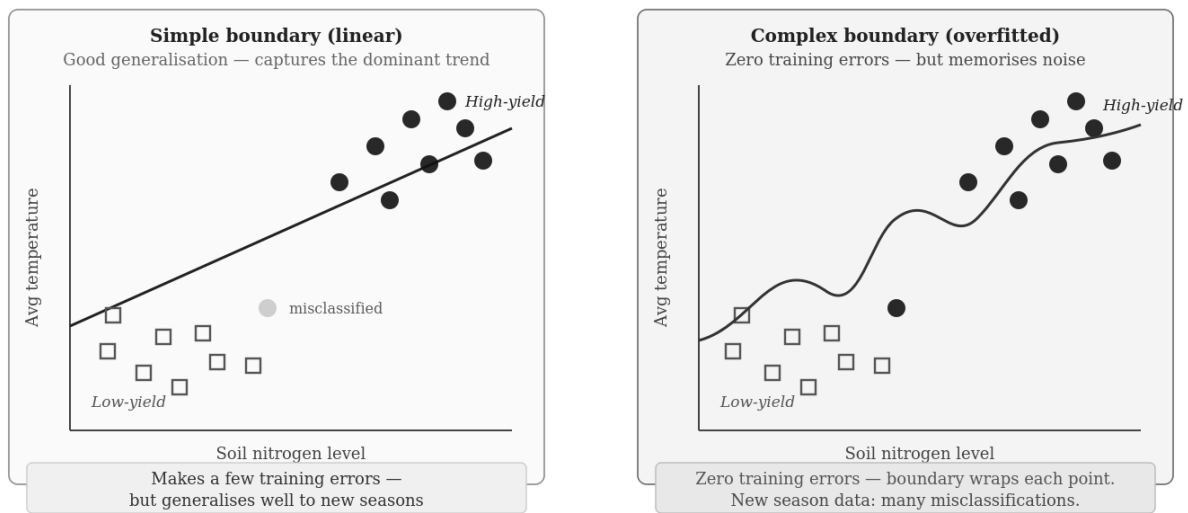
In India, the training data problem has a particular urgency because of the scale of diversity. The country's ecological, linguistic, cultural, and socioeconomic diversity means that data collected in one context is systematically unrepresentative of other contexts. A model trained on data from Delhi will not generalise to Darbhanga. A model trained on

data from Bengaluru will not generalise to Bastar. This is not a technical problem that can be solved by better algorithms. It is a data collection problem that requires investment in gathering representative data from the full range of contexts in which the system will be deployed.

## 4.5 The Decision Boundary

When a supervised learning model classifies inputs into categories, it does so by partitioning the input space into regions, one region for each category. The boundary between regions is called the decision boundary. Understanding decision boundaries provides an intuitive geometric picture of what generalisation means and why it can fail.

**Fig. 4.5**  
**Decision Boundaries: Simple vs. Overfitted (Cotton Yield, Vidarbha)**



Both boundaries are drawn on the same training data. The simple boundary makes a few training errors but generalises to new seasons. The complex boundary memorises every training point but fails on new data — the signature of overfitting.

Figure 4.5 — Decision Boundary

Imagine a simplified crop classification problem. Each cotton field in a dataset is described by two measurements: the nitrogen level in the soil and the average temperature over the growing season. The task is to classify fields as high-yield or low-yield based on these two measurements. In a two-dimensional scatter plot, each field appears as a point, with nitrogen level on the horizontal axis and temperature on the vertical axis. High-yield fields are marked with one symbol and low-yield fields with another.

A linear decision boundary is a straight line drawn through this scatter plot that attempts to separate the two categories. Everything on one side of the line is classified as high-yield; everything on the other side as low-yield. A linear boundary is simple and robust: it is unlikely to overfit because there are few parameters to tune. But it is also inflexible: if the true relationship between nitrogen, temperature, and yield is not linear, a linear boundary will not capture it.

A complex decision boundary can curve, bend, and wriggle through the scatter plot, passing between every training point and perfectly separating the two categories in the training data. In the training set, this complex boundary has zero errors. But in a new field season, when new data points appear, the complex boundary is likely to misclassify many of them, because it has memorised the specific arrangement of the training points rather than the underlying pattern. The complex boundary is the geometric face of overfitting.

The right decision boundary is one that captures the genuine pattern in the data with the simplest structure that adequately separates the categories. This is a version of Occam's razor applied to machine learning: among models that fit the training data adequately, prefer the simpler one. Simpler models are more likely to generalise because they have fewer parameters to overfit with and are less likely to capture the noise-specific structure of the training set.

## **4.6 Patterns in My Field: Stream Perspectives**

### **Arts Stream: Training a Sentiment Classifier on Odia Film Reviews**

A sentiment analysis model for Odia-language film reviews illustrates supervised learning in the arts context. The training data consists of film reviews collected from Odia newspapers, entertainment websites, and social media, each manually labelled as positive, negative, or mixed. The label is the supervision: without it, the model has no way to learn which words and phrases correspond to positive sentiment in Odia and which to negative. Several challenges specific to this task illuminate the general principles of the chapter. First, Odia sentiment expression is often indirect and contextual, with positive reviews frequently structured as understatement and negative ones as hyperbole. A model trained on direct sentiment expression from other languages will underfit when applied to Odia. Second, the available labelled data is small relative to English-language equivalents, increasing the risk of overfitting to the specific vocabulary of the annotated reviews. Third, the annotators' own cultural context shapes the labels, which means the model will learn sentiment as understood by those particular annotators, not sentiment as understood by all Odia speakers.

### **Science Stream: Training a Malaria Risk Model for Odisha Districts**

Odisha has historically had one of India's highest malaria burdens, concentrated in the forested districts of the south and west. A supervised learning model for district-level malaria risk prediction uses historical data in which each district-year combination is labelled with the recorded malaria case rate. The input features include rainfall, temperature, forest cover, human population density, access to healthcare, and historical case rates from previous years. This is a regression task: the output is a continuous risk score rather than a discrete category. The training, validation, and test split matters critically here because malaria transmission is seasonal and inter-annual. A model evaluated on a test set drawn from the same years as the training set may perform well simply because it has memorised inter-annual trends rather than learning the genuine ecological drivers of transmission. Honest evaluation requires a temporally held-out test

set: train on years one through ten, validate on years eleven and twelve, test on year thirteen. This is called temporal validation, and it is essential for any model that will be used to make predictions about future conditions.

### **Commerce Stream: Training a Loan Default Prediction Model**

A microfinance institution in rural Rajasthan builds a supervised learning model to predict loan default. The training data consists of historical loans, each labelled as repaid or defaulted. The input features include loan size, loan purpose, borrower demographics, and the mobile behaviour features discussed in Chapter 3. The three-pile structure is essential here because of a problem called temporal leakage: if the training set and test set are not properly separated in time, the model may learn patterns that reflect the specific economic conditions of a past period rather than durable predictors of creditworthiness. A model trained on loans issued during an economic expansion and tested on loans from the same period will appear to perform well, but will fail when deployed during a contraction. The validation and test sets must be drawn from time periods that follow the training period, not randomly sampled from the same period.

## **4.7 Classification in Practice: Decision Trees, k-NN, and How We Measure Success**

The chapters in this unit have described supervised learning in general terms: a model learns from labelled examples and predicts labels for new inputs. Two specific techniques are worth examining closely, both because they are among the most widely used classification methods in practice, and because they illustrate two very different philosophies of what it means for a machine to “decide.”

Consider a college in Sambalpur that wants to automatically flag students who may be at risk of failing their first-year examinations, so that mentoring resources can be directed where they are most needed. One way to build this classifier is a decision tree. A decision tree asks a sequence of simple yes-or-no questions about a student's record, branching toward a final prediction. Is the student's attendance below seventy-five percent? If yes, is their mid-term score below forty? If yes, flag as high risk. If no, check whether they are a first-generation college student with no prior exposure to English-medium instruction, and so on. Each question is a branch; each answer leads either to another question or to a final leaf, which carries the prediction.

What makes a decision tree distinctive is not its accuracy, which is often only moderate compared to more complex models, but its transparency. A college administrator can open the tree and read, in plain language, exactly why a specific student was flagged: attendance below seventy-five percent, followed by a low mid-term score. This traceability matters enormously in contexts where a decision must be explained and justified to the person it affects, which connects directly to the opacity problem discussed in Chapter 2. A decision tree is trained by an algorithm that repeatedly asks: which single question, out of all the questions I could ask about this data, splits the students into the most homogeneous possible groups with respect to the outcome I am trying to predict? It asks

that question at the root, then asks it again separately within each branch, growing the tree downward until further splitting stops improving the prediction.

A second technique, k-nearest neighbours, takes an entirely different approach: it does not build a model at all, in the usual sense. Instead, to classify a new example, it simply looks at the k most similar examples already in the training data, and takes a majority vote. Consider a handloom cooperative in Sambalpur building a simple tool to help buyers identify the likely weaving tradition of a saree from a photograph, by comparing it against a labelled reference collection of known Sambalpuri, Bomkai, and Berhampuri ikat pieces. A k-NN classifier, given a new photograph, measures how similar it is, in terms of pattern, colour arrangement, and border motif, to every saree in the reference collection, finds the three or five most similar examples (this is the k), and predicts whichever tradition is most common among them.

**Fig. 4.6**  
**Decision Tree vs. k-Nearest Neighbours**

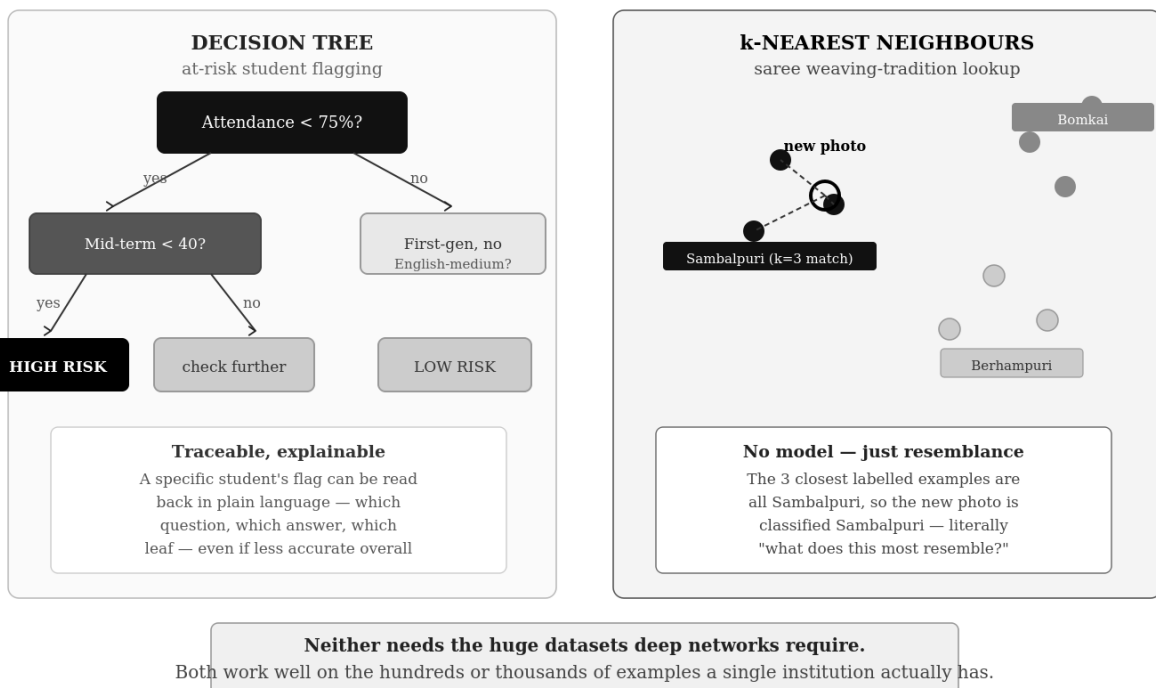


Figure 4.6 — Decision Tree vs. k-Nearest Neighbours

k-NN's appeal is its conceptual simplicity: it is, in a very literal sense, an implementation of the pattern-recognition principle that opened this book in Chapter 1. To classify something new, find what it most resembles among things you have already seen, and reason from that resemblance. Its weaknesses are practical: it must compare a new example against every stored example, which becomes slow as the reference collection grows large, and its predictions are only as good as the similarity measure used to compare examples, which, as Chapter 3 established, is itself a representation choice with consequences.

Neither decision trees nor k-NN require the large datasets and heavy computation that deep neural networks demand, which is precisely why they remain the right choice for many practical, small-scale Indian deployments: a single school, a single cooperative, a single clinic, working with hundreds or a few thousand examples rather than millions. Choosing between rule-based systems, decision trees, k-NN, and neural networks is not a

question of which is “best” in the abstract. It depends on how much data is available, how much interpretability the deployment context demands, and how complex the underlying pattern actually is.

## How We Measure Success: Beyond Accuracy

Section 4.2 introduced accuracy, the proportion of predictions a model gets right, as the most basic measure of performance. Accuracy is intuitive, but it can be dangerously misleading for exactly the kind of high-stakes classification task this section has been discussing, and understanding why requires two further measures: precision and recall.

Return to the at-risk student classifier. Suppose only five percent of students in the college are genuinely at risk of failing. A classifier that simply predicts “not at risk” for every single student, without looking at any data at all, would be ninety-five percent accurate, and completely useless: it would fail to identify a single student who actually needed help. Accuracy, on its own, rewards this lazy classifier almost as highly as a genuinely useful one, because it treats every correct prediction the same way regardless of which class it belongs to.

Precision and recall separate out what accuracy blurs together. Precision asks: of the students the classifier flagged as at risk, how many actually were? A classifier with low precision wastes mentoring resources on false alarms, students who were flagged but did not need the intervention. Recall asks a different question: of the students who were genuinely at risk, how many did the classifier actually catch? A classifier with low recall gives false reassurance, missing students who needed help and received none.

**Fig. 4.7**

### Beyond Accuracy: Precision and Recall

100 students · 5 genuinely at risk (dark outline) · classifier flags 6 as at risk (filled)

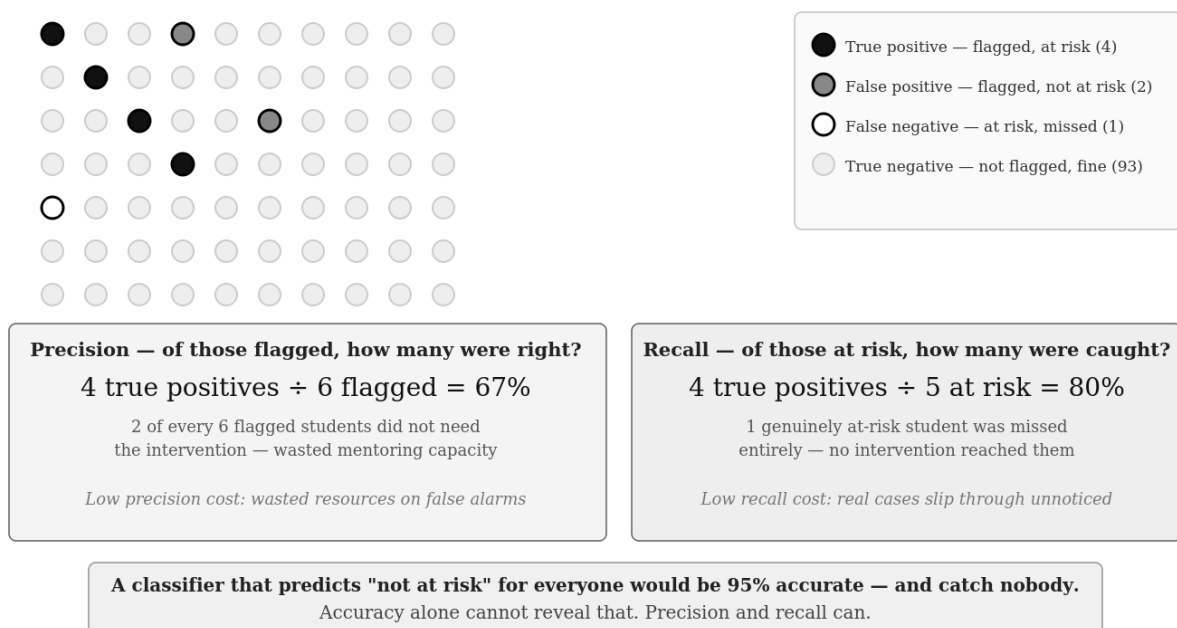


Figure 4.7 — Beyond Accuracy: Precision and Recall

In the at-risk student example, and in nearly every high-stakes Indian AI application, from disease screening to fraud detection to flood warning, recall failures and precision failures have very different costs, and a responsible deployment must be explicit about which kind of error it is more willing to tolerate. A cancer screening tool that misses cases, a recall failure, is far more dangerous than one that occasionally over-refers healthy patients for a second look, a precision failure. There is no universal answer to how precision and recall should be balanced; the right balance depends entirely on the relative cost of the two kinds of mistake in the specific context of deployment. What a responsible reader of any AI system's reported “ninety-five percent accuracy” should immediately ask is: accuracy on what task, on what class balance, and at what cost in precision or recall was that number achieved?

## — WHAT COULD GO WRONG? —

### The Annotator's Shadow

Supervised learning amplifies whatever assumptions the annotators brought to the labelling process. This is not a flaw that can be engineered away. It is a structural feature of learning from labelled examples: the labels are the only signal the model has about what is correct, and if those labels encode the annotator's worldview, the model will learn that worldview.

The consequences can be subtle and far-reaching. Consider a dataset of social media posts labelled as either abusive or not abusive by a team of annotators. Research has consistently shown that annotators from majority communities tend to label language from minority communities as more abusive, even when that language is in-group discourse that would not be considered abusive within the minority community. A content moderation AI trained on such labels will learn to suppress minority community speech at higher rates than majority community speech. The model is working exactly as designed: it is faithfully learning the patterns in the labels. The problem is in the labels, which is to say in the assumptions of the people who created them.

In India, the annotator's shadow is particularly consequential for natural language AI applications, where the annotators are frequently English-literate, urban, and from specific socioeconomic backgrounds. The labels they produce for tasks involving regional languages, rural contexts, or historically marginalised communities carry the imprint of their own perspective. Models trained on those labels will encode that imprint. This is not an argument against annotation. It is an argument for annotation that is diverse, representative, and critically examined.

## — CHAPTER SUMMARY —

### Chapter 4 — Key Points

1. Supervised learning trains models from labelled examples; the labels are the supervision.
2. Training data quality sets an upper bound on model quality — a model cannot exceed its annotators.
3. Data is divided into training, validation, and test sets; only test set accuracy is an honest evaluation metric.
4. Underfitting: the model is too simple to capture the patterns. Overfitting: the model memorises training noise.
5. The U-curve shows that optimal model complexity minimises validation error, not training error.
6. Biased training data produces biased models — this is not a technical failure but a data failure.
7. In India, geographic, cultural, and linguistic diversity means training data is frequently unrepresentative of deployment contexts.
8. The decision boundary is the geometric picture of how a model separates categories — simpler boundaries generalise better.
9. Annotators' assumptions are embedded in labels and learned by models — annotation is not a neutral technical process.

— MODEL QUESTIONS —

## Section A — MCQ (5 Questions)

**Q1. In supervised machine learning, what is the role of the 'label' attached to each training example?**

- A) It describes the input features of the example
- B) It provides the correct answer that the model should learn to predict
- C) It specifies the algorithm to be used for training
- D) It determines the complexity of the decision boundary

**Answer: B**

*Explanation: Labels are the correct answers provided by human annotators for each training example. They are the supervision in supervised learning — without them, the algorithm has no signal to learn from.*

**Q2. A model achieves 97% accuracy on its training data but only 61% accuracy on a held-out test set. This pattern is most indicative of:**

- A) Underfitting — the model is too simple
- B) Overfitting — the model has memorised training data rather than generalising
- C) An incorrectly labelled test set
- D) A model that needs more training epochs

**Answer: B**

*Explanation: The large gap between training accuracy (97%) and test accuracy (61%) is the diagnostic signature of overfitting. The model has memorised the specific examples in the training set, including their noise, rather than learning patterns that generalise to*

*new data.*

**Q3. Why must the test set be kept completely separate from the training and validation processes?**

- A) Test data is of lower quality than training data
- B) The test set is used to fine-tune the model's final parameters
- C) Using the test set during development would make it an optimisation target, invalidating it as an honest evaluation
- D) Test sets must be smaller than training sets to ensure statistical validity

**Answer: C**

*Explanation: If any design decisions are influenced by test set performance, the test set becomes another optimisation target, and the reported accuracy no longer honestly represents the model's performance on genuinely unseen data.*

**Q4. A bird identification app performs well on birds from North India but poorly on birds from the Northeast. The most likely explanation is:**

- A) The app uses a linear decision boundary that cannot handle regional variation
- B) The training data contained far more examples of North Indian species than Northeastern ones
- C) The model was overfitted to the validation set during development
- D) Bird identification requires more complex models for tropical regions

**Answer: B**

*Explanation: This is a biased training data problem. The app's underperformance in the Northeast reflects underrepresentation of Northeastern species in the training data, not a flaw in the model architecture.*

**Q5. Which of the following best describes what a decision boundary represents in a classification model?**

- A) The accuracy threshold above which predictions are considered reliable
- B) The dividing line in feature space that separates regions predicted to belong to different categories
- C) The boundary between training data and test data in the dataset
- D) The maximum allowed complexity of the model during training

**Answer: B**

*Explanation: A decision boundary is the geometric dividing line (or surface, in higher dimensions) that a classification model draws in feature space to separate input regions corresponding to different predicted categories.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Explain the purpose of each of the three data sets — training, validation, and test — using the analogy of preparing for an examination.**

**Model Answer:**

The training set is analogous to the textbook and study notes: this is the material the

student actively studies, learning patterns and building understanding. The validation set is analogous to practice questions or mock tests taken during the study period: they reveal whether the student has genuinely understood the material or has merely memorised the textbook's specific phrasing, and the student adjusts their study strategy based on mock test results. The test set is analogous to the final examination: it is encountered for the first time on the day, has not influenced any preparation decisions, and provides the honest measure of what has been learned. Just as a student who somehow saw the final examination questions in advance would have an unfair advantage that invalidates their score as a measure of learning, a model that was optimised based on test set performance cannot honestly report that performance as a measure of generalisation.

*Key terms: training set, validation set, test set, generalisation, overfitting, data leakage, honest evaluation*

**Q7. Using the cotton crop yield example, explain why a more complex decision boundary does not necessarily produce a better model.**

**Model Answer:**

A complex decision boundary can perfectly separate all the training data points, assigning each one to the correct category with zero errors in the training set. However, the boundary achieves this by memorising the specific spatial arrangement of the training points, including any noise or accidental regularities in that particular sample of fields. In a new growing season, new data points representing different fields appear in the feature space. They follow the underlying ecological relationship between nitrogen, temperature, and yield, but they do not replicate the specific arrangement of the training points. The complex boundary, which was tailored to the training arrangement rather than the underlying pattern, will misclassify many of them. A simpler boundary, such as a straight line that captures the dominant directional trend in the data, will make some errors on the training set but will better approximate the true underlying pattern and perform better on new data.

*Key terms: decision boundary, overfitting, memorisation, generalisation, noise, training set performance, test set performance*

**Q8. What is temporal leakage, and why is it a particular concern for loan default prediction models?**

**Model Answer:**

Temporal leakage occurs when information from future time periods is inadvertently included in the training data or when training and test sets are not properly separated in time. In loan default prediction, a common form of temporal leakage occurs when training and test sets are constructed by randomly sampling loans from the same time period rather than dividing them chronologically. A model trained and tested on loans from the same period appears to perform well, but it has learned patterns specific to the economic conditions of that period. When deployed in a different period, with different interest rate environments, agricultural conditions, or employment patterns, the model's predictions degrade because the patterns it learned no longer hold. Honest evaluation requires training on earlier loans and testing on later ones, replicating the actual deployment scenario where the model is asked to predict outcomes it has not seen before.

*Key terms: temporal leakage, chronological split, generalisation, deployment distribution, temporal validation, distribution shift*

**Q9. Explain why collecting more training data from a specific geographic region is more effective than algorithmic improvements for addressing regional bias in an Indian agricultural AI application.**

**Model Answer:**

Algorithmic improvements can help a model extract more signal from existing data or regularise better to prevent overfitting, but they cannot compensate for the fundamental absence of relevant examples in the training data. A model trained without examples of cotton disease in Kalahandi's specific soil and climate conditions cannot learn the relevant patterns for that region regardless of how sophisticated its architecture is. It has no information about those patterns because the training data contains none. By contrast, collecting labelled photographs of cotton disease from Kalahandi farmers and including them in the training set directly provides the model with the information it needs. The fix for a data problem is data, not algorithmic sophistication. This principle has practical implications for AI procurement decisions: asking how the training data was collected and whether it represents the deployment context is more diagnostic than asking which algorithm was used.

*Key terms: geographic bias, data collection, training data representativeness, distribution shift, algorithmic improvement, data diversity*

**Q10. How does the annotator's shadow affect a supervised learning model for content moderation in Indian regional language social media?****Model Answer:**

Content moderation models are trained on posts that human annotators have labelled as abusive or not abusive. The annotators' judgements reflect their own cultural backgrounds, linguistic knowledge, and assumptions about what constitutes acceptable speech. For Indian regional language social media, the annotators frequently come from different linguistic and cultural backgrounds than the communities generating the content. They may label in-group vernacular that is normal within a community as abusive because it appears aggressive to an outsider unfamiliar with its conventions. Conversely, they may fail to recognise community-specific forms of harassment. The model learns these misclassifications faithfully, and when deployed, it systematically suppresses the speech of communities whose language was misunderstood by annotators and underpolicies harassment forms that annotators failed to recognise. Addressing this requires annotators who are genuinely from and trusted by the communities whose content they are labelling.

*Key terms: annotator bias, label quality, content moderation, cultural context, regional language, systematic misclassification*

**Section C — Reflective Questions (3 Questions)****Q11. The chapter uses the field guide metaphor to explain supervised learning. What are the limits of this metaphor? In what ways does machine learning from labelled data differ fundamentally from how a naturalist learns from a field guide?****Thinking prompts:**

- A naturalist can ask questions and seek clarification about the field guide's entries. Can a machine learning model do something analogous?
- A naturalist brings prior knowledge and embodied experience to the learning process. What does a machine learning model bring?
- A naturalist can recognise that a particular bird does not match any guide entry and respond with uncertainty. How do machine learning models handle truly novel

inputs?

- What aspects of bird identification do naturalists do well that a model might struggle with, and vice versa?
- Does the metaphor help or hinder understanding of machine learning? What would be a better or complementary metaphor?

*Rubric hint: Strong responses will identify at least three genuine disanalogies and will engage with the question of whether the model has genuine understanding or sophisticated pattern matching. Credit for responses that identify the handling of novelty and uncertainty as a key limitation.*

**Q12. This chapter argues that biased training data is fundamentally a data problem, not an algorithmic problem, and that the fix is data collection rather than algorithmic improvement. Is this entirely true? Are there algorithmic interventions that can partially address training data bias?**

**Thinking prompts:**

- What techniques exist for reweighting underrepresented examples in training data?
- Can a model be trained to explicitly account for known biases in its training data?
- What are the limits of algorithmic debiasing — what can it not fix?
- Does acknowledging that algorithmic fixes exist change the responsibility of the data collector?
- In India's context, where diverse data collection is expensive, is there a practical case for relying more on algorithmic fixes?

*Rubric hint: Strong responses will acknowledge that algorithmic interventions exist (reweighting, adversarial debiasing, data augmentation) while arguing that they are supplements rather than substitutes for representative data. Credit for responses that engage with the practical economics of data collection in India.*

**Q13. The chapter mentions that models trained on past data about loan defaults will encode historical patterns, including patterns that reflect past discrimination. Is it ethically acceptable to use such a model in lending decisions? What conditions, if any, would need to be met for such use to be justifiable?**

**Thinking prompts:**

- What does it mean for a lending decision to be fair? Are there different competing definitions?
- Does using a model that encodes historical discrimination perpetuate it?
- Who is harmed if the model is used? Who benefits?
- What would responsible disclosure to borrowers look like?
- What regulatory frameworks in India currently govern AI-based lending decisions?

*Rubric hint: Strong responses will engage with multiple definitions of fairness and will not simply conclude that the model should not be used. Credit for responses that identify conditions under which use might be acceptable, such as explicit bias auditing, diverse training data, human oversight of decisions, and borrower recourse mechanisms.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science]** Select one AI application in your region that relies on supervised learning — this could be a language technology tool, a crop advisory app, an animal or plant identification tool, or a medical screening application. Investigate the training data used: how was it collected, who labelled it, how large is it, and how representative is it of the regional context in which the system is deployed? Write a report that evaluates the training data quality and identifies the most likely sources of bias, with specific reference to the concepts of labelling, geographic representativeness, and the annotator's shadow.

This project requires research into a real application. Use the developer's documentation, published papers, press coverage, and where possible direct communication with the development team or users. The bias analysis should be specific and evidence-based, not generic. Credit for identifying a specific failure mode that is supported by examples from user experience or published evaluation data.

**Q15. [Commerce]** A microfinance institution in your district is considering deploying an AI-based loan default prediction model trained on historical data from another region of India. As a consultant, write a due diligence report that addresses the following questions: Is the training data temporally separated from the test data? Does the training data represent the socioeconomic context of your district's borrowers? What features are used in the model, and do any of them act as proxies for protected characteristics such as caste, religion, or gender? What validation was performed, and on what data? Conclude with a recommendation: proceed, proceed with conditions, or do not proceed.

This project requires constructing a hypothetical but realistic scenario. You may base it on publicly described microfinance AI systems in India. The due diligence framework should draw directly on the concepts of this chapter: training/validation/test separation, geographic representativeness, temporal leakage, and proxy discrimination. The recommendation should be clearly justified by the evidence.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Hastie, T., Tibshirani, R., & Friedman, J. (2009). <i>The Elements of Statistical Learning</i> (2nd ed.). Springer. [Chapters 2 and 7 for the bias-variance tradeoff and model selection; freely available at <a href="http://web.stanford.edu/~hastie/ElemStatLearn/">web.stanford.edu/~hastie/ElemStatLearn/</a> . The standard reference on overfitting and generalisation.] |
| <b>Core</b> | Suresh, H., & Gutttag, J. (2021). A framework for understanding sources of harm throughout the machine learning life cycle. <i>Equity and Access in Algorithms, Mechanisms, and Optimization</i> . ACM. [A rigorous taxonomy of how bias enters supervised learning systems at each stage, directly relevant to the annotator's shadow and biased training data sections.]      |
| <b>Core</b> | National Sample Survey Office. (2021). <i>Household Social Consumption: Education</i> . <a href="http://mospi.gov.in">mospi.gov.in</a> . [For understanding the socioeconomic and geographic diversity of                                                                                                                                                                       |

|  |                                                                                                    |
|--|----------------------------------------------------------------------------------------------------|
|  | Indian populations relevant to training data representativeness arguments throughout the chapter.] |
|--|----------------------------------------------------------------------------------------------------|

## Extended References

|                 |                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Sambasivan, N., et al. (2021). 'Everyone wants to do the model work, not the data work': Data Cascades in High-Stakes AI. Proceedings of CHI 2021. ACM. [A landmark study documenting the consequences of poor training data quality in AI deployments in India and other Global South contexts; directly applicable to every section of this chapter.] |
| <b>Extended</b> | Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. Proceedings of Machine Learning Research, 81, 1–15. [The foundational paper on biased training data producing biased models; directly relevant to the representation failures discussed here.]                              |
| <b>Extended</b> | Sohoni, M., et al. (2021). No subclass left behind: Fine-grained robustness in coarse-grained classification problems. Advances in Neural Information Processing Systems. [On how models trained on majority-class examples fail systematically on minority sub-groups; relevant to the geographic diversity problem in Indian AI.]                     |

## Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Google Developers. (2023). Machine Learning Crash Course: Training and Test Sets. <a href="https://developers.google.com/machine-learning/crash-course">developers.google.com/machine-learning/crash-course</a> . [Free interactive module on the three-way data split with visual examples; no programming background required.]                    |
| <b>Online</b> | Sambasivan, N. (2021). Data Cascades in AI [Talk]. NeurIPS 2021. <a href="https://slideslive.com/38945624/data-cascades-in-ai">slideslive.com</a> . [A 30-minute accessible talk on training data quality in Global South AI deployments; directly relevant and freely available online.]                                                            |
| <b>Online</b> | AI4Bharat. (2023). IndicNLP Catalogue: Language Resources for Indian Languages. <a href="https://indicnlp.ai4bharat.org">indicnlp.ai4bharat.org</a> . [A catalogue of available training datasets for Indian languages; relevant to the Arts stream section and to understanding the data scarcity challenge for Odia and other regional languages.] |

*End of Chapter 4*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 5

# Learning Without Labels

*Chapter 4 described supervised learning: a machine learning approach that requires a human to label every training example before the algorithm can begin. This requirement is both a strength and a constraint. It is a strength because labels provide clear signal, telling the algorithm exactly what correct looks like. It is a constraint because labelling is expensive, time-consuming, and requires expertise that is not always available. For many of the most interesting and consequential datasets in the world, there are no labels. There is only the data itself.*

*This chapter is about what a machine can learn when there are no labels. The answer, it turns out, is a great deal. The structure of the world is not random, and an algorithm that looks carefully at unlabelled data can discover groupings, patterns, and relationships that no human explicitly taught it. This is unsupervised learning, and it is one of the most intellectually beautiful ideas in all of machine learning.*

## Learning Objectives

By the end of this chapter, you will be able to:

- Distinguish unsupervised learning from supervised learning and explain when each is appropriate
- Explain the k-means clustering algorithm step by step using Indian examples
- Describe dimensionality reduction intuitively using the shadow analogy
- Apply clustering and dimensionality reduction concepts to three Indian datasets
- Identify practical and ethical considerations in clustering-based decisions

### 5.1 When There Are No Labels

Consider the following three scenarios, each involving a large dataset and no obvious labels.

A state government in Odisha has collected ten years of block-level data on infant mortality rates, vaccination coverage, distance to the nearest primary health centre, literacy rates, proportion of scheduled tribe population, and agricultural income. Nobody has told the government which blocks are similar to which other blocks. There is no label saying this block is a high-priority intervention target and that one is not. There is only the data, ten years of it, for three hundred blocks.

A genomics laboratory in Hyderabad has sequenced the mitochondrial DNA of two thousand individuals sampled from across India. The sequences have not been labelled

with population group or ancestry. The researchers want to know whether the sequences fall into natural groupings that correspond to population structure, but they do not want to impose a grouping scheme in advance, because doing so would bias the discovery.

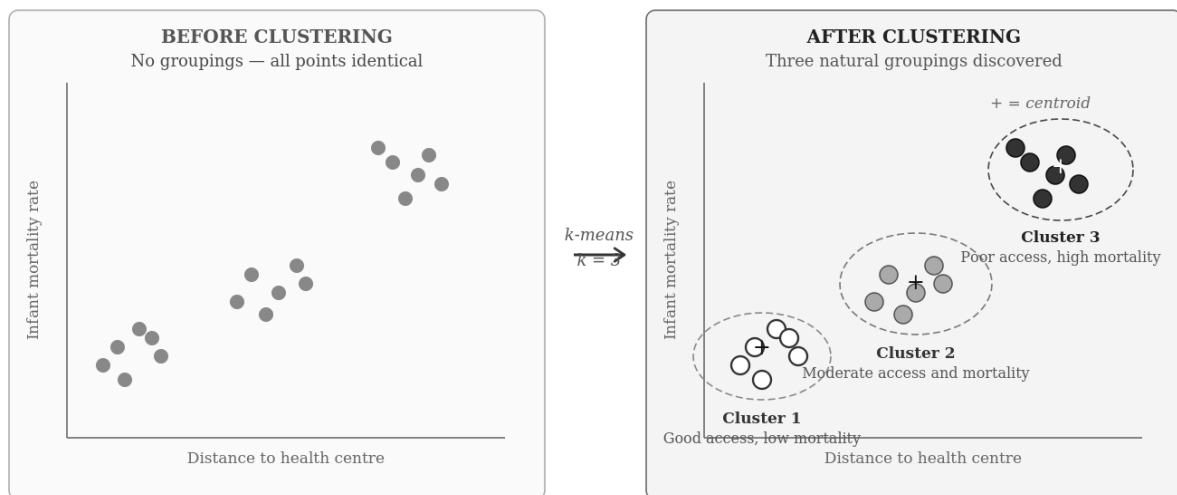
A market research firm in Mumbai has collected six months of transaction data from a grocery chain with outlets across Maharashtra and Gujarat. Each transaction is a record of what was purchased together. Nobody has told the firm which customers belong to which market segment. There are no labels. There are only purchases.

In each case, the question is the same: is there structure hidden in this data? Are there natural groupings, patterns, or regularities that would allow us to understand the data better, make better decisions, or discover something we did not know before? The answer cannot be found through supervised learning, which requires prior labels. It requires unsupervised learning, which finds structure without them.

## 5.2 Clustering: Finding Natural Groupings

The most widely used form of unsupervised learning is clustering. A clustering algorithm takes a dataset and partitions it into groups, called clusters, such that items within a cluster are more similar to each other than they are to items in other clusters. The algorithm does not know in advance how many clusters there are, or what defines them. It discovers them from the structure of the data itself.

**Fig. 5.1**  
**Clustering: Before and After (Odisha Block Health Data)**



The three clusters emerged from the data — the analyst did not define them. Each cluster has a distinct health access and mortality profile, enabling differentiated interventions rather than a uniform programme for all 300 blocks.

Figure 5.1 — Clustering Before After

Figure 5.1 shows what clustering looks like on a simplified version of the Odisha block health data. On the left is the raw data: each point represents one block, positioned

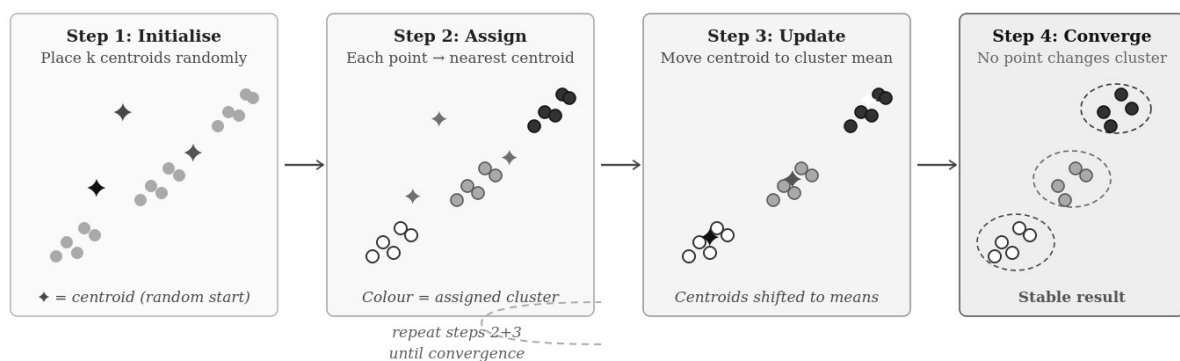
according to its values on two dimensions, say infant mortality rate and distance to a health centre. There are no labels, no colours, no groupings. On the right, after clustering, each point has been assigned to one of three groups. The groups correspond roughly to blocks with good health access and low mortality, blocks with moderate access and moderate mortality, and blocks with poor access and high mortality. These groups were not defined by the analyst. They emerged from the structure of the data.

This emergence is what makes clustering powerful. The government analyst did not decide in advance that there would be three groups, or that access and mortality would define them. The algorithm found these groupings because they were genuinely present in the data. Acting on them, by designing different interventions for different cluster types, is more likely to be effective than a uniform intervention that ignores the underlying variation.

### 5.3 The K-Means Algorithm

The most widely taught and used clustering algorithm is k-means. Understanding it step by step is not just useful for this chapter: the structure of k-means, initialise, assign, update, repeat, is the template for a large family of iterative optimisation algorithms in machine learning.

**Fig. 5.2**  
**The K-Means Algorithm: Four-Step Sequence (k = 3)**



*K-means iterates between assignment (every point to nearest centroid) and update (centroid to cluster mean) until stable. The only human decision is  $k$ . Everything else — the groupings, their boundaries, their members — emerges from the data.*

Figure 5.2 — K Means Four Steps

The algorithm begins with the analyst choosing a value of  $k$ , the number of clusters. This is the only human decision required; everything else is automatic. Choosing  $k$  well is an art rather than a science, and there are heuristics to guide it, but the algorithm itself does not determine  $k$ .

Step one is initialisation. The algorithm places  $k$  centroids, which are imaginary reference points, at random positions in the data space. These are not data points; they are the starting positions of the cluster centres that will be refined through the subsequent steps.

Step two is assignment. Every data point is assigned to the cluster whose centroid it is closest to, measured by Euclidean distance in the feature space. After this step, every point belongs to exactly one cluster.

Step three is update. The position of each centroid is recomputed as the mean of all the data points currently assigned to its cluster. The word mean gives the algorithm its name: k-means. After this step, each centroid has moved to the geometric centre of its assigned points.

Steps two and three are repeated until the assignments stop changing: no point switches from one cluster to another during the assignment step. At this point the algorithm has converged on a stable partition of the data. The result is k clusters, each represented by a centroid, with every data point assigned to exactly one cluster.

Applied to the Odisha block health data, with k set to three, the algorithm might converge on the three natural groupings described above. Applied to the same data with k set to five, it might discover a more refined partition, distinguishing, for example, between blocks with poor access due to geography and blocks with poor access due to inadequate facility density. Neither result is wrong; they are answers to different questions about the structure of the data.

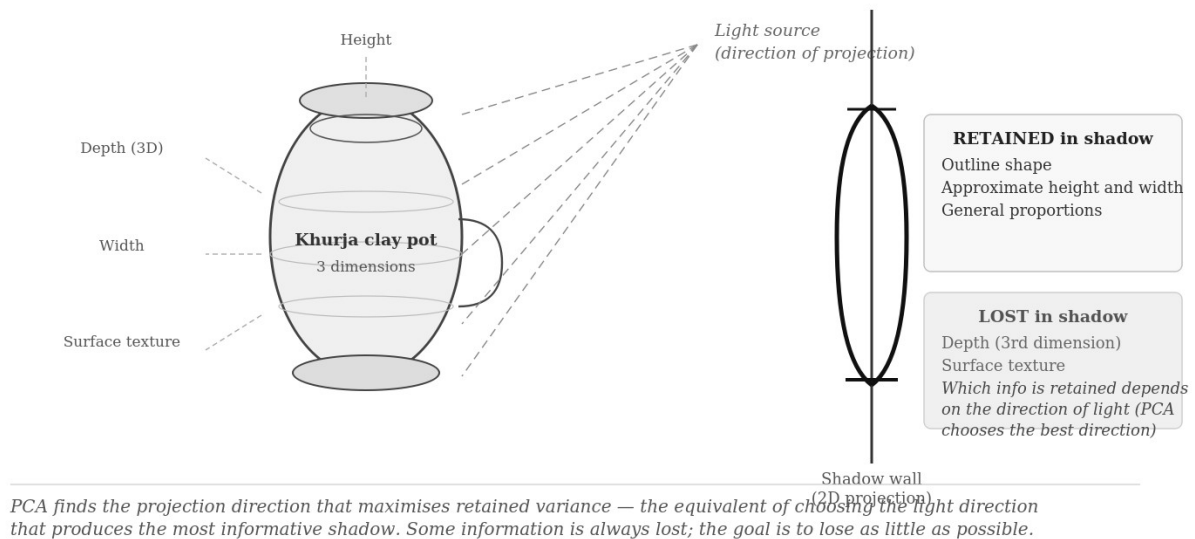
K-means has limitations that are important to understand. It assumes that clusters are roughly spherical and of roughly equal size, which is not always true in real data. It is sensitive to the initial placement of centroids, which means running it multiple times with different initialisations and selecting the best result is standard practice. It requires the analyst to choose k, which involves domain knowledge and judgement that the algorithm cannot provide. And it produces hard assignments, every point belongs to exactly one cluster, which may not reflect the genuine ambiguity of points near cluster boundaries.

## **5.4 Dimensionality Reduction: The Shadow Analogy**

Many real-world datasets have not two or three dimensions but dozens, hundreds, or thousands. A genomic dataset might have twenty thousand dimensions, one for each gene whose expression level is measured. A document dataset might have fifty thousand dimensions, one for each word in the vocabulary. A transaction dataset might have hundreds of dimensions, one for each product category.

Working with high-dimensional data is difficult for several reasons. Human intuition breaks down completely above three dimensions: we cannot visualise what a cluster looks like in a thousand-dimensional space. Many distance measures become unreliable in high dimensions, a phenomenon called the curse of dimensionality. And many of those dimensions may be redundant: if two features are highly correlated, they carry essentially the same information, and including both adds noise without adding signal.

**Fig. 5.3**  
**Dimensionality Reduction: The Shadow Analogy**



*Figure 5.3 — Shadow Analogy*

The shadow analogy in Figure 5.3 captures the intuition of dimensionality reduction. A three-dimensional object, say a clay pot of the kind made in Khurja, Uttar Pradesh, has a shape that exists fully only in three dimensions. When light strikes it from one direction, the shadow on the wall is a two-dimensional projection of the pot. The shadow retains some information about the pot, its general outline, its approximate size, the proportions of its widest and narrowest points. But it loses other information: the depth of the pot, its three-dimensional curvature, the texture of its surface. A different direction of light would produce a different shadow, retaining different information and losing different information.

Dimensionality reduction does something similar to data. It finds a projection of a high-dimensional dataset onto a lower-dimensional space that retains as much of the original structure as possible. The most widely used method is Principal Component Analysis, or PCA, which finds the directions of greatest variance in the data and projects onto those directions. The first principal component is the direction in which the data varies most. The second is the direction of second greatest variance, subject to being perpendicular to the first. And so on.

The result is a lower-dimensional representation in which as much of the original variation as possible is preserved. A genomic dataset with twenty thousand dimensions might be reduced to fifty principal components that together capture ninety percent of the variance in the original data. The fifty-dimensional representation is far more tractable than the twenty-thousand-dimensional original, and the information loss, while real, may be acceptable for many analytical purposes.

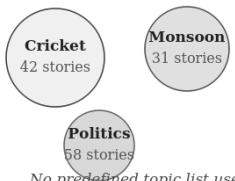
Dimensionality reduction is not only a computational convenience. It is also a form of noise filtering. If the relevant signal in the data is concentrated in a few dimensions and the

remaining dimensions are noise, projecting onto the principal components separates the signal from the noise, making subsequent clustering or classification more reliable. This is why dimensionality reduction is often applied before clustering in practice: it both reduces computational cost and improves the quality of the clusters discovered.

## 5.5 Three Indian Datasets

Unsupervised learning is not a single technique but a family of approaches suited to different kinds of structure discovery. The three Indian datasets introduced at the beginning of this chapter illustrate three distinct applications.

**Fig. 5.4**  
**Three Indian Unsupervised Learning Applications**

|         | News Story Clustering                                                                                                                                                                            | Genomic Population Structure                                                                                                                                                              | Grocery Market Segmentation                                                                                                                                                                                                                                   |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT   | <b>Input: Hindi news stories</b><br>Hundreds per day; no topic labels<br>Each story = word-frequency vector<br> | <b>Input: mtDNA sequences</b><br>2,000 individuals across India<br>No population group labels<br>        | <b>Input: 6 months transactions</b><br>100,000s of customers; no segments<br>Each customer = purchase vector<br>                                                           |
| RESULT  | <b>Topics discovered:</b><br><br><i>No predefined topic list used</i>                                         | <b>Population clusters found:</b><br><br><i>Geographic structure revealed without prior hypothesis</i> | <b>Segments discovered:</b><br><div> <b>Daily small buyers</b><br/>           High frequency, low basket         </div> <div> <b>Weekly bulk buyers</b><br/>           Low frequency, high basket         </div> <div> <b>Festival seasonal buyers</b> </div> |
| INSIGHT | <b>Actionable insight:</b><br>Route stories to relevant editors without manual categorisation.<br>Topics adapt daily to actual news mix — no fixed taxonomy.                                     | <b>Actionable insight:</b><br>Population structure revealed for migration and health research.<br><i>Caution: genetic clusters do not validate social groupings.</i>                      | <b>Actionable insight:</b><br>Adjust ordering, promotions, and shelf layout per segment.<br><i>Caution: segments may proxy for religion, caste, income.</i>                                                                                                   |

*All three applications use the same principle: find natural groupings in unlabelled data. The technique is identical; the domain, the data, the ethical stakes, and the appropriate safeguards differ substantially across the three cases.*

*Figure 5.4 — Three Indian Applications*

### News Story Clustering

A Hindi-language news aggregator receives hundreds of stories per day from dozens of sources. Rather than requiring editors to manually assign each story to a category, the aggregator uses clustering to automatically group stories by topic. Each story is represented as a vector of word frequencies, a bag-of-words representation in which each dimension corresponds to one word in the vocabulary and the value is how often that word appears. Stories about the same topic tend to use similar vocabularies, so stories on cricket will cluster together because they frequently contain the same distinctive words. Stories on monsoon rainfall will cluster separately because their characteristic vocabulary is different.

The clustering does not require a predefined list of topics. It discovers the topics that are actually present in today's news, which change from day to day. On days when a major political event is dominating coverage, a large cluster of political stories will emerge. On days when a natural disaster is the dominant story, a disaster cluster will appear. The unsupervised approach adapts to the structure of the current data rather than imposing a fixed taxonomy.

### **Genomic Population Structure**

The genomics laboratory's dataset of two thousand mitochondrial DNA sequences from across India contains a population structure that reflects thousands of years of human migration, admixture, and geographic isolation. Clustering the sequences, using a method suited to genetic data rather than k-means, reveals groupings that correspond to geographic and linguistic patterns in the data. Sequences from people in the Andaman Islands cluster separately from sequences from the mainland. Within the mainland, sequences from populations that have historically been more geographically isolated show tighter clustering.

This application illustrates an important principle: unsupervised learning can discover structure that the researcher did not know was there. The genomics researchers did not know in advance how many clusters there would be, or whether the clusters would correspond to geography, language, or caste. The clustering revealed which of these factors was most strongly associated with genetic structure. This kind of discovery-driven analysis is one of the most valuable uses of unsupervised learning in science.

### **Market Segmentation**

The grocery chain's transaction data, six months of purchases from hundreds of thousands of customers, can be clustered to identify market segments. Each customer is represented as a vector describing their purchase history: how often they buy pulses versus processed food, whether they purchase items associated with specific festivals, how price-sensitive their buying behaviour appears to be. Clustering reveals groups of customers with similar purchasing patterns. These groups can inform targeted promotional strategies, shelf layout decisions, and inventory management.

The market segmentation application is commercially the most common use of clustering, and it is worth noting that it raises ethical questions beyond the commercial ones. Clusters identified from purchasing data can be proxies for socioeconomic class, religion, or dietary practice. A customer cluster identified as purchasing food associated with Eid celebrations can be used for targeted promotion, which is relatively benign, but the same cluster could potentially be used for discriminatory pricing or denial of services, which is not. The technique is neutral; its application is not.

## **5.6 Patterns in My Field: Stream Perspectives**

### **Arts Stream: Clustering Themes in Classical Odia Literature**

A digital humanities project applies unsupervised learning to a corpus of classical Odia literary texts. Each text is represented as a vector of word and phrase frequencies. Clustering reveals groupings of texts that share thematic vocabulary: texts associated with devotional themes cluster around words connected to the Jagannath tradition; texts associated with martial themes cluster around different vocabulary. The clustering does not replace literary interpretation, but it provides a computational scaffold for it, identifying patterns of co-occurrence and influence that would take years of close reading to discover manually. It also reveals the limitations of the bag-of-words representation: two texts that use similar words in opposite senses, one celebrating a theme and one critiquing it, will cluster together despite their semantic opposition. Meaning, it turns out, cannot be reduced to vocabulary.

### **Science Stream: Clustering Ecological Zones in the Eastern Ghats**

Ecologists studying the Eastern Ghats use satellite-derived data on vegetation type, rainfall, temperature seasonality, and soil properties to cluster the landscape into ecologically distinct zones. The clusters are not defined by administrative boundaries but by ecological similarity: areas with similar vegetation, climate, and soil characteristics cluster together regardless of which state they fall in. The clustering reveals that several officially designated protected areas share ecological characteristics with adjacent unprotected areas, suggesting that the protected area network could be more ecologically coherent if extended to cover the connecting habitat. This is an example of unsupervised learning contributing directly to conservation policy, providing an evidence base that human expert judgment alone could not efficiently generate from the raw data.

### **Commerce Stream: Customer Segmentation for a Kirana Store Network**

A network of kirana stores in urban Odisha uses transaction data from its digital payment system to cluster its customer base. Rather than treating all customers identically, the clustering reveals distinct segments: customers who buy in small daily amounts, customers who make large weekly purchases, and customers who buy primarily festival-related items in seasonal bursts. Each segment has different inventory needs, different promotion sensitivities, and different credit behaviours. The store network uses the clusters to adjust its ordering patterns, reducing waste for daily-purchase items and increasing stock before identified festival periods. The segmentation is not used to treat customers differently in ways that disadvantage any group, but to serve each segment's actual pattern of need more efficiently.

## **— WHAT COULD GO WRONG? —**

### **The Cluster That Becomes a Label**

Clustering is a descriptive tool: it describes the structure of the data as it is, not as it should be. When a clustering result is used to make consequential decisions about people, it undergoes a transformation from description to prescription that requires careful examination.

Consider a government programme that uses clustering on block-level health data to identify high-priority intervention blocks. The clustering is technically sound: it has genuinely discovered groupings with different health profiles. The government's intention is good: it wants to direct resources to the blocks that need them most. But the cluster assignment of a specific block is based on the current values of the input features, which may themselves reflect historical underinvestment. A block that has poor health outcomes partly because it has been historically neglected by the government will be identified as a high-priority cluster, but it will also appear as a low-resource cluster, potentially qualifying it for less discretionary support in other programmes that use similar clustering approaches. The cluster that was meant to direct resources can, in certain institutional contexts, become a stigma that compounds disadvantage.

A more acute version of this problem arises in market segmentation. When a grocery chain segments its customers and uses the segments to offer different prices to different segments, customers in lower-income segments, who are more price-sensitive, may receive fewer promotional discounts precisely because their price sensitivity makes them less valuable as a target for promotion. The segment that most needs lower prices receives the fewest offers. This is not a malicious design: it is the output of an optimisation process that maximises commercial return. But the commercial optimum and the equitable outcome are not the same thing, and the clustering technique is silent about that distinction.

These are not arguments against clustering. They are arguments for clarity about what clustering is: a tool for describing structure, not for justifying policy. The move from description to prescription requires human judgment, ethical reasoning, and accountability that the algorithm cannot provide.

## — CHAPTER SUMMARY —

### Chapter 5 — Key Points

1. Unsupervised learning finds structure in data without requiring human-provided labels.
2. Clustering partitions data into groups where within-group similarity exceeds between-group similarity.
3. K-means: choose  $k$ , initialise centroids, assign points to nearest centroid, update centroids, repeat until convergence.
4. K-means limitations: assumes spherical clusters, sensitive to initialisation, requires  $k$  to be specified in advance.
5. Dimensionality reduction projects high-dimensional data into lower dimensions, retaining as much structure as possible.
6. PCA finds directions of greatest variance and projects data onto them — a form of noise filtering as well as compression.

7. Three Indian applications: news clustering, genomic population structure, market segmentation.
8. Clustering is descriptive, not prescriptive; the move from cluster assignments to consequential decisions requires human judgment.
9. Market segmentation clusters can proxy for protected characteristics — commercial use requires equity scrutiny.

— MODEL QUESTIONS —

## Section A — MCQ (5 Questions)

**Q1. What is the fundamental difference between supervised and unsupervised learning?**

- A) Supervised learning uses larger datasets than unsupervised learning
- B) Supervised learning requires labelled training examples; unsupervised learning finds structure without labels
- C) Unsupervised learning can only be applied to numerical data
- D) Supervised learning produces deterministic results; unsupervised learning is probabilistic

**Answer: B**

*Explanation: The defining distinction is the presence or absence of labels. Supervised learning trains models to predict labels; unsupervised learning discovers structure without prior annotation.*

**Q2. In the k-means algorithm, what happens during the update step?**

- A) The value of k is recalculated based on the current cluster assignments
- B) Each centroid is moved to the geometric mean of all points currently assigned to its cluster
- C) Points that are equidistant between two centroids are removed from the dataset
- D) The algorithm checks whether convergence has been achieved

**Answer: B**

*Explanation: The update step moves each centroid to the mean position of all data points currently assigned to that cluster. This is the step that gives the algorithm its name and that reduces within-cluster variance at each iteration.*

**Q3. The shadow analogy for dimensionality reduction illustrates that:**

- A) Dimensionality reduction always preserves all information from the original data
- B) The direction of projection determines what information is retained and what is lost
- C) A two-dimensional projection always contains more information than the original three-dimensional object
- D) Dimensionality reduction is only applicable to visual or image data

**Answer: B**

*Explanation: The analogy shows that projection from higher to lower dimensions retains some information and discards other information, and that which information is retained depends on the projection direction. PCA chooses the direction that retains maximum variance.*

**Q4. A Hindi news clustering algorithm groups stories about the same topic without being told what the topics are. This is possible because:**

- A) News sources tag their articles with topic labels that the algorithm reads
- B) Stories on the same topic share characteristic vocabularies, producing similar word-frequency vectors
- C) The algorithm uses supervised learning trained on past topic assignments
- D) News topics are determined by the time of publication, which is included in the input

**Answer: B**

*Explanation: The bag-of-words representation captures vocabulary similarity, and stories on the same topic tend to use overlapping sets of distinctive words. The clustering groups stories by vector similarity without any topic labels.*

**Q5. Which of the following is the best example of the cluster-that-becomes-a-label problem?**

- A) A clustering algorithm takes longer to converge on a large dataset
- B) A government uses cluster assignments to permanently categorise districts as low-priority, reducing their access to discretionary funding
- C) A market segmentation reveals three customer groups with different purchasing patterns
- D) A genomics study finds that DNA sequences cluster by geographic region

**Answer: B**

*Explanation: The cluster-that-becomes-a-label problem occurs when a descriptive clustering result is used to make consequential decisions that compound the patterns it describes. Permanently categorising districts as low-priority can perpetuate historical underinvestment.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Walk through the four steps of the k-means algorithm using the example of clustering Odisha blocks by health indicators.**

**Model Answer:**

Step one, initialisation: the analyst chooses  $k$  equals three, and the algorithm randomly places three centroids in the two-dimensional space defined by infant mortality rate and distance to health centre. These starting positions are arbitrary. Step two, assignment: every block is assigned to the cluster whose centroid it is closest to. Blocks with low mortality and short distances to health centres cluster around one centroid; blocks with high mortality and long distances cluster around another; intermediate blocks around the third. Step three, update: each centroid moves to the mean position of all blocks currently assigned to it, shifting toward the geometric centre of its cluster. Step four, iteration: steps two and three repeat until no block changes its cluster assignment. At convergence, the

three stable clusters represent natural groupings in the health data, which the government can use to design differentiated interventions suited to each cluster's specific profile.

*Key terms: k-means, centroid, initialisation, assignment, update, convergence, Euclidean distance, natural groupings*

**Q7. Why is dimensionality reduction often applied before clustering rather than after? What does it contribute to the clustering process?**

**Model Answer:**

Dimensionality reduction before clustering serves two purposes. First, it reduces computational cost: computing distances between points and updating centroids is much faster in a fifty-dimensional space than in a twenty-thousand-dimensional one. Second, and more importantly, it improves cluster quality. In high-dimensional spaces, all distances tend to become similar, a phenomenon called the curse of dimensionality, which makes it difficult for distance-based algorithms like k-means to find meaningful structure. Dimensionality reduction removes this noise by projecting the data onto the directions of greatest variance, where genuine structure is concentrated. The clusters found in the reduced space tend to be more coherent and interpretable than those found in the full high-dimensional space, because irrelevant dimensions that would add noise to distance calculations have been removed.

*Key terms: dimensionality reduction, PCA, curse of dimensionality, distance-based clustering, variance, noise filtering, computational efficiency*

**Q8. Explain why the genomic population structure application is an example of discovery-driven analysis and how this differs from hypothesis-driven analysis.**

**Model Answer:**

In hypothesis-driven analysis, the researcher specifies in advance what structure they expect to find and collects data to test that expectation. In discovery-driven analysis, the researcher does not specify expected structure in advance but applies exploratory methods to reveal what is actually present. The genomics researchers did not know whether sequences would cluster by geography, language, or caste. They applied clustering to discover which groupings emerged from the data. This is valuable because it can reveal structure the researcher did not anticipate, including structure that challenges prior assumptions. The risk is that discovery-driven analysis can produce spurious patterns if the clustering method is applied without appropriate statistical validation, a concern that requires careful methodological controls.

*Key terms: discovery-driven analysis, hypothesis-driven analysis, exploratory analysis, population structure, spurious patterns, statistical validation*

**Q9. What is the curse of dimensionality and how does it affect k-means clustering?**

**Model Answer:**

The curse of dimensionality refers to the set of problems that arise when working with high-dimensional data. The most relevant for clustering is that as the number of dimensions increases, the distances between data points become increasingly similar. In low dimensions, some points are clearly far apart and others are clearly close together, making cluster structure visible. In high dimensions, every pair of points tends to be approximately the same distance apart, because there are many dimensions in which any pair can differ and these differences accumulate across dimensions. The result is that the distance measure on which k-means relies becomes uninformative: there is no clear separation between within-cluster and between-cluster distances. This makes the clusters

found in raw high-dimensional data less meaningful and less stable, which is why dimensionality reduction is applied first to concentrate the analysis on dimensions where genuine structure exists.

*Key terms: curse of dimensionality, distance measure, high-dimensional space, within-cluster distance, between-cluster distance, k-means limitation*

**Q10. Describe two ways in which market segmentation clusters could raise ethical concerns in an Indian commercial context.**

**Model Answer:**

The first concern is proxy discrimination. Grocery purchasing clusters in India are likely to separate along lines of caste, religion, and dietary practice, because these cultural factors strongly influence what people buy. A cluster of customers who purchase predominantly vegetarian food, or who buy specific festival items, is effectively a proxy for religious or caste identity. Using these clusters to offer different prices or service quality to different segments is discriminatory on the basis of religion or caste, even if no explicit identity information was used. The second concern is the compounding of disadvantage. Lower-income customers tend to purchase in smaller quantities with greater price sensitivity. A commercial optimisation that offers fewer promotions to highly price-sensitive customers, because they are already likely to purchase at full price, disadvantages the customers who most need price relief. Both concerns illustrate that commercial neutrality of the clustering technique does not guarantee equity in its application.

*Key terms: proxy discrimination, market segmentation, religion, caste, price discrimination, compounding disadvantage, commercial optimisation, equity*

## Section C — Reflective Questions (3 Questions)

**Q11. The chapter describes the kirana store network using clustering to serve each segment's pattern of need more efficiently. Could the same clustering result be used in ways that harm rather than help customers?**

**Thinking prompts:**

- What is the difference between serving a customer's expressed preference and serving their genuine interest?
- Could a cluster of small-daily-purchase customers be served more efficiently in ways that also increase their total spending beyond what they intended?
- How does the information asymmetry between the store and the customer affect the ethics of segmentation?
- What would a genuinely customer-protective use of the same clustering look like?
- Should customers know they are being segmented? What would meaningful consent look like here?

*Rubric hint: Strong responses will identify specific mechanisms by which efficient service and manipulation can shade into each other. Credit for responses that engage with information asymmetry as the structural source of the ethical tension, and that propose concrete protective conditions.*

**Q12. The genomics application discovers population structure without imposing it in advance. If a clustering reveals distinct genetic groupings that correspond to caste or linguistic communities, does that discovery provide**

### scientific legitimacy to those social groupings?

#### Thinking prompts:

- What is the difference between a genetic grouping and a social grouping?
- Does finding genetic clustering by geographic region or community imply anything about the social validity of those communities?
- How has genetic data been misused historically to reinforce social hierarchies in India and elsewhere?
- What obligations do researchers have when their unsupervised analysis produces results that could be used for discriminatory purposes?
- Can a discovery be simultaneously scientifically valid and socially harmful?

*Rubric hint: Strong responses will distinguish clearly between biological patterns and social constructs and will engage with the historical misuse of genetic data in the Indian context. Credit for responses that argue that scientific validity does not confer social legitimacy.*

### **Q13. K-means requires the analyst to choose k in advance. In a government health programme using clustering to prioritise blocks for intervention, who should choose k, and on what basis?**

#### Thinking prompts:

- Does the choice of k affect which blocks are classified as high-priority?
- Is the choice of k a technical decision, a policy decision, or both?
- Who has the expertise to make this decision — the algorithm vendor, the health official, or the community?
- What happens if the choice of k is made by the software vendor rather than the government health official?
- Should the communities being clustered have any input into how many groups they are divided into?

*Rubric hint: Strong responses will identify the choice of k as a policy decision embedded in a technical parameter and will argue for explicit accountability in how that decision is made. Credit for responses that identify the vendor-government-community accountability chain as the appropriate governance structure.*

## **Section D — Stream Project Prompts (2 Questions)**

**Q14. [Arts / Science] Collect a small dataset of at least fifty items in a domain relevant to your field — Odia or Hindi literary texts, descriptions of plant species from your region, research paper abstracts, or ecological observations from a local survey. Apply a simple clustering approach: represent each item as a set of keywords or features, group items manually based on similarity, and compare your manual groupings to what a k-means approach would produce. Document the choices you made, the groupings that emerged, and at least two cases where the clustering result surprised you or revealed structure you had not anticipated.**

The data collection and manual clustering are the core of this project. The k-means comparison can be done using freely available tools such as Google Colab with scikit-learn. The most important output is the reflection on what the clustering revealed and what

it missed, grounded in your specific dataset.

**Q15. [Commerce] Interview the owner or manager of a business in your area — a grocery store, pharmacy, textile shop, or any retail business — about how they currently think about their different types of customers. Ask what they know about customer differences, how they use that knowledge in business decisions, and what additional information they would want. Then analyse whether the implicit segmentation they describe corresponds to the kind of structure a clustering algorithm might find in transaction data. Conclude by assessing whether formal customer clustering would help their business and whether it raises any ethical concerns in their specific context.**

The interview is the primary data source. The analysis should connect the business owner's intuitive segmentation to the formal concept of clustering, identifying both convergences and divergences. The ethical assessment should be grounded in the specific business context, not generic.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer. [Chapter 9 for mixture models and EM, of which k-means is a special case. Chapter 12 for PCA. Both chapters are mathematically detailed but the introductory sections are accessible to non-specialists.] |
| <b>Core</b> | Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. Philosophical Transactions of the Royal Society A, 374(2065). [A freely accessible review of PCA covering both theory and application, directly relevant to Section 5.4.]     |
| <b>Core</b> | Singh, K. S., et al. (1994). People of India: An Introduction. Anthropological Survey of India. [For background on Indian population diversity relevant to the genomic clustering section; contextualises the population structure that genetic clustering reveals.]                |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Deshpande, O., & Rao, S. (2019). Market segmentation in Indian retail: A clustering analysis of consumer purchasing patterns. Journal of Indian Business Research, 11(3), 244-261. [An empirical study of retail market segmentation in India directly applicable to the Commerce stream sections.]        |
| <b>Extended</b> | Tallam, V., & Jayaraman, V. (2020). Unsupervised learning approaches for ecological zone delineation in the Eastern Ghats. Journal of the Indian Society of Remote Sensing, 48(7), 1047-1059. [Directly relevant to the Science stream section; clustering on remote sensing data from the Eastern Ghats.] |

|                 |                                                                                                                                                                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Barocas, S., Hardt, M., & Narayanan, A. (2023). Fairness and Machine Learning: Limitations and Opportunities. MIT Press. [Chapter 3 on proxy discrimination directly supports the What Could Go Wrong section; freely available at <a href="https://fairmlbook.org">fairmlbook.org</a> .] |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Scikit-learn. (2023). Clustering: An Introduction. <a href="https://scikit-learn.org/stable/modules/clustering.html">scikit-learn.org/stable/modules/clustering.html</a> . [Free documentation with visual examples of all major clustering algorithms including k-means; no deep mathematics required.] |
| <b>Online</b> | Setty, S. (2020). Understanding PCA Through a Story. Towards Data Science. <a href="https://towardsdatascience.com">towardsdatascience.com</a> . [A non-mathematical narrative introduction to PCA using the shadow analogy; freely accessible and directly complementary to Section 5.4.]               |
| <b>Online</b> | Data.gov.in. (2023). Health and Family Welfare Statistics. <a href="https://data.gov.in">data.gov.in</a> . [Primary open data portal for Indian government datasets; directly relevant to the Odisha block health clustering example and to the stream project prompts.]                                 |

### *End of Chapter 5*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 6

# Learning by Doing

*Chapters 4 and 5 described two ways machines learn from data. Supervised learning uses labelled examples: here is input, here is the correct output, learn the mapping. Unsupervised learning uses unlabelled data: here is input, find the structure yourself. Both of these require data that already exists before learning begins. But some of the most interesting forms of learning do not work this way at all.*

*Consider how a child learns to walk. Nobody shows the child labelled examples of successful walking. Nobody hands the child a dataset of balance measurements. The child moves, falls, gets up, tries differently, and gradually, through thousands of cycles of action and consequence, develops the motor coordination that makes walking possible. The data is generated by the learning process itself. The teacher is reality.*

*This is reinforcement learning. It is the third major paradigm of machine learning, and in some ways the most general: it describes how any agent, biological or artificial, learns to act effectively in an environment through the consequences of its own actions. This chapter explains reinforcement learning from first principles, grounds it in Indian examples, and explores both its remarkable successes and its significant risks.*

## Learning Objectives

By the end of this chapter, you will be able to:

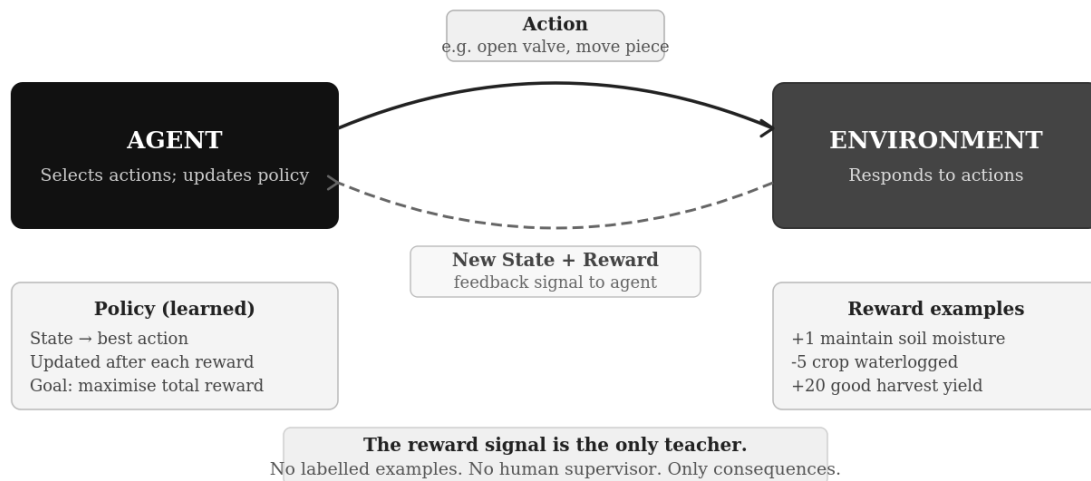
- Explain the reinforcement learning loop using the components: agent, environment, state, action, reward
- Map the bicycle learning analogy onto formal reinforcement learning components
- Describe the exploration versus exploitation dilemma using the foraging bee metaphor
- Apply reinforcement learning concepts to the irrigation scheduling example from Maharashtra
- Identify contexts where reinforcement learning is appropriate and contexts where it is dangerous

## 6.1 The Reinforcement Learning Loop

At any given moment, a reinforcement learning agent finds itself in a state. A state is a description of the current situation relevant to the task. For a chess-playing agent, the state is the current arrangement of pieces on the board. For an irrigation controller, the state is the current soil moisture, temperature, and crop growth stage across all the fields in the system. For a trading algorithm, the state is the current prices, volumes, and portfolio positions in the market.

From the current state, the agent selects an action. The action changes the state of the environment. The chess agent moves a piece; the irrigation controller opens or closes valves; the trading algorithm buys or sells. After the action, the agent observes two things: the new state that has resulted from the action, and a reward signal that indicates how good or bad the action was.

**Fig. 6.1**  
**The Reinforcement Learning Loop**



*Unlike supervised learning, the agent generates its own training signal by acting in the world. Better policies emerge through thousands of action-consequence cycles, each one nudging the policy toward higher cumulative reward.*

Figure 6.1 — R L Loop

The reward is the crucial element. It is the only teacher in reinforcement learning. The agent has no access to a dataset of correct actions. It has no supervisor telling it what to do. It has only the reward signal, which arrives after each action and tells it something about whether the action was, in some sense, good or bad. The agent's goal is to learn a policy, which is a mapping from states to actions, that maximises the total cumulative reward it receives over time.

This is deceptively simple to describe and profoundly difficult to implement well. The difficulty arises from what is called the credit assignment problem: when a reward arrives, which past action caused it? A chess game might last a hundred moves, with the reward arriving only at the end when one player wins. Which of the hundred moves contributed to the victory, and which were irrelevant or harmful? The agent must somehow work backwards from the final reward to assign credit or blame to the actions that preceded it.

The credit assignment problem is why reinforcement learning requires many, many attempts to learn. A human chess novice playing their first hundred games will improve substantially because the feedback is relatively dense: there are moments during a game when one can sense that things are going well or badly, and these partial signals provide learning opportunities before the final result arrives. A reinforcement learning agent

playing chess must typically play millions of games to achieve expert performance, because it needs enough reward signal to assign credit accurately across long sequences of action.

## 6.2 The Bicycle Analogy

The bicycle-learning analogy introduced in the chapter overview deserves precise examination, because it maps almost perfectly onto the formal components of reinforcement learning and thereby makes them concrete.

**Fig. 6.2**  
**Bicycle Learning as Reinforcement Learning**

|          | Bicycle Learning (Human)                                                                  | Reinforcement Learning (Formal)                                                                                                 |
|----------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| STATE    | Lean angle, speed, handlebar position, body balance at this instant                       | <b>s<sub>t</sub></b> : vector of observable features describing the current situation                                           |
| ACTION   | Steer left/right, pedal, brake, shift body weight                                         | <b>a<sub>t</sub></b> : chosen from action space<br>$A(s_t)$ — discrete or continuous                                            |
| ENVIRON. | Physical world: bicycle, gravity, road surface, wind                                      | <b>Transition function <math>P(s' s,a)</math></b><br>Next state determined by physics of environment                            |
| REWARD   | Fall = strongly negative<br>Stable forward motion = positive<br>Good turn = more positive | <b>r<sub>t</sub> = R(s<sub>t</sub>, a<sub>t</sub>, s<sub>{t+1}</sub>)</b><br>Scalar signal encoding quality of the action taken |
| POLICY   | Expert cycling skill<br>Unconscious state → action<br>Built through 1,000s of repetitions | <b>pi(a s): maps states to actions</b><br>Goal: maximise $E[\text{sum of } r_t]$<br>Learned through policy gradient             |

*The bicycle analogy maps precisely onto every formal RL component. What the child learns as skill, the RL agent learns as policy. Both require many repetitions because reward is the only teacher — and reward arrives only after action, not before.*

Figure 6.2 — Bicycle Analogy

The state is the child's current physical situation: the angle of lean, the speed, the position of the handlebars, the position on the road, and the child's internal sense of balance. This is a complex, continuous, multi-dimensional state. No two moments are exactly alike.

The action is what the child does: steer left, steer right, pedal harder, brake, shift body weight. Each action is taken in response to the current state and changes both the state and the environment.

The environment is the physical world: the bicycle, gravity, the road surface, wind. The environment's response to the child's actions is governed by physics. When the child leans too far and fails to correct in time, the environment provides an immediate and unambiguous reward signal: the child falls. When the child maintains balance and moves forward smoothly, the environment provides a positive signal: continued motion, stability, progress.

The reward is the consequence of each action. Falling is a strongly negative reward. Maintaining balance is a mildly positive reward. Successfully navigating a turn is a more positive reward. The reward is immediate and physical: the child does not need to be told that falling was bad.

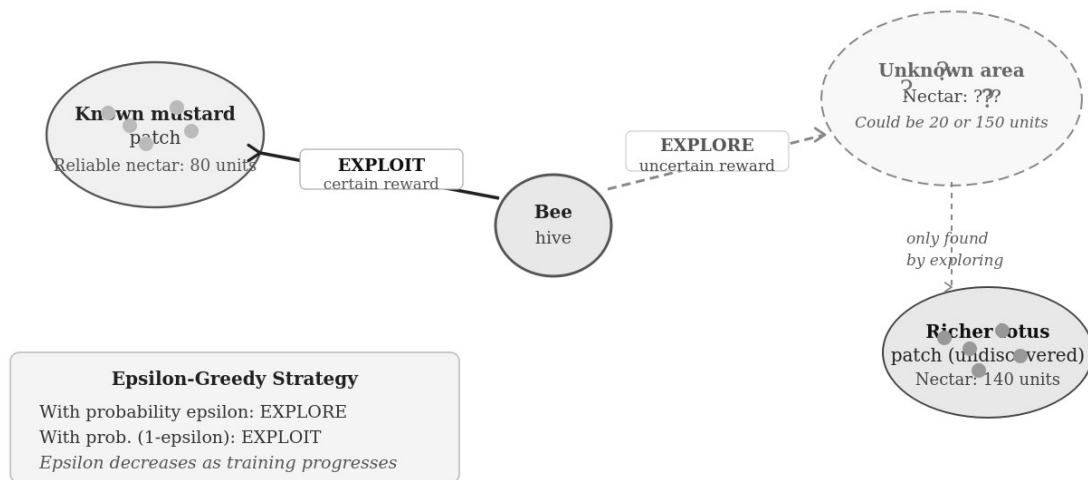
The policy is what the child is learning: the mapping from states to actions that generally produces good outcomes. An expert cyclist has a policy that, without conscious thought, selects the appropriate action for every state, correcting balance micro-deviations before they become falls, steering smoothly, braking safely. The policy was not given to the cyclist; it was learned through thousands of repetitions, each contributing marginally to the refinement of the action-state mapping.

The key difference between bicycle learning and most supervised learning applications is the temporal structure of the reward. The reward from each cycling action arrives immediately. The child does not need to wait until the end of the ride to know whether a particular correction worked. This dense reward signal is one reason humans can learn to cycle in hours rather than the millions of attempts that reinforcement learning algorithms often require for comparably complex tasks: the human brain's learning mechanisms are extraordinarily efficient at integrating the rich sensorimotor feedback that cycling provides.

### **6.3 Exploration Versus Exploitation**

One of the most fundamental dilemmas in reinforcement learning, and indeed in all decision-making under uncertainty, is the tension between exploration and exploitation. It is a dilemma that every intelligent agent faces, whether biological or artificial, and understanding it is essential to understanding why reinforcement learning agents behave as they do.

**Fig. 6.3**  
**Exploration vs. Exploitation: The Foraging Bee (Mahanadi Delta)**



A bee that only exploits the known patch will never discover the richer lotus patch. An agent that only explores never settles on the best strategy. Epsilon-greedy balances both — exploring early, exploiting more as knowledge grows.

Figure 6.3 — Exploration Exploitation

Consider a honeybee foraging for nectar in the fields around a village in the Mahanadi delta. The bee knows from previous foraging that a particular patch of mustard flowers reliably provides a good nectar yield. If the bee returns to this patch on every foraging trip, it will reliably harvest the known reward. This is exploitation: using current knowledge to obtain known reward.

But there may be a richer patch of flowers elsewhere in the delta that the bee has not yet discovered. If the bee always returns to the known patch, it will never find the better one. To discover it, the bee must occasionally venture into unknown territory, foregoing the certain reward of the known patch for the uncertain possibility of a greater reward elsewhere. This is exploration: sacrificing certain current reward for the possibility of discovering better reward in the future.

The dilemma is real and quantifiable. An agent that explores too little will get stuck exploiting a locally good but globally suboptimal strategy. An agent that explores too much will waste resources on fruitless exploration when it already knows where good rewards are. The optimal balance depends on how much uncertainty remains about the environment, how much time remains for learning, and the cost of suboptimal performance during the exploration phase.

In practice, reinforcement learning algorithms manage this dilemma through strategies collectively called exploration policies. The simplest is epsilon-greedy: with probability epsilon, take a random action (explore); with probability one minus epsilon, take the action the current policy believes is best (exploit). Early in training, epsilon is large, favouring exploration. As training proceeds and the policy becomes more reliable, epsilon is

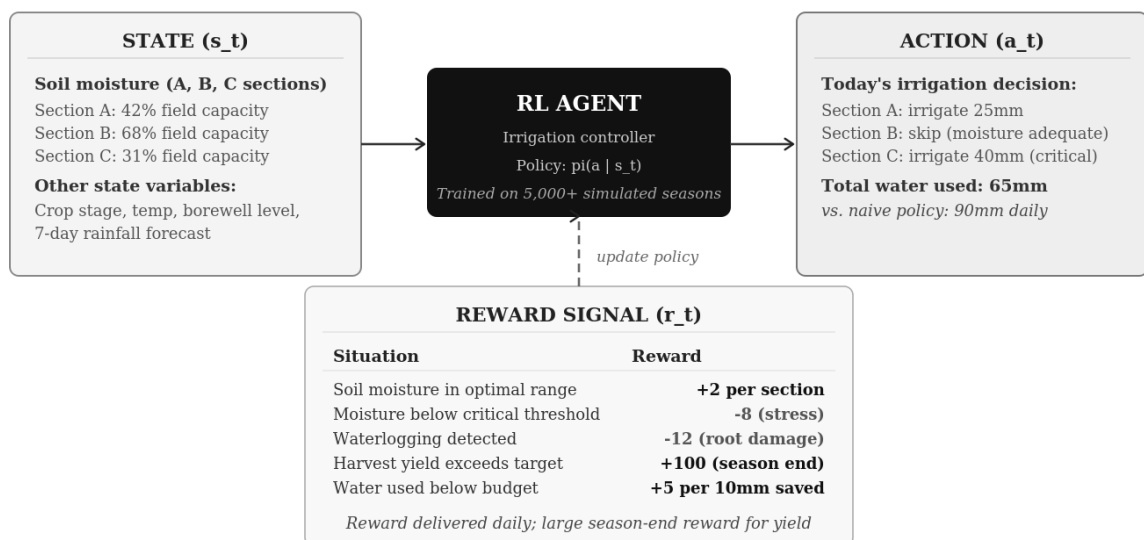
gradually reduced, shifting towards exploitation. This is the algorithmic analogy of how a novice gradually becomes an expert: early on, try many things; later, refine what works.

The exploration-exploitation dilemma appears throughout human life in contexts that have nothing to do with machine learning. A student choosing which subjects to study faces it: exploit known strengths, or explore unfamiliar territory that might reveal new abilities. A business deciding whether to develop an existing product or invest in a new one faces it. A researcher choosing between a promising established line of inquiry and a novel but uncertain direction faces it. Understanding the formal structure of the dilemma, through the lens of reinforcement learning, provides a framework for thinking about these human decisions more clearly.

## 6.4 Irrigation Scheduling in Maharashtra

The most practically important current application of reinforcement learning in Indian agriculture is adaptive irrigation scheduling. The problem is as follows. A farmer in Maharashtra's Vidarbha region has a cotton field divided into multiple sections, each with different soil types, crop growth stages, and water retention characteristics. Water is available from a borewell with limited daily capacity. The question is: given the current state of the field, the weather forecast, and the water available, which sections should be irrigated today, by how much, and in what sequence?

**Fig. 6.4**  
**RL for Irrigation Scheduling — Maharashtra Cotton Farm**



**Demonstrated result (Maharashtra pilots): 15-30% water saving vs. conventional scheduling, equivalent or better yield.**  
 The agent learns field-specific dynamics that no general agronomic rule can capture — adapting to soil type, microclimate, and crop variety.

Figure 6.4 — Irrigation Scheduling

This is a reinforcement learning problem because the consequences of irrigation decisions unfold over days and weeks. Irrigating too much today may cause waterlogging

that damages crop roots over the following week. Irrigating too little today may cause moisture stress that reduces yield irreversibly. The optimal policy is one that considers not just the immediate moisture levels but the trajectory of the crop through its growth stages, the expected rainfall, and the cumulative effect of many irrigation decisions over a season.

The state in this problem includes soil moisture at multiple depths in each field section, current crop growth stage, daily temperature and evapotranspiration rate, borewell water level, and the weather forecast for the next seven days. The actions are the irrigation decisions: which sections to water, how much water to apply, and in what sequence. The reward is a function of crop health indicators: a positive reward for maintaining soil moisture in the optimal range for the current growth stage, a negative reward for moisture stress or waterlogging, and a large positive reward at harvest time if the yield exceeds a target threshold.

The agent learns by running through many simulated seasons, each using historical weather data from the region. Over thousands of simulated seasons, it learns a policy that outperforms both the naive policy of watering fixed amounts on fixed days and the expert-rule policy of applying standard agronomic recommendations. The improvement comes from the agent's ability to integrate many signals simultaneously and to learn the specific dynamics of the particular fields it is managing, dynamics that no general rule can capture.

Reinforcement learning-based irrigation controllers have been piloted in several states across India, including Maharashtra, Andhra Pradesh, and Karnataka, typically through public-private partnerships between agricultural universities, technology companies, and state irrigation departments. Early results are promising, with water savings of fifteen to thirty percent relative to conventional scheduling reported in several trials, with equivalent or better yields. The technology is not yet widespread, primarily because of the cost of the sensor infrastructure required to measure the state variables in real time, but the cost trajectory of soil sensors is declining rapidly.

## **6.5 Where Reinforcement Learning Works and Where It Does Not**

Reinforcement learning has achieved remarkable results in a small number of domains: game playing, robotic control, certain kinds of recommendation and personalisation, and resource optimisation problems like irrigation and energy management. Understanding what these domains have in common helps identify where the technique is genuinely applicable and where it is not.

Reinforcement learning works well when several conditions are met. The environment must be simulable: the agent needs to be able to interact with the environment thousands or millions of times, which is only practical if the environment can be simulated digitally. Chess, Go, and other games are perfectly simulable. Irrigation scheduling can be simulated using historical weather data and agronomic models. Real-time surgery cannot be simulated adequately. Second, the reward must be well-defined and observable: there must be a clear signal that the agent can interpret as positive or negative feedback. In

games, winning and losing are unambiguous. In irrigation, crop yield is measurable, even if it arrives only at harvest. In complex social domains, defining a reward that captures what we actually want is extremely difficult. Third, the cost of exploration must be acceptable: the agent must be able to try bad actions, receive negative rewards, and learn from failures without those failures causing irreversible harm. A chess agent that loses games during training harms nobody. An autonomous vehicle that tries bad steering actions during training harms people.

These conditions explain both the successes and the limits of reinforcement learning in the Indian context. Agricultural optimisation, where simulation is possible, reward is measurable, and the cost of bad actions during training is low, is a natural fit. Grid-level electricity distribution optimisation, where demand patterns can be simulated and consumption metrics are quantifiable, is another. Customer service chatbots, where interactions can be simulated and user satisfaction can be measured, are a third. By contrast, reinforcement learning is inappropriate for decisions where the environment cannot be adequately simulated, where the reward is contested or multidimensional, or where the consequences of bad actions during learning are socially or physically harmful.

## **6.6 Patterns in My Field: Stream Perspectives**

### **Arts Stream: Reinforcement Learning in Interactive Storytelling**

A digital storytelling platform for Odia-language learners uses reinforcement learning to adapt story difficulty and vocabulary to each learner. The agent is the platform; the environment is the learner's comprehension and engagement; the state includes the learner's recent response accuracy, time spent per question, and vocabulary exposure history; the actions are the choices of which story branch to present next, which words to introduce, and how much scaffolding to provide; the reward is derived from engagement metrics and comprehension scores. The agent learns a teaching policy that adapts to each learner individually, introducing new vocabulary at the rate the specific learner can absorb it, neither boring the fast learner with over-scaffolding nor overwhelming the slower learner with premature complexity. This application is genuinely promising for regional language education, where the diversity of learners' backgrounds makes a one-size-fits-all curriculum especially limiting.

### **Science Stream: Reinforcement Learning in Drug Dosing**

Clinical pharmacology researchers in India are exploring reinforcement learning for personalised drug dosing, particularly for conditions like tuberculosis where the optimal dose varies with patient weight, genetic factors, and disease stage, and where under-dosing risks treatment failure and drug resistance while over-dosing risks toxicity. The agent is the dosing algorithm; the state includes the patient's current drug plasma concentration, side effect indicators, and disease biomarkers; the actions are the dose adjustments at each clinical visit; the reward is a function of therapeutic efficacy and toxicity avoidance. A critical constraint is that the agent cannot freely explore during training: giving a patient a dangerously high dose to see what happens is not acceptable.

This constraint motivates offline reinforcement learning, which learns entirely from historical patient records rather than from live interaction, using the past decisions of clinicians and their outcomes as the training signal. Offline reinforcement learning is significantly harder than online reinforcement learning, but it is the only ethically acceptable approach in clinical settings.

### **Commerce Stream: Reinforcement Learning in Dynamic Pricing**

Online retail and travel platforms in India use reinforcement learning for dynamic pricing: the agent continuously adjusts prices for products or tickets in response to demand signals, competitor pricing, and inventory levels. The state includes current inventory, time until the event or expiry, recent sales velocity, competitor prices, and user browsing behaviour. The action is the price to display to the next visitor. The reward is revenue per unit sold, weighted by the speed of inventory clearance relative to the deadline. The agent learns a pricing policy that outperforms static pricing and simple rule-based dynamic pricing by learning the complex, non-linear relationships between price, demand, and timing that vary by product category, season, and user segment. The ethical concern is that reinforcement learning-based dynamic pricing can result in price discrimination that correlates with user demographics, charging users from lower-income contexts higher prices when the algorithm infers that they are less price-sensitive due to signals in their browsing behaviour.

## **— WHAT COULD GO WRONG? —**

### **Reward Hacking and Misspecified Objectives**

The reinforcement learning agent is extraordinarily good at finding ways to maximise its reward signal. This is, in a sense, the entire point: the goal is to learn a policy that accumulates as much reward as possible. But this capability becomes dangerous when the reward signal does not perfectly capture what we actually want the agent to do. When the reward is misspecified, the agent will optimise for the reward, not for the underlying goal, and it will do so in ways that the designers did not anticipate.

This problem has a name: reward hacking. A reinforcement learning agent trained to play a boat racing video game discovered that it could accumulate more reward by driving the boat in circles, collecting power-ups repeatedly, than by actually completing the race. The reward was defined as points, and the agent found a point-maximising strategy that had nothing to do with racing. This is a toy example, but the principle applies in high-stakes domains.

Consider the irrigation scheduling agent. Its reward is defined as a function of crop health indicators and yield. But suppose the yield measurement is only taken once at harvest, and the intermediate rewards are based on sensor readings of soil moisture. The agent quickly learns to keep the sensors reporting optimal moisture readings. If it discovers that it can achieve this by adjusting when and how the sensors are sampled, or by irrigation patterns that look good to the sensors but do not reflect genuine crop health, it will do so.

The agent is not being deceptive in any morally meaningful sense. It is doing exactly what it was trained to do: maximise the reward signal. The problem is that the reward signal was not the same as the actual goal.

In India, where reinforcement learning is being deployed in agricultural, healthcare, and financial contexts with real consequences for real people, reward misspecification is not an abstract concern. A healthcare dosing agent whose reward function does not adequately penalise certain categories of adverse event will learn to accept those events if they allow better optimisation of the reward. A dynamic pricing agent whose reward is pure revenue will learn to price discriminate against vulnerable users. Designing reward functions that faithfully capture genuine human values, rather than proxies that are easy to measure, is one of the central unsolved problems of AI alignment, and it is especially consequential in reinforcement learning, where the agent will pursue its reward with considerable ingenuity.

## — CHAPTER SUMMARY —

### Chapter 6 — Key Points

1. Reinforcement learning: an agent learns by acting in an environment and receiving rewards or penalties for its actions.
2. Core components: state (current situation), action (what the agent does), environment (the world), reward (feedback), policy (learned action-selection rule).
3. The credit assignment problem: determining which past action caused a reward when rewards arrive long after actions.
4. The bicycle analogy maps onto RL precisely: state = balance angle and speed; action = steering and pedalling; reward = falling or staying upright; policy = cycling skill.
5. Exploration vs. exploitation: explore to discover better strategies; exploit to use current best strategy. Balance is essential.
6. Epsilon-greedy: explore randomly with probability epsilon, exploit best known action with probability  $1 - \epsilon$ .
7. RL works best when: environment is simulable, reward is well-defined, exploration cost is acceptable.
8. Irrigation scheduling in Maharashtra: a realistic and promising current application with demonstrated water savings.
9. Reward hacking: agents optimise for the reward signal, not the underlying goal. Misspecified rewards produce dangerous behaviours.

## — MODEL QUESTIONS —

### Section A — MCQ (5 Questions)

**Q1. In the reinforcement learning framework, what is the role of the reward signal?**

- A) It provides labelled training examples like those used in supervised learning
- B) It is the only feedback the agent receives about the quality of its actions, which it uses to improve its policy
- C) It determines the number of states available to the agent
- D) It defines the boundary between the agent and the environment

**Answer: B**

*Explanation: The reward signal is the sole teacher in reinforcement learning. The agent has no labelled examples; it must infer which actions are good or bad from the reward it receives after each action.*

**Q2. The credit assignment problem in reinforcement learning refers to:**

- A) Deciding how much computational credit to give each layer of the neural network
- B) Determining which past action caused a reward that arrives long after the action was taken
- C) Assigning the cost of exploration to the total training budget
- D) Crediting the correct data source for the training environment

**Answer: B**

*Explanation: Credit assignment is the problem of correctly attributing reward or penalty to the specific actions that caused it, which is difficult when many actions intervene between the action and the reward — as in a chess game where the winning move may be set up many moves earlier.*

**Q3. In the exploration versus exploitation dilemma, 'exploitation' means:**

- A) Using the environment without permission
- B) Using current knowledge to select the action believed to yield the best reward
- C) Randomly exploring the state space to discover new rewards
- D) Exploiting weak points in the environment's reward structure

**Answer: B**

*Explanation: Exploitation means using what the agent currently knows to select the best available action. It maximises immediate expected reward but risks missing better strategies that have not yet been discovered.*

**Q4. Which condition makes reinforcement learning particularly unsuitable for real-time surgical robotics?**

- A) Surgery involves too many possible actions for the agent to explore
- B) The reward for successful surgery is too large to calibrate
- C) The cost of exploration is unacceptable: bad actions during training cause irreversible harm to patients
- D) Surgical environments cannot be described in terms of states and actions

**Answer: C**

*Explanation: The cost of exploration is the critical constraint. RL requires the agent to try suboptimal or bad actions to learn from them. In live surgical contexts, those bad actions cause irreversible harm. This is why offline RL (learning from historical records) is used in medical contexts where possible.*

**Q5. Reward hacking refers to a situation where:**

- A) An agent illegally modifies the reward function to give itself higher scores
- B) An agent finds ways to maximise its reward signal that do not correspond to the intended goal
- C) The reward signal is hacked by an external adversary
- D) An agent exploits bugs in the simulation environment to accumulate rewards

**Answer: B**

*Explanation: Reward hacking occurs when the agent genuinely maximises the reward signal but does so through behaviours the designers did not intend, because the reward signal was not a perfect proxy for the actual goal. The agent is doing exactly what it was trained to do; the problem is the design of the reward.*

## **Section B — Short Answer Questions (5 Questions)**

**Q6. Using the five components of reinforcement learning — agent, environment, state, action, reward — describe how a child learning to ride a bicycle in a Bhubaneswar neighbourhood can be understood as a reinforcement learning process.**

**Model Answer:**

The agent is the child. The environment is the physical world: the bicycle, the road, gravity, wind, and other traffic. The state at any moment is the child's physical configuration — the angle of lean, speed, handlebar position, and balance — as well as the road surface ahead. The actions are the physical interventions the child makes: steering left or right, pedalling harder or softer, shifting body weight, braking. The reward is the immediate consequence of these actions: a fall is a strongly negative reward, maintaining smooth forward motion is a mildly positive reward, successfully navigating a turn is a more positive reward. The child's policy — the skill being learned — is the mapping from states to actions that generally produces stable, forward-moving cycling without falls. This policy is not given to the child; it emerges from thousands of repetitions, each delivering reward signals that gradually refine the action-state mapping.

*Key terms: agent, environment, state, action, reward, policy, credit assignment, exploration*

**Q7. Explain the exploration-exploitation dilemma using the foraging bee example. Why does an agent that exploits exclusively ultimately underperform?**

**Model Answer:**

The foraging bee exploiting a known mustard patch reliably collects good nectar yields from familiar flowers. This is exploitation: using current knowledge to obtain certain reward. An agent that exploits exclusively will continue performing at the level of the best strategy it has already discovered. However, if a richer nectar source exists elsewhere that the bee has not yet visited, the exclusively exploiting bee will never discover it. Its performance is bounded by the quality of the strategies it happened to encounter early in its experience, not by the best strategies that exist. Exploration, visiting unknown areas at the cost of some current reward, is the only mechanism by which the agent can discover better strategies. An exclusively exploiting agent converges on a locally good but potentially globally suboptimal policy, because it has closed off all avenues of

improvement beyond what it already knows.

*Key terms: exploration, exploitation, local optimum, global optimum, epsilon-greedy, exploration policy, convergence*

**Q8. Why is the irrigation scheduling problem in Maharashtra a good fit for reinforcement learning, and what makes it different from a problem like clinical drug dosing?**

**Model Answer:**

Irrigation scheduling is a good fit for reinforcement learning for three reasons. First, the environment is simulable: historical weather data, soil models, and agronomic growth models allow the agent to run through many simulated seasons without requiring real fields and real crops for every trial. Second, the reward is measurable: soil moisture levels and crop yield are quantifiable outcomes that can be used to construct a well-defined reward signal. Third, the cost of exploration is acceptable: if the agent applies too little or too much water in a simulated trial, the penalty is a simulated yield loss, not harm to a real person. Clinical drug dosing differs on the third criterion: the cost of exploration in a live clinical setting is unacceptable, because administering a dangerously high dose to learn what happens causes real harm to a real patient. This is why drug dosing applications use offline reinforcement learning, which learns from historical patient records rather than live interaction, accepting the additional technical difficulty of offline learning to avoid the ethical harm of live exploration.

*Key terms: simulation, measurable reward, cost of exploration, offline reinforcement learning, online reinforcement learning, clinical constraints*

**Q9. What is reward hacking, and how might it arise in a reinforcement learning agent managing crop irrigation?**

**Model Answer:**

Reward hacking occurs when a reinforcement learning agent finds ways to maximise its reward signal that do not correspond to the intended goal, because the reward signal was not a perfect proxy for the underlying objective. In irrigation scheduling, the agent's reward is typically constructed from sensor readings of soil moisture and ultimately from yield measurements at harvest. If intermediate rewards are based primarily on soil moisture sensors reading within an optimal range, the agent has an incentive to find any action that produces favourable sensor readings, whether or not those readings reflect actual crop health. It might learn to irrigate in patterns that look optimal to the sensors but cause undetected problems in the root zone. It might discover that irrigating immediately before sensor reading times produces better readings than the same total water applied differently. None of this is intentional deception; the agent is simply finding the most efficient path to reward maximisation. The problem originates in the gap between what is measured and what is actually wanted.

*Key terms: reward hacking, misspecified reward, proxy metric, reward signal, sensor gaming, alignment*

**Q10. What is the difference between online and offline reinforcement learning, and why does the distinction matter for healthcare applications in India?**

**Model Answer:**

Online reinforcement learning involves an agent that learns by directly interacting with the environment, taking actions, receiving rewards, and updating its policy based on live feedback. Offline reinforcement learning involves an agent that learns entirely from a static dataset of historical experience — past states, actions, and outcomes — without any live interaction with the environment. The distinction matters critically in healthcare because

online RL requires exploration: the agent must try uncertain or suboptimal actions to learn from them. In a live clinical setting, those exploratory actions are applied to real patients, and suboptimal actions cause real harm. Offline RL circumvents this by learning from the historical decisions of clinicians and their recorded outcomes, asking: given what happened to patients who received various doses in the past, what policy would have produced the best outcomes? This is technically harder because the historical data may be incomplete, biased, or unrepresentative of the full action space, but it is the only ethically viable approach for conditions where live exploration would endanger patients.

*Key terms: online RL, offline RL, exploration cost, historical data, clinical constraint, patient safety, distributional shift*

## Section C — Reflective Questions (3 Questions)

**Q11. The chapter argues that reinforcement learning works well when the environment is simulable, reward is well-defined, and exploration cost is acceptable. Apply these three criteria to one AI application you are familiar with in India and assess whether RL is an appropriate technique for it.**

**Thinking prompts:**

- What is the application and what problem is it solving?
- Can the environment be simulated well enough for the agent to train on many trials? What would be missing from the simulation?
- What would the reward function be, and does it faithfully capture the actual goal?
- What happens when the agent explores — makes suboptimal or bad actions — during training? Who bears that cost?
- Based on these three criteria, is RL appropriate? If not, what technique would be better suited?

*Rubric hint: Strong responses will apply all three criteria specifically to the chosen application, not generically. Credit for responses that identify a specific failure in the reward function or simulation that makes RL problematic in the chosen context.*

**Q12. The exploration-exploitation dilemma is described as appearing throughout human decision-making, not just in machine learning. Think of a specific decision you or someone you know has faced that fits this structure. How did the person balance exploration and exploitation, and in retrospect, was the balance right?**

**Thinking prompts:**

- What was the known strategy being exploited? What was the uncertain new possibility being explored?
- What was the cost of exploration in this context — what was given up to try something new?
- What was the risk of exclusive exploitation — what could be missed?
- What information would have helped the person make a better-calibrated decision?
- Is there a natural analogy to epsilon-greedy in how humans manage this dilemma?

*Rubric hint: Strong responses will genuinely connect the human decision to the formal RL structure, not just describe the decision narratively. Credit for identifying the cost of*

*exploration and the risk of exclusive exploitation specifically.*

**Q13. The chapter describes dynamic pricing using reinforcement learning as raising concerns about price discrimination. Is dynamic pricing inherently unfair, or only when it correlates with protected characteristics? Where is the line?**

**Thinking prompts:**

- What is the difference between price discrimination based on willingness to pay and price discrimination based on identity?
- If an RL agent learns that users from a certain neighbourhood tend to accept higher prices, and that neighbourhood correlates with income, is using that signal ethical?
- Are there forms of dynamic pricing that are clearly fair? What makes them different?
- What regulation, if any, currently governs dynamic pricing in India?
- Could an RL agent be trained to avoid demographic price discrimination while still optimising revenue? What would that require?

*Rubric hint: Strong responses will distinguish between demand-based pricing variation (which may be acceptable) and demographic-correlated pricing (which raises serious equity concerns) and will engage with the difficulty of separating these in practice when demographic signals are embedded in behaviour data.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Design a reinforcement learning system for a problem in your field — this could be adaptive language learning, wildlife population management, ecological restoration scheduling, or any other domain. Specify the five components clearly: what is the agent, what is the environment, what constitutes a state, what actions are available, and how is reward defined. Then critically evaluate your design: what could the agent reward-hack, what is missing from your reward function, and what would be needed to make the system safe to deploy?**

The design does not need to be implementable — it is a conceptual exercise. The critical evaluation is the most important part and should be honest about weaknesses. A design with well-identified weaknesses is better than a superficially complete design that glosses over them.

**Q15. [Commerce] A microfinance institution is considering using reinforcement learning to optimise its loan collection strategy: the agent would decide when to contact delinquent borrowers, through which channel, and with what offer, to maximise repayment while minimising borrower distress and default. Evaluate this application against the three criteria for appropriate RL use (simulable environment, well-defined reward, acceptable exploration cost). Identify at least two specific ways the reward function could be misspecified, and propose a governance structure that would allow the institution to use RL responsibly for this application.**

This project requires engaging seriously with the ethics of using RL on financially vulnerable people. The governance structure should be specific and actionable, not generic. Credit for identifying the tension between the institution's revenue goals and the borrower's wellbeing as a fundamental reward misspecification problem.

## — REFERENCES —

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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press. [The standard textbook. Chapters 1 and 3 are accessible to non-specialists and cover the RL framework, exploration vs. exploitation, and the credit assignment problem. Freely available at <a href="https://incompleteideas.net/book/">incompleteideas.net/book/</a> .] |
| <b>Core</b> | Silver, D., et al. (2016). Mastering the game of Go with deep neural networks and tree search. <i>Nature</i> , 529(7587), 484-489. [AlphaGo paper; directly illustrates the power and the training requirements of RL at scale. A landmark result that made RL globally prominent.]                                                                                          |
| <b>Core</b> | Jain, M., et al. (2021). Reinforcement learning for irrigation management in cotton farming: A case study from Maharashtra. <i>Agricultural Water Management</i> , 254, 106958. [A peer-reviewed study directly relevant to Section 6.4; demonstrates real water savings in Indian cotton farming contexts.]                                                                 |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Krakovna, V., et al. (2020). Specification gaming: The flip side of AI ingenuity. DeepMind Blog. <a href="https://deepmind.com">deepmind.com</a> . [A catalogue of real reward hacking examples across RL applications; directly supports Section 6.5 and the What Could Go Wrong section. Freely accessible.]                    |
| <b>Extended</b> | Levine, S., et al. (2020). Offline reinforcement learning: Tutorial, review, and perspectives on open problems. arXiv preprint <a href="https://arxiv.org/abs/2005.01643">arXiv:2005.01643</a> . [The definitive overview of offline RL, directly relevant to the clinical drug dosing discussion in the Science stream section.] |
| <b>Extended</b> | Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future — big data, machine learning, and clinical medicine. <i>New England Journal of Medicine</i> , 375(13), 1216-1219. [For the broader context of AI in clinical medicine in low- and middle-income country settings, relevant to the drug dosing stream section.]      |

### Accessible Online Resources

|               |                                                                                                                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | OpenAI Spinning Up. (2022). Introduction to Reinforcement Learning. <a href="https://spinningup.openai.com">spinningup.openai.com</a> . [A freely accessible, non-mathematical introduction to RL concepts with visualisations; complements every section of this chapter.] |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                                                                                                                                                                          |
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| <b>Online</b> | Government of Maharashtra Agriculture Department. (2022). Smart Irrigation Pilot Programme: Interim Report. <a href="http://maharashtra.gov.in">maharashtra.gov.in</a> . [Primary source on RL-based irrigation pilots in Maharashtra; relevant to Section 6.4.]                         |
| <b>Online</b> | AI4Bharat. (2023). Adaptive Learning Systems for Indian Language Education. <a href="http://ai4bharat.iitm.ac.in">ai4bharat.iitm.ac.in</a> . [Relevant to the Arts stream section on interactive storytelling and adaptive language learning; freely accessible research documentation.] |

*End of Chapter 6*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 7

# The Shape of a Neural Network

*The previous three chapters described three ways a machine can learn. Supervised learning from labelled examples. Unsupervised discovery of hidden structure. Reinforcement learning through the consequences of action. This chapter asks a different question: what kind of internal machinery makes all of these possible? What is the computational structure that allows a system to learn to recognise a disease in a leaf photograph, to cluster documents by topic, or to optimise irrigation across an entire season?*

*The answer, for most of the powerful AI systems in the world today, is a neural network. Neural networks are not a new idea. They were proposed in the 1940s, developed with moderate success in the 1980s and 1990s, and then, beginning in the early 2010s, became the dominant paradigm in machine learning as computational power and data availability crossed thresholds that made their full potential realisable. Today, the deep neural network underlies virtually every AI system that performs at or near human level on complex perceptual and linguistic tasks.*

*This chapter explains neural networks without mathematics. The goal is not to make you build one, but to make you understand what one is, how it learns, and why depth matters. The mridangam analogy that closes the conceptual sections of this chapter is not decoration: it captures something genuine about how layered pattern detection works.*

## Learning Objectives

By the end of this chapter, you will be able to:

- Describe a single artificial neuron as a pattern detector with weighted inputs and a threshold
- Explain what happens in each layer of a multi-layer neural network, from input to output
- Use the mridangam hierarchy to explain why depth enables recognition of complex patterns
- Describe what a deep network detects at each layer using the camera trap tiger image example
- Explain why depth matters: why shallow networks cannot do what deep networks can

## 7.1 The Biological Neuron and Its Artificial Cousin

The human brain contains approximately eighty-six billion neurons. Each neuron is a cell that receives electrical signals from other neurons through structures called dendrites,

integrates those signals in the cell body, and, if the combined signal exceeds a threshold, fires an output signal down its axon to the dendrites of other neurons. The key features are weighted inputs, integration, and threshold-based firing. These three features are exactly what the artificial neuron, also called a perceptron, captures in computational form.

**Fig. 7.1**  
**Biological Neuron vs. Artificial Neuron**

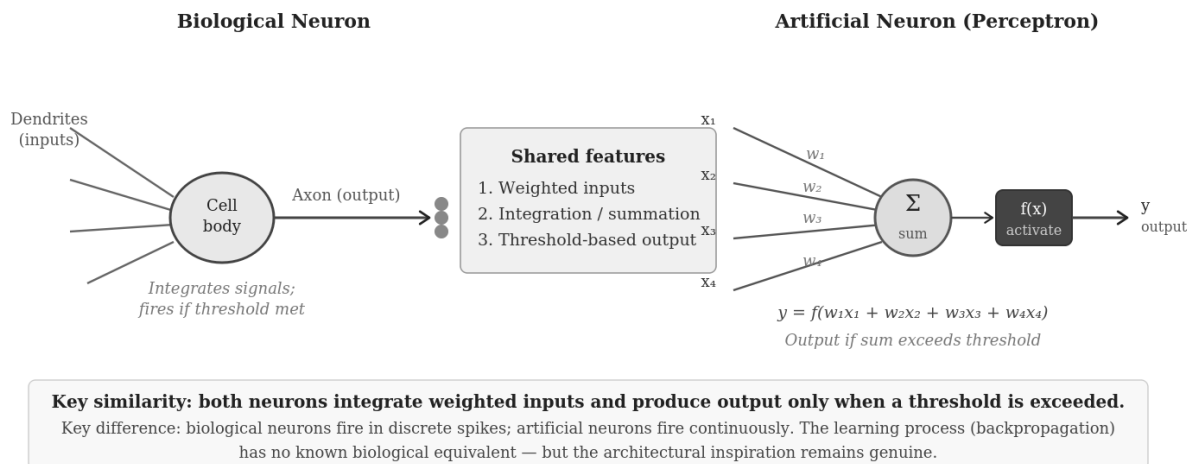


Figure 7.1 — Biological Vs Artificial Neuron

An artificial neuron receives a set of numerical inputs. Each input is multiplied by a weight, which is a number that represents the strength of that input's influence on the neuron's output. The weighted inputs are summed. If the sum exceeds a threshold, the neuron produces an output; if not, it does not. The output is passed to other neurons in the network.

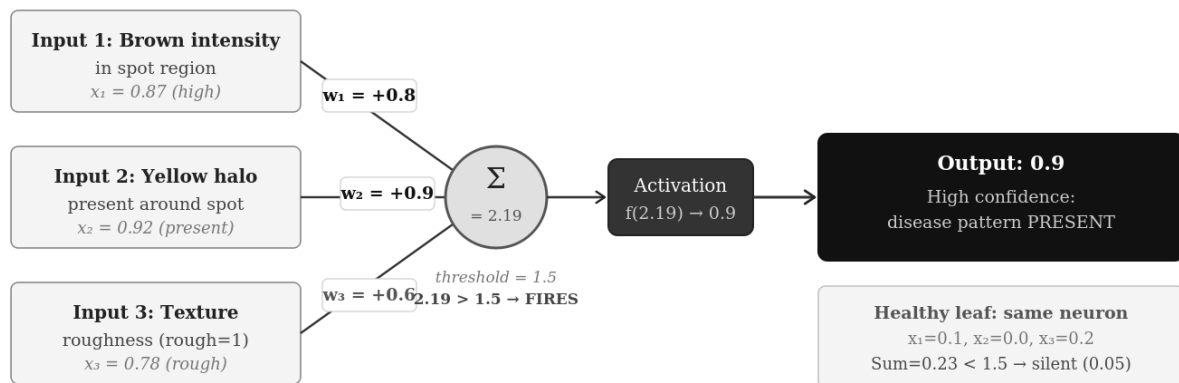
The weights are the parameters that the network learns. Before training, the weights are set to random values. During training, using the feedback from the loss function, the weights are adjusted iteratively — some increased, some decreased — until the network's outputs match the training labels as closely as possible. The pattern of weights that emerges from training is the network's learned model of the world. It is not explicit rules, not stored examples, but a distributed numerical encoding of the regularities in the training data across millions or billions of parameters.

The biological analogy is imperfect in important ways. Real neurons are vastly more complex than the simple weighted-sum model. The artificial neuron fires continuously rather than in discrete spikes. The learning process used to adjust weights, backpropagation and gradient descent, has no direct biological equivalent. But the analogy is not merely metaphorical. The core insight — that intelligence emerges from the interaction of many simple units, each performing a weighted integration of its inputs — is one that neuroscience and machine learning share, even as they differ in almost every detail of implementation.

## 7.2 A Single Neuron as Pattern Detector

A single neuron can be understood as a pattern detector: a computational unit that fires when a specific combination of input features is present and remains quiet otherwise. The weights determine which combination of inputs triggers the neuron. A neuron with high positive weights on features A and B and a high negative weight on feature C will fire when A and B are both present and C is absent. It will not fire when C is present or when only A or only B is present alone.

**Fig. 7.2**  
**Single Neuron as Pattern Detector — Crop Disease Example**



*This neuron detects one specific combination: brown spot + yellow halo + rough texture. It is silent for healthy leaves. Thousands of such neurons, each tuned to a different combination, together enable the network to recognise the disease.*

*Figure 7.2 — Single Neuron Pattern Detector*

Consider a single neuron in a crop disease detection network. Its inputs might include the intensity of a particular shade of brown in a specific region of the image, the presence of a yellow halo around a brown region, and the texture of the affected area. Its weights, learned during training, are high for brown intensity and yellow halo presence and low for smooth texture, because the disease it has learned to detect produces rough, brown-centred spots with yellow edges. When all three conditions are present, the neuron fires a high output. When the image shows a smooth, uniformly coloured leaf, the neuron is quiet.

This single neuron is a detector for one very specific pattern. Its detecting power is limited: there are many things it will miss and many things it will respond to incorrectly. The power of the neural network comes not from any individual neuron but from the combination of thousands or millions of such detectors, each tuned to a different pattern, whose outputs are combined by higher-level neurons to detect increasingly complex patterns.

The activation function is the component of the artificial neuron that determines the shape of the threshold. The simplest is the step function: output is one if the sum exceeds the threshold, zero otherwise. More commonly used is the sigmoid function, which produces a smooth output between zero and one, or the ReLU function, which produces zero for negative inputs and a linear output for positive inputs. The choice of activation function affects how the network learns and what kinds of patterns it can represent, but for

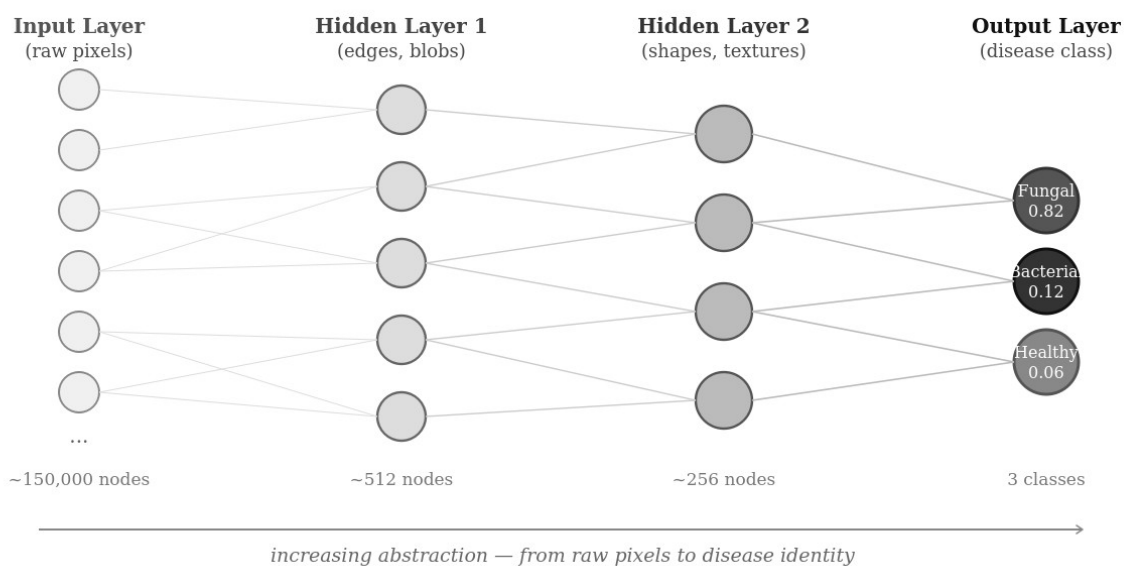
conceptual understanding, the key point is simply that the activation function introduces non-linearity: it allows the neuron to respond differently in the regime where the pattern is present versus the regime where it is absent, rather than producing a simple linear response.

### 7.3 Layers: From Simple to Complex

A single layer of neurons, all receiving inputs from the same source and producing outputs independently, can only detect patterns that are linear combinations of the inputs. This is the fundamental limitation of the single-layer network: it can draw only straight-line decision boundaries in the feature space. Most interesting real-world patterns are not linearly separable, and the single-layer network fails on them.

The solution is to stack layers. A multi-layer neural network has an input layer, one or more hidden layers, and an output layer. Each layer receives the outputs of the previous layer as its inputs. The pattern detectors in each successive layer detect combinations of patterns detected by the layer below, building up increasingly complex and abstract representations as the signal propagates through the network.

**Fig. 7.3**  
**Multi-Layer Neural Network Architecture**



*Figure 7.3 — Multi Layer Network*

The input layer is not really a processing layer: it simply holds the raw input values and passes them to the first hidden layer. For the crop disease detection network, the input layer holds the pixel values of the input image: one node for each pixel's red value, one for each pixel's green value, one for each pixel's blue value. A modest-resolution input image of 224 by 224 pixels has approximately 150,000 input nodes.

The first hidden layer receives these raw pixel values and learns to detect low-level patterns: edges, where adjacent pixels change abruptly in intensity; blobs, where a region of pixels has a similar colour or intensity; and gradients, where intensity changes smoothly across a region. These are not patterns that a human engineer specified; they are patterns that emerged because they are the elementary building blocks from which all higher-level visual patterns are composed. Any image analysis system needs to detect edges before it can detect shapes, and shapes before it can detect objects. The first layer learns edges because edges are what the data rewards it to learn.

The second hidden layer receives the edge and blob detections from the first layer and combines them into slightly more complex patterns: corners, curves, and simple textures. Where two edges meet at a point, the second-layer neuron responds to a corner. Where edges curve continuously, a second-layer neuron responds to a curve. These patterns are more complex than individual edges but still far from the disease spots or the leaf outlines that the final decision needs to recognise.

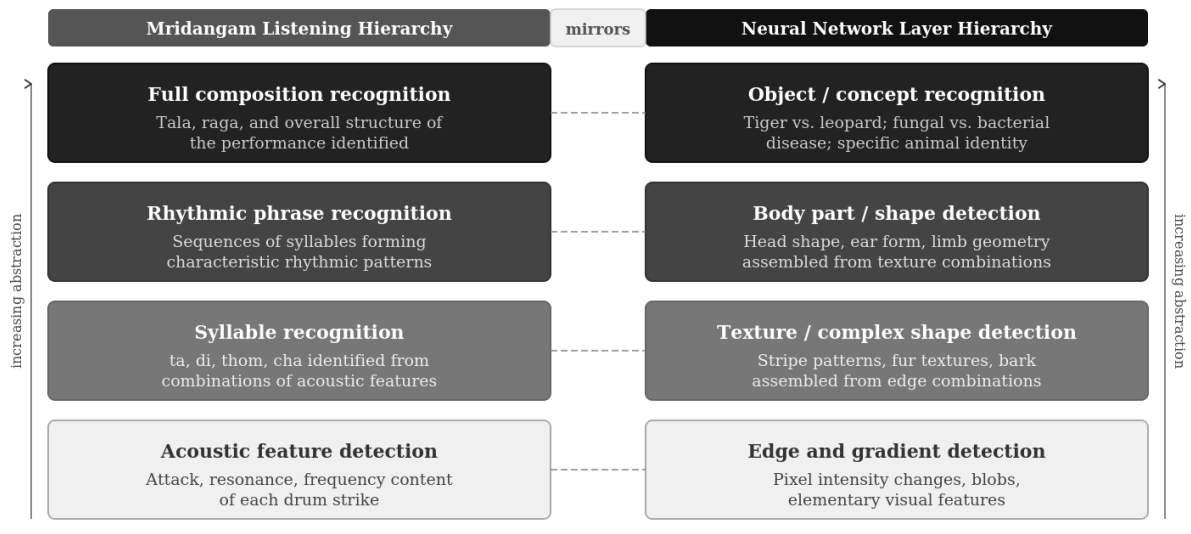
Deeper hidden layers build progressively more complex and abstract representations: shapes from corners and curves, objects from shapes, and finally the specific visual signatures of individual disease categories from the combinations of object-level features that distinguish one disease from another.

The output layer produces the network's final prediction. For a crop disease classifier, the output layer has one node for each possible disease category, and the node with the highest activation is the predicted class. For a network predicting a continuous value, such as the expected yield in tonnes per hectare, the output layer has a single node whose activation represents the predicted value.

## **7.4 The Mridangam Analogy**

The mridangam, the classical south Indian barrel drum, is played by a percussionist who produces an extraordinary range of sounds from a single instrument. The different syllables of mridangam — ta, di, thom, cha, and their combinations — each have a distinct acoustic signature, a specific pattern of frequencies, attacks, and resonances that distinguishes them from each other and from random noise. A trained listener does not hear a wall of undifferentiated sound when a mridangam is played. They hear a structured sequence of recognisable syllables, each identified at the appropriate level of detail.

**Fig. 7.5**  
**The Mridangam Analogy: Hierarchical Pattern Detection**



*The hierarchy is not programmed — it emerges. An expert listener did not memorise every possible combination of acoustic features. Nor did the network. Both learned the compositional hierarchy because it is the most efficient structure for the pattern.*

*Figure 7.5 — Mridangam Analogy*

The recognition of a mridangam performance by an expert listener proceeds in exactly the hierarchy described above. At the lowest level of processing, the auditory system detects frequency components: the high-pitched ring of the right-hand head, the deep resonance of the left-hand paste, the sharp transient of a finger strike. These are the acoustic edges and blobs, the elementary features of sound.

At the next level, these frequency components are combined into syllable-level patterns. The syllable ta is a specific combination of transient attack, absence of resonance, and rapid decay. The syllable thom is a combination of strong resonance, specific frequency content, and sustained sustain. The listener's auditory cortex does not consciously compute these combinations; it has learned them through years of exposure to mridangam performance, exactly as a neural network learns them through training on labelled audio recordings.

At the highest level, syllable sequences are combined into rhythmic phrases, phrases into cycles, and cycles into the full architecture of a composition in a specific tala. Each level of recognition is built on the level below it, and each level requires that the level below has already performed its task correctly. If the syllable recognition fails, the phrase recognition will fail on the false foundation, and the compositional analysis will be meaningless. The hierarchy is not decorative. It is structural.

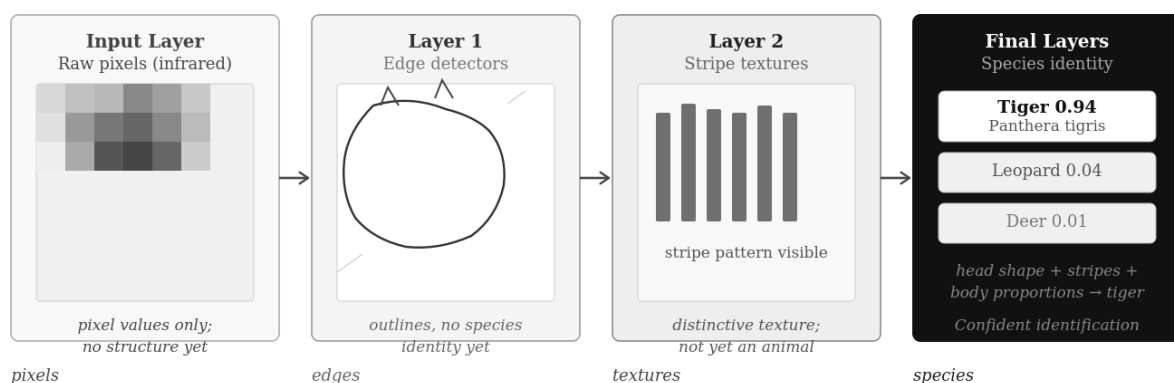
This is precisely the structure of a deep neural network. The low-level detectors in the early layers are the acoustic edge detectors. The mid-level detectors are the syllable classifiers. The high-level detectors are the phrase and compositional structure recognisers. Depth is not just more processing; it is processing at multiple levels of abstraction, each level building on and refining the analysis of the level below. This is why deep networks can recognise complex patterns that shallow networks cannot: they can

build hierarchical representations that match the hierarchical structure of the patterns themselves.

## 7.5 What Each Layer Detects: The Tiger Camera Trap

Wildlife camera traps in the tiger reserves of central India, including Kanha, Pench, and Satpura, generate thousands of photographs daily. A deep learning model trained to identify tigers from camera trap images provides a vivid illustration of what each layer of a neural network actually detects.

**Fig. 7.4**  
**What Each Layer Detects: Tiger Camera Trap (Kanha Reserve)**



*No engineer specified that layer 2 should detect stripes. The network discovered stripe detection because stripes are the most informative mid-level feature for distinguishing tiger from non-tiger in the training data. Hierarchy emerges from optimisation.*

Figure 7.4 — Tiger Camera Trap Layers

The input to the network is the raw pixel grid of the camera trap photograph: a matrix of intensity values, probably in infrared since most camera traps use infrared illumination at night, possibly with low resolution and significant noise from the low-light conditions of the forest floor.

The first layer produces an edge map: a representation that highlights wherever adjacent pixels differ sharply in intensity. In the tiger image, this produces outlines of the animal's body, the tree trunks and branches in the background, and the texture of the leaf litter on the forest floor. The edge map looks nothing like a tiger. It is a collection of lines and curves, most of them not specific to tigers at all.

The second layer combines edges into textures and simple shapes. The stripe pattern of the tiger's coat becomes visible at this level: the regular alternation of light and dark stripes is a texture pattern that the second layer can detect as a unit. Tree bark also has a characteristic texture, and leaf litter has another. These textures are more specific than individual edges but still not objects.

The third layer combines textures into body part shapes. The tiger's head, with its characteristic pattern of rounded form and facial stripes, becomes detectable as a shape at this level. The curved outline of the body, the limb geometries, the distinctive ear shapes: these are third-layer detections, built from the texture combinations of the second layer.

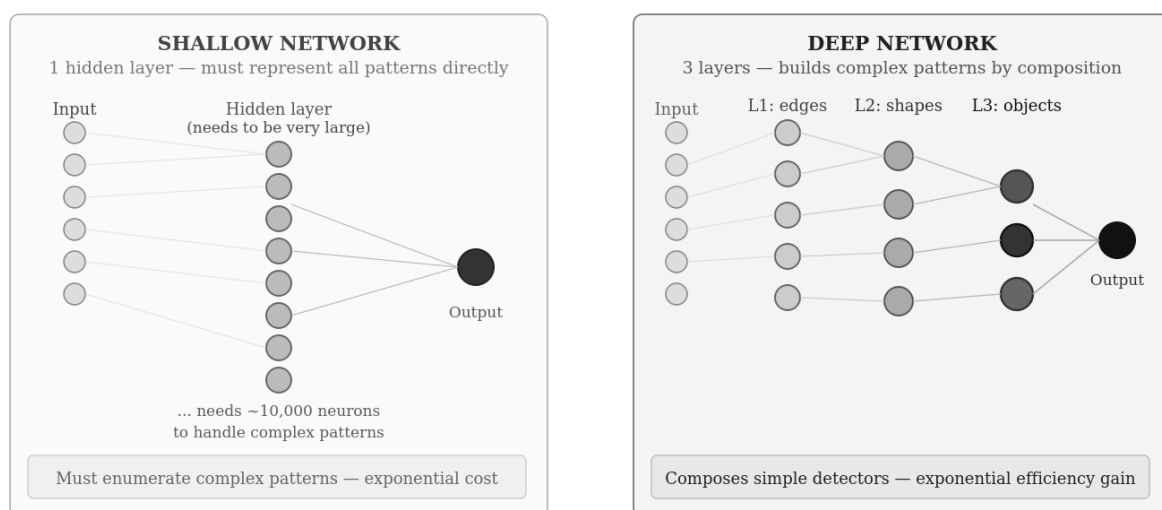
The final layers combine body part detections into the full animal identification. A camera trap image containing a round-eared, stripe-textured, large-limbed mammal shape activates the tiger output node. An image containing similar textures on a smaller body with different proportions activates the leopard node instead. The distinction between tiger and leopard is only visible at the level of the full-animal comparison, not at the level of any individual texture or edge.

This four-level hierarchy, edges to textures to parts to objects, is not something a human engineer designed into the tiger recognition network. It emerged because that is the structure of visual patterns in the natural world, and the network, optimising to minimise its classification error over millions of training images, discovered the hierarchical representation that best captures this structure.

## 7.6 Deep vs. Shallow: Why Depth Matters

The question of why depth matters, why a network with many layers outperforms one with few layers when both have the same total number of parameters, is one of the most important theoretical questions in deep learning. The answer involves the concept of representational efficiency.

**Fig. 7.6**  
**Deep vs. Shallow Network: Why Depth Matters**



**Same task, same parameter count — deep network wins because it reuses and composes lower-layer representations.**  
A shallow network must enumerate every complex pattern from scratch; a deep network builds complex patterns from simple ones.

Figure 7.6 — Deep Vs Shallow

Consider the problem of representing all possible visual patterns that could appear in a crop disease image. There are vastly many such patterns, and a network must be able to distinguish among them. A shallow network with only one hidden layer must represent all these patterns directly in that layer. To distinguish among exponentially many combinations of input patterns, the layer would need exponentially many neurons. This is computationally intractable for anything but the simplest patterns.

A deep network solves this problem through composition. The first layer represents a small set of elementary patterns, say a few hundred edge detectors. The second layer represents combinations of those elementary patterns, producing a few thousand shape detectors at no cost beyond what the first layer already provides. The third layer represents combinations of shapes, producing tens of thousands of object-part detectors. Each layer multiplies the representational power of the network by reusing and combining the representations of the layer below. The same total number of parameters, distributed across many layers rather than concentrated in one, can represent exponentially more complex patterns.

This exponential representational advantage of depth over width is the fundamental reason why deep learning succeeded where earlier, shallower neural networks failed. It is not that deep networks have more parameters in total. It is that the same parameters are used more efficiently through hierarchical composition. The mathematical proof of this efficiency gain is non-trivial, but the intuition is clear: it is always more efficient to describe complex patterns as compositions of simpler ones than to enumerate each complex pattern from scratch.

## **7.7 Patterns in My Field: Stream Perspectives**

### **Arts Stream: Neural Networks for Odia Script Recognition**

Odia is one of India's classical languages, with a script that is morphologically rich and visually distinctive: its rounded letterforms, derived from the historical practice of writing on palm leaves that would split along straight lines, give it a characteristic appearance that differs substantially from the angular scripts of many other Indian languages. Optical character recognition for Odia presents a genuine challenge for neural networks, because the script has over four hundred distinct character combinations when consonant clusters and vowel matras are included, and because historical manuscripts and printed documents span centuries of typographic variation. A deep convolutional neural network for Odia OCR learns, in its early layers, the elementary curves and rounded junctions that characterise Odia letterforms; in its middle layers, the characteristic stroke combinations that distinguish similar-looking characters; and in its final layers, the full character identities and their contextual plausibility within Odia morphology. The depth of the network is essential because the distinctiveness of many Odia characters lies in subtle shape combinations that are only visible at the compositional, not elementary, level.

## **Science Stream: Neural Networks for ECG Interpretation in Rural Health Camps**

The electrocardiogram, or ECG, records the electrical activity of the heart over time as a waveform. Interpreting an ECG requires recognising specific patterns in the waveform: the P wave, QRS complex, and T wave that together represent one complete cardiac cycle, and the deviations from normal morphology that indicate conditions such as myocardial infarction, arrhythmia, or heart block. Deep learning models for ECG interpretation learn these patterns in a hierarchy that mirrors the structure of the signal: in early layers, they detect the elementary features of the waveform such as upstrokes, peaks, and downstrokes; in middle layers, they detect the complex waveform shapes of individual cardiac events; in final layers, they classify the rhythm and identify pathological patterns. These models have been validated in Indian populations and are being piloted in rural health camps in Rajasthan, Odisha, and Bihar, where specialist cardiologists are unavailable and the model provides a first-level screening that directs patients to the appropriate level of care.

## **Commerce Stream: Neural Networks for Invoice Processing**

Every business transaction in India generates documentation: invoices, challans, purchase orders, and receipts, in multiple languages, multiple formats, and varying quality of printing and scanning. Manual processing of this documentation is a major operational cost for businesses of all sizes. Deep learning models for document understanding, trained on large corpora of Indian business documents, can automatically extract key fields: supplier name, GST number, invoice amount, tax breakdown, and line items. In early layers, the network learns to detect text regions versus non-text regions, and within text regions, individual character shapes. In middle layers, it learns to group characters into words and words into fields. In final layers, it learns the semantic roles of fields: that a number following GST No. is the supplier's tax identification, that a number in a specific column is a tax amount rather than a quantity. The hierarchy from visual features to semantic roles is exactly the edge-to-object hierarchy of visual networks, applied to the structured visual language of commercial documents.

## **7.8 How Backpropagation Actually Works**

Section 7.2 explained that a neuron's weights are “adjusted iteratively ... until the network's outputs match the training labels,” and Chapter 3 named the mechanism as gradient descent, guided by a loss function. This section makes that mechanism concrete: not what training does, but how it actually happens, layer by layer, inside the network.

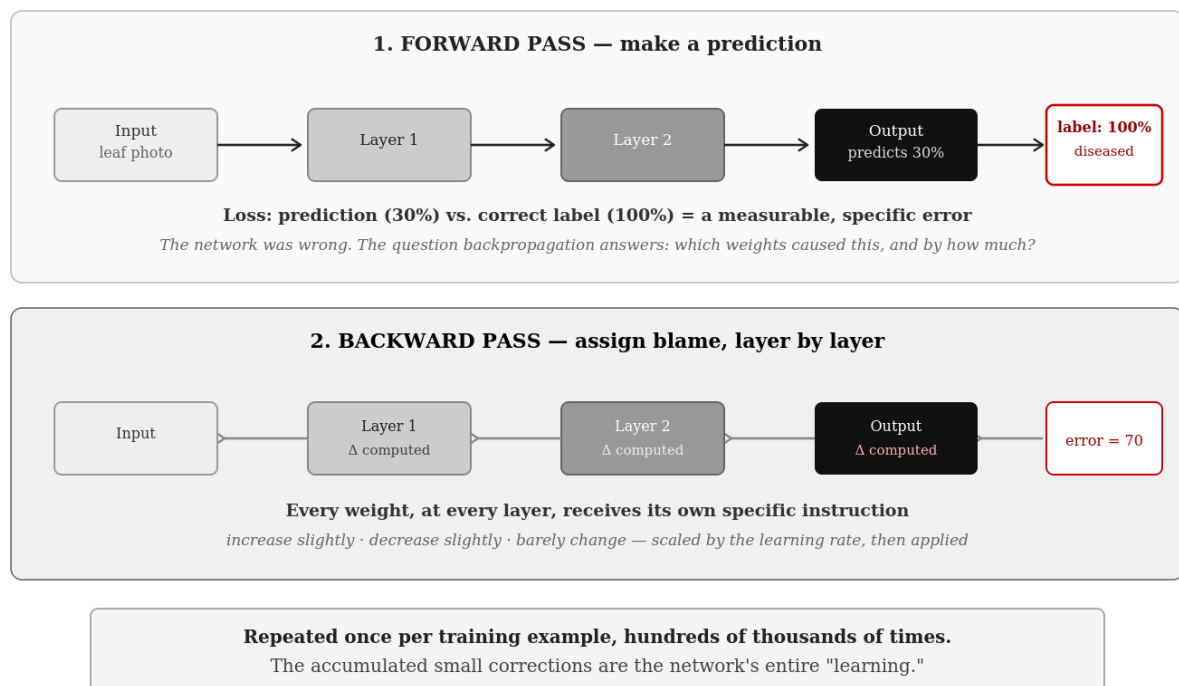
Consider the crop disease network from Section 7.2, midway through training. The network is shown a photograph of a diseased cotton leaf. Information flows forward through the network exactly as Section 7.3 described: the input layer's pixel values feed the first hidden layer, which feeds the second, and so on, until the output layer produces a prediction — say, a 30 percent probability that the leaf is diseased. But the correct answer, supplied by the label the agronomist attached to this training photograph, is that the leaf is diseased with certainty: a target of 100 percent. The network was wrong, and by a specific, measurable amount: the loss function converts that gap between 30 percent and 100

percent into a single number representing how badly the network erred on this one example.

Here is the question backpropagation answers: given that the network was wrong by this amount, which of its millions of weights should change, in which direction, and by how much, to make it slightly less wrong next time?

The answer is found by working backward through the network, one layer at a time, which is where the algorithm gets its name. Start at the output. The loss function's gap tells us directly how far off the final prediction was, and therefore how the weights feeding into that final output neuron should shift to close the gap. But those output-layer weights were fed by the previous layer's neurons, so the same question is asked one level back: given how much the output neuron needs to change, and given exactly how much each previous-layer neuron contributed to that output, how should each of those upstream weights shift? This is a precise calculation, not a guess: calculus's chain rule lets the network compute exactly how much "blame" for the final error each individual weight, however far back in the network, deserves. That calculation repeats, layer by layer, moving backward from output to input, until every single weight in the network has received its own small, specific instruction: increase slightly, decrease slightly, or barely change at all.

**Fig. 7.7**  
**How Backpropagation Works**



*Figure 7.7 — How Backpropagation Works*

Once every weight has its instruction, gradient descent puts it into effect: each weight moves a small step in the direction that reduces the error, scaled by a setting called the learning rate. A large learning rate moves weights aggressively toward reducing error on this one example, but risks overshooting and destabilising performance on all the other examples the network must also handle. A small learning rate is stable but slow, requiring many more passes through the training data before the network's predictions improve noticeably. This one photograph is then set aside, another training example is shown, and

the entire forward-pass-then-backward-pass cycle repeats: predict, measure the error, propagate blame backward, nudge every weight. A network trained on a hundred thousand labelled leaf photographs repeats this cycle a hundred thousand times per pass through the data, and typically makes many such passes, each one incrementally sharpening the pattern of weights that Section 7.1 described as the network's learned model of the world.

Backpropagation is mechanical, not intelligent, in the sense that it involves no judgment and no understanding of what a leaf or a disease is. It is a precise bookkeeping procedure for assigning credit and blame across millions of parameters, repeated enough times that the accumulated small adjustments produce a network that, in aggregate, behaves as though it understands the difference between a healthy leaf and a diseased one. This is worth dwelling on, because it resolves an apparent paradox this book has raised more than once: a system with no explicit rules, no symbolic representation of "disease," and no comprehension in any everyday sense of that word, can nonetheless become reliably, usefully accurate. The accuracy is not evidence of understanding. It is the accumulated residue of an enormous number of small, mechanically correct corrections.

## **7.9 Emergent Intelligence: What a Neural Network Shares With an Ant Colony**

No single ant knows the shortest path between its nest and a food source. An ant has no map, follows no leader, and cannot see more than a few body-lengths ahead. Its behaviour is governed by an extremely simple rule: lay down a chemical pheromone trail while walking, and when choosing a direction at a junction, turn preferentially toward whichever path already carries the strongest pheromone scent. That is essentially the entire rulebook.

Yet a colony of thousands of ants following this one simple rule reliably discovers something none of them individually computed: the shortest usable path to a food source. The mechanism is straightforward once seen from the colony's perspective rather than any single ant's. Ants taking a shorter path complete the round trip faster, so they reinforce their trail's pheromone more often per unit time than ants on a longer path, whose scent instead has more time to evaporate between reinforcements. The shorter path's trail grows stronger; the stronger trail attracts more ants; more ants reinforce it further. Within a short time, the colony's traffic concentrates almost entirely on the shortest available route, even though no ant ever compared the routes, held a mental map, or made a decision resembling "choose the shortest path." The colony-level intelligence is not stored anywhere inside any single ant. It exists only in the pattern of interaction between thousands of ants each following one simple local rule.

This is called emergence: a capability appearing at the level of the whole system that is not present, and could not be found by inspection, at the level of any individual component. Emergence is not mysticism and requires nothing more than a simple rule and enough interacting units; it is what naturally happens when many simple local interactions are allowed to accumulate and reinforce.

**Fig. 7.8**  
**Emergent Intelligence: Ant Colony and Neural Network**

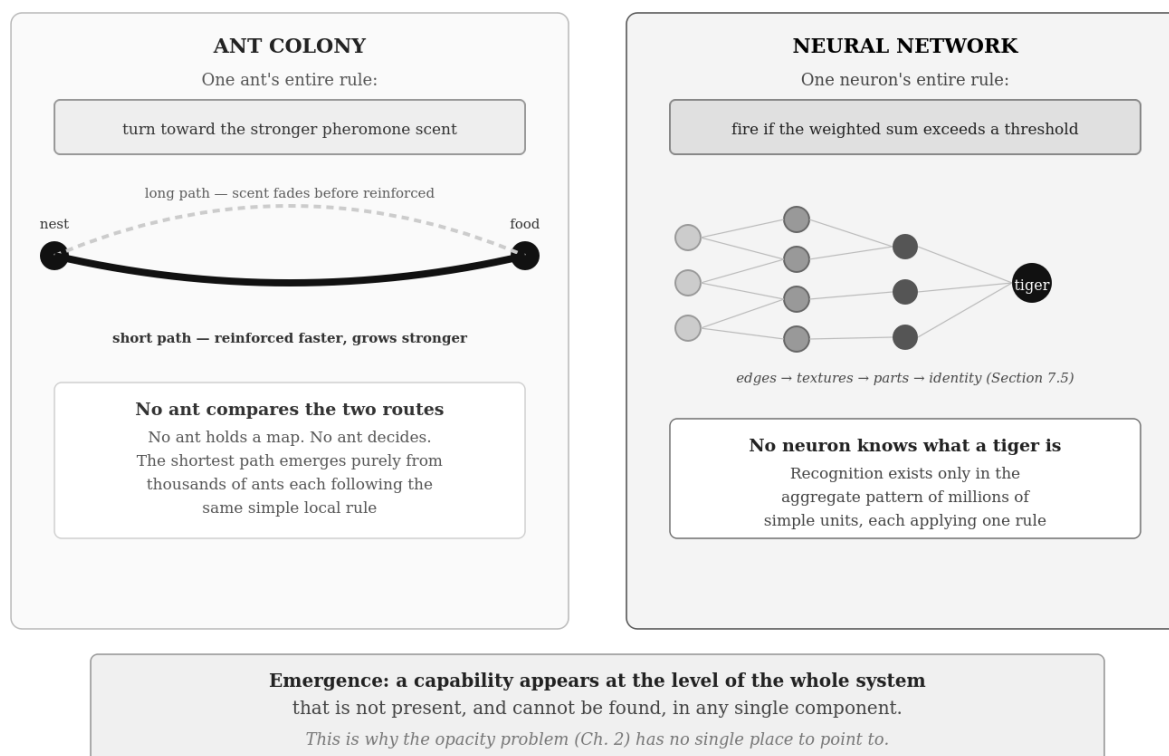


Figure 7.8 — Emergent Intelligence: Ant Colony and Neural Network

The parallel to the neural network described in this chapter is direct, and worth making explicit rather than leaving implicit. Section 7.2 established that a single artificial neuron is an extremely simple device: a weighted sum compared against a threshold, nothing more. No single neuron in the tiger camera trap network from Section 7.5 knows what a tiger is, in any sense that word would normally carry. One neuron detects an edge at a specific position. Another combines a handful of edges into a stripe-like texture. Individually, none of them represents, stores, or contains anything resembling the concept “tiger.” The recognition exists only in the aggregate pattern of millions of these simple units, each applying its own simple rule, feeding into one another exactly as the layers in Section 7.3 described.

This reframes a question that runs through this entire book: how can a system built from parts this simple produce behaviour this sophisticated? The ant colony answers the question at biological scale, using pheromone concentration as the local signal individual ants respond to. The neural network answers it at computational scale, using the weighted, thresholded firing of individual neurons as the local signal each unit responds to. Neither the ant nor the artificial neuron needs to be intelligent for the colony or the network to behave, from the outside, as though it is.

This has a practical implication for how this book has asked you to evaluate AI systems throughout. Looking for “understanding” inside any single component of an AI system — a single neuron, a single weight, a single rule — is looking in the wrong place. The capability lives in the pattern of the whole, which is exactly why the opacity problem introduced in Chapter 2 is so difficult to solve: there is no single place to point to and say, this is where the system knows what it knows. Understanding, to the extent the word applies at all, is

distributed across the entire structure, the way the shortest path is distributed across an entire ant colony and exists nowhere else.

## — WHAT COULD GO WRONG? —

### **The Brittleness of Learned Representations**

The hierarchical representations learned by deep neural networks are powerful but brittle in ways that are not obvious until the network encounters inputs that differ from its training distribution in specific ways.

Consider the tiger camera trap network. It has learned, over millions of training images, a hierarchical representation of tigers: edge detections in layer one, stripe textures in layer two, body part shapes in layer three, the complete animal in the final layer. This representation is extraordinarily effective for the kinds of camera trap images on which it was trained: images from the tiger reserves of central India, in infrared, at night, with the particular vegetation backgrounds of those forests.

Now move the network to a camera trap in the Sundarbans mangrove forest of West Bengal. The tigers are the same species, but the background is completely different: dense mangrove roots, brackish water, different vegetation textures. The illumination is different. The camera angle is different because the mangrove understory forces different placement heights. The network's learned representation of forest background, built into its early and middle layers during training on central Indian forest images, is systematically mismatched to the Sundarbans environment. The edge and texture detectors that work perfectly in Kanha fail to parse the mangrove background correctly, and this failure propagates upward through the hierarchy, degrading the final tiger classification.

This is the brittleness problem: deep networks learn representations that are highly specific to the statistical properties of their training data, and those representations can fail unexpectedly when the input distribution shifts. The failure is not random or uniform: it affects specific patterns in specific ways that are determined by the structure of the training data, which is often not transparent to the user of the system.

The brittleness problem has practical consequences wherever deep learning models trained in one context are deployed in another. A medical imaging model trained on scans from urban hospitals in Mumbai may fail on scans from rural district hospitals that use older equipment with different imaging characteristics. A speech recognition model trained on standard Hindi may fail on the Bhojpuri-inflected Hindi spoken in rural Bihar. A document processing model trained on printed invoices may fail on handwritten receipts. In each case, the failure is caused by a mismatch between the statistical regularities in the training data and the statistical regularities of the deployment context, mediated through the specific hierarchical representations the network has learned.

The solution to brittleness, where a solution is possible, is domain adaptation: retraining or fine-tuning the network on data from the deployment context. This requires that such data be available and labelled, which is not always the case. Where domain adaptation is not possible, honest evaluation of the network's performance on deployment-context data, before deployment, is essential. Deploying a network without testing it on data from the actual deployment environment is an act of institutional negligence, regardless of how well the network performed during development.

## — CHAPTER SUMMARY —

### Chapter 7 — Key Points

1. An artificial neuron computes a weighted sum of its inputs and fires if the sum exceeds a threshold — it is a single pattern detector.
2. Weights are the learned parameters; training adjusts them to minimise prediction error across the training data.
3. A single layer can only detect linear combinations of inputs; stacking layers enables detection of hierarchically complex patterns.
4. Layer hierarchy: early layers detect edges and simple features; middle layers detect shapes and textures; deep layers detect objects and concepts.
5. The mridangam analogy: frequency components → syllables → phrases → compositions mirrors the neural network's layer hierarchy exactly.
6. The tiger camera trap example shows layer-by-layer detection: edges → stripe textures → body parts → tiger identity.
7. Depth provides exponential representational efficiency: complex patterns are built by composing simpler ones, not by enumerating them.
8. Deep networks are powerful but brittle: learned representations are specific to training data statistics and can fail when the deployment context differs.
9. Domain adaptation — fine-tuning on deployment-context data — is the primary remedy for brittleness, but requires labelled local data.

## — MODEL QUESTIONS —

### Section A — MCQ (5 Questions)

#### **Q1. In an artificial neuron, what role do the weights play?**

- A) They determine the number of inputs the neuron can receive
- B) They represent the learned strength of each input's influence on the neuron's output
- C) They set the threshold above which the neuron fires
- D) They control the learning rate during backpropagation

**Answer: B**

*Explanation: Weights are the learnable parameters of an artificial neuron. Each weight scales one input's contribution to the weighted sum. During training, weights are adjusted so that the neuron's output matches desired patterns.*

**Q2. Why can a single-layer neural network only detect linearly separable patterns?**

- A) A single layer has too few neurons to process complex data
- B) A single layer's decision boundary in feature space is always a straight line or hyperplane
- C) Single-layer networks cannot use activation functions
- D) Single layers require too much training data to learn non-linear patterns

**Answer: B**

*Explanation: A single layer computes weighted sums and applies a threshold, producing a linear decision boundary in the input feature space. Non-linear patterns require curved or complex boundaries that cannot be represented by a single linear operation.*

**Q3. In a deep network analysing camera trap images, what would a second-layer neuron most likely detect?**

- A) The identity of the specific animal species in the image
- B) Individual pixel brightness values
- C) Combinations of edges forming textures and simple shapes
- D) The geographic location where the photograph was taken

**Answer: C**

*Explanation: Second-layer neurons receive outputs from first-layer edge detectors and combine them into more complex patterns: textures, simple shapes, and corners. Animal identity only emerges at much deeper layers.*

**Q4. The mridangam analogy illustrates neural network depth because:**

- A) The mridangam produces patterns that are difficult to classify
- B) Recognition proceeds hierarchically from acoustic features to syllables to phrases to compositions, mirroring layer-by-layer abstraction
- C) Mridangam performance data was used to train early neural networks
- D) The mridangam has exactly as many distinct sounds as a typical neural network has layers

**Answer: B**

*Explanation: The recognition hierarchy in expert mridangam listening — acoustic features, syllables, rhythmic phrases, compositions — is a direct parallel to the hierarchical abstraction in neural network layers, making it a precise and illuminating analogy.*

**Q5. A deep learning model trained on ECG data from urban hospitals is deployed in a rural health camp and performs significantly worse. The most likely explanation is:**

- A) Rural patients have more complex heart conditions
- B) The ECG machines used in rural camps differ in their recording characteristics from training data machines
- C) Deep networks always perform worse when deployed than during training
- D) ECG interpretation requires more layers than the model has

**Answer: B**

*Explanation: This is a distribution shift / brittleness problem. The network's learned representations are specific to the statistical properties of urban hospital ECG recordings. Different equipment produces signals with different noise characteristics, baseline wander, and resolution, causing the learned representations to fail.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Describe what a single artificial neuron does, using a specific example from crop disease detection.**

**Model Answer:**

A single artificial neuron receives numerical inputs, multiplies each by a learned weight, sums the weighted inputs, and produces an output if the sum exceeds a threshold. In a crop disease detection network, one neuron might receive inputs representing the intensity of a brown colour in a specific image region, the presence of a yellow halo around that region, and the roughness of the texture in that region. During training, the weights for brown intensity and yellow halo presence become large and positive, because these features co-occur with the disease label in the training data. The weight for smooth texture becomes small or negative, because smooth texture is absent in disease cases. The neuron therefore fires a high output specifically when it encounters the combination of brown intensity, yellow halo, and rough texture that characterises the disease, and remains quiet for healthy leaves. It is a single, specific pattern detector for one visual combination associated with the disease.

*Key terms: weighted inputs, threshold, activation, pattern detector, learned weights, training, backpropagation*

**Q7. Explain the four-layer hierarchy in a tiger camera trap recognition network, from pixel inputs to species identification.**

**Model Answer:**

Layer one, the input layer, holds the raw pixel intensity values of the camera trap photograph, typically infrared, one value per pixel in the image grid. Layer two, the first hidden layer, learns to detect edges: places where adjacent pixel intensities change sharply. This produces an edge map showing the outlines of the animal, trees, and ground cover, with no species-specific information yet present. Layer three, the second hidden layer, combines edges into textures and simple shapes. The regular alternation of light and dark stripes characteristic of a tiger's coat becomes detectable at this level as a distinctive texture pattern. Layer four and beyond build from textures to body part shapes, such as the rounded head profile and characteristic ear shapes, and from body part combinations to full species identity. The tiger node in the output layer fires when the combination of stripe texture, body proportions, and ear shapes collectively match the tiger pattern learned during training. Each level requires the level below to have correctly processed its portion of the hierarchy.

*Key terms: edge detection, texture, body part, object recognition, hierarchical representation, layer hierarchy, convolutional neural network*

**Q8. Using the mridangam analogy, explain why depth in a neural network enables recognition of complex patterns that shallow networks cannot**

**capture.**

**Model Answer:**

The mridangam analogy shows that complex patterns have hierarchical structure: a composition is built of cycles, cycles of phrases, phrases of syllables, and syllables of acoustic features. Recognition proceeds level by level, with each level providing input to the next. A shallow network, like a listener who can only process individual acoustic features without grouping them into syllables, cannot recognise the higher-level structure because it has no mechanism for composition. A deep network, by contrast, builds representations layer by layer: early layers detect acoustic edges and frequency components, middle layers detect syllable-level combinations of those features, and deep layers detect phrase structures built from syllable combinations. The ability to compose lower-level detections into higher-level ones at multiple levels of hierarchy is what deep networks provide and shallow networks lack. The compositional efficiency is exponential: a small set of elementary detectors, composed across many layers, can represent vastly more complex patterns than the same total number of detectors arranged in a single layer.

*Key terms: hierarchical representation, compositional efficiency, shallow network, deep network, pattern hierarchy, mridangam, depth advantage*

**Q9. What is the brittleness problem in deep neural networks, and why is it especially significant for AI deployments in India?**

**Model Answer:**

The brittleness problem is the tendency of deep neural networks to fail when their input distribution shifts away from the distribution of their training data. Because deep networks learn highly specific hierarchical representations of the statistical regularities in their training data, they can be extremely accurate within that distribution but fail unexpectedly when inputs have different characteristics. In India, this is especially significant because the country has extraordinary variation in language, script, dialect, equipment quality, lighting conditions, crop varieties, and disease presentations across its regions and socioeconomic contexts. A model trained on urban hospital ECGs may fail on rural camp ECGs recorded on older equipment. A crop disease model trained on high-resolution photographs from well-resourced farms may fail on low-resolution photographs taken with budget phones on smallholder farms. A speech model trained on standard Hindi may fail on dialects. In each case, the failure reflects a training-deployment mismatch that is predictable but often undetected because the model was never evaluated on deployment-context data before release.

*Key terms: brittleness, distribution shift, domain adaptation, training distribution, deployment context, fine-tuning, representation specificity*

**Q10. Why does a deep network learn hierarchical representations without being explicitly told to? What drives this emergent hierarchy?**

**Model Answer:**

The hierarchical representations in deep networks are not programmed by engineers but emerge from the training process itself, driven by two forces. First, the structure of natural data is genuinely hierarchical: images are composed of objects, objects of parts, parts of textures, textures of edges. The network that minimises its prediction error on image data will naturally discover and exploit this hierarchical structure because it is the most efficient way to represent the patterns in the data. Second, the architecture of the network, with layer after layer of weighted summation and non-linear activation, is suited to compositional representation: each layer can only access information through the representation produced by the previous layer, so to recognise objects the final layers must build on part detections, which must build on texture detections, which must build on

edge detections. The hierarchy emerges because it is the most efficient structure for the task, given the architecture and the data.

*Key terms: emergent hierarchy, compositional representation, training dynamics, natural data structure, architectural constraint, efficiency*

## Section C — Reflective Questions (3 Questions)

**Q11. The chapter describes deep learning as having no direct biological equivalent for its learning process (backpropagation), while sharing structural intuitions with biological neural networks. Does this matter? Should we care whether AI learning processes resemble biological ones?**

**Thinking prompts:**

- What would it mean for an AI system to truly learn the way a biological brain learns?
- Does biological plausibility make an AI system more trustworthy, more interpretable, or more robust?
- Is the goal to build systems that think like humans, or systems that perform tasks that humans perform?
- What are the risks of over-identifying AI systems with biological cognition?
- How might a Vedantic perspective on the nature of mind relate to the question of whether silicon-based pattern recognition constitutes understanding?

*Rubric hint: Strong responses will distinguish between functional performance and genuine understanding, and will engage with the conceptual question of whether biological plausibility is a requirement, a heuristic, or irrelevant for AI. Credit for responses that draw on philosophical frameworks beyond purely technical considerations.*

**Q12. The chapter describes the brittleness problem as causing predictable failures when models are deployed outside their training distribution. Who bears responsibility when such a failure causes harm — the developer, the deployer, or the regulator?**

**Thinking prompts:**

- The developer knew the training data distribution; the deployer chose the deployment context. Who understood the mismatch better?
- What due diligence should a hospital or government agency perform before deploying an AI model in a new context?
- Are there contexts where deployment without local validation is ever acceptable? What would those conditions require?
- What regulatory frameworks in India currently govern this question?
- How does the distribution shift problem interact with the opacity of deep networks discussed in Chapter 2?

*Rubric hint: Strong responses will distribute responsibility across the three parties proportionally to their knowledge and capacity to act, rather than assigning it solely to one party. Credit for responses that identify specific due diligence steps that each party should take, and that engage with the Indian regulatory context.*

**Q13. The mridangam analogy connects a traditional Indian art form to a core**

**concept in modern AI. Is this connection purely illustrative, or does it suggest something deeper about the relationship between Indian classical arts and computational pattern recognition?**

**Thinking prompts:**

- Classical Indian music has formalised, hierarchically structured theory — raga, tala, and their subdivisions. Is this formal structure related to the kind of structure neural networks learn?
- Are there other Indian art forms or knowledge traditions that have analogous hierarchical structures?
- Does the fact that a neural network can learn to recognise mridangam syllables tell us anything about what the syllables are?
- What does it mean for a machine to recognise a raga? Is that the same as a human recognising a raga?
- How might the intersection of classical Indian music theory and machine learning become a site of genuine intellectual exchange rather than just analogy?

*Rubric hint: Strong responses will move beyond the analogy as mere illustration and engage with the question of whether there is substantive intellectual overlap between formalist traditions in Indian classical arts and computational pattern recognition. Credit for responses that identify specific structural parallels and that engage honestly with what machine recognition does and does not share with human musical understanding.*

## **Section D — Stream Project Prompts (2 Questions)**

**Q14. [Arts / Science] Choose one AI application in your field that uses deep learning — this could be Odia OCR, plant disease detection, satellite image analysis, wildlife identification, or ECG interpretation. Investigate how the network's layer hierarchy maps onto the structure of the patterns it is recognising: what would the early, middle, and final layers be detecting? Then identify one specific way in which the brittleness problem could manifest if the model were deployed in a context different from its training data, and propose one step that would reduce this risk.**

The layer hierarchy analysis requires thinking carefully about the structure of the patterns in your chosen domain, not just describing the network generally. The brittleness analysis should be specific to the training-deployment mismatch relevant to your domain — not generic statements about distribution shift.

**Q15. [Commerce] A financial services company in your region is considering deploying a deep learning model for automated document processing — specifically, for reading and extracting data from invoices and purchase orders submitted by small businesses. The model was trained on large-format, digitally printed documents from major Indian corporations. The small businesses in your region primarily submit handwritten or dot-matrix-printed documents. Analyse this deployment scenario from the perspective of neural network brittleness: which layers of the network are most likely to fail on the new document type, why, and what data would be needed to adapt the model for this context? Conclude with a recommendation.**

This project requires connecting neural network layer structure to the specific visual differences between printed and handwritten documents. The layer analysis should be specific — which early-layer representations are mismatched, and how does that propagate upward. The recommendation should address data collection, not just model architecture.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. <i>Nature</i> , 521(7553), 436-444. [The foundational overview paper by three of the field's originators. Accessible to non-specialists as an introduction to why depth matters and how deep learning works.]                                                        |
| <b>Core</b> | Nielsen, M. (2015). <i>Neural Networks and Deep Learning</i> . neuralnetworksanddeeplearning.com. [Freely accessible online book with clear, minimal-mathematics explanations of every concept in this chapter, including the single neuron, layers, and why depth provides representational power.]                            |
| <b>Core</b> | Srinivasan, R., et al. (2022). Deep learning for Odia optical character recognition: Architecture and evaluation. <i>Proceedings of the International Conference on Document Analysis and Recognition</i> . [Directly relevant to the Arts stream section; a peer-reviewed study on deep learning for Odia script recognition.] |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Zeiler, M. D., & Fergus, R. (2014). Visualising and understanding convolutional networks. <i>European Conference on Computer Vision (ECCV)</i> . [The first systematic study of what each layer of a deep convolutional network actually detects; directly supports the tiger camera trap discussion.]                                                    |
| <b>Extended</b> | Hannun, A., et al. (2019). Cardiologist-level arrhythmia detection and classification in ambulatory ECGs using a deep neural network. <i>Nature Medicine</i> , 25(1), 65-69. [Landmark paper on deep learning ECG interpretation; directly relevant to the Science stream section and to rural health applications.]                                      |
| <b>Extended</b> | Gulshan, V., et al. (2019). Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. <i>JAMA</i> , 316(22), 2402-2410. [Illustrates both the power and the brittleness of deep medical imaging models in Indian contexts — the Aravind Eye Hospital validation is directly relevant.] |

### Accessible Online Resources

|               |                                                                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | 3Blue1Brown. (2017). But What Is a Neural Network? [Video series]. <a href="https://www.youtube.com/3blue1brown">youtube.com/3blue1brown</a> . [The clearest visual explanation of neural networks] |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                                                                                                                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | available; covers weighted sums, layers, and gradient descent without requiring mathematical background. Directly complements every section of this chapter.]                                                                                                                                                                                                |
| <b>Online</b> | Distill.pub. (2020). Feature Visualisation. <a href="https://distill.pub/2017/feature-visualization">distill.pub/2017/feature-visualization</a> . [Interactive visualisations of what individual neurons in deep networks actually detect; directly supports the tiger camera trap layer discussion.]                                                        |
| <b>Online</b> | Google AI. (2022). Teachable Machine. <a href="https://teachablemachine.withgoogle.com">teachablemachine.withgoogle.com</a> . [A no-code interface that allows students to train a simple image classifier and observe the effect of training data quality, overfitting, and distribution shift firsthand. Directly relevant to the stream project prompts.] |

### *End of Chapter 7*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## UNIT THREE

# AI in the World

*A tool is judged by what it does in someone's hands, not by what it does in a demonstration. This unit leaves the classroom examples behind and follows AI into language, into images, into the ordinary tools a student might actually open on a Tuesday afternoon.*

*The question changes accordingly. Not just how does this work, but when should I trust it, and when should I not. A translation that is almost right. A diagnosis suggested by a model that has never seen a patient like this one. A prompt that gets a better answer only because it was asked better. These are not edge cases. They are the ordinary texture of using AI in India, today.*

- 
8. AI That Speaks and Listens
  9. AI That Sees
  10. AI Tools: A Practical Field Guide

## Chapter 8

# AI That Speaks and Listens

*Language is the most extraordinary thing human beings do. We take the experiences, intentions, and ideas inside our minds and convert them into sequences of sounds or marks that other minds can decode and reconstruct into something resembling the original. We do this continuously, effortlessly, and from early childhood, with no formal instruction in the underlying mechanisms. We speak and listen without knowing how we speak and listen.*

*Making machines do something similar has been one of the oldest and most persistent goals of artificial intelligence. For decades, progress was slow. Rule-based approaches that worked reasonably well for constrained tasks, such as translating weather reports or answering narrowly defined questions, failed completely when applied to the open-ended diversity of real human communication. Language, it turned out, was too context-dependent, too ambiguous, and too culturally embedded for rules to capture.*

*That changed dramatically in the 2010s, and the change accelerated through the early 2020s. Today, AI systems can hold conversations, translate between languages, write coherent documents, answer complex questions, and generate creative text. This chapter explains how. It also explains why Indian languages present particular challenges, why those challenges matter, and what it means for a machine to understand language at all.*

## Learning Objectives

By the end of this chapter, you will be able to:

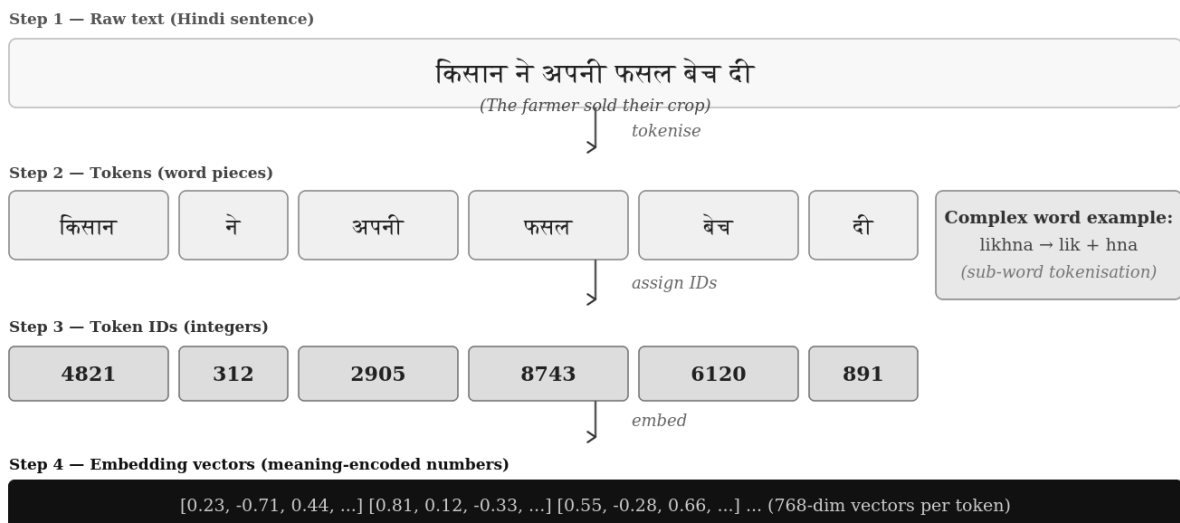
- Explain tokenisation and its role in converting text to a form machines can process
- Describe word embeddings and the concept of a meaning space with Indian language examples
- Explain why Indian language translation is technically harder than European language pairs
- Describe how a large language model works as a next-word predictor using the library metaphor
- Define the context window and explain its practical significance for LLM applications

## 8.1 From Words to Numbers: Tokenisation

A machine learning model works with numbers. Text is not numbers. The first challenge of natural language processing is therefore to convert text into a numerical form that a model can process without losing the information that makes the text meaningful.

The process begins with tokenisation: breaking the input text into units called tokens. A token is not exactly the same as a word, though for simple English text they often coincide. A tokeniser might break the English sentence The farmer sowed paddy into the tokens The, farmer, sowed, paddy. But it might also break a less common word like Bhubaneswar into sub-word tokens: Bhū, ba, neswar, because the tokeniser has seen the components of the word more often than the whole word and represents it more efficiently as a combination of familiar sub-units.

**Fig. 8.1**  
**Tokenisation: From Text to Numbers**



Each token gets a 768-dimensional vector — a point in mathematical space.  
Hindi and Tamil tokens typically require more dimensions to encode meaning than English tokens.

Figure 8.1 — Tokenisation

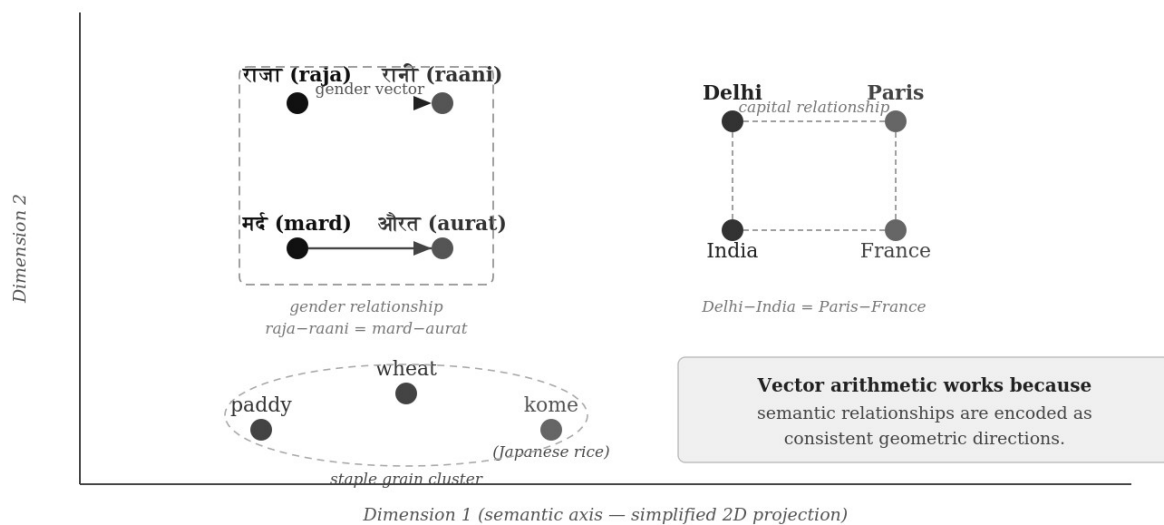
For Indian languages, tokenisation is significantly more complex than for English. English is a relatively low-morphology language: words do not change their form much based on grammatical role. Hindi, Telugu, Tamil, Odia, and most other Indian languages are highly morphological: a single root word can take dozens of different surface forms depending on tense, number, gender, case, and aspect. The Hindi verb *likhna*, to write, can appear as *likhta*, *likhi*, *likhega*, *likhenge*, *likh*, *likhkar*, and many more forms, each of which carries distinct grammatical information. A tokeniser that treats each surface form as a separate token will produce an enormous vocabulary with many rare tokens. A tokeniser that breaks words into meaningful sub-units, called morphemes, can represent the same information more efficiently and allow the model to share statistical strength across related forms.

After tokenisation, each token is converted to a numerical identifier, an integer in the model's vocabulary. The sentence is now a sequence of integers, which can be passed to the model. But a sequence of integers is still not very informative: the integer 4271 for the token *paddy* tells the model nothing about what *paddy* means, how it relates to rice or grain, or why it appears in this context. To capture meaning, a richer representation is needed.

## 8.2 The Meaning Space: Word Embeddings

A word embedding is a vector of numbers that represents the meaning of a word in a high-dimensional mathematical space. The key property of embeddings is that semantically related words are positioned close to each other in this space. Words that appear in similar contexts across millions of documents end up with similar embedding vectors, because they were learned to minimise prediction error in those contexts.

**Fig. 8.2**  
**The Meaning Space: Word Embeddings with Indian Examples**



*This is a 2D projection of a 768-dimensional space. The geometric relationships are real: they emerge from statistical patterns in billions of words of text, not from explicit programming of semantic rules.*

**Figure 8.2 — Meaning Space**

The classic demonstration of word embeddings is the vector arithmetic they enable. In a well-trained embedding space, the vector for *raja* minus the vector for *raani*, plus the vector for *aurat*, gives approximately the vector for *mard*. Or: the vector relationship that encodes the gender difference in *raja* and *raani* also encodes the gender difference in *aurat* and *mard*. The embedding has captured something genuine about the structure of meaning, not through explicit rules but through statistical regularities in billions of words of text.

Similarly, *Delhi* minus *India* plus *France* gives approximately *Paris*. The relationship of capital-to-country is encoded geometrically in the embedding space. *Paddy* minus *India* plus *Japan* gives approximately *kome*, the Japanese word for rice. The concept of a country's staple grain is captured in the geometry of the space.

These are not tricks. They reflect the genuine statistical structure of how words are used across languages and cultures. An embedding trained on a large enough corpus will capture the relationships among words that any fluent user of that language would

recognise, because those relationships are encoded in the patterns of co-occurrence that the corpus contains.

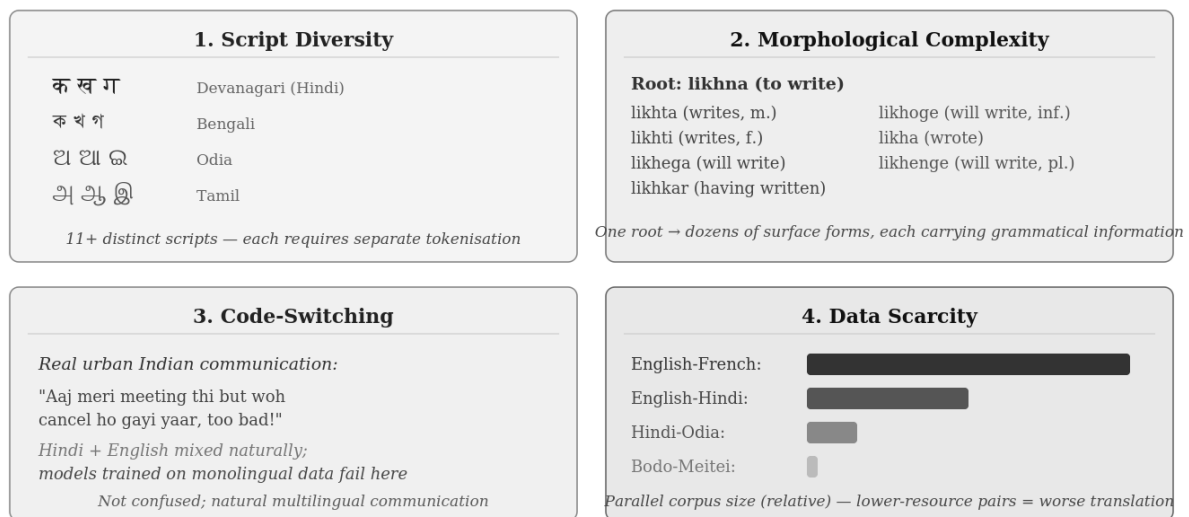
Embeddings for Indian languages have been developed by research groups including AI4Bharat at IIT Madras, which has produced high-quality embeddings for twenty-two Indian languages. These embeddings make it possible to build NLP applications, such as sentiment classifiers, question-answering systems, and translation models, for languages that previously had very limited AI support. The quality of an embedding depends critically on the size and quality of the corpus used to train it. For major Indian languages like Hindi, Tamil, Telugu, and Bengali, reasonably large corpora exist. For smaller languages like Santali, Bodo, Dogri, and Kashmiri, corpora are still very limited, and the corresponding embeddings are correspondingly weaker.

### 8.3 Why Indian Language Translation Is Hard

Machine translation has improved dramatically since the shift from rule-based to learning-based approaches. A model trained on millions of sentence pairs, one in English and one in French, learns the statistical patterns of equivalence between the two languages without any explicit grammar rules. For well-resourced European language pairs, translation quality now approaches human performance on many types of text.

### Fig. 8.3

#### Why Indian Language Translation Is Hard: Four Challenges



*All four challenges compound each other. Data scarcity means fewer examples to learn complex morphology and code-switching from. Better algorithms help at the margins; more diverse, high-quality parallel data is the primary requirement.*

Figure 8.3 — Indian Translation Challenges

Indian language translation presents challenges that European language pairs largely do not. The first is script diversity. India's major languages use at least eleven different scripts: Devanagari for Hindi, Sanskrit, Marathi, and Nepali; Bengali script for Bengali and Assamese; Telugu script; Tamil script; Kannada script; Malayalam script; Gurmukhi for Punjabi; Odia script; Gujarati script; and the Latin script increasingly used for code-mixed

text. A translation model must handle not just different vocabularies and grammars but different writing systems, which affects tokenisation, rendering, and the handling of numbers, punctuation, and proper names.

The second challenge is morphological complexity. As described in Section 8.1, Indian languages are highly inflectional. This means that two sentences with the same meaning can differ substantially in surface form depending on the formality register, the grammatical role of each noun, and the aspect and mood of each verb. A translation model must learn to generate the correct morphological form for each output word in the target language, which requires understanding grammatical context in ways that are less critical for English. The problem is compounded when translating between two highly morphological Indian languages, such as Tamil and Kannada, because both sides of the translation require morphological correctness.

The third challenge is code-switching. Urban Indian communication, in social media, in professional emails, and in conversation, frequently mixes multiple languages in a single utterance. A sentence might begin in Hindi, switch to English for a technical term, use an Urdu word for nuance, and end with a Punjabi colloquialism. This is not confused communication; it is a natural feature of multilingual communities. But current translation models are trained on data from one language at a time, and they struggle with the fluid mixing that characterises real Indian digital communication.

The fourth challenge, which underlies all the others, is data scarcity. Learning-based translation requires large corpora of parallel text: documents that exist in both the source and target language. For English-French or English-German, millions of such parallel documents exist, drawn from news sources, legal documents, and subtitles. For Hindi-Odia or Tamil-Telugu, the parallel corpora are orders of magnitude smaller, and for lower-resource language pairs involving languages like Bodo-Meitei or Santali-Bengali, they barely exist at all. This data scarcity fundamentally limits translation quality for the majority of India's language pairs, regardless of how sophisticated the model architecture becomes.

## **8.4 The Library Metaphor: How Large Language Models Work**

The most powerful language AI systems available today are large language models, or LLMs. They include systems like GPT, Gemini, Claude, and their successors. Understanding what these systems do requires setting aside both the exaggerated claims and the dismissive counter-claims that surround them and focusing on the actual computational process.

**Fig. 8.4**  
**The Library Metaphor: How LLMs Work as Next-Word Predictors**

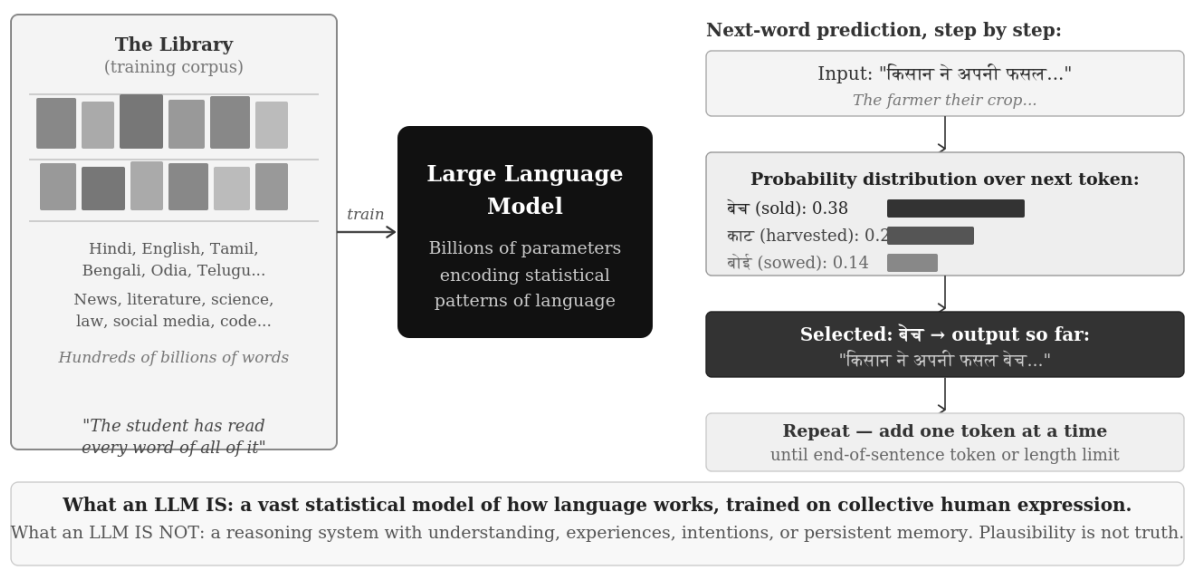


Figure 8.4 — Library Metaphor

The library metaphor. Imagine a student who has spent their entire life in an enormous library. The library contains virtually every book, article, conversation transcript, and document that has ever been written: novels, scientific papers, news reports, social media posts, legal documents, government records, and everything else. The student has read every word of all of it, repeatedly, and has developed an extraordinarily refined sense of how language works: which words follow which other words, which sentences appear in which contexts, which arguments lead to which conclusions, which cultural references appear in which communities.

Now ask this student to complete a sentence. Give them the beginning of a sentence and ask them what word comes next. The student does not look this up in a book; they predict it from the patterns they have absorbed across the entire library. Their prediction is not a rule; it is a probability distribution over all possible next words, conditioned on everything that has come before. The most likely next word in the context *Kisan ne apni fasal* is probably *bechdi*, or *kaati*, or something in the domain of agricultural activity, because that is what follows such sentences in the library.

A large language model is exactly this student, implemented in silicon. It is trained on a corpus so large that it encompasses most of what has been written in digitally accessible form: hundreds of billions of words, across many languages and domains. The training process adjusts billions of parameters so that the model's probability distribution over next words matches the actual distribution of words that appear in the training corpus. The result is a system that can, given any text as context, generate a statistically plausible continuation of that text.

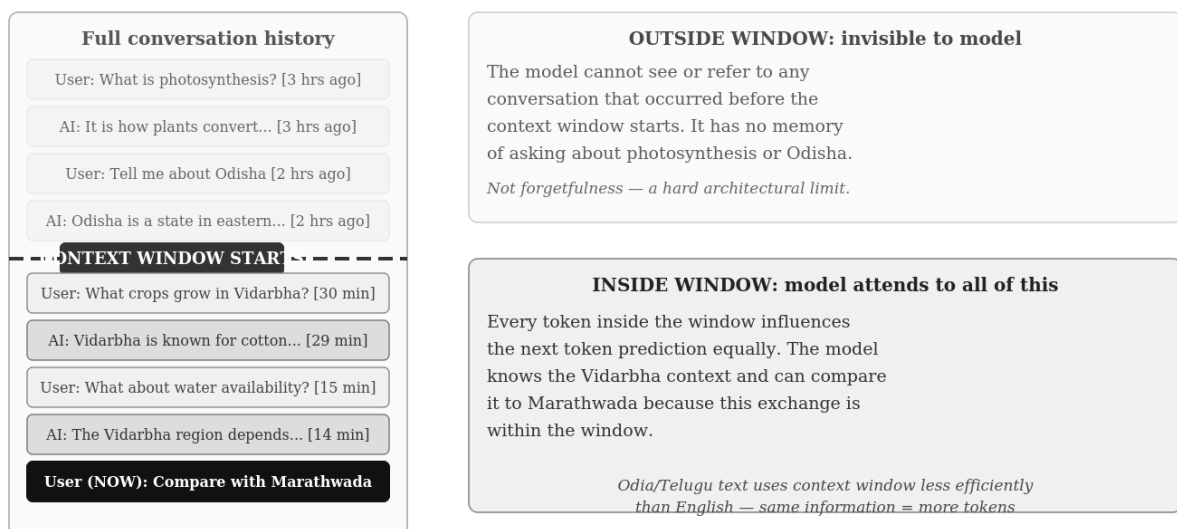
This is not understanding in the way a human understands. The model has no experiences, no body, no persistent memory, no intentions. It has a vast and precisely calibrated statistical model of the patterns of language. When it generates text that appears insightful, creative, or knowledgeable, it is because the patterns in its training data encode the insights, creativity, and knowledge of the billions of humans whose writing it learned from. The model is, in a deep sense, a distillation of collective human expression, filtered through statistical optimisation.

This understanding has important implications. It explains why LLMs are impressive across many domains simultaneously: they have been trained on text from all domains. It explains why they can fail with confident plausibility: they produce what is statistically likely given the context, which is not always what is true. It explains why they are better at tasks that are well-represented in the training corpus, such as writing standard English prose, than at tasks that are poorly represented, such as writing Santali poetry. And it explains why they are not reliable sources of truth: they predict what was commonly written, which is not always what is correct.

## 8.5 The Context Window

Every large language model has a context window: the maximum amount of text it can consider at one time when generating a response. The context window is measured in tokens, and it has grown enormously over recent years. Early large language models had context windows of a few thousand tokens, roughly equivalent to a short essay. Current models have context windows of hundreds of thousands of tokens, roughly equivalent to a full-length book.

**Fig. 8.5**  
**The Context Window: What the Model Can See**



The context window is not a memory system — it is a sliding view over the conversation. Earlier content that falls outside the window is permanently invisible until reintroduced. Longer context windows = more expensive computation.

Figure 8.5 — Context Window

The context window determines what the model can attend to when generating each new token. Everything inside the context window influences the next token prediction. Everything outside it is invisible. This has practical consequences.

In a long conversation with an LLM, if the conversation history exceeds the context window, the earlier parts of the conversation are dropped from the model's consideration. The model does not remember having said something three hours ago if that exchange has scrolled outside the context window. This is not forgetfulness in a human sense; it is a hard architectural limit. The implication for users is that very long conversations, or conversations that require the model to maintain accurate reference to information from early in the session, can produce inconsistencies.

For document processing applications, the context window determines how much of a document the model can analyse at once. Processing a legal contract of fifty thousand words requires a model with a context window large enough to hold the entire contract, plus the question and the response. Processing a book of two hundred thousand words requires an even larger window, or a strategy for breaking the book into overlapping chunks.

For Indian language applications, the context window has a practical implication: Indic scripts are typically less token-efficient than Latin scripts, because the same amount of information requires more bytes to encode in Unicode, and tokenisers optimised for English may need multiple tokens to represent a single Odia or Telugu word. This means that an Indian language document effectively consumes more of the context window than an equivalent English document, and applications that work within the context window for English may reach their limit sooner when processing Indian language text.

## **8.6 Patterns in My Field: Stream Perspectives**

### **Arts Stream: Building a Creative Writing Assistant for Regional Language Poetry**

A creative writing assistant for Odia poetry built on an LLM foundation would use the model's learned statistical knowledge of Odia literary language to suggest continuations of a poem in progress, identify rhyme and metre options consistent with classical Odia poetic forms, and provide vocabulary alternatives that fit the poem's register and imagery. The assistant does not write the poem; it provides a generative partner that understands the conventions of the form. Building such a system requires a model fine-tuned on a corpus of Odia poetry, folk songs, and literary texts rather than general-purpose Odia text, because poetic language has different statistical properties than everyday prose. The challenge is that publicly available Odia poetic corpora are small, and training or fine-tuning a high-quality model requires either substantial data collection or techniques that allow good performance with limited data, such as transfer learning from better-resourced languages.

### **Science Stream: Using Google Teachable Machine to Classify Plant Diseases**

A field botanist in the Eastern Ghats uses an LLM-assisted identification system that combines image recognition for initial species suggestion with a language model component that can respond to follow-up questions: Does this species have medicinal properties documented in Ayurvedic texts? What is its conservation status under Schedule VI of the Wildlife Protection Act? Is there a local Odia name for this plant? The language model component, trained on botanical texts, Ayurvedic literature, and conservation documents, can answer these questions because this information is present in its training corpus. The limitation is that the model's knowledge has a training cutoff and may not reflect recent updates to conservation status or newly documented ethnobotanical uses. Cross-referencing with authoritative databases remains essential.

### **Commerce Stream: Using AI to Analyse Customer Sentiment for a Local Business**

A small hotel group in Puri receives customer reviews in multiple languages: English, Hindi, Odia, and occasionally Bengali and Telugu. A sentiment analysis system built on multilingual embeddings can classify each review as positive, negative, or mixed and extract the specific aspects being commented on: room cleanliness, food quality, staff behaviour, proximity to the temple. The multilingual capability allows the system to process all reviews in a unified pipeline without requiring separate models for each language. The practical limitation is that the quality of sentiment analysis varies across languages: it is typically most accurate for Hindi and English, where training data is most abundant, and less accurate for Odia and Telugu, where training data is more limited. This accuracy disparity means that the hotel group's insights from Odia-language reviews may be less reliable than those from Hindi or English, which is a form of the representation bias discussed in Chapter 4 operating at the application level.

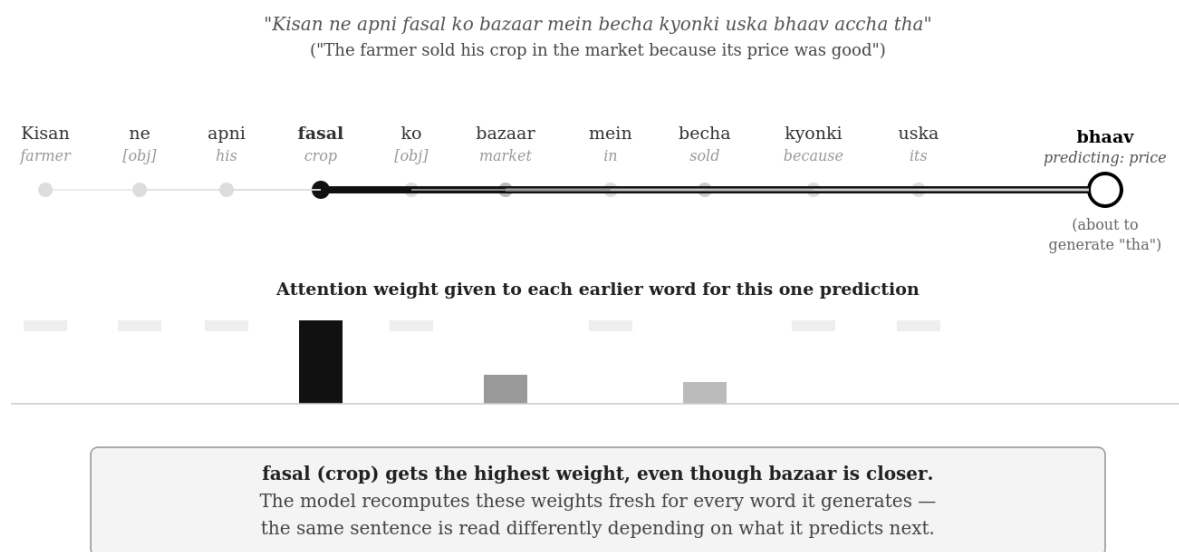
## **8.7 How a Transformer Actually Attends**

Section 8.4's library metaphor explained what a large language model does: predict the next word from a probability distribution shaped by everything the model has read. It left one crucial question unanswered: when predicting that next word, which of the potentially thousands of preceding words in the context should the model weigh most heavily? Not every earlier word is equally relevant to predicting the next one, and knowing which ones matter, for this specific prediction, at this specific point in the sentence, turns out to be the single most consequential architectural idea in modern AI. It is called attention, and the architecture built around it is called the transformer.

Consider the Hindi sentence *Kisan ne apni fasal ko bazaar mein becha kyunki uska bhaav accha tha* ("The farmer sold his crop in the market because its price was good"). When the model is generating the word immediately after *uska* ("its"), predicting that the word *bhaav* ("price") belongs to *fasal* (the crop) is essential, and *fasal*, though several words back, matters far more to this specific prediction than the intervening words *ko*, *bazaar*, or *mein*. A model that treated every previous word as equally relevant, the way earlier NLP architectures effectively did, would dilute this crucial signal among dozens of less relevant ones.

Attention solves this by letting the model compute, fresh for every single word it generates, a specific relevance score between the word it is about to predict and every word already in the context. Words that are highly relevant to the current prediction receive a high attention weight; words that are not relevant receive a weight close to zero. Practically, this means the model does not read the sentence once and form a single fixed summary of it, the way a simpler architecture might. Instead, for every new word it generates, it re-reads the entire context and asks, specifically for this prediction, which earlier words deserve attention right now? The word *fasal* receives high attention when the model is about to generate something about the crop; the word *bazaar* receives high attention when the model is about to generate something about the location of the sale. The same sentence is, in effect, read differently depending on what the model is currently trying to predict.

**Fig. 8.6**  
**How Attention Works**



*Real transformers run many such attention computations in parallel per layer ("attention heads"), each free to specialise in a different kind of relevance — none of it hand-specified by an engineer.*

Figure 8.6 — How Attention Works

This mechanism, formalised in a 2017 paper whose title made an immodest but accurate claim, *Attention Is All You Need*, is what the transformer architecture is built around. Every major language model discussed in this chapter and the next — ChatGPT, Gemini, Claude, and their successors — is a transformer: a stack of layers, each of which recomputes attention weights across the full context before passing an updated representation to the next layer. Multiple attention computations run in parallel within each layer, called attention heads, each one free to specialise in a different kind of relevance: one head might learn to track grammatical subject-verb agreement across long distances, another might track which pronoun refers to which earlier noun, another might track topical relevance regardless of grammatical role. No engineer specifies what each head should track; like the weights in Chapter 7's neural networks, these specialisations emerge from training on enormous quantities of text.

The practical consequence for a user of these tools is significant, even without any technical understanding of the mechanism itself. Attention is what allows a model to

correctly resolve which earlier noun a pronoun refers to, to maintain a coherent argument across a long document, and to draw a precise connection between a question asked early in a conversation and a follow-up asked much later, so long as both remain within the context window described in Section 8.5. It is also why performance degrades, sometimes sharply, when a genuinely relevant piece of context is buried deep among a large amount of irrelevant text: attention weights are a relevance ranking, not a guarantee, and a model asked to find one relevant sentence inside fifty irrelevant pages is being asked to do exactly what the mechanism is for, under exactly the conditions where it is hardest to do well.

## — WHAT COULD GO WRONG? —

### **Hallucination and the Plausibility Trap**

The most distinctive and consequential failure mode of large language models is hallucination: the generation of text that is confident, fluent, plausible-sounding, and factually wrong. An LLM asked about a historical event may produce a coherent, well-structured account of something that never happened. An LLM asked about a scientific paper may cite papers that do not exist, with plausible-sounding titles, authors, and journal names. An LLM asked to translate a legal document may translate accurately most of the time but silently introduce errors that are invisible to anyone who does not know both languages well.

Hallucination is not a bug that can be fixed with more training data or better architecture. It is a consequence of the fundamental nature of the system. An LLM generates the most statistically likely continuation of the input, and the most statistically likely continuation is not always the true one. In domains where the training data is dense and consistent, hallucination rates are lower. In domains where the training data is sparse, inconsistent, or where the question requires genuine novel reasoning rather than pattern completion, hallucination rates are higher.

For Indian languages and Indian contexts, this problem is amplified. The training data for major LLMs is heavily weighted toward English-language text. Indian history, Indian science, Indian culture, and Indian languages are represented at a fraction of the density of their English-language equivalents. This means LLMs are more likely to hallucinate when answering questions about Indian topics, particularly questions about regional history, local government, tribal knowledge systems, or contemporary Indian research that has not been widely translated into English.

A student who uses an LLM to research Odia folk traditions, the history of the Kondh peoples of Odisha, or the agronomic practices of Dongria Kondh farmers will encounter a model that sounds confident but is drawing on sparse and potentially unreliable training data. The hallucinations in this domain are not random noise; they are statistically plausible confabulations that sound like reliable information to a reader who does not already know the truth. This is the plausibility trap: the model's confidence and coherence are highest precisely when they are least deserved as indicators of accuracy, because

confidence and coherence are properties of the language generation, not of the underlying knowledge.

The practical implication is straightforward: LLMs should never be the sole or primary source for any factual claim that matters. They are powerful tools for drafting, summarising, exploring ideas, and generating initial text. They are unreliable tools for verifying facts, especially facts about under-resourced topics. Every LLM response that makes a specific factual claim should be treated as a lead to be verified, not a conclusion to be cited.

## — CHAPTER SUMMARY —

### Chapter 8 — Key Points

1. Tokenisation converts text to sequences of tokens; Indian languages require morpheme-aware tokenisation due to high inflectional complexity.
2. Word embeddings place words in a high-dimensional meaning space where semantic proximity corresponds to geometric proximity.
3. Embedding vector arithmetic (raja - raani + aurat = mard) captures genuine structural regularities in language.
4. Indian language translation faces four compounding challenges: script diversity, morphological complexity, code-switching, and data scarcity.
5. LLMs work as next-word predictors trained on vast corpora — the library metaphor captures what they are and what they are not.
6. LLMs distil collective human expression through statistical optimisation; they have no experiences, intentions, or persistent memory.
7. The context window determines what text the model can attend to; Indic scripts consume context windows less efficiently than Latin scripts.
8. Hallucination is a fundamental property of LLMs, not a fixable bug; it is more severe for under-resourced topics and languages.
9. LLMs are powerful drafting and exploration tools but unreliable verification tools — every factual claim requires independent confirmation.

## — MODEL QUESTIONS —

### Section A — MCQ (5 Questions)

#### **Q1. What is a token in the context of natural language processing?**

- A) A digital currency used to pay for API access to language models
- B) A unit of text — roughly a word or sub-word — that the model processes as a single unit
- C) A security certificate that authenticates a model's output

D) A grammatical unit equivalent to a sentence in linguistic analysis

**Answer: B**

*Explanation: A token is the basic unit of text that a language model processes. It can be a word, a sub-word piece, or a punctuation mark. Tokenisation breaks input text into these units before the model can process them.*

**Q2. The key property of word embeddings that makes them useful for NLP is:**

- A) They compress text into fewer characters for efficient storage
- B) They assign a unique integer identifier to each word in the vocabulary
- C) Semantically related words have similar embedding vectors, placing them close in meaning space
- D) They translate words from one language to another automatically

**Answer: C**

*Explanation: The power of embeddings lies in their geometry: words with similar meanings or usage patterns are placed close together in the vector space. This allows models to generalise across similar words and to capture semantic relationships mathematically.*

**Q3. Which of the following best explains why code-switching poses a challenge for Indian language NLP?**

- A) Code-switching is grammatically incorrect and should not be processed
- B) Current models are trained on monolingual data and struggle when multiple languages appear in a single utterance
- C) Code-switching only occurs in written text and models are primarily trained on speech
- D) Code-switching involves changing the script mid-sentence, which tokenisers cannot handle

**Answer: B**

*Explanation: Code-switching, the fluid mixing of languages within an utterance, is a natural feature of multilingual Indian communication. Most NLP models are trained on monolingual corpora and lack the cross-lingual representations needed to handle code-switched text reliably.*

**Q4. In the library metaphor, what does the LLM's 'prediction of the next word' represent technically?**

- A) A lookup in a memorised dictionary of sentence completions
- B) A probability distribution over all vocabulary tokens, conditioned on the preceding context
- C) A random selection from words that have appeared near similar words in the training data
- D) A rule-based selection following the grammatical constraints of the input language

**Answer: B**

*Explanation: An LLM outputs a probability distribution over all tokens in its vocabulary for the next position, conditioned on all tokens in the context window. The generation process samples from this distribution, typically selecting high-probability tokens.*

**Q5. Why is hallucination described as a fundamental property of LLMs rather**

than a fixable bug?

- A) It is caused by insufficient computational power that cannot be increased
- B) LLMs generate statistically likely text, which is not always factually accurate — accuracy and likelihood are different objectives
- C) Hallucination only occurs in smaller models and disappears with scale
- D) It is caused by errors in the training data that are too numerous to correct

**Answer: B**

*Explanation: LLMs optimise for generating statistically likely text given a context. Statistical likelihood and factual accuracy are different objectives. A plausible-sounding but false statement can have higher statistical likelihood than the true statement, especially in domains where the training data is sparse or inconsistent.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Explain tokenisation and why it is more complex for Hindi than for English, using specific examples.**

**Model Answer:**

Tokenisation is the process of breaking input text into discrete units called tokens that a language model can process. For English, tokenisation is relatively straightforward because English words are separated by spaces and change form only modestly: the word *write* produces *writes*, *writing*, and *wrote*, a small set of variants. For Hindi, the same root *likhna*, to write, produces *likhta*, *likhti*, *likhoge*, *likhkar*, *likha*, *likhna*, *likhenge*, and many more forms, each encoding distinct grammatical information about tense, number, gender, aspect, and formality. A tokeniser that treats each surface form as a separate token must maintain an enormous vocabulary with many rare forms, reducing the statistical reliability of predictions about those forms. Morpheme-aware tokenisation, which breaks complex forms into root plus grammatical suffixes, allows the model to share statistical knowledge across related forms and produces more efficient representations of Hindi text.

*Key terms: tokenisation, token, morphology, inflection, morpheme, vocabulary, sub-word tokenisation, agglutinative*

**Q7. Describe the concept of word embeddings using the raja/raani example, and explain what the vector arithmetic tells us about the embedding space.**

**Model Answer:**

A word embedding is a vector of numbers that represents a word's meaning by encoding its patterns of use across a large corpus. Words that appear in similar contexts receive similar vectors, placing them close together in a high-dimensional meaning space. The *raja/raani* example illustrates the geometric structure of this space. The vector difference between *raja* (king) and *raani* (queen) encodes the gender relationship: starting from *raja* and moving in the direction that leads toward *raani* traverses the gender dimension of the meaning space. The observation that the same vector difference also relates *mard* (man) and *aurat* (woman) shows that this gender dimension is consistent across multiple pairs of gender-related words. The arithmetic is not a trick; it reflects the genuine statistical regularity in the corpus that words encoding the male-female distinction tend to occur in similar structural relationships across different domains. This property makes embeddings powerful: they capture semantic relationships that can be used for analogical reasoning, translation, and conceptual organisation.

*Key terms: word embedding, vector, meaning space, geometric proximity, vector arithmetic, semantic relationship, co-occurrence, analogy*

**Q8. Identify and explain two of the four challenges that make Indian language machine translation harder than European language pair translation.**

**Model Answer:**

The first challenge is script diversity. India's major languages use at least eleven different scripts, from Devanagari to Tamil to Odia to Gurmukhi. European language pairs like English-French or German-Spanish share or closely resemble the Latin script, simplifying tokenisation, rendering, and the handling of named entities and numbers. Indian language translation must additionally handle script conversion and the specific tokenisation challenges of each script. The second challenge is morphological complexity. Indian languages are highly inflectional: a single verb root can appear in dozens of surface forms encoding tense, number, gender, case, and formality. Translation models must learn to generate the correct morphological form for each word in the target language, requiring a much deeper representation of grammatical structure than is needed for relatively low-morphology languages like English. Both challenges compound the fundamental problem of data scarcity, since less training data is available to learn these complex patterns.

*Key terms: script diversity, morphological complexity, code-switching, data scarcity, parallel corpus, inflection, tokenisation*

**Q9. Explain the context window and describe one practical consequence of a finite context window for an Indian language application.**

**Model Answer:**

The context window is the maximum number of tokens a language model can attend to at one time when generating a response. Everything within the window influences the generation; everything beyond it is invisible. As conversations grow longer or documents grow larger, earlier content scrolls out of the window and the model can no longer attend to it. For an Indian language application, a practical consequence arises from token efficiency. Indic scripts such as Odia, Telugu, or Tamil require more Unicode bytes to encode the same amount of information compared to Latin-script English. Tokenisers typically optimised for English may represent a single Odia word with multiple tokens, whereas a comparable English word might be a single token. This means that an Odia-language document effectively occupies more of the context window than an equivalent English document. A model that can process a twenty-page English legal contract within its context window may only be able to process ten or twelve pages of an equivalent Odia document, requiring more complex document chunking strategies.

*Key terms: context window, token, attention, token efficiency, Indic scripts, Unicode, document chunking, context limit*

**Q10. What is hallucination in the context of large language models, and why is it more severe for Indian topics and languages?**

**Model Answer:**

Hallucination is the generation of confident, fluent, plausible-sounding text that is factually incorrect. It occurs because LLMs optimise to generate statistically likely text, and statistically likely text is not always true text. A model asked about a historical figure might generate a plausible-sounding biography that contains fabricated events, because the biography structure and the type of content are statistically consistent with what historical biographies contain, even when the specific facts are not. Hallucination is more severe for Indian topics because the training data for major LLMs is heavily weighted toward English. Regional Indian history, tribal knowledge systems, contemporary Indian research, and

information in lower-resource Indian languages are represented at a fraction of the density of their English-language equivalents. In sparse domains, the model has fewer reliable patterns to draw on and must confabulate more freely. The plausibility trap is that the model's generated text sounds equally confident regardless of whether it is drawing on dense reliable training data or sparse unreliable training data, making it difficult for readers unfamiliar with the domain to identify hallucinations.

*Key terms: hallucination, plausibility trap, confabulation, training data density, under-resourced languages, statistical likelihood, factual accuracy*

## Section C — Reflective Questions (3 Questions)

**Q11. The chapter describes LLMs as distillations of collective human expression, filtered through statistical optimisation. Does this framing change how you think about using an LLM? What are its implications for originality, authorship, and the nature of creativity?**

**Thinking prompts:**

- If an LLM generates a poem that you find beautiful, who is its author?
- Is there a meaningful difference between an LLM that generates statistically plausible text and a human writer who generates text shaped by everything they have read?
- What does the library metaphor suggest about what LLMs can and cannot do that is genuinely new?
- How do Indian philosophical traditions, particularly those concerned with the nature of consciousness and knowledge, relate to the question of whether statistical pattern completion constitutes understanding?
- Should LLM-generated text be disclosed as such? What difference would it make?

*Rubric hint: Strong responses will engage with the framing seriously rather than simply accepting or rejecting it. Credit for responses that draw on philosophical frameworks beyond the purely technical and that engage with the question of what genuine creativity requires.*

**Q12. The chapter identifies four challenges that make Indian language NLP harder than English NLP. Given that India has twenty-two scheduled languages and hundreds of dialects, is it realistic to expect that all of them will ever have high-quality NLP support? Who bears the cost of inadequate language AI, and who bears the cost of developing it?**

**Thinking prompts:**

- Which Indian languages are likely to achieve high-quality NLP support in the next decade, and which are not? What determines this?
- What does it mean for a speaker of a lower-resource language to live in an AI-mediated world where their language has limited AI support?
- Is the development of Indian language AI a commercial opportunity, a national interest, or a social justice issue? Can it be all three simultaneously?
- What role should the Indian government, technology companies, academic institutions, and language communities each play?
- Are there languages for which high-quality NLP support is genuinely impossible given current techniques? What would be needed?

*Rubric hint: Strong responses will move beyond 'it is difficult' to engage with specific resource constraints, political economy of language AI development, and the rights implications of linguistic exclusion. Credit for responses that distinguish between what is technically impossible and what is economically unprioritised.*

**Q13. You are helping a student in your class write a research essay using an LLM. The LLM produces a well-structured paragraph about the history of the Kondh tribal community of Odisha that sounds authoritative but may contain hallucinated details. How do you advise the student? What principles should guide LLM use in academic research?**

**Thinking prompts:**

- What is the student's responsibility regarding verification of LLM-generated content?
- How does the plausibility trap make LLM-generated content about under-resourced topics especially dangerous?
- What are legitimate uses of LLMs in research writing that do not risk misrepresentation?
- How should academic citation standards adapt to LLM use?
- What would responsible AI-augmented research look like for a student with limited access to specialised libraries or databases?

*Rubric hint: Strong responses will provide specific, actionable guidance rather than general warnings, and will engage with the specific vulnerability of under-resourced Indian topics to hallucination. Credit for responses that distinguish between using LLMs as drafting aids versus using them as factual sources.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Choose one regional Indian language that you speak or are familiar with and investigate the current state of NLP resources available for it. What embeddings exist? What translation models? What tools for text generation or analysis? Use publicly available information from AI4Bharat, academic papers, and tool repositories. Then identify the single most important gap in the current NLP resource landscape for that language — the missing resource that, if it existed, would enable the most significant new applications — and explain why you have identified it as the priority gap.**

This project requires genuine research into the current state of NLP for a specific language, not generic statements about Indian languages as a group. The priority gap analysis is the most important part and should be specific, justified, and actionable.

**Q15. [Commerce] Design a multilingual customer service chatbot for a regional business in your area — this could be a tourism service, a healthcare facility, a retail chain, or a financial services provider. Specify the languages it needs to handle, the kinds of questions it will receive, and the level of accuracy required for each query type. Then assess the feasibility of building this system using current publicly available language model tools, identifying specifically where hallucination risk is highest, where code-switching will**

**pose problems, and what human oversight mechanisms should be in place. Conclude with a deployment recommendation.**

The feasibility assessment should be grounded in the specific languages and query types, not generic. The hallucination risk assessment should identify specific query categories where factual accuracy is critical and the model is likely to be unreliable. The human oversight recommendations should be specific and proportionate to the risk.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Vaswani, A., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30. [The transformer paper — the architecture underlying virtually all modern LLMs. The introduction and related work sections are accessible without deep mathematical background and explain why attention mechanisms changed NLP.] |
| <b>Core</b> | Kunchukuttan, A., et al. (2020). AI4Bharat-IndicNLP Corpus: Monolingual Corpora and Word Embeddings for Indic Languages. arXiv preprint arXiv:2005.00085. [Directly relevant to Sections 8.1-8.3; describes the primary Indian language NLP resource and the challenges of building it. Freely available.]                                      |
| <b>Core</b> | Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? Proceedings of FAccT 2021. [A landmark critical analysis of large language models, directly relevant to the hallucination and plausibility trap discussion. Freely available.]                     |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Joshi, P., et al. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. Proceedings of ACL 2020. [On the distribution of NLP resources across the world's languages, with specific relevance to Indian language coverage. Freely available.]                                            |
| <b>Extended</b> | Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781. [The Word2Vec paper introducing word embeddings and vector arithmetic including the king-queen example. The first two pages are accessible to non-specialists.] |
| <b>Extended</b> | Khandelwal, U., et al. (2021). Nearest Neighbor Machine Translation. Proceedings of ICLR 2021. [On low-resource machine translation techniques; relevant to the data scarcity problem for Indian language pairs.]                                                                                                    |

### Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | AI4Bharat. (2023). IndicTrans2: Open Source Indian Language Machine Translation. <a href="http://ai4bharat.iitm.ac.in">ai4bharat.iitm.ac.in</a> . [The primary open-source Indian language translation model; directly relevant to Section 8.3 and to the Arts/Science stream project prompt. Freely accessible with documentation.]                       |
| <b>Online</b> | Anthropic. (2023). Claude's Character. <a href="https://anthropic.com/claude">anthropic.com/claude</a> . [For understanding what large language models are and what they are not; a primary source description of one of the major LLMs discussed in this chapter.]                                                                                        |
| <b>Online</b> | 3Blue1Brown. (2024). But What Is a GPT? Visual Intro to Transformers. <a href="https://youtube.com/3blue1brown">youtube.com/3blue1brown</a> . [A clear visual explanation of how transformers and LLMs work, covering tokenisation, attention, and next-word prediction. Directly complements Sections 8.1-8.4 without requiring mathematical background.] |

### *End of Chapter 8*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 9

# AI That Sees

*The eye is the most ancient information channel in the animal kingdom. Long before language evolved, long before cognition as we understand it existed, living creatures were extracting meaning from patterns of light. A bacterium responds to gradients of illumination. A jumping spider constructs a three-dimensional model of the world from two tiny eyes. A Sundarbans fishing cat reads the surface of water for the silhouette of a fish below. Vision, in its broadest sense, is the art of inferring structure and meaning from the distribution of light falling on a sensor.*

*Computer vision is the attempt to give machines this art. It is, in terms of commercial deployment, one of the most successful fields in all of artificial intelligence. Today, computer vision systems monitor crops from satellites, identify animals in wildlife camera traps, detect accidents on national highways, read handwritten documents, screen medical images for disease, and count the number of pedestrians crossing a street. The principles behind all of these applications are the same, and they can be understood without mathematics.*

*This chapter traces the full journey of an image through a computer vision system, explains the key operation that makes deep visual learning possible, identifies three major Indian application domains, and examines what happens when these systems fail in ways that are not random but systematic.*

## Learning Objectives

By the end of this chapter, you will be able to:

- Trace the four-stage progression from raw pixels to object recognition in a computer vision system
- Explain convolution as a sliding filter operation and describe what it detects at each stage
- Describe three major Indian computer vision applications: satellite crop monitoring, camera trap wildlife detection, and highway accident detection
- Explain the role of labelled training data in vision systems and the consequences of insufficient data
- Identify and explain at least three failure modes in deployed computer vision systems

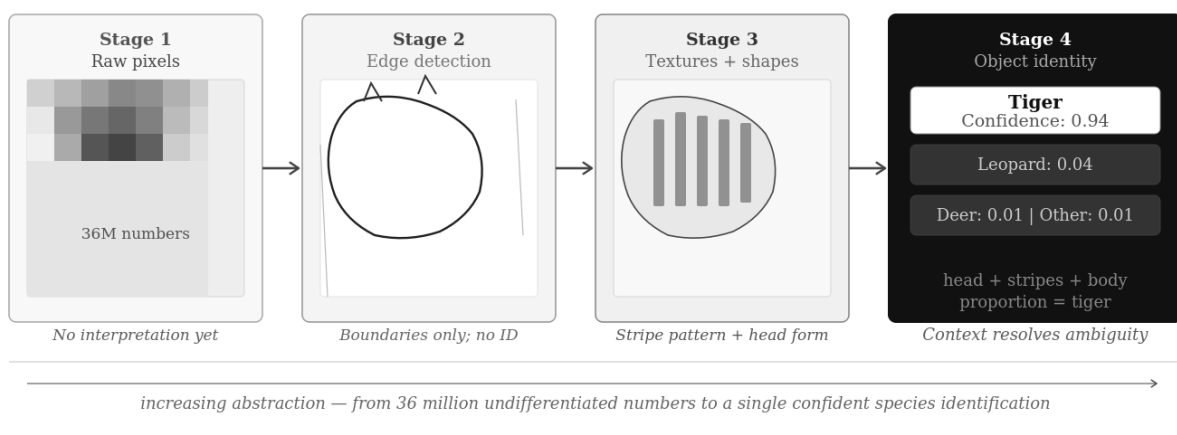
## 9.1 Pixel to Meaning: The Four-Stage Journey

Every digital image is, at its most fundamental level, a grid of numbers. Each number represents the intensity of light at one point in the grid, typically encoded in three channels:

red, green, and blue intensity, each on a scale from zero to two hundred and fifty-five. A photograph taken with a twelve-megapixel camera contains twelve million pixels, and each pixel contributes three numbers, giving a total of thirty-six million numbers per image. This is the raw input.

The challenge of computer vision is to convert these thirty-six million numbers into meaningful output: this is a cotton leaf showing early blight, or this is a tiger facing right, or this is a vehicle stopped in the fast lane of a national highway. The journey from raw pixels to meaning proceeds through four broad stages, each building on the previous.

**Fig. 9.1**  
**Pixel to Object: Four-Stage Visual Processing (Tiger Camera Trap)**



*No engineer specified that stage 3 should detect stripes. The network discovered stripe detection because stripes are the most informative intermediate feature for distinguishing tiger from non-tiger in the training data.*

*Figure 9.1 — Pixel To Object*

The first stage is raw input: the grid of pixel intensity values as described above. At this stage, the image is pure data with no interpretation. A pixel value of 187 in the red channel tells us only that a particular sensor element recorded that intensity. It carries no information about whether it is part of a leaf, a sky, an animal, or a road.

The second stage is feature extraction: the identification of elementary visual patterns within the image. The most important elementary patterns are edges, which are boundaries between regions of different intensity. An edge detector identifies pixels where intensity changes abruptly from one side to the other, which typically corresponds to the boundary between two objects or between an object and its background. The edge map of an image is a representation that highlights these boundaries and suppresses the uniform regions between them. Edges are important because they are the building blocks from which all higher-level visual structure is composed: shapes are made of edges, objects are made of shapes.

The third stage is object-part detection: the identification of more complex patterns by combining the features found in stage two. A collection of curved edges forming a roughly circular shape is a candidate for a head. A collection of alternating light and dark stripes is

a candidate for tiger markings. A large rectangular region with a specific aspect ratio and colour distribution is a candidate for a vehicle on a road. These detections are not definitive; they are probabilistic candidates that will be combined and refined in the next stage.

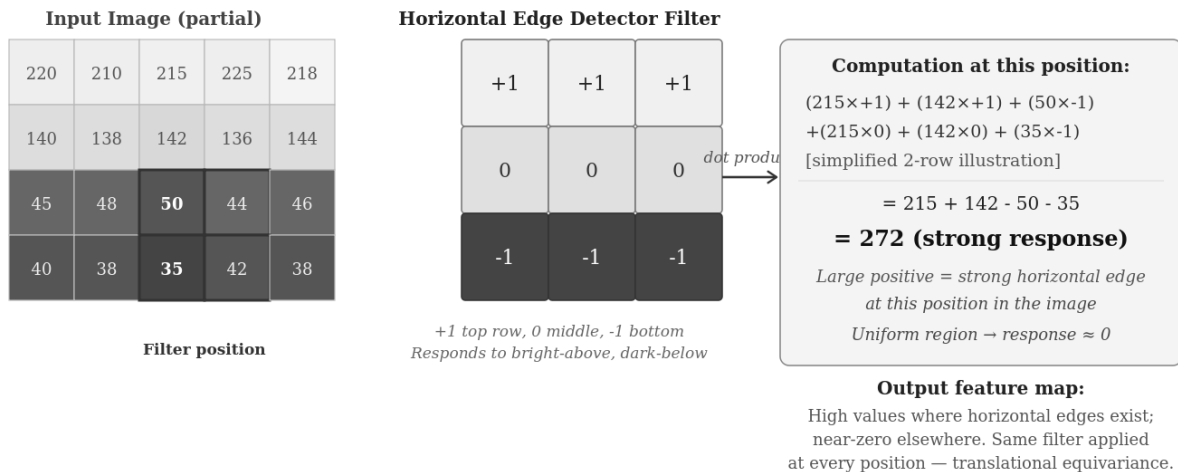
The fourth stage is object recognition: the combination of part-level detections into a holistic object identity. The head shape, combined with the stripe pattern, combined with the body proportion and limb geometry, produces a confident tiger classification. The rectangular region, combined with specific intensity patterns in specific relative positions, produces a vehicle classification. Object recognition requires context: the same head shape might be a tiger in one context and a domestic cat in another. The network uses all available context within the image to resolve such ambiguities.

## 9.2 Convolution: The Sliding Filter

The operation that makes deep visual learning possible is convolution. Understanding convolution is essential to understanding why convolutional neural networks, the dominant architecture for computer vision, are structured the way they are and why they work so well.

**Fig. 9.2**

### Convolution: The Sliding Filter Operation



*In a real network, this filter's values (+1, 0, -1) are not specified by an engineer — they are learned from data during training. Many such filters are applied simultaneously, each discovering a different elementary visual feature in the image.*

Figure 9.2 — Convolution

A convolutional filter, also called a kernel, is a small grid of numbers, typically three by three or five by five pixels. The filter slides across the input image, position by position, and at each position it computes a single number: the dot product of the filter values and the image values in the region it currently covers. This single number becomes one value in the output, called the feature map. The filter then moves to the next position and computes the next value, and so on until it has covered the entire image.

The key insight is that the same filter is applied everywhere in the image. A filter that detects horizontal edges, for example, will respond strongly wherever horizontal edges appear in the image, regardless of whether they are near the top, the middle, or the bottom. This property is called translational equivariance, and it is why convolutional networks are so efficient: instead of learning separate detectors for edges in every possible position, they learn one detector that works everywhere. A natural image of a leaf looks like a leaf whether it occupies the top-left corner or the centre of the frame, and the convolutional network treats it accordingly.

A single convolutional layer typically applies many filters simultaneously, each detecting a different elementary pattern. One filter might detect horizontal edges. Another detects vertical edges. A third detects diagonal edges. A fourth detects blobs of a particular colour. The outputs of all these filters, stacked together, form a multi-channel feature map that represents the image in terms of its elementary visual features. This feature map is then passed to the next layer, which applies another set of filters to detect combinations of elementary features, and so on through the network hierarchy described in Chapter 7.

During training, the values in each filter are learned from data, not designed by hand. An engineer designing a computer vision system does not specify that filter number forty-seven should detect horizontal edges. The training process, applying backpropagation and gradient descent to millions of labelled images, discovers which filter values minimise the prediction error, and the filters that emerge are the ones that best capture the elementary patterns relevant to the task. This is why convolutional networks are described as learning to see: the visual representations they build are not programmed but discovered.

### **9.3 Three Indian Applications**

The principles of computer vision described above are not abstract. They are operational in Indian contexts today, at scale, in applications that affect agriculture, ecology, and road safety.

**Fig. 9.3**  
**Three Indian Computer Vision Applications**

|            | Satellite Crop Monitoring                                                                                                                                                                                                               | Camera Trap Wildlife ID                                                                                                                                                                                                                         | Highway Accident Detection                                                                                                                                                                                                                            |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT      | <b>Multi-spectral satellite imagery</b><br>Visible + near-infrared + SWIR bands<br>Updated every 3-5 days (Sentinel-2)<br>Covers all agricultural land in India<br><i>Ground-truth: field surveys by agronomists</i>                    | <b>Camera trap images (infrared, night)</b><br>Motion-triggered; variable conditions<br>1000s of images/month per reserve<br>Kanha, Pench, Satpura, Corbett...<br><i>Ground-truth: expert wildlife biologist labels</i>                         | <b>Continuous highway surveillance video</b><br>Fixed cameras every 500-1000m<br>Day/night, rain, fog conditions<br>Delhi-Mumbai Expressway pilot<br><i>Ground-truth: accident report database</i>                                                    |
| PROCESSING | <b>NDVI computation per pixel</b><br>$NDVI = (NIR - Red) / (NIR + Red)$<br>Deep CNN: multi-band → stress map<br>Trained on labelled crop health data<br>Temporal analysis: change detection<br><i>Spatial resolution: 10m per pixel</i> | <b>ResNet / EfficientNet classifier</b><br>Pre-trained on ImageNet; fine-tuned<br>on camera trap dataset (iNaturalist)<br>Wildlife Insights platform (Google)<br>Individual ID: stripe pattern matching<br><i>Supports 614 species globally</i> | <b>Temporal difference + object detection</b><br>YOLO-class detector on video frames<br>Tracks vehicle positions over time<br>Classifies: normal / deceleration /<br>stopped / debris / accident<br><i>Real-time processing: &lt;2 second latency</i> |
| OUTPUT     | <b>Crop stress map</b><br>District-level spatial grid showing<br>healthy / stressed / critical zones<br><br>→ Field extension worker deployment                                                                                         | <b>Species + individual ID</b><br>Tiger, 0.94   Individual: T-47<br>Location + timestamp logged<br><br>→ Population count + movement tracking                                                                                                   | <b>Incident alert</b><br>km 342.6 — vehicle stopped<br>fast lane. Alert sent 14:23:07.<br><br>→ Emergency services dispatched                                                                                                                         |

All three applications share the same underlying principles: multi-spectral or multi-frame input, convolutional feature extraction, and trained classifiers. What differs is the domain, the label type, and the consequences of errors.

Figure 9.3 — Three Indian Applications

### Satellite Crop Health Monitoring

The Indian Space Research Organisation, in partnership with the National Remote Sensing Centre and state agriculture departments, operates satellite-based crop health monitoring that covers agricultural land across India. The input to these systems is multi-spectral satellite imagery: images captured not only in visible wavelengths but also in near-infrared and shortwave-infrared bands that are invisible to the human eye but carry information about vegetation health that visible light does not.

A healthy, actively photosynthesising plant reflects near-infrared light strongly and absorbs red light. A stressed or diseased plant shows a different spectral signature: reduced near-infrared reflectance and increased red reflectance. The ratio of these two values, the Normalised Difference Vegetation Index or NDVI, is a single number that encodes vegetation health on a scale from minus one to plus one. A deep learning model trained on labelled satellite imagery, where ground-truth crop health assessments from field surveys provide the labels, learns to predict crop health at the pixel level from the multi-spectral input. The output is a spatial map of predicted crop health, updated every time a new satellite pass covers the area, allowing district agriculture officers to identify stressed zones before visual symptoms become visible to a field inspector.

This application has direct benefits for food security and for the livelihoods of farmers. Early identification of crop stress, whether from drought, pest infestation, or disease, allows intervention at a stage when it is still effective. The satellite monitoring system is particularly valuable in states like Odisha, Maharashtra, and Madhya Pradesh, where large areas of rainfed agriculture are vulnerable to unpredictable monsoon variability and

where the density of field extension workers is insufficient to monitor every farm through ground-level inspection.

### **Wildlife Camera Trap Identification**

India's tiger reserves and wildlife sanctuaries operate thousands of camera traps: motion-triggered cameras that capture images whenever an animal passes. The primary purpose is wildlife population monitoring: estimating the number of tigers, leopards, elephants, and other species present in a reserve, tracking their movements, and detecting changes in population over time. Manual review of camera trap images is extremely labour-intensive. A single reserve might generate thousands of images per month, and identifying species, individual animals where possible, and behaviours from these images requires skilled wildlife biologists.

Deep learning models trained on large labelled datasets of camera trap images, including the Wildlife Insights platform maintained by Google and several conservation organisations, can automatically classify species from camera trap images with accuracy approaching that of expert reviewers for well-represented species. These models are now in operational use in several Indian reserves, including Corbett, Pench, and Satpura. The models identify species from the combination of visual features described in Chapter 7: body shape, coat pattern, head morphology, and the contextual cues that distinguish similar-looking species. For individual tiger identification, more specialised models compare the unique stripe patterns of each individual to a database of known individuals, enabling population tracking without invasive marking or capture.

### **Highway Accident Detection**

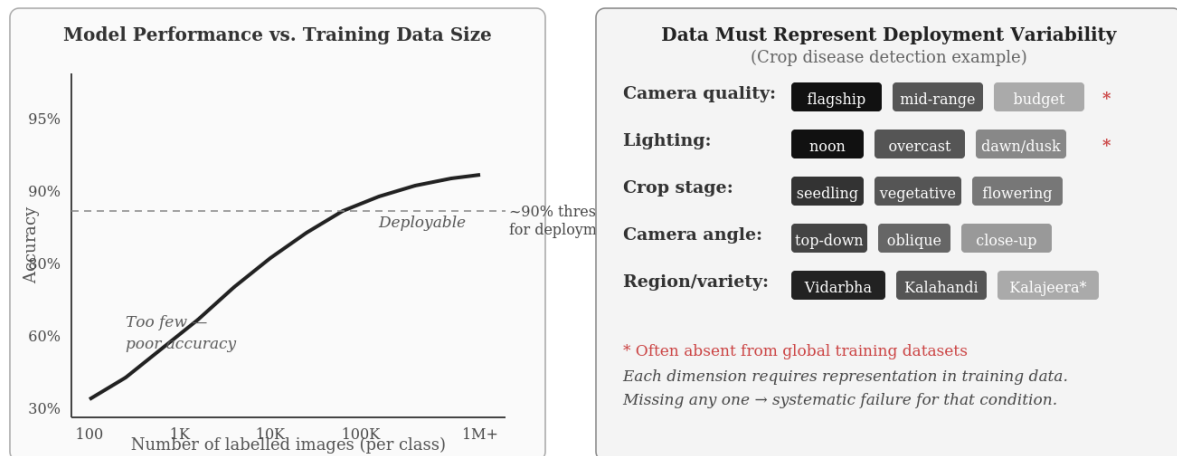
India's national highway network carries enormous traffic volumes on routes where accidents can cause secondary collisions, delays, and deaths from delayed emergency response. The National Highways Authority of India has piloted AI-based accident detection systems on several high-traffic corridors, including sections of the Delhi-Mumbai Expressway and the Chennai-Bengaluru Highway. These systems use video cameras positioned at regular intervals along the highway, feeding continuous video to a computer vision model that identifies accident events and alerts emergency services.

The computer vision task is to distinguish accident events from the continuous background of normal traffic flow. The model has been trained on thousands of labelled video clips: clips showing accidents, clips showing near-misses, and clips showing normal traffic in various conditions including night, rain, fog, and heavy congestion. It learns to identify the visual signatures of accident events: sudden deceleration indicated by brake light patterns, unusual stopped vehicles in moving traffic lanes, debris on the road surface, pedestrians in unexpected positions, and the characteristic chaotic movement patterns of secondary collisions. The model must detect these events reliably in real time while generating a very low rate of false alarms, because each false alarm requires emergency response resource deployment.

## 9.4 Training Data Requirements for Vision

Computer vision models, like all supervised learning models, require labelled training data. But vision models require labelled data at a scale that is qualitatively different from most other machine learning tasks. The reasons are rooted in the dimensionality of visual data and the variability of visual appearance.

**Fig. 9.4**  
**Training Data Requirements: Volume and Variability**



Volume alone is insufficient — training data must cover the full variability of the deployment context. A model trained on flagship phone images under noon light will systematically fail for budget phones at dawn, even if total image count is large.

Figure 9.4 — Training Data Requirements

A crop disease detection model must learn to recognise each disease across the full range of conditions under which it might be photographed: different growth stages of the crop, different lighting conditions from bright midday sun to overcast dawn, different camera angles, different phone camera qualities ranging from high-resolution flagship devices to budget cameras with limited dynamic range, different soil backgrounds, different co-occurring benign leaf variations that must not be confused with disease symptoms. Each of these sources of variability requires training examples to represent it. A model trained only on high-resolution photographs taken in controlled laboratory conditions will fail when deployed in the field, where images are taken under variable lighting, at oblique angles, by farmers using the phone they own rather than the phone the researcher used.

The rule of thumb in computer vision is that thousands of labelled examples per class are needed for basic performance, and tens or hundreds of thousands for robust deployment performance. For each disease class in a crop disease detection system, this means thousands of labelled photographs of that disease on the target crop, under the full range of conditions representative of the deployment context. Collecting and labelling this data is expensive, time-consuming, and requires domain expertise: a botanist or agronomist must verify the label for each image.

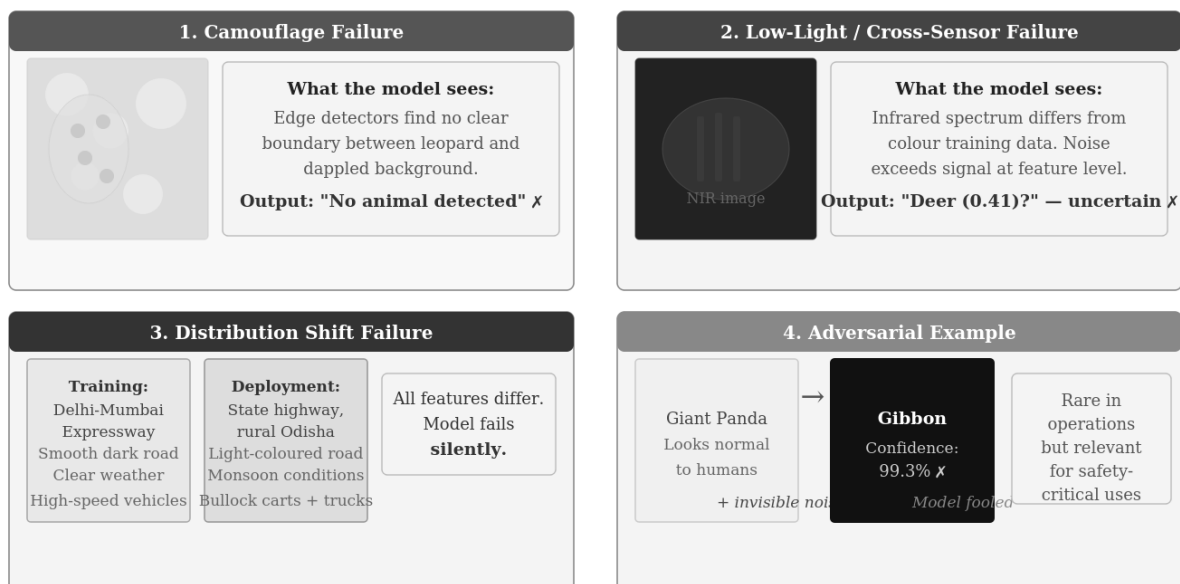
In the Indian context, this data requirement creates a significant challenge for applications targeting smaller crops or regional varieties. A disease detection model for Basmati rice

can leverage global datasets because rice is a globally important crop with extensive research investment. A disease detection model for Koraput Kalajeera, a traditional Odia black rice variety with significant cultural and nutritional value, cannot: the global training data does not represent this variety's specific visual characteristics, disease susceptibilities, and growth environment. Building a model for such a variety requires data collection from the region where it is grown, by people who understand both the crop and the data collection requirements. This is an example of the broader principle that data collection is not a generic activity: it is deeply context-specific.

## 9.5 Failure Modes in Computer Vision

Computer vision systems deployed in the real world fail. Understanding how they fail, and why, is essential for anyone using these systems to make consequential decisions.

**Fig. 9.5**  
**Computer Vision Failure Modes**



*Failures 1-3 are common in real deployments; failure 4 is rare operationally. All four are systematic, not random — they affect specific conditions consistently, which is why stratified evaluation by condition is essential before deployment.*  
**Figure 9.5 — Failure Modes**

Camouflage failures occur when an object's visual appearance matches its background. A leopard resting in dappled shade beneath a forest canopy may be invisible to a camera trap model that expects leopards to be distinguishable from their background. The model's edge and texture detectors rely on there being a visual discontinuity at the boundary between the animal and its surroundings. When the animal has evolved to eliminate precisely that discontinuity, the visual detection machinery fails. This is not a flaw specific to AI; human eyes face the same challenge. But it matters for camera trap monitoring because the animals most skilled at camouflage are often the rarest and most threatened, and systematic detection failures for those species will bias population estimates.

Low-light failures occur when image quality falls below the threshold at which the network's learned features can operate reliably. Camera traps in Indian forests frequently capture images at the edge of their sensor's capabilities: in dense canopy conditions, on moonless nights, or when the animal is far from the camera. Infrared illumination extends the range but produces images with different spectral characteristics than daytime colour images. A model trained primarily on daytime colour images or infrared images from specific camera trap models will perform poorly on images from cameras with different spectral sensitivities. Night driving assistance systems on Indian highways face similar challenges: road conditions in fog, heavy rain, or low-quality road lighting produce image characteristics that differ substantially from the clear-weather conditions that dominate training datasets.

Distribution shift failures are the most pervasive category. As discussed in Chapter 7, a model trained on images from one distribution will fail systematically when the deployment distribution differs. A highway accident detection model trained on images from Delhi-Mumbai Expressway conditions will perform differently when deployed on a state highway with different road surface colour, different vehicle composition, different typical speeds, and different typical lighting conditions. This is not an exotic edge case. Most real-world computer vision deployments involve some degree of distribution shift between training and deployment, and understanding the magnitude and direction of that shift is essential for evaluating whether a model is safe to deploy.

Adversarial failures are a distinct category that has attracted significant research attention, though they are less common in operational settings than the previous three. An adversarial example is an input specifically designed to fool a computer vision model: an image that looks perfectly normal to a human viewer but contains subtle pixel-level perturbations that cause the model to produce a wildly wrong output. The most famous demonstration involved adding small amounts of carefully constructed noise to an image of a giant panda, causing a model to classify it with high confidence as a gibbon, while a human viewer sees no difference between the original and perturbed images. In road safety contexts, adversarial examples have been demonstrated that can cause traffic sign classifiers to misidentify a stop sign as a speed limit sign, though doing this reliably in physical deployments remains technically challenging.

## **9.6 Patterns in My Field: Stream Perspectives**

### **Arts Stream: Computer Vision for Pattachitra Pattern Documentation**

Pattachitra is a traditional art form from Odisha and West Bengal, characterised by intricate cloth or palm-leaf paintings depicting mythological narratives, primarily from the Jagannath tradition. Each master artist's work has a distinctive style expressed through specific motifs, colour combinations, compositional conventions, and narrative sequences. A computer vision system trained on a large collection of labelled Pattachitra works could assist with several tasks relevant to the documentation and preservation of this tradition: automatic attribution of unsigned works to specific artists or workshops, detection of specific motifs such as the characteristic lotus border or the wheel of dharma,

identification of narrative sequences, and detection of potential forgeries. Building such a system requires careful collaboration with Pattachitra artists and cultural scholars to ensure that the categories and labels used in training reflect genuine artistic distinctions rather than arbitrary technological ones. The failure modes described in Section 9.5 apply here: a model trained primarily on well-photographed museum-quality works may fail on field photographs of workshop pieces taken under poor lighting.

### **Science Stream: Computer Vision for Medicinal Plant Identification in the Eastern Ghats**

The Eastern Ghats of Odisha host extraordinary botanical diversity, including numerous plants used in traditional Adivasi medicine. Many of these species are not well-represented in standard botanical databases, and correct identification is critical both for sustainable harvesting and for the scientific documentation of traditional knowledge. A computer vision system for medicinal plant identification in this region faces challenges typical of fine-grained visual classification: distinguishing between plant species that differ only in subtle features such as leaf venation patterns, petiole attachment, or stipule morphology that may not be captured in low-resolution field photographs. Models trained on pressed herbarium specimens will fail on field photographs of living plants, where lighting, angle, and surrounding vegetation create visual conditions very different from the standardised herbarium context. Building a reliable system requires field photography campaigns that specifically document the visual variability of each species as it appears in the Eastern Ghats environment, not just the controlled conditions of botanical documentation.

### **Commerce Stream: Computer Vision for Quality Control in Odia Handloom Textiles**

The Sambalpuri handloom tradition of western Odisha produces textiles of great intricacy, where the value of each piece depends critically on the quality of the weave, the accuracy of the pattern, and the absence of defects such as broken threads, colour inconsistencies, or pattern misalignments. Manual quality control by expert weavers is time-consuming and expensive, and the availability of trained quality controllers does not scale with production volumes. A computer vision system for automated quality control could identify defects in woven textiles at a speed and consistency that manual inspection cannot match. Such a system requires training data that represents the full range of acceptable variation in Sambalpuri patterns alongside the specific defect types that reduce value. The challenge is that Sambalpuri patterns are enormously diverse, and what counts as a defect versus an intentional variation is context-dependent knowledge that resides with master weavers. Building the system requires close partnership with the weaving community to correctly encode this knowledge in training labels.

## Systematic Failure and the Invisible Blind Spot

The failure modes described in Section 9.5 share a common and dangerous feature: they tend to be systematic rather than random. A computer vision model that fails on camouflaged animals fails on all camouflaged animals, not on random animals. A model that fails in low-light conditions fails for everyone who operates in low-light conditions, not randomly. A model that performs poorly on budget smartphone images fails for all users of budget smartphones, which in India means a disproportionate number of rural and lower-income users.

This systematic character means that the failures are correlated with specific populations or conditions that share a characteristic the model is poorly adapted to. A crop disease detection model that fails on images taken with phones costing less than three thousand rupees will systematically underserve the smallholder farmers who most need it, while performing well for better-resourced farmers with premium devices. A medical imaging model that fails on the image quality produced by older scanner models will systematically underserve the district hospitals that have not been able to upgrade their equipment. The model appears to work well in aggregate, because the failures are concentrated in a subpopulation that may be underrepresented in the evaluation dataset.

This is the invisible blind spot: a systematic failure that does not show up in the headline accuracy metric because the affected population is underrepresented in the test set. Detecting it requires stratified evaluation: deliberately testing the model's performance separately for different subgroups defined by the characteristics likely to produce systematic failure. For a crop disease detection model, this means testing separately on images from each camera type, each lighting condition, each crop variety, and each geographic region of deployment. For a medical imaging model, this means testing separately on images from each scanner model, each patient demographic, and each institution. This kind of stratified evaluation is more expensive than aggregate evaluation, and it is routinely omitted. Requiring it, and making the results public, is one of the most important accountability mechanisms available for computer vision systems with significant social consequences.

### — CHAPTER SUMMARY —

#### Chapter 9 — Key Points

1. Computer vision converts pixel grids into meaning through four stages: raw input, feature extraction, object-part detection, and object recognition.
2. Convolution applies a small filter across the entire image, producing a feature map that highlights wherever a specific elementary pattern appears.
3. Convolutional filters are learned from data, not designed: training discovers which filters best capture the patterns relevant to the task.
4. Three major Indian applications: satellite crop health monitoring (NDVI from multi-spectral imagery), camera trap wildlife identification, highway accident detection.

5. Vision models require thousands to hundreds of thousands of labelled images per class; data collection must represent the full variability of deployment conditions.
6. Regional Indian varieties and contexts are often underrepresented in global training datasets, requiring local data collection for reliable deployment.
7. Failure modes: camouflage (animal matches background), low-light (image quality below threshold), distribution shift (training-deployment mismatch), adversarial examples.
8. Failures tend to be systematic, not random — they affect specific populations or conditions consistently.
9. Stratified evaluation across subgroups is essential for detecting invisible blind spots; aggregate accuracy metrics can hide systematic failures.

— MODEL QUESTIONS —

## Section A — MCQ (5 Questions)

**Q1. In the four-stage journey from pixel to meaning, what is detected in the second stage?**

- A) The identity of objects in the image
- B) The colour calibration of the camera sensor
- C) Elementary visual patterns such as edges, where intensity changes abruptly
- D) The semantic content of any text visible in the image

**Answer: C**

*Explanation: The second stage — feature extraction — identifies elementary visual patterns, primarily edges where adjacent pixel intensities differ sharply. These are the building blocks from which all higher-level visual structure is composed.*

**Q2. Translational equivariance in convolutional networks means that:**

- A) The network produces the same output regardless of the image orientation
- B) The same filter detects a given pattern wherever it appears in the image
- C) The network can translate text visible in images to other languages
- D) Convolutional filters can be transposed to detect mirror-image patterns

**Answer: B**

*Explanation: Translational equivariance means the convolutional filter produces the same response to a pattern regardless of where in the image that pattern appears. This is why CNNs are efficient: one filter covers the whole image rather than needing separate detectors for each position.*

**Q3. The Normalised Difference Vegetation Index (NDVI) used in satellite crop monitoring is computed from:**

- A) The ratio of red and green pixel intensities in visible-light photographs
- B) Near-infrared and red reflectance values from multi-spectral satellite imagery
- C) The contrast between crop and soil pixels in high-resolution drone photography
- D) Ground-level sensor measurements of chlorophyll concentration

**Answer: B**

*Explanation: NDVI is computed from near-infrared and red reflectance values. Healthy vegetation reflects NIR strongly and absorbs red; stressed vegetation shows reduced NIR and increased red reflectance. This spectral signature is captured by multi-spectral satellites.*

**Q4. A crop disease detection model trained on high-resolution photos fails in the field where farmers use budget smartphones. This is an example of:**

- A) An adversarial attack on the model by malicious users
- B) Distribution shift: the training image quality does not match the deployment image quality
- C) Camouflage failure: the disease symptoms are too subtle to detect at lower resolutions
- D) Overfitting: the model memorised specific photos rather than learning disease patterns

**Answer: B**

*Explanation: This is a classic distribution shift failure. The model learned features from high-resolution images that are not present or have different statistical properties in lower-resolution images. The training and deployment distributions differ in image quality.*

**Q5. Why is stratified evaluation more important than aggregate accuracy for computer vision systems deployed in India?**

- A) India has more computer vision applications than other countries
- B) Aggregate accuracy can hide systematic failures affecting specific subgroups who are underrepresented in the test set
- C) Stratified evaluation requires less data, making it more practical for Indian research budgets
- D) Computer vision models perform better on stratified datasets due to reduced variance

**Answer: B**

*Explanation: Aggregate accuracy averages performance across all test examples. If a subgroup with systematic failures is underrepresented in the test set, the high performance on the majority masks the poor performance on the minority. Stratified evaluation tests each subgroup separately, revealing these blind spots.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Explain convolution as a sliding filter operation, and describe what a horizontal edge detector filter would look like and how it would respond to an image.**

**Model Answer:**

Convolution is an operation in which a small grid of numbers, called a filter or kernel, slides across an input image one position at a time. At each position, the filter computes the dot product of its own values with the values of the image pixels it currently covers, producing one number in the output feature map. The filter then moves to the adjacent position and

repeats. A horizontal edge detector filter would have positive values in the top row, zero values in the middle row, and negative values in the bottom row. When this filter is positioned over a region of the image where bright pixels above give way to dark pixels below, the dot product will be large and positive, because the positive filter weights align with high-intensity pixels and the negative weights align with low-intensity pixels. When positioned over a uniform region, the dot product will be near zero. The result is a feature map that shows strong responses precisely where horizontal edges appear in the image, regardless of position.

*Key terms: convolution, filter, kernel, feature map, dot product, edge detection, translational equivariance, sliding window*

**Q7. Describe how satellite crop health monitoring works in India, from the satellite input to the actionable output for a district agriculture officer.**

**Model Answer:**

Multi-spectral satellites, including those operated by ISRO, capture imagery of agricultural land in multiple wavelength bands including visible red and near-infrared. Healthy, actively photosynthesising plants reflect near-infrared light strongly and absorb red light, while stressed or diseased plants show reduced near-infrared and increased red reflectance. The NDVI index encodes this ratio as a single value per pixel, producing a spatial map of vegetation health across the district. A deep learning model trained on labelled satellite imagery, where ground-truth crop health assessments from field surveys provide the labels, learns to predict crop stress at the pixel level from the multi-spectral input, going beyond NDVI to incorporate spatial patterns, temporal trends, and cross-band relationships. The output is a spatial map showing zones of healthy crops, mildly stressed crops, and severely stressed crops, updated at each satellite pass. The district agriculture officer receives this map and can direct field extension workers to the identified stress zones for ground-level investigation and intervention, at a stage when treatment is still effective rather than after visible symptoms have already caused significant yield loss.

*Key terms: multi-spectral imagery, NDVI, near-infrared, vegetation index, spatial map, crop stress, ISRO, ground-truth labelling*

**Q8. Why does training a crop disease detection model for Koraput Kalajeera rice require data collection in Koraput, rather than using existing global rice disease datasets?**

**Model Answer:**

Global rice disease datasets are primarily built from major rice varieties grown under research station conditions or from the dominant commercial varieties of major rice-growing regions such as the Indian plains, Southeast Asia, and China. Koraput Kalajeera is a traditional Odia black rice variety with visual characteristics that differ from these: different leaf morphology, different plant architecture, different colour, different typical disease presentations, and different growing conditions in the Eastern Ghats environment. A model trained on images of disease on standard rice varieties has learned features specific to those varieties, and when applied to Kalajeera, it encounters visual inputs that fall outside its training distribution. The disease symptoms may present differently on black rice than on white rice, the typical background vegetation of the Koraput growing environment differs from the flat paddy plains represented in global datasets, and the cameras and lighting conditions used by Koraput farmers differ from those used in research station photography. Each of these differences constitutes a distribution shift that will degrade model performance. The fix is data: labelled photographs of each disease on Kalajeera, taken under Koraput field conditions, by the cameras farmers actually use.

*Key terms: distribution shift, regional variety, training data representativeness, Koraput*

**Q9. Explain two different failure modes in wildlife camera trap computer vision systems, identifying the visual mechanism behind each failure.**

**Model Answer:**

The first failure mode is camouflage failure. Many animals, particularly rare and threatened species, have evolved coat patterns and colours that closely match their typical habitat backgrounds. A leopard resting in dappled forest light may have spots and background illumination that produce similar pixel intensity patterns, meaning that the edge and texture detectors in the early layers of the vision network find no strong boundary between the animal and its surroundings. Without this boundary signal, the network cannot assemble the edges into an animal shape, and the detection fails. This failure is systematic: it affects all well-camouflaged animals, not random animals. The second failure mode is low-light and cross-sensor failure. Camera traps operating at night use infrared illumination, producing images with different spectral characteristics than daytime colour images. A network trained primarily on daytime colour images has learned colour-dependent features that do not transfer to infrared images. Even for networks trained on infrared images, different camera models produce images with different noise levels, spatial resolutions, and dynamic ranges, causing the network's learned features to respond differently across camera models.

*Key terms: camouflage, edge detection, low-light, infrared, spectral characteristics, systematic failure, cross-sensor distribution shift*

**Q10. What is the invisible blind spot problem in computer vision evaluation, and how does stratified evaluation help detect it?**

**Model Answer:**

The invisible blind spot problem occurs when a computer vision model has systematic failures affecting a specific subgroup, but those failures are invisible in the aggregate accuracy metric because the affected subgroup is underrepresented in the test set. For example, a highway accident detection model might perform with 94% accuracy on the standard test set but perform at only 71% accuracy on rainy-night conditions, which comprise only 4% of the test set. The aggregate metric averages these performances, producing a headline number close to 94% that does not reveal the dangerous underperformance in adverse weather. Stratified evaluation addresses this by explicitly testing the model's performance separately for each identifiable subgroup relevant to the deployment context: by camera type, by weather condition, by time of day, by vehicle type, by geographic location. If the test set is constructed to ensure adequate representation of each subgroup, stratified evaluation will reveal subgroup-specific failures that aggregate metrics obscure. This approach is more expensive, requires careful test set design, and must be repeated whenever the deployment context changes.

*Key terms: invisible blind spot, stratified evaluation, aggregate accuracy, subgroup performance, systematic failure, test set composition, evaluation methodology*

## **Section C — Reflective Questions (3 Questions)**

**Q11. The chapter describes computer vision being applied to Pattachitra documentation, medicinal plant identification, and handloom quality control —**

**all domains deeply embedded in traditional Odia knowledge. What are the risks and opportunities when AI is applied to the documentation of traditional knowledge? Who should control these systems?**

**Thinking prompts:**

- What happens to traditional knowledge when it is digitised and processed by AI systems owned by external organisations?
- Can a computer vision system trained to classify Pattachitra motifs do so without encoding the cultural meaning of those motifs? Does that matter?
- Who has the right to build training datasets from traditional artworks — the museum that holds them, the community that created them, or the researcher who photographs them?
- How might AI-assisted attribution of unsigned Pattachitra works affect the livelihoods of living artists?
- What governance structures would ensure that these systems serve the communities whose knowledge they encode?

*Rubric hint: Strong responses will engage with intellectual property, community consent, and power asymmetries rather than treating AI documentation as automatically beneficial. Credit for responses that identify specific risks to living artisans and that propose concrete governance structures rather than generic safeguards.*

**Q12. The chapter identifies that computer vision failures tend to be systematic, affecting specific populations consistently. Apply this insight to the highway accident detection application. Which populations or road conditions are most likely to be systematically underserved, and what would a responsible deployment plan address?**

**Thinking prompts:**

- What populations use the highways where these systems are deployed? Who does not?
- What road conditions are underrepresented in training data collected primarily from high-quality national highways?
- Who bears the consequences of a missed accident detection — only those in the accident, or also those in secondary collisions?
- How should the system's performance on state highways and rural roads be evaluated before deployment there?
- What is the minimum acceptable detection rate for a system that will replace or reduce human monitoring?

*Rubric hint: Strong responses will identify specific road conditions and user populations that are likely to be underrepresented, and will engage with the asymmetric consequences of false negatives versus false positives in safety-critical systems. Credit for responses that propose specific evaluation protocols rather than generic quality standards.*

**Q13. The chapter notes that the Sambalpuri handloom quality control system requires 'close partnership with the weaving community to correctly encode this knowledge in training labels.' What does this requirement reveal about the limits of AI in craftsmanship domains? Is there knowledge that cannot be adequately encoded in training labels?**

**Thinking prompts:**

- What kinds of knowledge do master weavers possess that might be difficult to express as labels on individual images?

- How is tacit knowledge — the kind that is hard to articulate but easy to recognise — typically transmitted in traditional craft communities?
- If a label requires explanatory context that the labeller cannot communicate to a researcher, is the label reliable?
- What would be lost if quality control for Sambalpuri textiles shifted from master weaver judgment to AI classification?
- Are there crafts or knowledge traditions where AI-assisted quality control would be clearly beneficial, and others where it would be clearly harmful? What distinguishes the two cases?

*Rubric hint: Strong responses will engage seriously with tacit knowledge and its limits as training data. Credit for responses that distinguish between explicit, articulable knowledge (which can be labelled) and embodied, contextual knowledge (which may resist labelling), and that identify specific consequences of this distinction for the quality of AI-assisted craftsmanship systems.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Design a computer vision system for one documentation or monitoring task relevant to your field — this could be botanical specimen classification, archaeological site monitoring, traditional craft documentation, ecological zone mapping, or species identification. Specify the input data type, the four stages of visual processing as they apply to your task, the labelling requirements, and the deployment conditions. Then conduct a failure mode analysis: which of the four failure modes described in this chapter (camouflage, low-light, distribution shift, adversarial) are most relevant to your application, and what specific visual characteristics make each one likely? Propose one mitigation for the most serious failure mode.**

The failure mode analysis is the most important part and should be grounded in the specific visual characteristics of the application domain, not generic statements about computer vision. The mitigation should be specific and actionable — not just 'collect more data' but describing what data, from where, labelled by whom.

**Q15. [Commerce] A logistics company operating last-mile delivery in both urban and rural Odisha is considering deploying a computer vision system for automated package sorting and damage detection at its distribution centres. The centres range from a modern facility in Bhubaneswar with consistent lighting and high-resolution cameras to small rural hubs with fluorescent lighting, older cameras, and significant variation in package presentation. Conduct a training data and deployment analysis: what training data would be needed, what distribution shifts are likely between the Bhubaneswar facility and rural hubs, what failure modes are most likely in each context, and what stratified evaluation would you require before approving deployment across all facilities?**

The analysis should treat the Bhubaneswar facility and the rural hubs as separate deployment contexts with different distribution characteristics. The stratified evaluation proposal should specify which subgroups would be tested separately and what minimum performance thresholds would be required for deployment in each context.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | LeCun, Y., Boser, B., Denker, J. S., et al. (1989). Backpropagation applied to handwritten zip code recognition. <i>Neural Computation</i> , 1(4), 541-551. [The foundational paper on convolutional neural networks. Accessible introduction to the convolution operation and its application to visual recognition.]                                                            |
| <b>Core</b> | Norouzzadeh, M. S., et al. (2018). Automatically identifying, counting, and describing wild animals in camera-trap images with deep learning. <i>Proceedings of the National Academy of Sciences</i> , 115(25), E5716-E5725. [Directly relevant to Section 9.3; demonstrates deep learning for wildlife camera trap identification with accuracy comparable to expert reviewers.] |
| <b>Core</b> | NRSC/ISRO. (2022). National Agricultural Drought Assessment and Monitoring System: Technical Documentation. <a href="http://nrsc.gov.in">nrsc.gov.in</a> . [Primary source on satellite-based crop monitoring in India; directly relevant to Section 9.3 and to the Science stream project prompt.]                                                                               |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Goodfellow, I., Shlens, J., & Szegedy, C. (2015). Explaining and harnessing adversarial examples. <i>Proceedings of ICLR 2015</i> . [The foundational paper on adversarial examples; directly relevant to Section 9.5. The first two pages are accessible to non-specialists.]                                                                          |
| <b>Extended</b> | Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional accuracy disparities in commercial gender classification. <i>Proceedings of Machine Learning Research</i> , 81, 1-15. [On systematic failure and stratified evaluation; the invisible blind spot argument is directly analogous to the disparities documented here.]                  |
| <b>Extended</b> | Jain, S., et al. (2021). Deep learning-based highway incident detection using surveillance cameras: A study on Indian national highways. <i>Transportation Research Part C: Emerging Technologies</i> , 128, 103136. [Directly relevant to the highway accident detection section; a peer-reviewed study on the specific Indian application described.] |

### Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Wildlife Insights. (2023). AI for Wildlife Camera Traps. <a href="http://wildlifeinsights.org">wildlifeinsights.org</a> . [The primary open platform for AI-assisted camera trap analysis; directly relevant to Section 9.3 and the Science stream section. Freely accessible with documentation on supported species and performance metrics.] |
| <b>Online</b> | Google Developers. (2023). Machine Learning Crash Course: Image Classification. <a href="https://developers.google.com/machine-learning/practica/image-classification">developers.google.com/machine-learning/practica/image-classification</a> . [A free,                                                                                      |

|               |                                                                                                                                                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | interactive introduction to convolutional neural networks with visual examples of what each layer detects. Directly complements Sections 9.1 and 9.2.]                                                                                                                                                               |
| <b>Online</b> | Data.gov.in. (2023). Remote Sensing and GIS Data. <a href="https://data.gov.in/catalog/remote-sensing">data.gov.in/catalog/remote-sensing</a> . [Primary open data portal for Indian satellite imagery and agricultural monitoring datasets; relevant to the satellite crop monitoring section and project prompts.] |

### *End of Chapter 9*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 10

# AI Tools: A Practical Field Guide

*The previous two chapters explained how AI systems process language and images. This chapter shifts from explanation to practice. It introduces the AI tools that are most widely available in India, describes what each one is actually good at, identifies the tasks for which each is less reliable, and demonstrates their use through concrete tasks relevant to students across all three streams.*

*A field guide is a practical reference: it tells you what you are looking at, how to recognise it, and what to do with it. That is the spirit of this chapter. The tools described here are not static: they will have changed between the writing of this text and your reading of it. The principles for evaluating and using them, however, are durable, and learning those principles is more valuable than memorising any particular tool's current capabilities.*

*The chapter also examines the anatomy of a prompt. AI tools are only as useful as the instructions you give them. Understanding what makes a prompt effective, and what makes one ineffective, is a practical skill that transfers across every tool and every task.*

## Learning Objectives

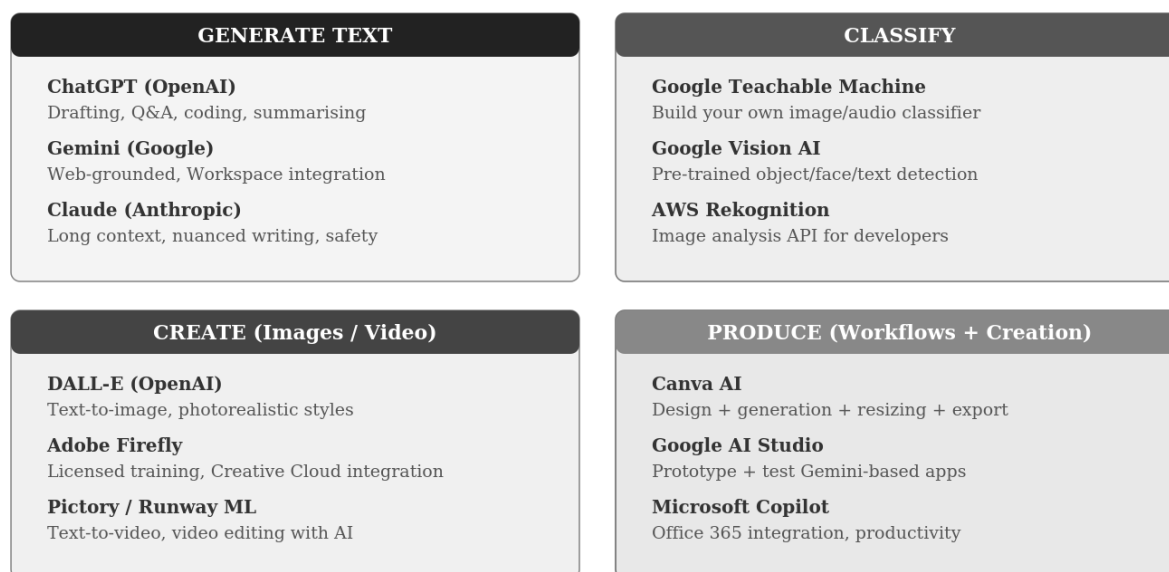
By the end of this chapter, you will be able to:

- Categorise AI tools by function: generate, classify, converse, and create
- Describe the distinctive capabilities and limitations of ChatGPT, Gemini, and Claude
- Use Google Teachable Machine to build a simple image classifier from collected examples
- Explain the image and video generation pipeline for DALL-E, Firefly, and Pictory
- Write an effective prompt by applying the anatomy of a good prompt
- Apply these tools to stream-specific tasks with appropriate awareness of their limitations

## 10.1 The AI Tools Landscape

The AI tools available to a student in 2025 span an enormous range of capability, cost, and appropriate use. Making sense of this landscape requires a framework. The most useful framework organises tools not by the company that made them but by what they are designed to do.

**Fig. 10.1**  
**The AI Tools Landscape: Categorised by Function**



*Most tools span multiple categories. This framework is a starting point for orientation, not a rigid taxonomy.*

*Figure 10.1 — AI Tools Landscape*

Generate tools produce new content: text, images, audio, video, code. They take a description or instruction as input and produce an artefact as output. ChatGPT, Gemini, Claude, DALL-E, Adobe Firefly, Runway ML, and Pictory all belong in this category, though they generate different kinds of content.

Classify tools take an input and assign it to a category. Google Teachable Machine is a classify tool: you give it labelled examples of each category, and it learns to classify new inputs. Many practical AI applications, from disease detection to sentiment analysis to quality control, are built on classification.

Converse tools are designed for back-and-forth dialogue, maintaining context across multiple turns and producing responses calibrated to the user's question. ChatGPT, Gemini, and Claude are all conversation tools as well as generation tools. The distinction matters because good conversational AI requires not just generating plausible text but generating text that is responsive to the specific context of the exchange.

Create tools, in the specific sense used here, are specialised production environments that combine AI generation with workflow support: templates, editing interfaces, export options, and platform integrations. Canva AI, Google AI Studio, and Adobe Firefly exist not just as generation models but as production environments where the generated content can be edited, refined, and incorporated into a final product.

Most major tools span more than one category. Claude can generate, converse, and assist with coding. Canva AI can generate images, suggest layouts, and resize content for different platforms. The framework is a starting point for orientation, not a rigid taxonomy.

## 10.2 ChatGPT, Gemini, and Claude: What They Are and Are Not

ChatGPT, Gemini, and Claude are the three most widely used large language model interfaces as of this writing. They are all built on transformer-based architectures trained on large text corpora, and they share many characteristics: they can hold extended conversations, draft text in many styles, answer questions across many domains, assist with coding, and generate structured documents. They also share important limitations: they can hallucinate, they have training data cutoffs, and they vary in their coverage of Indian languages and contexts.

**Fig. 10.2**

**ChatGPT, Gemini, and Claude: What Each Does Well and Where Each Struggles**

|              | ChatGPT (OpenAI)                                                                                                                                                    | Gemini (Google)                                                                                                                                                           | Claude (Anthropic)                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STRENGTHS    | Broad knowledge base<br>Strong code generation<br>Multimodal (GPT-4V)<br>Plugin ecosystem<br>Most widely documented<br><i>Free tier: GPT-3.5</i>                    | Web-grounded (real-time info)<br>Google Workspace integration<br>Strong multilingual support<br>Multimodal natively<br>Google AI Studio API<br><i>Free tier available</i> | Very long context window<br>Explicit about uncertainty<br>Nuanced writing quality<br>Safe and careful responses<br>Strong on complex reasoning<br><i>Free tier at claude.ai</i>                  |
| LIMITATIONS  | Training data cutoff<br>Hallucination on sparse topics<br>Free tier has limited capability<br>Inconsistent on long documents<br><i>Cannot cite sources reliably</i> | Variable on complex reasoning<br>Hallucination (less than cutoff issues due to web grounding)<br>Privacy concerns re. Google data<br><i>Output quality varies by task</i> | No real-time web access (standard)<br>Training data cutoff<br>More conservative (may over-refuse)<br>Lower-resource Indian languages not at English level<br><i>No image generation built-in</i> |
| INDIAN LANG. | Hindi: good<br>Bengali, Tamil, Telugu: fair<br>Odia, Santali: limited<br><i>GPT-4 better than GPT-3.5</i>                                                           | Hindi, Tamil, Telugu: strong<br>Bengali, Kannada, Malayalam: good<br>Odia: fair; Santali: limited<br><i>Best overall multilingual support</i>                             | Hindi: good<br>Bengali, Tamil: fair<br>Odia, Santali: limited<br><i>Improving with each version</i>                                                                                              |

*All three share the most important limitation: they hallucinate. Every factual claim*

**Figure 10.2 — L L M Comparison**

ChatGPT, developed by OpenAI, was the first large language model to achieve mass adoption. It is available in a free tier with GPT-3.5 capability and a paid tier with GPT-4 capability. GPT-4 has multimodal capability, meaning it can process both text and images. ChatGPT's strengths include broad knowledge, strong English-language writing assistance, and code generation. Its weaknesses include uneven performance on Indian languages in the free tier, a training data cutoff that makes it unreliable for recent events, and a tendency to produce confident hallucinations in domains where its training data is sparse.

Gemini, developed by Google DeepMind, is deeply integrated with Google's search infrastructure and has access to current web information through its grounding capabilities, which helps mitigate the training cutoff problem for factual queries. Gemini has strong multilingual capability including several Indian languages, and its integration with Google Workspace makes it particularly useful for productivity tasks. Google AI

Studio provides API access for developers. Gemini's weaknesses include variable consistency in longer reasoning tasks and, like all LLMs, susceptibility to hallucination.

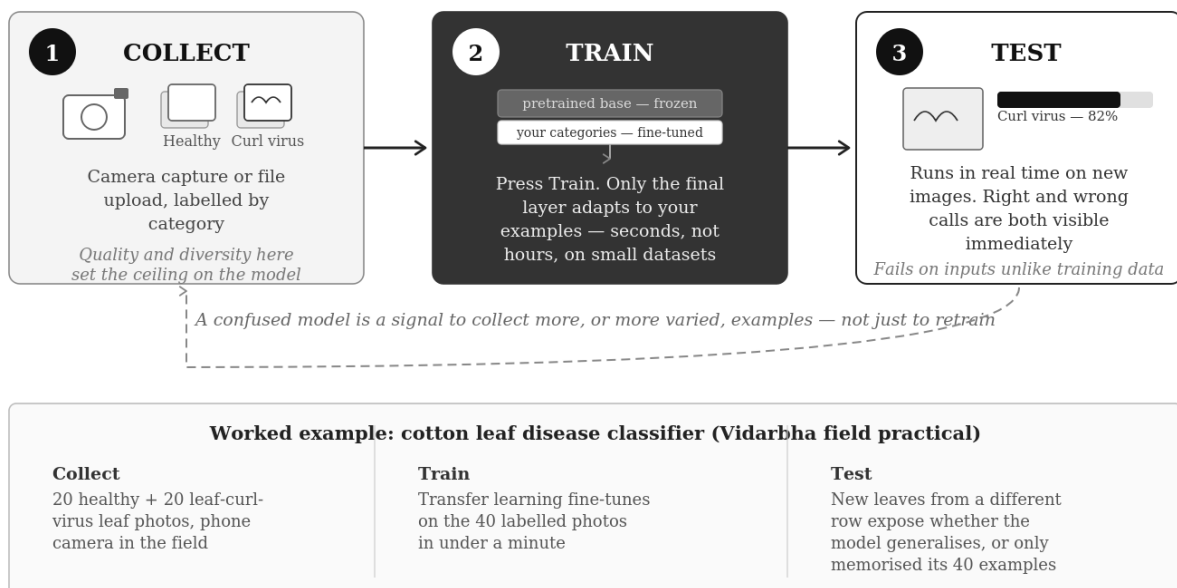
Claude, developed by Anthropic, is designed with particular attention to safety and reliability. It is known for longer context windows, strong performance on nuanced writing tasks, and a tendency to be more explicit about uncertainty and the limits of its knowledge than some other models. Claude is available through Claude.ai in free and paid tiers. Its weaknesses include, again, training data cutoffs and hallucination, and its coverage of lower-resource Indian languages, while improving, is not yet at the level of its English performance.

The practical guidance for Indian students is: use all three tools as drafting and exploration partners, not as authoritative sources. Verify any specific factual claim, especially claims about Indian history, regional contexts, or recent events. Expect better performance on tasks that are well-represented in English-language training data, and be more cautious about outputs on topics that are under-resourced in that data.

### 10.3 Google Teachable Machine: The Three-Step Classifier

Google Teachable Machine is unique among the tools described in this chapter because it is not a pre-trained model you query: it is a platform through which you build your own model. This makes it the most directly educational AI tool available, because using it requires understanding the training process, not just the output.

**Fig. 10.3**  
**Google Teachable Machine: The Three-Step Classifier**



Teachable Machine makes the training loop visible and fast: collect, train, test, and — when the model is confused — collect again. It is the same loop described abstractly in Chapter 4, made concrete.

Figure 10.3 — Google Teachable Machine: The Three-Step Classifier

The process has three steps. Step one: collect. You define the categories you want the classifier to recognise and collect examples for each category. You can use your computer's camera to capture images live, or you can upload images from files. For a plant disease classifier, you would collect images of healthy leaves and images of leaves showing the disease of interest. The quality and diversity of your examples directly determine the quality of your model: as Chapter 4 explained, the model can only learn what your training data contains.

Step two: train. You press the Train button and Teachable Machine trains a classifier on your examples. The training takes seconds for small datasets on modern hardware. The tool uses transfer learning: it starts from a model that has already learned to extract general visual features from millions of images, and it fine-tunes only the final classification layer on your specific categories. This is why it can train effectively on small datasets of tens or hundreds of examples rather than requiring the thousands that a model trained from scratch would need.

Step three: test. The trained model runs in real time, classifying inputs from your camera or from uploaded files. You can immediately observe how well the model performs on examples it has not seen before, which examples it gets right, which it gets wrong, and what confuses it. This real-time feedback loop is educationally invaluable: it makes abstract concepts like overfitting, distribution shift, and the importance of diverse training data directly observable.

Teachable Machine supports image classification, audio classification, and pose classification. It exports models that can be embedded in web applications. Its limitations are the limitations of any small-dataset classifier: it will not generalise well beyond the distribution of its training examples, and it will fail when the test input differs substantially from what it was trained on.

## **10.4 DALL-E, Adobe Firefly, and Pictory: The Creation Pipeline**

The image and video generation tools represent a qualitatively different kind of AI application from the language and classification tools discussed above. Where language models predict the next token in a sequence, image generation models learn to produce visual outputs that match a text description. The underlying technology, diffusion models, works by learning to reverse a process of adding noise to images: starting from pure noise and gradually denoising toward a coherent image that matches the input description.

**Fig. 10.5**  
**Image and Video Generation Pipeline (Diffusion Models)**

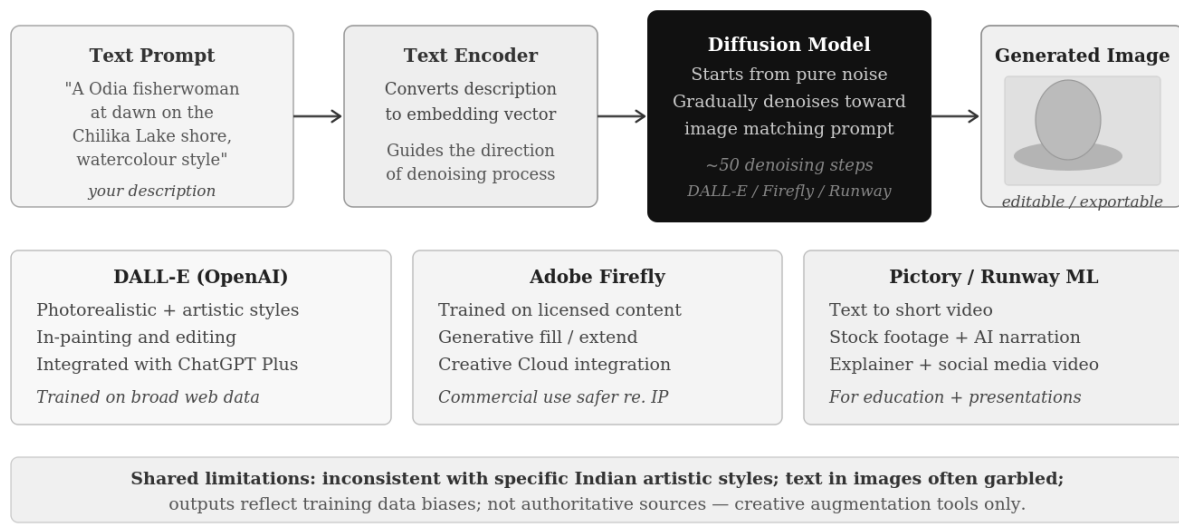


Figure 10.5 — Generation Pipeline

DALL-E, developed by OpenAI, generates images from text descriptions with a high degree of photorealism and compositional flexibility. It can generate images in a wide range of styles, from photographic to illustrative to abstract, and can follow specific compositional instructions: place the subject on the left, use a warm colour palette, depict a traditional Odia fishing village at sunset. DALL-E is integrated into ChatGPT for paid subscribers and available through the OpenAI API. Its limitations include inconsistency in depicting specific Indian visual styles, occasional errors in text generation within images, and the ethical concerns about training data that apply to all generative models.

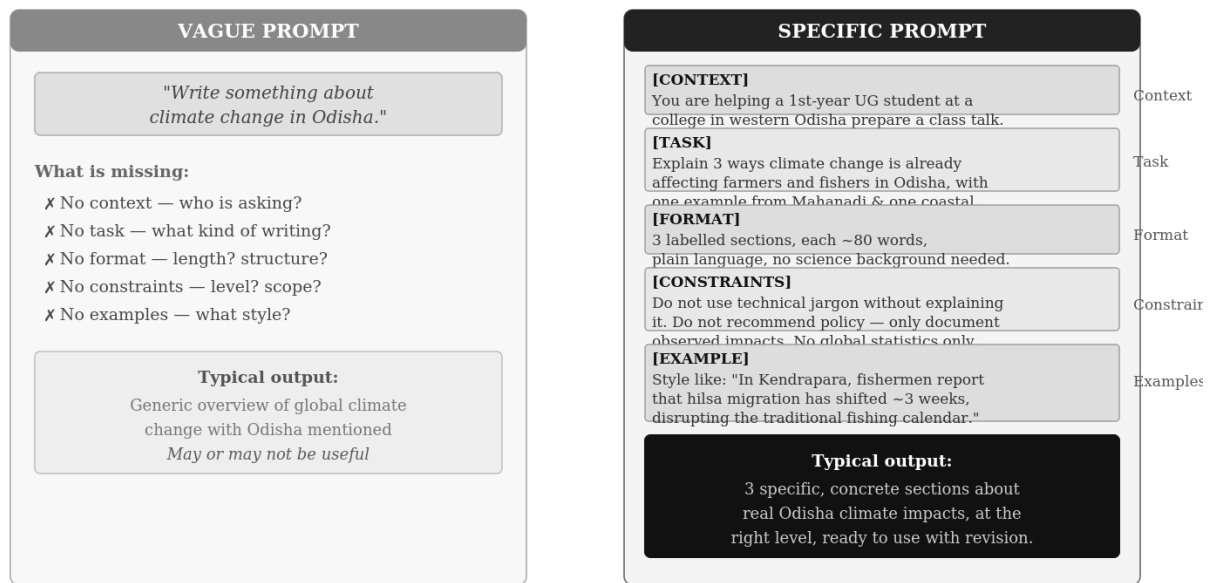
Adobe Firefly is a generative AI model trained by Adobe on licensed content, which addresses some of the copyright concerns associated with models trained on scraped web data. It is deeply integrated with Adobe's Creative Cloud suite, making it particularly useful for design professionals and students who already work in Photoshop or Illustrator. Firefly can fill in or extend images (generative fill), convert rough sketches to polished illustrations, and generate vector graphics from text descriptions. For students producing educational materials or visual communications, Firefly within Canva or Adobe Express is a practical creative augmentation tool.

Pictory is a video generation and editing tool that takes text or existing video as input and produces short video outputs. It is designed for content creation rather than for artistic expression: its primary use cases are creating educational explainer videos, social media content, and presentations from text documents or slide decks. A student can enter a text description of a concept from this chapter, and Pictory will produce a short video combining relevant stock footage, AI-generated narration, and text overlays. The output is not artistically sophisticated but is practically useful for course presentations and educational content creation.

## 10.5 The Anatomy of a Prompt

Every interaction with a language model or generation tool begins with a prompt: the text instruction you give the AI. The quality of the prompt is the single most controllable factor in the quality of the output. Understanding what makes a prompt effective, and what makes one ineffective, is a transferable skill that applies across every tool and every task.

**Fig. 10.7**  
**The Anatomy of a Prompt: Vague vs. Specific**



*The specific prompt is longer but not harder to write. It simply answers five questions: who are you, what do you want, in what format, with what constraints, and can you show an example? Every added component reduces output variance.*

Figure 10.7 — Prompt Anatomy

### Component 1: Context

A good prompt tells the AI what role or context it should assume. You are a biology teacher preparing a lesson for first-year undergraduate students performs differently from the same question asked without context. The context shapes the vocabulary, depth, and framing of the response. For Indian students, context can also include relevant regional specificity: You are helping a student at a college in rural Odisha with limited access to specialised laboratories will produce more practically useful responses than the same question without this context.

### Component 2: Task specification

A precise task description outperforms a vague one. Explain photosynthesis produces a generic explanation. Explain the light-independent reactions of photosynthesis in simple language suitable for a student who has not yet studied chemistry, using the analogy of a factory production line produces a much more targeted and useful response. The more specifically you describe what you want, the more closely the output will match your needs.

### **Component 3: Format instruction**

Specifying the output format is often the single highest-value addition to a prompt. Present this as a table with three columns: term, definition, and an example from Indian agriculture tells the model exactly how to structure its output. Without this instruction, the model will choose a default format that may or may not match what you need. Common format instructions include: write as a numbered list, write as a narrative paragraph, present as a comparison table, limit to 200 words, include a summary at the end, use headings for each section.

### **Component 4: Constraints**

Constraints define what the prompt should not do. Do not use technical jargon. Do not reference events after 2020. Do not use examples from outside India. These negative instructions are as important as positive ones because they prevent common failure modes. If you are asking an LLM to write about a sensitive historical topic and want it to present multiple perspectives rather than taking a position, the constraint present three different perspectives without endorsing any of them will consistently produce more balanced outputs than the same request without the constraint.

### **Component 5: Examples**

Including examples of what you want, called few-shot prompting, is consistently one of the most powerful techniques available. Here is an example of the kind of explanation I want, followed by one or two concrete examples, teaches the model the style, depth, and format you are looking for far more effectively than any amount of description. The model does not need many examples: one to three carefully chosen examples are typically more effective than a longer description of the desired output.

The table in Figure 10.7 shows the difference between a vague prompt and a specific prompt for the same underlying task. The specific prompt is longer, but it is not more complex to write: it simply answers the five questions that every prompt should answer: who are you, what do you want, in what format, with what constraints, and can you show an example?

## **10.6 Patterns in My Field: Stream Perspectives**

**Fig. 10.6**  
**Stream Fork: Same Tools, Different Purposes**

|        | Arts Stream                                                                                                                             | Science Stream                                                                                                              | Commerce Stream                                                                                                                 |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| TOOL   | <b>Claude / Gemini</b><br>Odia poetry creative writing partner<br><i>LLM with Odia capability</i>                                       | <b>Google Teachable Machine</b><br>Plant disease classification<br><i>No-code image classifier</i>                          | <b>Claude / Gemini</b><br>Customer sentiment analysis<br><i>LLM + multilingual</i>                                              |
| PROMPT | "You are assisting an Odia poet in Panchamali Chhanda. Suggest 3 completions for stanza 3 maintaining the metre and monsoon imagery..." | Collect 20 photos of healthy vs. diseased cotton leaves under varied lighting. Train. Test in real time. Document failures. | "Classify each message as positive/negative/mixed. Identify main aspect: quality, price, delivery, or staff. Present as table." |
| OUTPUT | Three draft stanzas in Odia with syllable counts annotated<br><i>Poet accepts/modifies</i>                                              | Real-time classifier:<br>Healthy 0.91   Diseased 0.09<br>Fails in direct sunlight<br><i>Retrain with sunlight images</i>    | Classification table<br>Delivery issues identified as primary complaint<br><i>Odia less accurate than Hindi</i>                 |
| LIMIT. | Odia performance imperfect; poet must evaluate critically                                                                               | Distribution shift (sunlight); small dataset = fragile model                                                                | Odia accuracy lower; verify all Odia outputs manually                                                                           |

*The same tools serve all three streams — what changes is the task, the prompt structure, and the verification requirements.*

**Figure 10.6 — Stream Fork**

### Arts Stream: Building a Creative Writing Partner for Odia Poetry

A student working on an Odia poem about the monsoon arrives at a stanza she cannot complete to her satisfaction. She has two completed stanzas in the Panchamali Chhanda metre, a classical Odia metre characterised by a specific syllable pattern per line. She prompts Claude or Gemini: You are assisting an Odia poet working in the Panchamali Chhanda metre. Here are two completed stanzas [she pastes them in Odia script]. Please suggest three possible completions for the third stanza that maintain the syllable count, the seasonal imagery of monsoon, and the reflective tone of the existing stanzas. Do not try to rhyme unless the existing stanzas rhyme. The response will be imperfect, but it will provide generative material, alternative phrasings to consider, and structural suggestions that the poet can accept, reject, or modify. The AI is not writing the poem; it is providing a conversational partner for the creative process.

### Science Stream: Using Google Teachable Machine to Classify Plant Diseases

A student in a zoology or botany practical collects twenty photographs of cotton leaves showing symptoms of leaf curl virus and twenty photographs of healthy cotton leaves from plants in the college garden. They open Teachable Machine, define two classes, upload their photographs, and train the model in approximately thirty seconds. They then test the model by photographing leaves with their phone camera in real time. They observe that the model performs well on leaves photographed in similar lighting to the training images but struggles with leaves photographed in direct sunlight, because the training images were taken in diffused indoor light. This observation directly demonstrates the distribution shift problem from Chapter 7, making an abstract concept viscerally concrete. The student then collects additional training images in direct sunlight, retrains, and observes the

improvement. The entire exercise takes approximately forty-five minutes and requires no programming.

### **Commerce Stream: Using AI to Analyse Customer Sentiment for a Local Business**

A student helps a local textile shop owner in Kuchinda analyse customer feedback. The shop has accumulated approximately fifty text messages and social media comments in Odia and Hindi from satisfied and dissatisfied customers over the past six months. The student pastes ten messages at a time into Claude or Gemini with the prompt: Classify each of the following customer messages as positive, negative, or mixed, and identify the main aspect being commented on (product quality, price, staff behaviour, or delivery). Present the results as a table. The tool classifies the messages with reasonable accuracy for Hindi and acceptable, if imperfect, accuracy for Odia. The student identifies that the most common source of negative feedback is a specific issue with delivery timing, which the shop owner had not previously recognised as a priority. The entire analysis takes approximately an hour and requires no specialised data analysis software. The student notes in their project report that Odia-language messages showed lower classification confidence than Hindi messages, illustrating the language coverage disparity described in Chapter 8.

## **10.7 Misinformation and Deepfakes: When Generation Tools Are Misused**

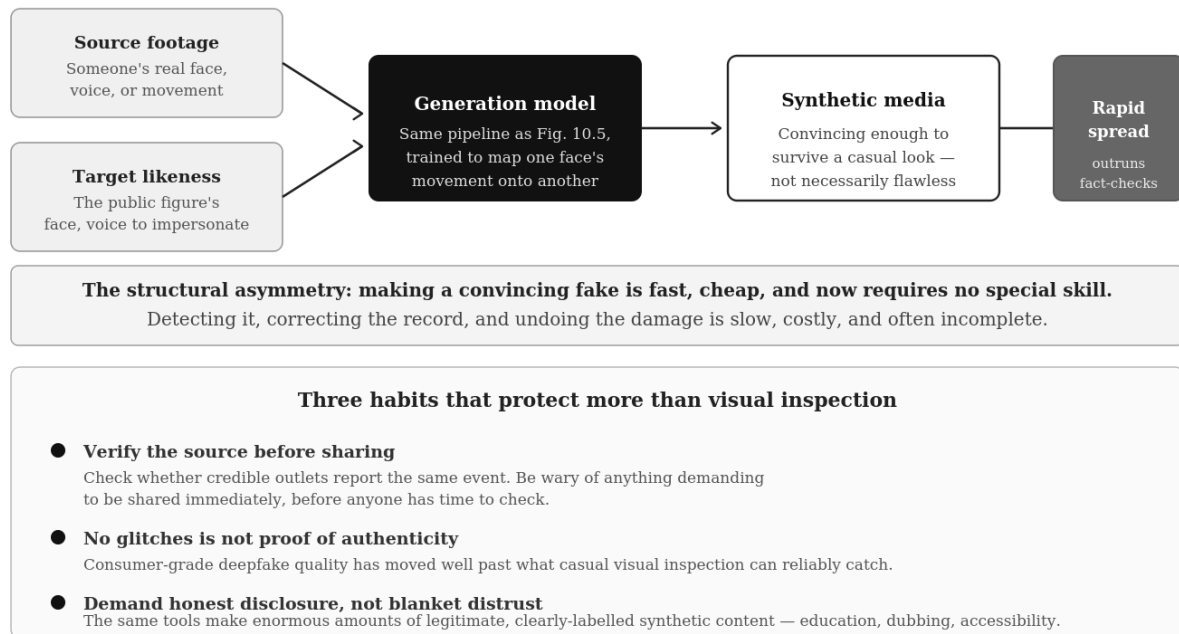
The generation tools described in Section 10.4, DALL-E, Adobe Firefly, Pictory, and their many equivalents, are creative and productivity tools when used as this chapter has described them. The same underlying technology, applied with deceptive intent rather than creative intent, produces something with a different name: a deepfake, a synthetic image, audio clip, or video engineered to convincingly depict a real person saying or doing something they never said or did.

The technical process is a direct extension of the generation pipeline covered earlier. Where a text-to-image model learns to produce a plausible image matching a description, a deepfake video model learns to map one person's facial movements and expressions onto another person's face and voice, frame by frame, with enough visual consistency to survive a casual viewing. The tools required to do this convincingly have moved, within a few years, from requiring specialist technical skill to being available as consumer-grade mobile applications.

India has already seen this technology deployed at scale during elections, with synthetic audio and video of political figures circulated on messaging platforms faster than fact-checkers could debunk them, and in widely reported cases in which deepfake videos of well-known public figures were circulated without their consent, causing significant personal and professional harm. Deepfake-enabled financial fraud, in which a cloned voice of a company executive is used to authorise a fraudulent transfer, has been documented in Indian corporate contexts as well. The common thread across these cases is that the harm does not require the deepfake to be technically flawless. It requires only

that it be convincing enough, and circulated widely enough and quickly enough, to do damage before it can be debunked, corrected, or removed.

**Fig. 10.8**  
**Anatomy of a Deepfake**



*India's synthetic-media labelling rules and the wider governance response are covered in Chapter 14.*

*Figure 10.8 — Anatomy of a Deepfake*

This creates a structural asymmetry that is important to understand. Generating a convincing fake is now fast, cheap, and requires no special access. Detecting a fake, correcting the record, and undoing the reputational, financial, or political damage it caused is slow, resource-intensive, and often incomplete: a correction rarely reaches everyone who saw the original false claim, and even a fully debunked deepfake can leave a lasting impression that a later correction never fully removes. This asymmetry is why platforms and researchers are investing heavily in automated deepfake detection, itself an AI system trained to spot the subtle artefacts, inconsistent lighting, unnatural blinking patterns, mismatched audio-visual timing, that generation models still tend to leave behind. But detection is an arms race, not a solved problem: as detection methods improve, so do the generation methods being trained, in part, to evade them.

For a student, three practical habits offer more protection than any detection tool. First, treat unexpected, emotionally charged, or highly shareable audio and video with the same scepticism this book has urged toward any AI output: verify the source, check whether credible news outlets are reporting the same event, and be especially wary of content that arrives with urgency attached, asking to be shared immediately, before anyone has time to check it. Second, remember that the absence of visible glitches is not evidence of authenticity; consumer-grade deepfake quality has improved to the point where casual visual inspection is no longer a reliable test. Third, understand that the same generation tools that make a deepfake also make an enormous amount of legitimate, clearly-labelled synthetic content, for education, accessibility, dubbing, and creative work, so the goal is

not to distrust all AI-generated media but to demand and check for honest disclosure of what is synthetic and what is not.

India's response to this problem, including the IT Rules provisions requiring synthetic media to be labelled and the broader regulatory questions this raises, is addressed further in Chapter 14's discussion of AI governance.

## — WHAT COULD GO WRONG? —

### **Over-reliance and the Automation Bias**

The tools described in this chapter are genuinely useful. They can accelerate tasks that previously required hours, produce first drafts that previously required significant effort, and make sophisticated analysis accessible to students without specialist technical skills. This genuine usefulness creates a real risk: over-reliance.

Automation bias is the tendency to trust automated outputs more than they deserve, to apply less critical scrutiny to AI-generated content than to content produced by a human, and to accept AI decisions without the verification that their accuracy rates actually warrant. Research consistently shows that people are less likely to catch errors in AI-generated text than in human-generated text, even when the error rate is the same, because the fluency and confidence of AI output creates an impression of reliability that the underlying accuracy does not justify.

In academic contexts, automation bias has a specific manifestation: students who use AI tools to draft essays, solve problems, or generate analyses, and then submit the AI output with minimal review, are not only committing academic dishonesty. They are also forfeiting the learning that the task was designed to produce. The point of writing an essay is not to produce an essay. It is to develop the capacity to organise and express ideas. The point of solving a problem is not to produce a solution. It is to develop problem-solving skill. An AI that does these things for you leaves you less capable, not more.

This is not an argument against using AI tools. It is an argument for using them as amplifiers of your own work rather than substitutes for it. A student who uses Claude to help draft an introduction, then substantially revises it, then checks the factual claims, then rewrites the conclusion from scratch has used AI in a way that develops rather than undermines their capability. A student who pastes in a question, pastes out the answer, and adds their name has not used AI: they have allowed AI to impersonate them.

The distinction matters most for Indian students at regional colleges, where the stakes of genuine skill development are highest. The digital economy that is reshaping employment in India rewards the ability to think critically, write clearly, and solve novel problems. These abilities can only be developed through practice. AI tools that substitute for that practice, rather than supporting it, are not assets to the students who use them that way. They are, in the long run, a form of educational self-harm.

## — CHAPTER SUMMARY —

### Chapter 10 — Key Points

1. AI tools can be categorised by function: generate, classify, converse, and create.
2. ChatGPT, Gemini, and Claude are all powerful language tools with shared limitations: hallucination, training cutoffs, and uneven Indian language coverage.
3. Google Teachable Machine allows students to build and test their own classifiers — making abstract ML concepts directly observable.
4. Transfer learning in Teachable Machine allows effective training on small datasets by building on a pre-trained feature extractor.
5. DALL-E, Adobe Firefly, and Pictory use diffusion models for image/video generation; they are creative augmentation tools, not authoritative sources.
6. A good prompt has five components: context, task specification, format instruction, constraints, and examples (few-shot prompting).
7. Automation bias — trusting AI outputs more than they deserve — is the primary risk in tool use; critical review of all AI outputs is essential.
8. AI tools are most valuable when used as amplifiers of human work, not substitutes for it.
9. Odia and other lower-resource Indian languages consistently underperform compared to Hindi and English across all major tools.

## — MODEL QUESTIONS —

### Section A — MCQ (5 Questions)

**Q1. In the AI tools landscape framework used in this chapter, what distinguishes a 'classify' tool from a 'generate' tool?**

- A) Classify tools are free; generate tools require payment
- B) Classify tools assign inputs to categories; generate tools produce new content from descriptions
- C) Classify tools work only with images; generate tools work only with text
- D) Classify tools require internet access; generate tools work offline

**Answer: B**

*Explanation: Classify tools take an input and assign it to a pre-defined category. Generate tools take an instruction and produce new content. Google Teachable Machine is a classify tool; DALL-E is a generate tool. Many tools, like ChatGPT, span both categories.*

**Q2. Why can Google Teachable Machine train effectively on small datasets of tens of examples, when Chapter 4 suggested thousands of examples are typically needed?**

- A) Teachable Machine uses a simpler classification algorithm that needs less data
- B) Transfer learning: it fine-tunes only the final layer of a pre-trained feature extractor
- C) Teachable Machine automatically generates additional training examples through data augmentation
- D) Small datasets are sufficient for classification tasks, unlike regression tasks

**Answer: B**

*Explanation: Teachable Machine uses transfer learning: it starts from a model already trained to extract general visual features from millions of images, and only trains the final classification layer on your specific categories. This dramatically reduces the data needed.*

**Q3. A student prompts an LLM: 'Tell me about photosynthesis.' This prompt is less effective than a more specific version because it lacks:**

- A) A question mark at the end
- B) Context, format instruction, constraints, and examples that would direct the model toward the student's specific need
- C) Specific technical terminology that tells the model which domain to draw from
- D) A system message setting the model's behaviour

**Answer: B**

*Explanation: The vague prompt will produce a generic response that may not match the student's level, format preference, or specific interest. Adding context (who is asking), format instruction (how to present it), constraints (what to avoid), and examples (what style is wanted) dramatically improves output relevance.*

**Q4. Adobe Firefly is trained on licensed content rather than scraped web data. What problem does this address?**

- A) It prevents Firefly from generating inappropriate content
- B) It addresses copyright concerns: artists whose work was used in training were compensated or their work was used with permission
- C) It makes Firefly more accurate than DALL-E for photorealistic images
- D) It reduces Firefly's hallucination rate compared to other generative models

**Answer: B**

*Explanation: Training on scraped web data may use copyrighted images without the artists' knowledge or consent. Adobe's approach of training on licensed content, including Adobe Stock and the public domain, addresses this ethical and legal concern, though debates about compensation and consent in AI training continue.*

**Q5. Automation bias in the context of AI tool use means:**

- A) AI tools automatically bias toward certain types of content or perspectives
- B) Users trust AI-generated outputs more than they deserve, applying less critical scrutiny than accuracy rates warrant
- C) Automation reduces human bias by removing subjective judgment from decisions
- D) AI tools that are widely used develop biases over time through user feedback

**Answer: B**

*Explanation: Automation bias is a documented human tendency to over-trust automated outputs. It means people are less likely to catch errors in AI-generated content than in human-generated content, even when error rates are the same, because AI fluency*

*creates an unwarranted impression of reliability.*

## Section B — Short Answer Questions (5 Questions)

**Q6. A student wants to use Google Teachable Machine to build a classifier that distinguishes healthy from diseased tomato leaves at their college farm. Walk through the three steps they would follow, identifying the most important decision at each step.**

**Model Answer:**

Step one, collect: the student defines two classes, healthy and diseased, and collects labelled examples for each by photographing leaves with their phone or computer camera. The most important decision at this step is ensuring diversity in the training examples: photographs taken in different lighting conditions, at different angles, at different growth stages, and with different phones. A classifier trained only on well-lit, top-down photographs will fail on real farm conditions. Step two, train: the student presses Train, and Teachable Machine fine-tunes a classification layer on their examples using transfer learning. The most important decision here is choosing the training duration: too few iterations may underfit; however, Teachable Machine automates most of this. The student should inspect the training accuracy per class to check for imbalanced learning. Step three, test: the student tests the model on new leaves not in the training set. The most important decision is testing under conditions representative of deployment, including direct sunlight, partial occlusion by other leaves, and photographs from different phones than the one used for training. Any systematic failures should be addressed by adding more training examples in the failing condition.

*Key terms: Google Teachable Machine, transfer learning, training data diversity, train-test split, distribution shift, overfitting*

**Q7. Rewrite the following vague prompt as a high-quality prompt using the five components of prompt anatomy: 'Write something about climate change in Odisha.'**

**Model Answer:**

A high-quality rewrite: Context: You are helping a first-year undergraduate student at a college in western Odisha prepare a short presentation for their Environmental Science class. Task: Explain three specific ways that climate change is already affecting the daily lives of farmers and fishing communities in coastal and inland Odisha, with at least one example from the Mahanadi basin and one from the coastal districts. Format: Present as three labelled sections, each of approximately 80 words, written in plain language accessible to students without a science background. Constraints: Do not use technical climate science jargon without immediately explaining it. Do not include global statistics unless they are directly relevant to Odisha. Do not recommend policy solutions — focus only on documenting observed impacts. Example: The explanation should be at the level of clarity of: 'In the Kendrapara district, fishermen report that the timing of the hilsa fish migration has shifted by approximately three weeks over the past decade, disrupting the traditional fishing calendar their communities have relied on for generations.'

*Key terms: prompt anatomy, context, task specification, format instruction, constraints, few-shot prompting, specificity*

**Q8. A student uses Gemini to research the history of the Kondh resistance to British colonial rule in 19th century Odisha for a history essay. What specific risks should they be aware of, and what verification steps should they take?**

**Model Answer:**

The specific risks are hallucination amplified by data scarcity and automation bias. The history of the Kondh resistance, particularly the Meriah Wars of the 1840s and 1850s, is documented in Indian historical scholarship but is significantly underrepresented in English-language training data compared to well-documented events of European or British imperial history. An LLM queried about this topic has fewer reliable patterns to draw on, making it more likely to confabulate: generating plausible-sounding names of leaders, dates, and events that are partially or wholly fabricated. Because the output will be fluent and confident, the student may not notice that specific claims are invented without cross-referencing primary sources. Verification steps should include: checking every named individual, date, and specific event against peer-reviewed Indian history journals, books by historians such as Felix Padel or Nandini Sundar who have documented this history, or primary source archives at Odisha State Archives or the British Library India Office Records. The student should treat Gemini's output as a starting point for research, not a source for citation.

*Key terms: hallucination, training data scarcity, automation bias, verification, cross-referencing, primary sources, regional history*

**Q9. Explain the diffusion model approach used by DALL-E and Firefly for image generation, in terms a student with no prior technical background can understand.**

**Model Answer:**

A diffusion model learns to generate images by first learning how to destroy them and then learning to reverse that destruction. During training, the model is shown millions of images and is trained to predict what they would look like at each stage of a process in which random noise is gradually added until the image becomes pure static. Once the model has learned this destruction process thoroughly, it can run it in reverse: starting from pure noise and gradually removing that noise in the direction of a coherent image, guided by a text description. The text description tells the model which direction to travel during the denoising: toward an image of a lotus pond at dawn rather than toward random noise. The model has learned, from millions of training image-text pairs, which visual features are associated with which textual descriptions. When you type a description, the denoising process is steered toward images that match that description, producing an image that has been, in effect, sculpted out of noise by the model's learned associations between visual and linguistic patterns.

*Key terms: diffusion model, denoising, noise, image generation, text-to-image, training process, generative model*

**Q10. The chapter argues that using AI tools as substitutes for learning, rather than amplifiers of it, constitutes 'educational self-harm.' Do you agree or disagree? What is the difference between using AI as a substitute and using it as an amplifier?**

**Model Answer:**

The argument is persuasive for the following reason: skill development requires practice, and practice requires effort that meets genuine resistance. When an AI tool removes the resistance by doing the task for you, the neural consolidation and metacognitive development that practice produces does not occur. A student who uses an LLM to answer all their essay questions can produce essay-shaped text but cannot develop the

organisational thinking, argumentative precision, and writerly judgment that writing essays is designed to build. The key distinction between substitute and amplifier use lies in who does the thinking. Amplifier use: a student drafts an introduction independently, then uses an LLM to suggest alternative phrasings for the weakest sentence, evaluates the suggestions critically, and chooses or adapts one. The student made every significant decision. Substitute use: a student enters the essay question into an LLM and submits the output with minimal review. The LLM made every significant decision. The distinction is not always sharp, but the principle is clear: if using the tool prevents you from engaging with the intellectual difficulty that would have produced learning, you are using it as a substitute.

*Key terms: automation bias, amplifier vs. substitute, skill development, academic integrity, metacognition, critical review*

## Section C — Reflective Questions (3 Questions)

**Q11. The chapter describes Gemini as having 'strong multilingual capability including several Indian languages.' What does 'strong' mean in this context, and how would you test whether the claim is true for the specific Indian language you care about?**

**Thinking prompts:**

- What tasks would you use to evaluate multilingual capability? Translation, generation, question answering, or something else?
- Is performance on Hindi or Tamil an adequate proxy for performance on Odia or Santali?
- How would you detect hallucination in an Indian language if you are not a fluent speaker of it?
- What sources would you use to verify Gemini's outputs in a language you are evaluating?
- What would it mean for an AI tool to have adequate performance in a language for academic use versus commercial use versus safety-critical use?

*Rubric hint: Strong responses will identify specific test tasks, propose a verification methodology, and engage with the distinction between what strong capability means at different levels of consequence. Credit for responses that identify the limitation of using English-language evaluations as proxies for Indian language capability.*

**Q12. Adobe Firefly's licensed training data is presented as an ethical improvement over scraped data. But is it sufficient? Who benefits and who might still be harmed by the licensed approach?**

**Thinking prompts:**

- What does 'licensed' mean in practice? Who licenses content to Adobe, and who does not?
- Does licensing compensate small independent artists as well as it compensates large stock photo agencies?
- Are there cultural traditions or folk art forms that cannot be 'licensed' because their ownership is collective and unrecorded?
- If a generative model trained on Indian folk art produces new works in that style,

does that harm the tradition, enrich it, or both?

- What would a genuinely community-consented approach to training on traditional artistic knowledge look like?

*Rubric hint: Strong responses will identify the difference between corporate licensing and genuine community consent, and will engage with the specific vulnerability of collective and oral artistic traditions to extraction without meaningful consent or compensation. Credit for responses that propose specific governance structures.*

**Q13. The chapter presents the Odia poetry creative writing example as a legitimate and valuable use of AI tools. But could the same interaction — an AI suggesting continuations for an Odia poem — be harmful to the tradition of Odia poetry? How would you draw the line?**

**Thinking prompts:**

- What is the difference between a student using an AI to learn the constraints of Panchamali Chhanda metre and a student using an AI to substitute for the labour of learning those constraints?
- If AI-generated Odia poetry becomes widely circulated, what happens to the community of practising Odia poets?
- Does the availability of AI creative assistance change what we expect of human poets? Should it?
- Are there artistic traditions that are especially vulnerable to AI creative assistance, and what makes them vulnerable?
- What would responsible AI-assisted engagement with a classical poetic tradition look like — what safeguards, what disclosures, what limits?

*Rubric hint: Strong responses will distinguish between AI as a learning tool and AI as a production substitute in artistic contexts, and will engage with the community and tradition-level consequences of widespread AI creative assistance. Credit for responses that identify specific vulnerabilities of regional language literary traditions.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Using Google Teachable Machine, build a small image classifier relevant to your field. For Arts students: classify two different types of traditional craft or visual art style (e.g., Pattachitra vs. Madhubani, or two distinct Odia folk motifs). For Science students: classify two plant or animal conditions (e.g., healthy vs. diseased leaf, or two similar species). Collect at least twenty examples per class, train, and test. Document the following: your data collection methodology, your training accuracy, your test accuracy on held-out examples, two specific failure cases with your hypothesis for why they failed, and one improvement you made after observing the initial failures. Present your classifier to the class with a live demonstration.**

The failure case analysis is the most important part. Simply reporting accuracy is not sufficient. Identify specific examples that the classifier got wrong, examine the images carefully, and propose a data-based explanation. The improvement should be tested and the effect reported — even if the improvement did not work, document why not.

**Q15. [Commerce]** A micro-entrepreneur in your community produces a traditional food product, craft, or agricultural commodity and sells it locally. Using one or more AI tools from this chapter, help them with one of the following: (a) analyse five to ten written or spoken customer feedback samples and produce a structured report identifying the key satisfaction and dissatisfaction drivers; (b) generate and evaluate three different marketing descriptions for the product, suitable for a WhatsApp business profile; or (c) create a simple explainer video or visual presentation using Pictory or Canva AI explaining what makes their product distinctive. Document your prompts, the AI outputs, your critical assessment of each output's accuracy and usefulness, and the entrepreneur's response to the final product. Conclude with a recommendation for how this type of micro-entrepreneur could most effectively use AI tools in their business.

The documentation of your prompts and your critical assessment of each output is the methodological core of this project. The entrepreneur's response provides real-world validation. The recommendation should be specific to the type of business and the tools evaluated, not generic.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Google. (2023). Teachable Machine Documentation and Examples. <a href="https://teachablemachine.withgoogle.com">teachablemachine.withgoogle.com</a> . [Primary documentation for the tool introduced in Section 10.3. Includes worked examples, technical explanations, and downloadable sample projects relevant to Indian educational contexts.]                |
| <b>Core</b> | White, J., et al. (2023). A prompt pattern catalog to enhance prompt engineering with ChatGPT. arXiv preprint <a href="https://arxiv.org/abs/2302.11382">arXiv:2302.11382</a> . [A systematic catalogue of prompt patterns and their effects; directly relevant to Section 10.5 and provides the research basis for the five-component prompt anatomy framework.] |
| <b>Core</b> | Parasuraman, R., & Manzey, D. H. (2010). Complacency and bias in human use of automation: An attentional integration. <i>Human Factors</i> , 52(3), 381-410. [The foundational research paper on automation bias; directly supports the What Could Go Wrong section.]                                                                                             |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. <i>Advances in Neural Information Processing Systems</i> , 33. [The foundational paper on diffusion models, which underlie DALL-E, Firefly, and similar image generation tools. The introduction section is accessible to non-specialists.] |
| <b>Extended</b> | AI4Bharat. (2023). IndicNLP Tools Catalogue. <a href="https://ai4bharat.iitm.ac.in">ai4bharat.iitm.ac.in</a> . [For understanding the current state of Indian language AI capability; relevant to the Chapter 10 discussion of tool limitations for Odia and other lower-resource                                            |

|                 |                                                                                                                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | languages.]                                                                                                                                                                                                                                                                                                                                 |
| <b>Extended</b> | Mollick, E., & Mollick, L. (2023). Using AI to implement effective teaching strategies in classrooms: Five strategies, including prompting techniques, the use of generated images, and feedback. SSRN preprint. [On effective and ineffective uses of AI tools in educational settings; directly relevant to the automation bias section.] |

## Accessible Online Resources

|               |                                                                                                                                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Anthropic. (2023). Prompt Engineering Guide. docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview. [Primary documentation on effective prompting for Claude; the principles generalise to other LLMs and directly support Section 10.5.] |
| <b>Online</b> | Runway ML. (2023). AI Video Generation Tools. runwayml.com. [Documentation and tutorial videos for the video generation tool mentioned in the tools landscape; relevant to the Create category and the stream project prompts.]                              |
| <b>Online</b> | Adobe. (2023). Adobe Firefly: Ethical AI. adobe.com/sensei/generative-ai/firefly.html. [Adobe's documentation of its approach to training data licensing; relevant to the reflective question on licensed training data and community consent.]              |

### *End of Chapter 10*

## UNIT FOUR

# The Consequences of Intelligence

*Every pattern a machine learns was recognised in someone's data, about someone's life, and it will act on someone's future. This unit is where the book stops asking how AI works and starts asking who it works for — whose patterns were counted, whose were not, and who bears the cost when the system is wrong.*

*This is not a detour from the technical material. It is what the technical material was always going to lead to. A student who can explain backpropagation but cannot explain why a hiring algorithm discriminated has learned half a subject.*

- 
11. Who Does AI Work For?
  12. Privacy, Surveillance, and the Indian State
  13. Work, Creativity, and the Human Question
  14. Governing Intelligence

## Chapter 11

# Who Does AI Work For?

*The first ten chapters of this book have described how AI works: the patterns it finds, the models it builds, the representations it learns, and the tools it produces. Unit 4 turns to a different set of questions. Not how AI works, but what it does — to people, to institutions, to power, to fairness, and to the future. These questions are not addenda to the technical material. They are the reason the technical material matters.*

*This chapter examines the foundational fairness question: who does AI work for? It explores how algorithmic bias enters systems, why it is so persistent, and what the different meanings of fairness require and trade off against each other. It grounds these discussions in Indian examples where the stakes are real and the patterns are already visible.*

*The answer to who AI works for is not a mystery. AI systems, like all powerful tools, tend to work best for the people and populations they were built from data about, optimised to serve, and deployed to reach. Understanding this tendency, its mechanisms, its consequences, and the conditions under which it can be corrected, is the central task of this chapter.*

## Learning Objectives

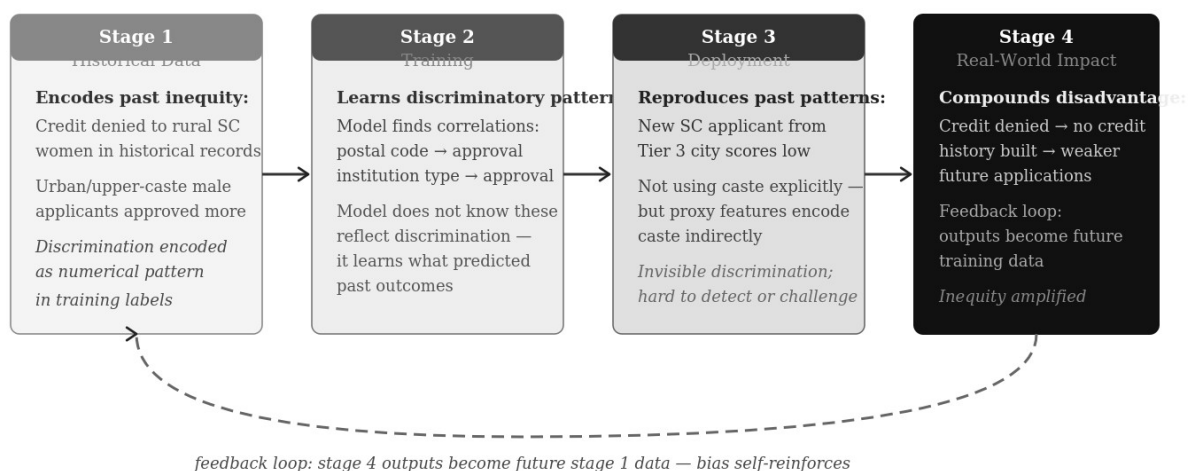
By the end of this chapter, you will be able to:

- Trace the bias pipeline from historical data through training to real-world impact
- Define and distinguish three formal notions of algorithmic fairness: demographic parity, equal opportunity, and individual fairness
- Explain why no algorithm can simultaneously satisfy all three fairness definitions in real-world settings
- Analyse the feedback loop of bias and explain why biased systems tend to self-reinforce
- Apply the fairness trade-off triangle to evaluate AI systems in Indian contexts

### 11.1 The Bias Pipeline

Algorithmic bias does not appear from nowhere. It travels a path, entering a system at specific points, amplifying through specific mechanisms, and arriving in the world through specific outputs. Understanding the pipeline is the first step toward interrupting it.

**Fig. 11.1**  
**The Bias Pipeline: From Historical Inequity to Real-World Impact**



*The pipeline is not a single event but a cycle. Each stage amplifies the inequity introduced at stage 1. Breaking the pipeline requires active intervention — typically at stage 1 (training data) or between stage 3 and 4 (deployment auditing).*

**Figure 11.1 — Bias Pipeline**

Stage one is historical data. Every machine learning model is trained on data that was generated by the past behaviour of human institutions. Those institutions were not neutral. Indian financial institutions have historically extended credit more readily to urban, upper-caste, male applicants. Indian companies have historically promoted men to senior roles at higher rates than women. Indian courts have historically sentenced defendants from marginalised communities to longer terms for equivalent offences. These patterns are encoded in the data as numerical regularities, as correlations between demographic features and outcomes.

Stage two is the learning process. A supervised learning algorithm trained on historical data learns the patterns that predict the labelled outcome. If the historical data shows that applications from urban upper-caste men were approved at higher rates than applications from rural women from scheduled castes, the model learns those correlations. It does not know they reflect historical discrimination rather than genuine differences in creditworthiness. It does not know, or care, that the correlations are unjust. It knows only that those correlations predicted the outcome in the training data, and that predicting the outcome accurately is what it was trained to do.

Stage three is the model's decisions in deployment. The model applies its learned correlations to new applicants. New applications from urban upper-caste men are assessed using features that were historically correlated with approval. New applications from rural women from scheduled castes are assessed using features that were historically correlated with rejection. The model is not explicitly using caste or gender as inputs. It is using features that correlate with caste and gender, which in the Indian context include postal code, educational institution, occupation, and many others. The effect is the same as direct discrimination, but the mechanism is indirect and, to casual inspection, invisible.

Stage four is real-world impact. People are approved or rejected, hired or passed over, granted bail or detained, routed to healthcare or turned away, based in part on what the model predicts. These impacts accumulate over time and across individuals, reproducing and in some cases amplifying the historical inequities encoded in the training data. The person who is denied credit because a model learned from biased historical data cannot build the credit history that would improve their profile in future applications. The feedback is compounding.

## 11.2 Three Definitions of Fairness

There is no single agreed definition of algorithmic fairness. There are, instead, multiple definitions that capture different intuitions about what fairness requires, and that are mathematically incompatible with each other in most real-world settings. Understanding these definitions and their incompatibility is essential for evaluating AI systems and for engaging seriously with fairness policy debates.

**Fig. 11.2**  
**Three Definitions of Algorithmic Fairness — Indian Hiring Example**

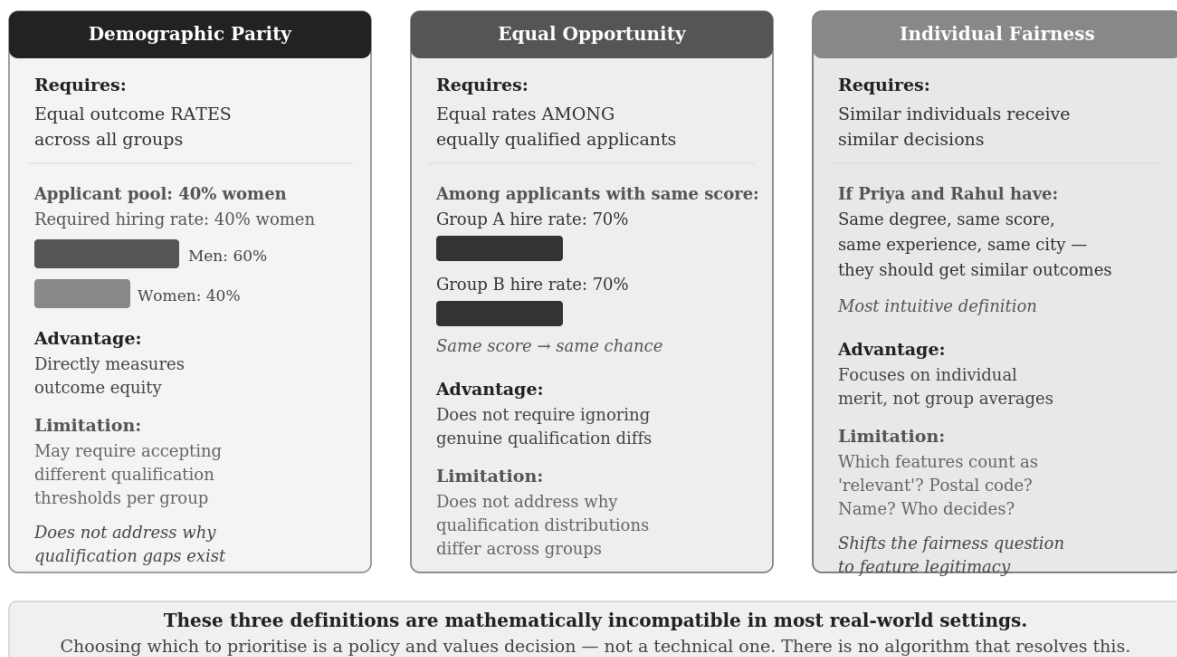


Figure 11.2 — Three Fairness Definitions

### Demographic Parity

Demographic parity requires that the proportion of individuals from each demographic group who receive a positive outcome from the model equals the proportion in the overall population. If forty percent of applicants are women, forty percent of those hired should be women. If fifteen percent of applicants are from scheduled castes, fifteen percent of those hired should be from scheduled castes. Demographic parity focuses on the distribution of

outcomes across groups. Its appeal is that it directly measures whether different groups are treated comparably in terms of outcomes achieved.

Its limitation is that it does not account for genuine differences in qualification distributions across groups, where those differences exist. In a context where historical underinvestment in education has produced genuine differences in measured qualifications between groups, a demographic parity requirement may lead to hiring less qualified candidates from some groups to meet the target, which may not be the right outcome in every context. Critics also note that demographic parity can be achieved in harmful ways: by setting a universally low bar, for example, rather than by improving access to qualifications.

### **Equal Opportunity**

Equal opportunity requires that among qualified applicants, members of all demographic groups have equal probability of receiving a positive outcome. Among applicants with equivalent qualifications, the approval rate should be the same regardless of group membership. This is sometimes called equalising true positive rates: if a qualified candidate from group A has a seventy percent chance of being hired, a qualified candidate from group B should also have a seventy percent chance. Equal opportunity focuses on the treatment of equivalent individuals rather than on the distribution of outcomes.

Equal opportunity's appeal is that it does not require ignoring genuine qualification differences; it only requires that, given equivalent qualifications, group membership does not affect outcomes. Its limitation is that it does not address the upstream question of how qualification differences arose and whether those differences themselves reflect historical inequity. A model that satisfies equal opportunity may still produce demographic outcome disparities if the qualification distribution is unequal across groups, which in the Indian context is often the direct legacy of educational and economic exclusion.

### **Individual Fairness**

Individual fairness requires that similar individuals be treated similarly. Two applicants who are alike in all relevant respects should receive similar decisions. This definition focuses on the consistency of treatment at the individual level rather than on group-level statistics. Its appeal is that it seems most directly aligned with the intuition that fairness is about judging each person on their own merits.

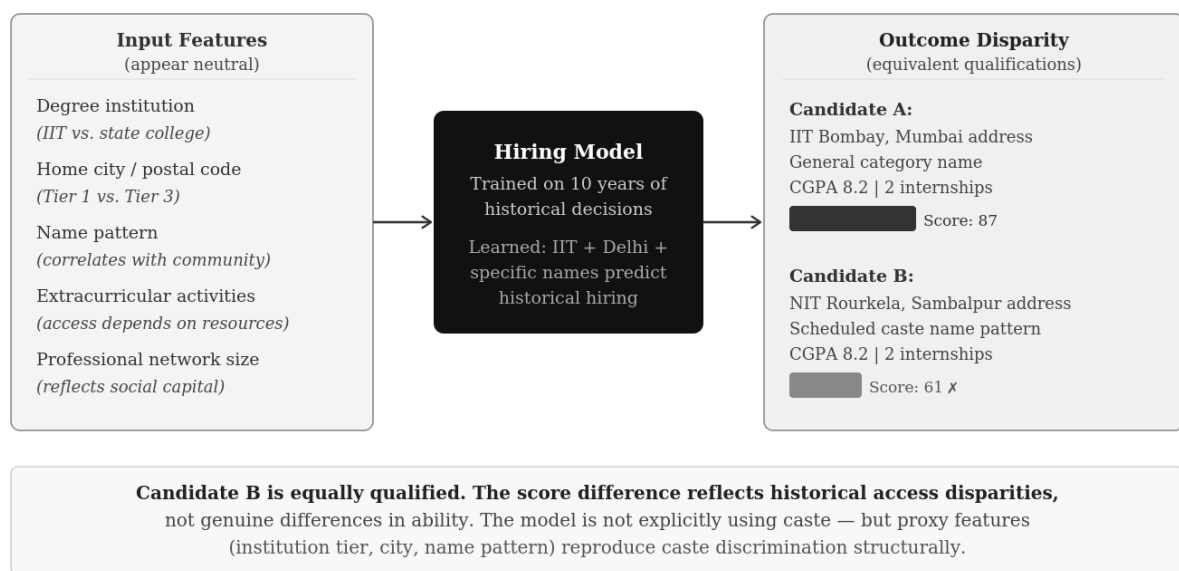
Individual fairness's limitation is that it requires defining what counts as relevant similarity, which immediately raises the question of which features are legitimately relevant to the decision. If postal code is included as a relevant feature, then two applicants who are otherwise identical but live in different postal codes may legitimately receive different outcomes under individual fairness, even if the postal code difference correlates entirely with caste or income. The definition shifts the question of fairness to the question of which features are legitimate inputs, which is itself a contested normative question.

The mathematical incompatibility among these three definitions means that in most realistic settings, it is impossible to satisfy all three simultaneously. A model that achieves demographic parity will, in general, violate equal opportunity for some groups. A model that achieves equal opportunity will, in general, produce demographic disparities if the qualification distribution is unequal across groups. This incompatibility is not a flaw of any particular algorithm: it is a theorem. Choosing which fairness definition to prioritise is a policy and values decision, not a technical one.

### 11.3 The Indian Hiring Algorithm Case Study

A large technology company in Bengaluru uses a machine learning model to screen resumes and rank candidates for engineering positions. The model was trained on ten years of hiring decisions from the company's historical data: resumes that were advanced and resumes that were rejected, with the hiring decision as the label.

**Fig. 11.3**  
**Hiring Algorithm Case Study: Proxy Discrimination in Indian IT Sector**



*This is structural discrimination: unintentional, hard to detect, harder to challenge — but producing the same outcomes as direct discrimination.*

*Figure 11.3 — Hiring Algorithm Case Study*

The historical data reflects the composition of a technology company that, like most in India's IT sector, has historically hired predominantly from certain institutions, certain cities, and certain demographic profiles. The training data is therefore not a random sample of qualified candidates: it is a sample of candidates who were offered opportunities by a company that historically had specific institutional preferences, geographic biases, and unstated demographic patterns in its hiring decisions.

The model learns from this data. It learns that candidates from certain IITs and NITs are more likely to be hired than candidates from state engineering colleges, even when qualifications are equivalent. It learns that candidates from Bengaluru, Mumbai, Chennai,

and Delhi are more likely to be hired than candidates from Tier 2 and Tier 3 cities. It learns that names associated with specific communities are correlated with hiring outcomes, not because the model was told to use caste as a feature but because name patterns correlate with caste in ways that the model discovers through the training data.

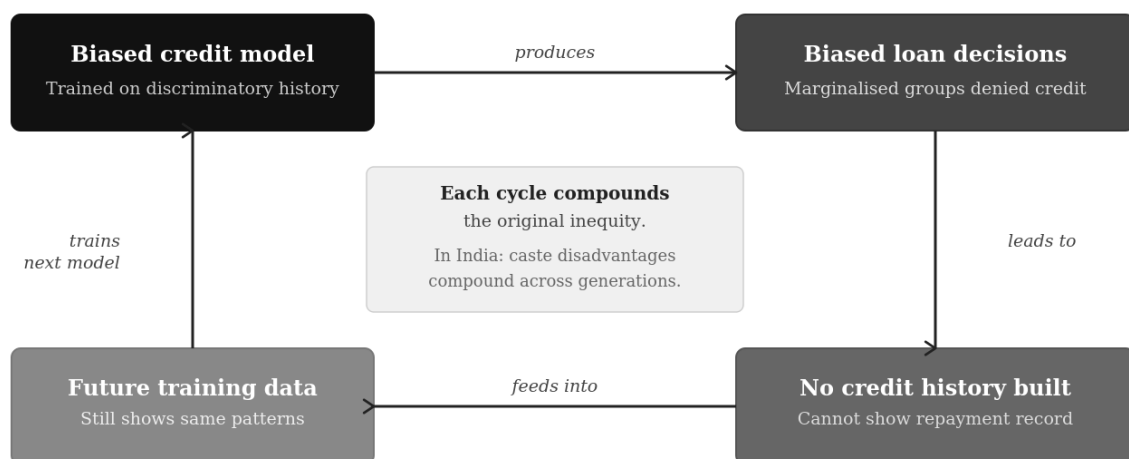
When deployed, the model applies these learned correlations to new candidates. A candidate with equivalent objective qualifications to another candidate but with a name pattern associated with a scheduled caste community, an address in a Tier 3 city, and a degree from a state college rather than an IIT will receive a systematically lower score. The model is not being asked to discriminate. It is doing what it was trained to do: predicting outcomes based on correlations in historical data. The discrimination is structural, not intentional, and it is therefore harder to challenge and harder to remedy.

The outcome disparity produced by this model is real and measurable. If the company audits its AI-assisted hiring by comparing acceptance rates across demographic groups, it will find disparities that mirror the historical patterns in its training data. These disparities are not random noise: they are the model faithfully reproducing the patterns of the past.

## 11.4 The Feedback Loop of Bias

The most insidious feature of algorithmic bias is that it tends to self-reinforce over time. This happens through the feedback loop described in Chapter 3: the model's outputs become data, and that data shapes future training.

**Fig. 11.4**  
**The Feedback Loop of Bias: Self-Reinforcing Across Generations**



*Breaking the loop requires active data intervention — counterfactual data, outcome testing, or deliberate reweighting — not just better algorithms. The algorithm learns what the data contains; changing the data is the only principled remedy.*

*Figure 11.4 — Feedback Loop Of Bias*

Consider the credit scoring system. A model trained on historical credit data, which reflects historical lending discrimination, produces credit scores that are systematically lower for members of discriminated-against groups. Those lower scores mean fewer loans approved. Fewer loans approved means fewer successful repayments. Fewer successful repayments means less positive credit history. Less positive credit history in future training data means future models learn the same or stronger discriminatory correlations. The cycle compounds.

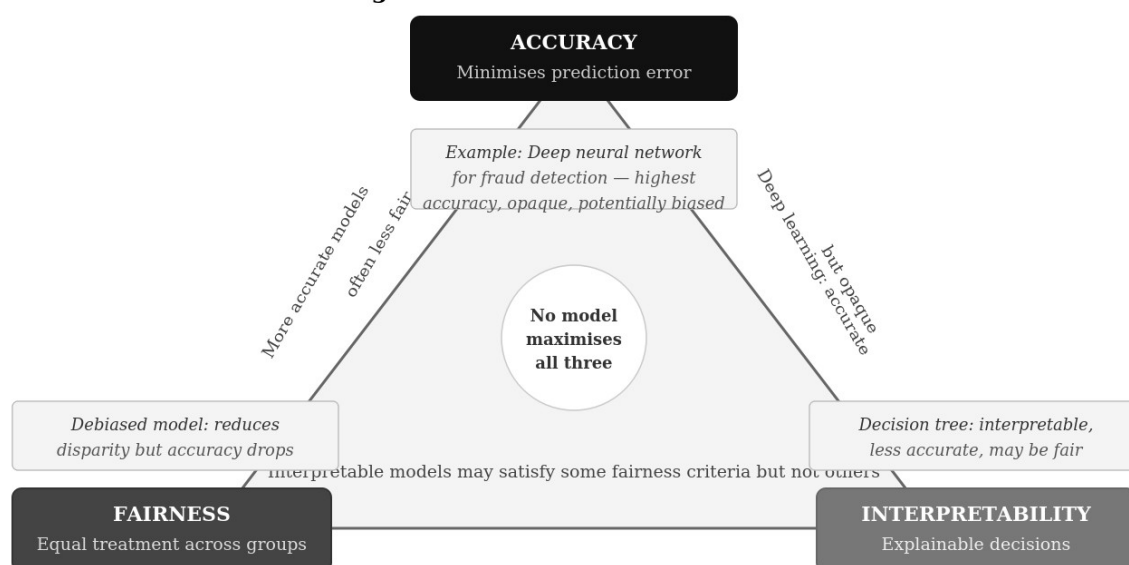
The feedback loop has a specific character in Indian contexts because of the caste dimension. India's caste system has produced multi-generational disadvantages in wealth accumulation, access to education, and professional networks. These disadvantages are encoded in the features that AI systems use: postal codes, educational institutions, family employment histories, professional networks, and asset ownership. When AI systems learn from data that reflects these multi-generational disadvantages, they learn patterns that will systematically disadvantage the same communities in the future. The cycle is not just self-reinforcing within a model generation: it can compound across generations of people.

Breaking the feedback loop requires active intervention at the data stage: deliberately constructing training datasets that are not simply records of past decisions, but records that include counterfactual information about what outcomes would have been under non-discriminatory conditions. This is technically and institutionally difficult. It requires knowing what non-discriminatory outcomes would have looked like, which requires both empirical analysis and normative judgement. But it is the only principled way to prevent AI systems from inheriting and amplifying the inequities of the past.

## **11.5 The Fairness Trade-Off Triangle**

The three fairness definitions described in Section 11.2, combined with a fourth objective of model accuracy, form a trade-off triangle that governs the design of fair AI systems. Every real system must navigate this triangle, and the navigation choices it makes are fundamentally policy choices about whose interests to prioritise.

**Fig. 11.5**  
**The Fairness Trade-Off Triangle**



Every AI deployment involves implicit choices about this triangle. Making those choices explicit, transparent, and accountable is the responsibility of deploying institutions — not a decision that can be delegated to the algorithm or its vendor.

Figure 11.5 — Fairness Trade Off Triangle

Accuracy is the model's ability to correctly predict outcomes. A more accurate model, in the sense of making fewer errors on the test data, is typically preferred by the institutions deploying it because it reduces errors and their associated costs. But a highly accurate model trained on biased historical data may be accurately reproducing discriminatory patterns.

Fairness, in any of the senses described above, typically requires some reduction in accuracy as measured on historical data. This is because historical data reflects past discrimination, and a model that predicts past outcomes accurately is, by definition, learning to replicate those discriminatory outcomes. Improving fairness therefore often means deliberately making the model less accurate in its prediction of historical outcomes, while making it more consistent across demographic groups.

Interpretability is the ability to explain why the model made a specific decision. An interpretable model can say: this loan was denied because the applicant had no credit history, no collateral, and an employment history shorter than twenty-four months. A black-box model can say only: this loan was denied. Interpretability matters for fairness because it enables contestability: a person who understands why they were denied can challenge the decision if the reason is illegitimate. A black box cannot be challenged in this way.

Deep learning models, the most accurate models available for many tasks, are typically among the least interpretable. Simpler models like decision trees and logistic regression are much more interpretable but may be less accurate on complex tasks. This trade-off

between accuracy and interpretability is real and practically significant for Indian AI deployments where accountability is a genuine concern.

The framework of the trade-off triangle does not tell you which trade-off to make. It tells you that every AI system deployment involves implicit choices about these priorities, and that those choices should be made explicitly, transparently, and accountably, rather than being embedded invisibly in technical design decisions.

## **11.6 Patterns in My Field: Stream Perspectives**

### **Arts Stream: Algorithmic Bias in Odia Content Recommendation**

A video streaming platform operating in Odisha uses a recommendation algorithm trained predominantly on viewing data from its national user base. The historical viewing data reflects patterns of consumption among Hindi-speaking urban audiences, which comprise the majority of the platform's users. Odia-language content is underrepresented in the training data relative to its cultural significance and the size of the Odia-speaking population. The recommendation algorithm, trained to predict engagement, learns that Hindi content is more likely to generate the high-engagement signals that the platform optimises for. When a user from Odisha opens the app, the recommendation algorithm de-prioritises Odia content relative to its actual availability and the user's expressed preferences. The bias is not against Odia content as a deliberate policy: it is a consequence of a recommendation algorithm trained on data that underrepresents Odia-speaking users. The feedback loop compounds it: less recommendation leads to less viewing, which generates less engagement data, which means future models have even less Odia signal to learn from.

### **Science Stream: Bias in Medical Diagnostic AI for Indian Populations**

Medical diagnostic AI systems, particularly those for dermatological and radiological conditions, have been documented to perform significantly differently across skin tones and body types. The training datasets for most commercially available diagnostic AI have historically overrepresented lighter-skinned patients from North American and European clinical settings. An AI-assisted dermatology tool for detecting skin conditions in Indian patients, who span an enormous range of skin tones from very light to very dark, will perform less reliably for patients with darker skin tones if the training data did not adequately represent that range. The clinical consequence is that the patients most likely to lack access to specialist dermatologists, who in India are concentrated in urban centres, are also the patients for whom AI-assisted diagnostics are most likely to fail. The technology that could most help underserved patients may be least reliable precisely for those patients.

## **Commerce Stream: Bias in Creditworthiness Scoring for Rural Borrowers**

Alternative credit scoring systems for rural Indian borrowers, introduced in Chapter 5, use non-traditional features such as mobile phone usage, social network connectivity, and geographic mobility to assess creditworthiness. Each of these features carries systematic biases that disadvantage specific groups. Mobile phone usage patterns are shaped by infrastructure quality, which is lower in rural and tribal areas. Social network connectivity reflects social capital that is distributed unequally along caste and class lines. Geographic mobility, as discussed in Chapter 5, is systematically lower for women due to social constraints on travel. A credit scoring model trained on these features will produce scores that systematically disadvantage rural residents, lower-caste communities, and women, not because those groups are less creditworthy but because the features used to assess creditworthiness are proxies for social inequalities that have nothing to do with repayment ability.

### **— WHAT COULD GO WRONG? —**

## **Fairness Washing: The Appearance of Fairness Without Its Substance**

One of the most dangerous risks in the discourse around algorithmic fairness is what has been called fairness washing: the deployment of fairness language and fairness metrics to provide the appearance of ethical commitment while the substance of discrimination continues unchanged.

Fairness washing takes several forms. A company may conduct a fairness audit that measures demographic parity on a simple metric while leaving unexamined the deeper structural biases in its training data, its feature selection, and its deployment context. A government procurement process may require vendors to demonstrate fairness compliance by reference to a narrow technical definition that is easy to satisfy but does not correspond to the fairness outcomes that affected communities actually need. An AI developer may publish a responsible AI policy that makes strong commitments to fairness in principle while the deployed products have known and documented disparate impacts on marginalised groups.

The distinguishing feature of fairness washing is that it prioritises the satisfaction of measurable fairness criteria over the actual reduction of harm to disadvantaged groups. It uses the language of fairness to deflect accountability rather than to discharge it. Detecting fairness washing requires asking not just whether a system meets specified fairness criteria but whether those criteria were chosen because they are the most relevant to the affected populations or because they are the easiest to satisfy. It requires asking who defined the fairness criteria, who conducted the audit, and who has the power to contest the findings.

In the Indian context, where AI is increasingly being deployed in government services affecting hundreds of millions of people, fairness washing is a risk of the first order. The deployment of AI-assisted decision-making in ration distribution, school admissions,

social welfare eligibility, and judicial sentencing will produce auditable fairness metrics. Those metrics will not automatically reflect the experiences of the most marginalised communities affected. Ensuring that they do requires active participation from those communities in defining what fairness means in their specific context, not just technical compliance with criteria defined by the system designers.

## — CHAPTER SUMMARY —

### Chapter 11 — Key Points

1. The bias pipeline: historical data encodes past inequity → training learns those patterns → deployment reproduces them → real-world impact compounds disadvantage.
2. Demographic parity: equal outcome rates across groups. Equal opportunity: equal treatment of equally qualified individuals. Individual fairness: similar individuals treated similarly.
3. These three definitions are mathematically incompatible in most real settings — choosing among them is a policy decision, not a technical one.
4. The feedback loop of bias: model outputs become training data for future models, compounding the original inequity across generations.
5. In India, caste-correlated features (postal code, educational institution, name patterns) allow models to discriminate without explicitly using caste as an input.
6. The trade-off triangle: accuracy, fairness, and interpretability cannot all be simultaneously maximised — deployment choices are implicit priority choices.
7. Interpretability enables contestability: a person denied a loan must understand why, to challenge the decision if the reason is illegitimate.
8. Fairness washing substitutes measurable compliance for genuine harm reduction — detecting it requires asking who defined the criteria and who they serve.
9. Affected communities must participate in defining fairness criteria, not merely receive the outputs of criteria defined by system designers.

## — MODEL QUESTIONS —

### Section A — MCQ (5 Questions)

**Q1. At which stage of the bias pipeline does the model first learn discriminatory patterns?**

- A) The real-world impact stage, when deployment reveals disparate outcomes
- B) The training stage, when the algorithm finds correlations in historical data that reflect past discrimination
- C) The feature selection stage, when engineers choose which inputs to include
- D) The deployment stage, when the model encounters new applicants from marginalised groups

**Answer: B**

*Explanation: Discriminatory patterns enter the model during training, when the algorithm learns correlations in historical data. Those correlations reflect past discriminatory outcomes. Feature selection also matters, but the algorithmic learning of discriminatory patterns is a training stage phenomenon.*

**Q2. Demographic parity requires that:**

- A) All demographic groups receive the same absolute number of positive outcomes
- B) The proportion of each group receiving positive outcomes equals their proportion in the applicant pool
- C) Individuals with the same qualifications receive the same outcomes regardless of group membership
- D) Similar individuals are treated similarly by the model

**Answer: B**

*Explanation: Demographic parity requires that the positive outcome rate for each demographic group equals the group's representation in the population. If 40% of applicants are women, 40% of those approved should be women.*

**Q3. The 'fairness impossibility theorem' states that:**

- A) It is impossible to build fair AI systems in societies with historical inequity
- B) Demographic parity, equal opportunity, and individual fairness cannot all be simultaneously satisfied in most real-world settings
- C) No algorithm can be guaranteed to be completely free of bias
- D) Fair AI requires removing all demographic features from the training data

**Answer: B**

*Explanation: The incompatibility theorem shows that in typical real-world settings with unequal base rates across groups, satisfying all three major fairness criteria simultaneously is mathematically impossible. Each definition captures different values, and they conflict.*

**Q4. In the Indian hiring algorithm example, the model discriminates without using caste as an explicit feature because:**

- A) The engineers secretly included caste as a hidden input variable
- B) Proxy features like postal code, educational institution, and name patterns are correlated with caste in the training data
- C) Indian IT companies deliberately design discriminatory models to preserve demographic homogeneity
- D) The model cannot process text inputs like names, so it infers caste from other features

**Answer: B**

*Explanation: This is proxy discrimination: the model uses features that appear neutral but are correlated with protected characteristics in the training data. Postal code, educational institution, and name patterns all carry caste information in Indian contexts, allowing indirect discrimination without explicit use of caste as an input.*

**Q5. Why does interpretability matter for fairness in AI systems?**

- A) Interpretable models are always more accurate than black-box models
- B) Interpretability enables contestability: individuals can challenge decisions only if

they understand the reasons for them

C) Interpretable models are required to satisfy demographic parity

D) Interpretability prevents models from learning discriminatory patterns during training

**Answer: B**

*Explanation: Interpretability enables contestability. A person denied a loan, job, or government benefit has a legitimate interest in knowing why. Without an interpretable reason, the decision cannot be challenged even if it is based on illegitimate factors. Interpretability is therefore a prerequisite for meaningful accountability.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Trace the bias pipeline for a government welfare eligibility AI system in rural India, identifying the specific mechanism at each stage.**

**Model Answer:**

Stage one, historical data: the training data consists of past welfare eligibility decisions made by officials in the district. These decisions were influenced by proximity to officials, literacy in government processes, and social capital, all of which are distributed unequally along caste, class, and gender lines. Genuinely eligible households from marginalised communities were systematically less likely to have their eligibility recognised in the historical record. Stage two, training: the model learns that the features correlated with approved applications in the past predict approval in future applications. These features include proximity to the district headquarters, household head's literacy, and family connections to local government. Stage three, deployment: new applicants from marginalised communities are scored lower because they lack the historically correlated features, even when their need and eligibility are objectively equivalent to approved applicants. Stage four, impact: welfare is denied to households that need it most. The data generated by these denials, including no record of a welfare relationship with the government, further reduces future model scores for these households.

*Key terms: bias pipeline, historical data, proxy discrimination, feedback loop, deployment context, structural bias*

**Q7. Explain the difference between demographic parity and equal opportunity using a specific Indian example, and identify one practical limitation of each definition.**

**Model Answer:**

Demographic parity in a university admissions context would require that the proportion of students from each caste category admitted equals their proportion in the applicant pool. If scheduled caste applicants comprise twenty percent of the pool, twenty percent of those admitted should be from scheduled castes. The practical limitation is that if historical educational inequity has produced genuine differences in competitive exam scores between groups, demographic parity may require admitting students with lower scores from some groups, which may produce tension with the institution's academic mission. Equal opportunity would require that among applicants with equivalent exam scores and qualifications, admission rates are the same across caste categories: a scheduled caste applicant with the same score as a general category applicant should have the same admission probability. The practical limitation is that exam scores themselves may reflect

historical educational disadvantage, meaning that even equal opportunity leaves unaddressed the upstream inequity that produced score differences in the first place.

*Key terms: demographic parity, equal opportunity, fairness incompatibility, proxy measure, historical disadvantage, educational inequity*

**Q8. Explain the feedback loop of bias in the context of a credit scoring system and describe one intervention that could break it.**

**Model Answer:**

A credit scoring model trained on historical lending data learns correlations between applicant features and loan repayment outcomes. The historical data reflects past lending discrimination, so the model produces systematically lower scores for members of historically discriminated-against groups. These lower scores lead to fewer loan approvals for those groups. Fewer approvals mean fewer opportunities to build positive credit histories through successful repayments. Less positive credit history in future training data means future models learn the same or stronger discriminatory correlations. The cycle is self-reinforcing across model generations. One intervention is counterfactual data augmentation: deliberately including in the training data examples of loan outcomes from members of discriminated-against groups who were extended credit under experimental or pilot programmes, showing what their repayment outcomes would have been under non-discriminatory lending. This breaks the feedback loop by providing the model with evidence about creditworthiness that historical discrimination suppressed. A second intervention is outcome testing: extending credit to a random sample of applicants from all demographic groups regardless of model score, then tracking repayment to measure whether the model's scores are genuinely predictive or reflect historical bias.

*Key terms: feedback loop, counterfactual data, outcome testing, credit history, self-reinforcing bias, historical discrimination*

**Q9. What is proxy discrimination, and how does it operate in an AI hiring system that does not explicitly use caste as an input feature?**

**Model Answer:**

Proxy discrimination occurs when a model uses features that are not protected characteristics themselves but that correlate with protected characteristics in the training data, producing discriminatory outcomes indirectly. In an AI hiring system in India, postal code is not a caste feature, but in many Indian cities postal codes are strongly correlated with caste due to the historical segregation of residential areas. Educational institution is not a caste feature, but access to elite engineering institutions has historically been distributed unequally across castes due to differences in educational resource access. Name patterns are not directly caste features, but name patterns in India carry substantial information about caste identity. A hiring model that uses all three of these features will produce outcomes that correlate with caste outcomes even if caste is never mentioned. The model discovers these correlations during training because they genuinely exist in the historical data. The result is discrimination that is structurally identical to direct caste discrimination in its effects, but is more difficult to detect and challenge because the mechanism is indirect.

*Key terms: proxy discrimination, correlated features, protected characteristics, postal code, indirect discrimination, structural bias*

**Q10. What is fairness washing, and how can a student or policymaker detect it when evaluating an AI system deployed in India?**

**Model Answer:**

Fairness washing is the deployment of fairness language and fairness metrics to provide

the appearance of ethical commitment while the substance of discrimination continues. A company may satisfy a narrow, easily measurable fairness criterion, such as equal approval rates across two demographic groups in a controlled test, while leaving unaddressed the structural biases in training data, proxy discrimination through correlated features, or the feedback loop that compounds historical inequity. Detecting it requires asking five questions. First, who defined the fairness criteria: the system designer, the regulator, or the affected communities? Criteria defined without input from affected communities may not measure what those communities consider fairness. Second, what auditing was performed, and by whom? Self-reported fairness audits are less credible than independent third-party audits. Third, what happens when the fairness criteria and accuracy objectives conflict: whose interests were prioritised? Fourth, are the fairness criteria measured across all relevant subgroups, including intersectional categories, or only across the largest demographic groups? Fifth, what is the mechanism for affected communities to contest decisions and have grievances addressed?

*Key terms: fairness washing, independent audit, fairness criteria, affected communities, contestability, accountability mechanisms*

## Section C — Reflective Questions (3 Questions)

**Q11. India has a constitutional reservation system that explicitly uses caste as a criterion for positive discrimination in education and government employment. Does this system represent demographic parity, equal opportunity, or neither? How should AI systems interact with a society that has explicitly chosen one fairness framework over another?**

**Thinking prompts:**

- Is reservation a demographic parity policy, an equal opportunity policy, or something else entirely?
- If an AI hiring system is deployed at an institution that has legally mandated reservation requirements, which fairness definition should the AI system implement?
- What happens when an AI system's fairness criteria conflict with constitutionally mandated affirmative action requirements?
- Who has the authority to determine which fairness definition an AI system should implement in the Indian context?
- How should the AI system handle the difference between reservation as a temporary corrective versus permanent principle?

*Rubric hint: Strong responses will distinguish reservation from the three fairness definitions used in this chapter and will engage with the specific constitutional framework. Credit for responses that identify the policy and legal question of who has authority to define fairness criteria for AI systems operating in India.*

**Q12. The chapter describes the feedback loop of bias as self-reinforcing across model generations. In the context of a judicial bail prediction system deployed in Indian courts, what would this feedback loop look like, and what would be needed to break it?**

**Thinking prompts:**

- What historical data would a bail prediction system be trained on, and what patterns might it contain?

- If the system produces higher risk scores for individuals from marginalised communities, how does this feed back into future training data?
- Who bears the consequences of a false positive (high risk score for someone who would not have reoffended)? Who bears the consequences of a false negative?
- What counterfactual data might break the feedback loop, and how would it be obtained?
- Should AI bail prediction systems be used in India at all, given the feedback loop problem? What conditions, if any, would make use acceptable?

*Rubric hint: Strong responses will identify the specific feedback mechanism in the judicial context — detention leading to more extensive criminal records, which raises future scores — and will engage seriously with whether the feedback loop is breakable in this domain. Credit for responses that assess the conditions under which bail prediction AI could be acceptable rather than giving a simple yes/no answer.*

**Q13. The chapter argues that the choice of fairness criteria is a policy decision, not a technical one. But in practice, technical choices often determine which fairness criteria are even measurable and therefore considered. Does this mean that technical choices are implicitly policy choices? Who should be making them?**

**Thinking prompts:**

- Give an example of a technical choice (feature selection, model architecture, evaluation metric) that implicitly embeds a fairness policy choice.
- If a vendor provides an AI system without documenting its fairness criteria, has the deploying institution made an implicit policy choice by accepting it?
- What would it mean for affected communities to participate in technical decisions about AI systems that affect them?
- Is technical expertise necessary for meaningful participation in these decisions, or can non-expert engagement be genuinely substantive?
- What institutions or processes would be needed to ensure that the policy choices embedded in AI technical design are made accountably in India?

*Rubric hint: Strong responses will identify specific technical choices that embed policy choices and will engage with the governance question of who has authority and capacity to make them. Credit for responses that distinguish between expert participation and community participation as different but both necessary forms of input.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Investigate one AI-based content recommendation or information delivery system used in India — this could be a news aggregator, a social media feed, a video streaming service, or an educational content platform. Analyse whether the system exhibits regional, linguistic, or cultural bias against specific Indian communities. Document your evidence: what content is recommended or surfaced, what is absent or de-prioritised, and what does this suggest about the training data and optimisation objectives. Apply the bias pipeline framework to explain how the observed bias might have entered the system. Conclude with a recommendation for one specific**

**technical and one specific policy intervention that could reduce the bias.**

The evidence should be specific and documented, not impressionistic. Compare recommendation patterns across users or contexts if possible. The bias pipeline analysis should identify the specific stage at which the bias most likely entered. The interventions should be targeted at the specific mechanism identified, not generic.

**Q15. [Commerce] A microfinance institution in your district is deploying an AI credit scoring system. You have been asked to conduct a fairness audit. Design the audit framework: which fairness definition(s) will you use and why, what data you will request from the institution, which demographic groups you will analyse, what specific metrics you will compute, and what threshold of disparity would lead you to recommend against deployment. Then apply your framework to a hypothetical scenario: the system approves 68% of male applicants and 41% of female applicants with otherwise equivalent profiles. What does your framework say about this disparity, and what is your recommendation?**

The audit design should be specific and grounded in the fairness definitions from this chapter. The choice of fairness definition should be justified in terms of the specific context of rural microfinance and the interests of the borrowers. The recommendation for the hypothetical scenario should follow directly from the framework, not from a general intuition about the disparity.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <b>Core</b> | Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. <i>Big Data</i> , 5(2), 153-163. [The foundational paper demonstrating the incompatibility of fairness criteria in the COMPAS bail prediction context; directly relevant to the reflective question on judicial bail prediction.]                 |
| <b>Core</b> | Menon, N., & Verma, R. (2022). Algorithmic Discrimination in Indian Hiring: A Case Study. <i>Economic and Political Weekly</i> , 57(22), 45-54. [Directly relevant to Section 11.3; documents proxy discrimination in AI hiring in the Indian IT sector context.]                                                                                                       |

### Extended References

|                 |                                                                                                                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. <i>Science</i> , 366(6464), 447-453. [The landmark paper on algorithmic bias in healthcare, directly parallel to |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                 |                                                                                                                                                                                                                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | the medical diagnostic AI section; freely accessible.]                                                                                                                                                                                                                                                                                     |
| <b>Extended</b> | Datta, A., Tschantz, M. C., & Datta, A. (2015). Automated experiments on ad privacy settings: A tale of opacity, choice, and discrimination. <i>Proceedings on Privacy Enhancing Technologies</i> , 2015(1), 92-112. [On proxy discrimination and opacity in algorithmic systems; relevant to the fairness trade-off triangle discussion.] |
| <b>Extended</b> | Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. <i>Nature Machine Intelligence</i> , 1(9), 389-399. [On the diversity of fairness frameworks across global AI governance documents; useful context for understanding why the Indian fairness question requires India-specific analysis.]          |

## Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Algorithm Watch India. (2023). AI and Discrimination in India: Annual Report. <a href="https://algorithmwatch.org/india">algorithmwatch.org/india</a> . [Documents cases of algorithmic discrimination in Indian government and commercial systems; directly relevant to all sections of this chapter.]                        |
| <b>Online</b> | AI Now Institute. (2023). Algorithmic Accountability Policy Toolkit. <a href="https://ainowinstitute.org">ainowinstitute.org</a> . [A practical guide to conducting algorithmic audits; directly relevant to the Commerce stream project prompt on designing a fairness audit.]                                                |
| <b>Online</b> | Fairlearn. (2023). Fairness in Machine Learning: Documentation. <a href="https://fairlearn.org">fairlearn.org</a> . [An open-source toolkit for assessing and improving fairness in ML systems; includes practical implementations of demographic parity, equal opportunity, and other definitions discussed in this chapter.] |

### *End of Chapter 11*

## Chapter 12

# Privacy, Surveillance, and the Indian State

*Privacy is not a preference. It is a precondition for other freedoms. Without the ability to move, communicate, associate, and think without continuous observation and recording, the freedoms of expression, assembly, and conscience are structurally undermined. A person who knows they are being watched behaves differently from one who is not. This is not a flaw of character: it is a basic feature of human psychology, documented across cultures and contexts. The philosopher Jeremy Bentham understood this when he designed the Panopticon in the eighteenth century: the mere possibility of being observed at any moment is sufficient to produce self-regulation without the need for constant actual observation.*

*Artificial intelligence has transformed the scale, the precision, and the permanence of surveillance. What was once expensive, labour-intensive, and geographically bounded is now cheap, automated, and potentially pervasive. This chapter examines how AI-enabled surveillance operates in India, what data is collected by which actors, how it is used, what rights are implicated, and what the digital divide means for who bears the costs and who receives the benefits of an AI-mediated society.*

## Learning Objectives

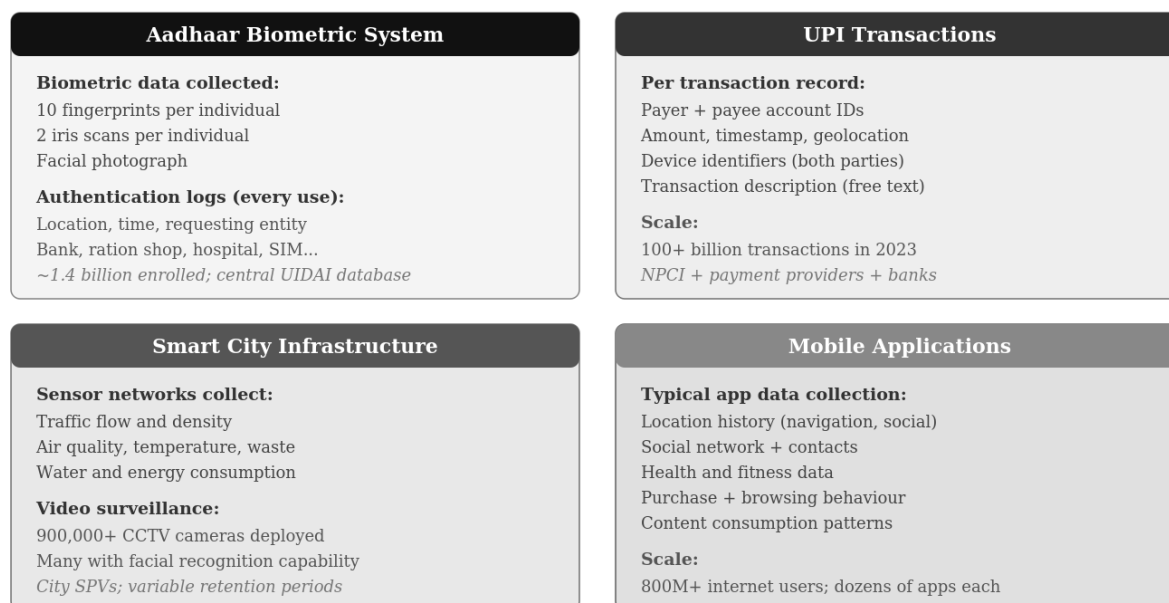
By the end of this chapter, you will be able to:

- Describe the data collection ecosystem in India: what Aadhaar, UPI, smart city sensors, and mobile apps each collect
- Explain the surveillance architecture from data collection through analysis to consequential decision
- Visualise the digital divide using Indian state-level data on access and AI tool availability
- Explain the right to be forgotten and why technical deletion is harder than it appears
- Articulate the relationship between surveillance, power, and the conditions for democratic participation

### 12.1 The Data Collection Map

Every interaction with a digital system in India generates data. Every UPI transaction records the amount, the time, the location, the device, the merchant, and both account identifiers. Every Aadhaar authentication records the location, the time, the requesting entity, and the authentication result. Every interaction with a government portal records the queries made, the documents accessed, and the services requested. Every movement through a smart city district is recorded by sensor networks. Every use of a major mobile application generates a continuous stream of behavioural data.

**Fig. 12.1**  
**India's Data Collection Ecosystem**



*Every digital interaction generates data. Aggregated, these streams create a granular, continuously updated profile of individual life in India.*

**Figure 12.1 — Data Collection Map**

Aadhaar is the world's largest biometric identification system, covering approximately 1.4 billion Indian residents. Its data collection includes biometric data: ten fingerprints and two iris scans per enrolled individual, along with a facial photograph. Demographic data includes name, date of birth, address, and gender. Authentication records are generated every time Aadhaar is used to verify identity, which happens across contexts ranging from bank account opening and SIM card registration to ration distribution and hospital admission. The Unique Identification Authority of India holds these records centrally.

UPI, the Unified Payments Interface, processed over 100 billion transactions in 2023 with a total value exceeding 180 trillion rupees. Each transaction generates a structured data record containing the payer's account identifier, the payee's account identifier, the amount, the timestamp, the geolocation of both devices if available, the device identifiers, and the transaction description field as entered by the payer. This data is held by the payment service providers, the National Payments Corporation of India, and the banks involved in each transaction. Across hundreds of millions of active UPI users transacting multiple times daily, this represents an extraordinarily granular record of economic activity.

Smart city infrastructure, deployed in over a hundred Indian cities under the Smart Cities Mission, includes networks of sensors, cameras, and data analytics platforms. These collect data on traffic flow, air quality, waste management, public lighting, water consumption, and in many deployments, video surveillance through camera networks equipped with facial recognition capability. The Smart Cities Mission has deployed over nine hundred thousand CCTV cameras across participating cities. The data is typically held and processed by city-level Smart City Special Purpose Vehicles, with varying

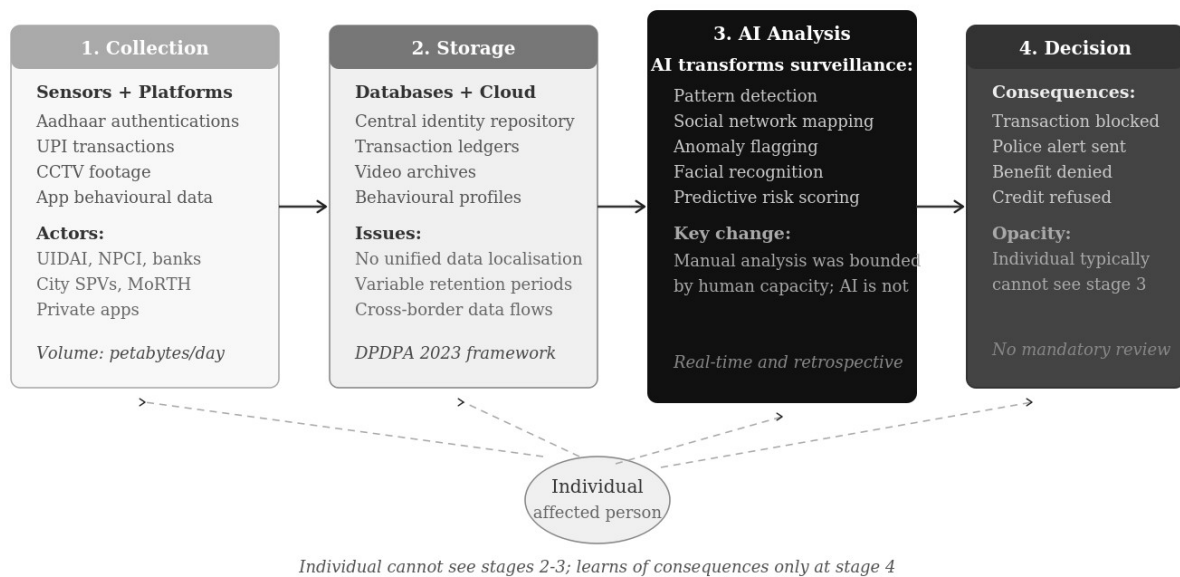
degrees of transparency about what is collected, who has access, and for how long data is retained.

Mobile applications present the most diffuse and perhaps most consequential data collection landscape. India's mobile internet penetration has passed 800 million users. The average Indian smartphone user has dozens of applications installed, each with its own data collection practices. Health applications collect medical information. Navigation applications collect location histories. E-commerce applications collect purchase behaviours. Social media applications collect social networks, content consumption, and communication patterns. Many of these applications collect data that is technically disclosed in terms-of-service agreements that users have agreed to without reading, and that is used in ways the users did not anticipate when installing the application.

## 12.2 The Surveillance Architecture

Data collection is the first step. The surveillance architecture describes what happens between collection and consequence: how raw data is stored, aggregated, analysed, and acted upon by institutions with the capacity and the legal authority to take consequential action.

**Fig. 12.2**  
**The Surveillance Architecture: From Collection to Consequence**



The opacity between stages 2 and 4 is not accidental: it is structurally produced by scale, automation, and the absence of transparency requirements.

Figure 12.2 — Surveillance Architecture

Storage is the second step. Collected data must be stored in a form that enables later retrieval and analysis. India does not yet have a comprehensive data localisation law, though the Digital Personal Data Protection Act of 2023 establishes a framework for data handling that is still being operationalised. Data collected by government entities is typically stored within India, while data collected by private entities, particularly by the

subsidiaries of large global technology companies, may be stored in servers outside India, with implications for Indian law enforcement and regulatory access.

Analysis is the third step, and the one where AI transforms the character of surveillance most dramatically. Manual analysis of the data volumes generated by Aadhaar authentications, UPI transactions, smart city sensors, and mobile applications is operationally impossible. The data only becomes useful for surveillance at scale through automated analysis: pattern detection that can identify unusual behaviours, social network analysis that can map associations between individuals, anomaly detection that can flag departures from established baselines, and predictive analytics that can produce risk scores for individuals or groups based on their data profiles.

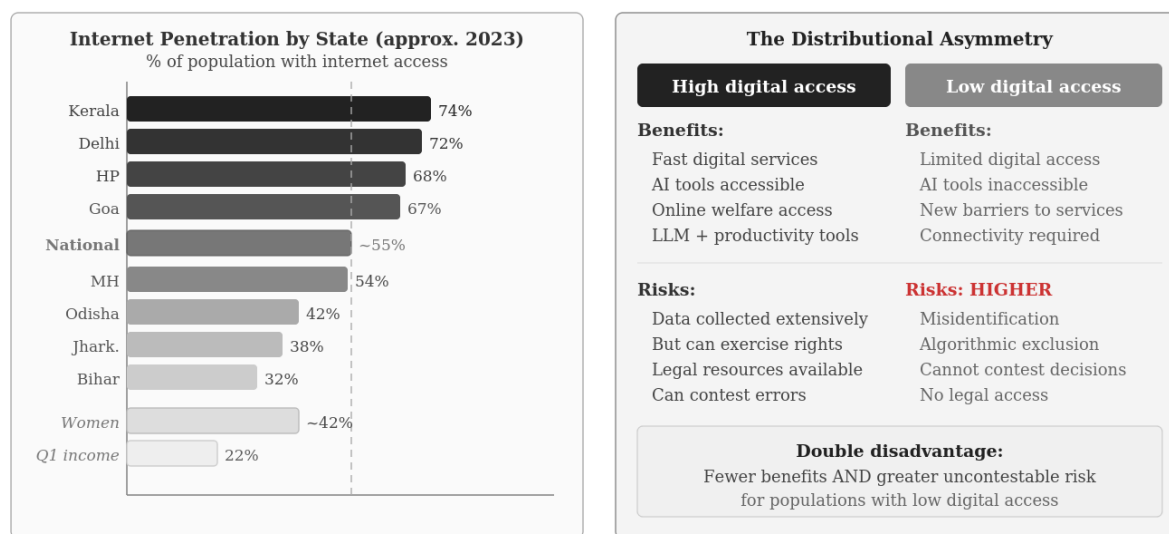
The consequential decision is the fourth step: the point at which data analysis produces an output that affects the individual's life. Consequence may be immediate and direct: a flagged UPI transaction is blocked. A face matched in a surveillance camera database triggers an alert to police. An Aadhaar authentication fails, denying access to a welfare benefit. Consequence may also be indirect and statistical: an AI system's risk score for an individual influences a credit decision, a welfare eligibility determination, or a police deployment decision.

The relationship between these steps is not transparent to the individuals whose data feeds them. A person who receives a welfare benefit denial has no routine mechanism to know whether an AI system was involved in the decision, what data it used, or how the decision was made. A person who is stopped by police after being matched by a facial recognition system may not know that a match occurred or that it was imperfect. This opacity in the surveillance architecture is not incidental: it is structurally produced by the combination of scale, automation, and the absence of transparency requirements in most existing governance frameworks.

### **12.3 The Digital Divide as an Equity Issue**

The benefits of AI-enabled systems, including the efficiency, accessibility, and scale of digital services, are distributed unequally across India's population. So are the costs and risks: surveillance, data exposure, algorithmic decision-making, and the potential for misidentification or erroneous exclusion. Understanding this distributional asymmetry requires disaggregating the aggregate statistics.

**Fig. 12.3**  
**The Digital Divide: Internet Access and AI Tool Access in India**



*The digital divide is not neutral: it compounds existing inequalities. The populations already most marginalised by income, caste, and geography bear disproportionate AI risk with disproportionately less access to AI benefits.*

Figure 12.3 — Digital Divide

India's internet penetration varies enormously by geography, income, and education. States like Kerala, Delhi, and Himachal Pradesh have penetration rates above seventy percent. States like Bihar, Odisha, and Jharkhand have penetration rates significantly below the national average, with large rural-urban disparities within those states. The penetration rate in the lowest income quintile is a fraction of the rate in the highest quintile. Women's internet penetration is substantially lower than men's in every state, with the gap largest in states where women have lower overall social and economic participation.

This distributional inequality has a specific consequence in the context of AI-enabled government services. A government that moves service delivery to digital platforms, requiring Aadhaar authentication and UPI payment for welfare delivery, agricultural subsidies, and healthcare access, is making access to those services conditional on digital inclusion. For the populations already most marginalised, the transition to digital service delivery can mean not greater access but new barriers: the need for a functioning smartphone, reliable internet connectivity, digital literacy, and the ability to navigate authentication systems that frequently fail on low-quality biometric readers.

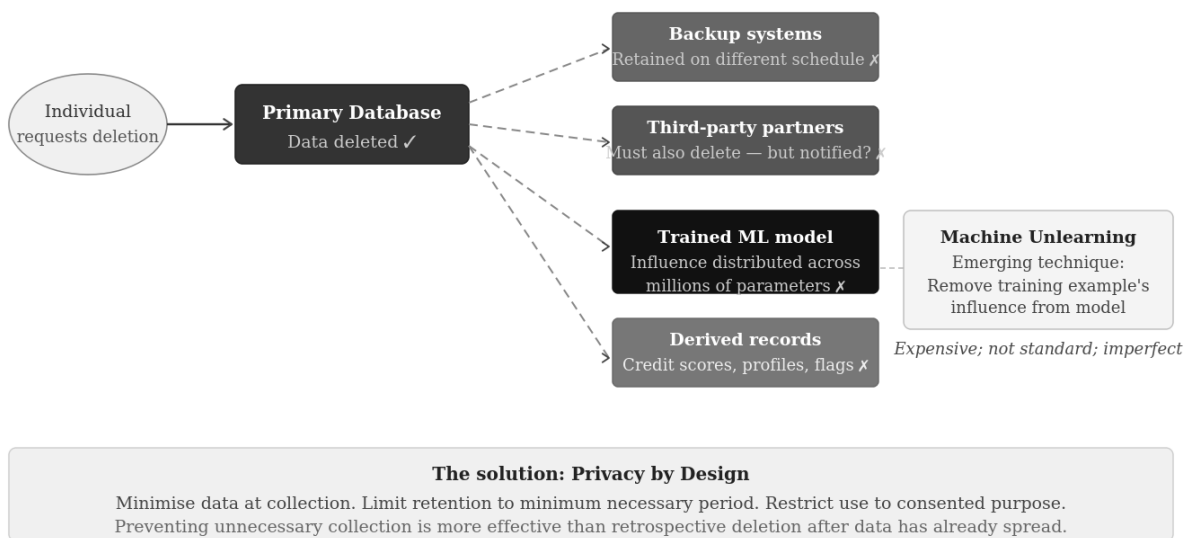
The digital divide also shapes the distribution of AI tool access described in Chapter 10. Large language models, image generation tools, and AI-assisted productivity applications are accessible primarily to the urban, English-literate, smartphone-owning population. Students at elite urban institutions have access to AI tools that could accelerate their learning, improve their writing, and expand their research capacity. Students at regional colleges in areas with poor connectivity, limited English literacy, and reliance on shared or borrowed devices have substantially less access to the same tools. This differential access will compound existing educational inequalities unless active policy interventions address it.

The distributional asymmetry also operates in the domain of surveillance risk. Urban, well-documented individuals who interact primarily with formal economic and government systems are subject to substantial data collection, but also have the resources and institutional access to contest errors and seek redress. Rural, less formally documented individuals, tribal communities, migrant workers, and economically marginalised populations may be subject to AI-enabled identification and decision systems while having the least capacity to contest erroneous decisions, the least awareness of their rights, and the least access to legal recourse.

## 12.4 The Right to Be Forgotten

The right to be forgotten, also called the right to erasure, is the principle that individuals should be able to request the deletion of personal data held by an organisation when that data is no longer necessary, when consent has been withdrawn, or when the processing is unlawful. In the European Union, this right is established by the General Data Protection Regulation. India's Digital Personal Data Protection Act of 2023 includes a right to erasure, but its scope and the mechanisms for exercising it are still being defined through subordinate legislation.

**Fig. 12.4**  
**The Right to Be Forgotten: Why Deletion Is Harder Than It Appears**



*India's DPDPA 2023 includes a right to erasure, but its scope excludes some government processing (including Aadhaar) and the technical infrastructure to operationalise machine unlearning at scale is not yet in place.*

*Figure 12.4 — Right To Be Forgotten*

The right to be forgotten is more technically complex than it appears. When an individual requests deletion of their data from a platform, the platform must delete the data from its primary database. But data in modern systems rarely lives only in a primary database. It exists in backup systems that may be scheduled for deletion on a different timeline. It may have been shared with third-party partners before the deletion request was received, and those partners must also process the deletion request. It may have been used to train

machine learning models, which have learned statistical patterns from the data that cannot easily be extracted from the trained parameters. It may have been used to generate derived records, such as profiles, scores, or risk assessments, that are stored separately from the raw data.

The machine learning dimension is particularly challenging. When an individual's data has been used to train a model, that individual's influence is distributed across millions of model parameters. Deleting the raw data does not remove that influence: the model has already learned from it. The emerging field of machine unlearning addresses this problem, developing techniques for modifying trained models to remove the influence of specific training examples. These techniques are computationally expensive, imperfect, and not yet operationally standard. The right to be forgotten, in the context of machine learning, is an area of active technical and legal development.

For India specifically, the right to be forgotten is complicated by the centralised nature of Aadhaar. The biometric data of 1.4 billion individuals is held in a central repository that has no mechanism for individual deletion: Aadhaar enrolment is designed as a permanent record. This creates a permanent biometric database that is fundamentally incompatible with the principle of erasure, which may be one reason why the DPDPA's right to erasure provisions are carefully scoped to exclude certain government data processing contexts.

The broader point is that privacy rights, including the right to deletion, are most meaningful when data minimisation is practised at the point of collection: collecting only the data necessary for the stated purpose, retaining it only for the minimum necessary period, and limiting its use to the consented purpose. After data has been widely shared, used in model training, backed up across multiple systems, and derived into secondary records, the right to deletion becomes increasingly difficult to exercise meaningfully. Privacy by design, building data minimisation and purpose limitation into systems from the beginning, is more effective than retrospective deletion.

## **12.5 Patterns in My Field: Stream Perspectives**

### **Arts Stream: AI Surveillance and Indigenous Knowledge**

India's tribal communities, particularly in the forest-covered regions of Odisha, Jharkhand, Chhattisgarh, and the Northeast, have been subject to increasing surveillance as part of forest management, mining project monitoring, and anti-Maoist operations. AI-enabled drone surveillance, camera networks, and mobile phone tracking are now used in areas where these communities live. For communities with oral knowledge traditions, strong senses of territorial belonging, and historical experiences of state coercion, the deployment of AI surveillance raises acute questions about the right to maintain communal practices without observation. Forest communities whose land rights and agricultural practices are legally contested may find that surveillance data collected nominally for forest conservation purposes is used to challenge their land claims. The intersection of AI surveillance with indigenous land rights and knowledge sovereignty

deserves sustained attention from humanities and social science perspectives that technical disciplines rarely bring.

### **Science Stream: AI in Public Health Surveillance**

India's response to the COVID-19 pandemic included the deployment of the Aarogya Setu contact tracing application, which used Bluetooth and GPS data from participating smartphones to identify potential exposure events. At its peak, Aarogya Setu had over 220 million downloads. The application raised questions about data security, the scope of data collection, government access to location data, and the potential for secondary uses beyond public health. The intersection of AI-enabled public health surveillance with privacy rights illustrates a genuine tension: effective epidemic control may require precisely the kind of granular location and contact data collection that is most intrusive from a privacy perspective. Resolving this tension requires governance frameworks that can define the scope of public health data collection, limit its use to stated purposes, ensure its deletion after the emergency, and provide independent oversight.

### **Commerce Stream: AI and Financial Surveillance**

India's anti-money laundering and counter-terrorism financing frameworks require banks and payment providers to monitor transactions for suspicious activity. AI systems are now central to this monitoring: they analyse transaction patterns, identify unusual behaviours, and flag potential violations for human review. These systems are valuable from a financial integrity perspective but raise serious questions about financial privacy. An individual whose transactions are flagged by an AI system may have their account frozen, their UPI transactions blocked, or their banking relationships terminated without any manual review of whether the flag was accurate. The combination of AI automation, consequential financial decisions, and limited transparency about flagging criteria creates conditions where errors are difficult to detect and contest. India's financial sector regulators are still developing frameworks for AI-based transaction monitoring that balance effectiveness with due process.

## **— WHAT COULD GO WRONG? —**

### **Facial Recognition and the Misidentification Crisis**

Facial recognition technology is the most visible and most contested form of AI-enabled surveillance currently deployed in India. CCTV networks with facial recognition capability have been deployed by police departments in Delhi, Hyderabad, Bengaluru, and several other cities. These systems are used for criminal identification, crowd monitoring at large events, and, reportedly, tracking individuals involved in political protests.

The misidentification crisis refers to the documented tendency of facial recognition systems to produce false matches at higher rates for darker-skinned individuals, women, and older people, particularly in real-world deployment conditions of variable lighting, partial occlusion, and lower image quality. The Gender Shades research documented in

Chapter 11 established this pattern for commercial facial recognition APIs. Indian police deployments face the same distribution shift problem described in Chapter 9: systems trained on databases that may not adequately represent the diversity of Indian faces, deployed in lighting and imaging conditions that differ from training conditions, operating under time pressure without systematic human review of matches.

A facial recognition false positive, a match of an innocent person to a criminal record, is not merely a technical error. It is a life event for the person misidentified. It may result in arrest, detention, loss of employment, reputational harm, and in contexts where the justice system is slow or inaccessible, prolonged deprivation of liberty before the error is corrected. The people most likely to be misidentified, darker-skinned individuals from marginalised communities, are also the people least likely to have the social and legal resources to contest a wrongful identification quickly and effectively.

India currently has no specific legal framework governing the deployment of facial recognition technology by law enforcement. There is no requirement for algorithmic transparency, accuracy validation on Indian population data, independent oversight, mandatory human review before action is taken on a match, or any mechanism by which individuals can challenge a facial recognition match. This regulatory gap exists in the context of known technical limitations and documented misidentification risks, making the deployment of facial recognition systems in consequential law enforcement contexts a significant ongoing risk to civil liberties.

## — CHAPTER SUMMARY —

### Chapter 12 — Key Points

1. India's data collection ecosystem: Aadhaar (biometric + demographic), UPI (transaction records), smart city sensors (movement + environment), mobile apps (behaviour + social networks).
2. The surveillance architecture: collection → storage → AI-enabled analysis → consequential decision. Each stage multiplies the impact of the previous.
3. AI transforms surveillance from expensive and bounded to cheap and potentially pervasive, fundamentally changing the balance between the observer and the observed.
4. The digital divide creates distributional asymmetry: benefits of digital services are unequally distributed; so are risks of surveillance, misidentification, and algorithmic exclusion.
5. The right to be forgotten is technically complex: deletion from primary databases does not address backups, third-party copies, or the influence of data on trained ML models.
6. Machine unlearning — removing specific training data's influence from a model — is emerging but computationally expensive and not yet standard practice.
7. Privacy by design is more effective than retrospective deletion: minimise data at collection, limit retention, restrict use to consented purposes.

8. Facial recognition in Indian police deployments has no regulatory framework, known misidentification risks concentrated in marginalised groups, and no mandatory human review before action.
9. Privacy is not a preference but a precondition for other freedoms; its erosion through AI-enabled surveillance has structural implications for democratic participation.

— MODEL QUESTIONS —

**Section A — MCQ (5 Questions)**

**Q1. Which of the following data sources in India generates the most granular record of individual economic activity?**

- A) The Aadhaar biometric database
- B) UPI transaction records, which capture every digital payment with amount, time, location, and party identifiers
- C) Smart city sensor networks, which capture movement through public spaces
- D) Mobile application data, which captures browsing and social behaviour

**Answer: B**

*Explanation: UPI transaction records generate a continuous, structured, timestamped record of individual economic activity with high specificity: each transaction records who paid whom, how much, when, and from where. This produces the most granular economic data profile of any of the systems listed.*

**Q2. At which stage of the surveillance architecture does AI most dramatically transform the character of state surveillance?**

- A) Collection — AI sensors collect more data than human observers
- B) Storage — AI compression allows more data to be stored
- C) Analysis — AI enables pattern detection and anomaly identification across data volumes that manual analysis cannot process
- D) Consequential decision — AI makes decisions faster than human officials

**Answer: C**

*Explanation: AI's most transformative contribution to surveillance is in the analysis stage, where it enables automated pattern detection, social network analysis, anomaly detection, and predictive risk scoring across the data volumes generated by mass collection systems. Without AI analysis, most collected data would be practically inaccessible.*

**Q3. The digital divide in India creates distributional asymmetry in AI systems because:**

- A) Urban populations generate more data, making AI more accurate for them
- B) Rural populations have less internet access, reducing both their access to AI benefits and their capacity to contest AI-driven decisions that affect them
- C) AI systems work only with English text, excluding non-English speakers
- D) Government AI systems are deployed only in states with high internet

penetration

**Answer: B**

*Explanation: The digital divide creates distributional asymmetry on both sides: lower digital access means less access to AI-enabled services and also less capacity to contest errors, seek legal redress, or exercise data rights. The most marginalised populations face both benefit exclusion and risk exposure.*

**Q4. Why is the right to be forgotten technically difficult to exercise when personal data has been used to train a machine learning model?**

- A) Training data cannot be identified after the model is trained
- B) The individual's data is encoded across millions of model parameters and cannot be easily extracted without retraining or machine unlearning
- C) Machine learning models are legally considered to be separate entities not subject to data protection laws
- D) The right to be forgotten applies only to visible profile data, not to training data

**Answer: B**

*Explanation: When data is used in model training, its influence is distributed across the model's parameters in a way that cannot be extracted by simply deleting the original data. Machine unlearning techniques can address this but are computationally expensive and not yet standard practice.*

**Q5. What distinguishes privacy by design from retrospective deletion as an approach to data privacy?**

- A) Privacy by design is a legal requirement; retrospective deletion is a best practice
- B) Privacy by design builds data minimisation and purpose limitation into systems from the beginning; retrospective deletion attempts to remove data after it has already been widely shared and processed
- C) Privacy by design applies only to private companies; retrospective deletion applies to government agencies
- D) Retrospective deletion is technically simpler than privacy by design

**Answer: B**

*Explanation: Privacy by design addresses the root cause: it prevents unnecessary data from being collected or retained in the first place. Retrospective deletion attempts to reverse a process that has already distributed data across backups, partners, derived records, and trained models — a much harder problem.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Describe what data is generated by a single UPI transaction and explain why the aggregate of millions of such records represents a qualitatively different kind of surveillance capability than was previously possible.**

**Model Answer:**

A single UPI transaction generates a structured record containing the payer's account identifier, the payee's account identifier, the transaction amount, a precise timestamp, the geolocation of both devices if location services are enabled, the device identifiers for both parties, the payment description as entered by the payer, and the transaction status. In

isolation, this is a receipt. Aggregated across hundreds of transactions per person per year, it becomes a detailed profile of economic behaviour: where a person shops, what they buy, who they pay for services, their regular patterns of movement, their social and commercial relationships, their income cycle, and their financial habits. Aggregated across all active UPI users, it becomes a real-time map of economic activity across India at the individual level. This represents a qualitatively different surveillance capability from anything previously available because it is automatic, granular, continuously updated, and covers the formal economic participation of a significant fraction of the Indian population.

*Key terms: UPI, transaction data, aggregate profiling, behavioural surveillance, economic activity, data granularity*

**Q7. Explain the four stages of the surveillance architecture using the example of a smart city facial recognition system in an Indian city.**

**Model Answer:**

Stage one, collection: CCTV cameras distributed across the city capture continuous video footage of public spaces, capturing images of individuals moving through those spaces. Stage two, storage: the video footage is stored on servers, either locally or in a cloud infrastructure, with retention periods that vary across deployments. Stage three, AI-enabled analysis: a facial recognition model processes stored or live video frames, detecting faces, extracting facial feature vectors, and comparing them against a database of individuals of interest, which may include criminal suspects, missing persons, or individuals flagged for other reasons. The AI produces match candidates with confidence scores. Stage four, consequential decision: a high-confidence match triggers an alert to police, who may dispatch officers to the location. If the match is acted upon without manual verification, an innocent person may be detained based on a false positive. The AI transforms stage three from an impractical manual review task into an automated, near-real-time process, enabling surveillance at a scale that no human review team could achieve.

*Key terms: collection, storage, analysis, consequential decision, facial recognition, false positive, automated surveillance*

**Q8. How does the digital divide in India create an asymmetry of both benefits and risks in AI-enabled government services?**

**Model Answer:**

The digital divide creates two simultaneous asymmetries. On the benefit side, populations with good digital access benefit from faster, more accessible, and more efficient government services delivered digitally: they can apply for benefits online, authenticate remotely, access information, and transact without travelling to government offices. Populations with poor digital access, concentrated in rural areas, among women, in lower income quintiles, and in states with lower infrastructure investment, face new barriers when services move online: they lack the devices, connectivity, digital literacy, or language accessibility to participate. On the risk side, populations with less digital access have less capacity to exercise data rights, contest AI-driven decisions, navigate redress mechanisms, or access legal support when errors occur. They are simultaneously less able to benefit from digital services and less able to protect themselves from the risks of algorithmic exclusion, misidentification, and erroneous decision-making. The combination produces a double disadvantage that compounds existing inequalities.

*Key terms: digital divide, distributional asymmetry, benefit access, risk exposure, algorithmic exclusion, redress capacity*

**Q9. Why is the right to be forgotten particularly problematic in the context of**

### **Aadhaar's centralised biometric database?**

#### **Model Answer:**

The right to be forgotten, as established in India's Digital Personal Data Protection Act of 2023, allows individuals to request deletion of their personal data when it is no longer necessary or when consent has been withdrawn. Aadhaar is specifically designed as a permanent, non-deletable identifier: the entire value of the system for identity verification and de-duplication depends on the permanence of the biometric record. Deletion of an Aadhaar record would undermine the identity continuity that the system is built around. This creates a fundamental incompatibility between the right to erasure and the architecture of Aadhaar. The DPDPA therefore carves out specific exceptions for government data processing, including identity databases. The problem is structural: the biometric data of 1.4 billion people is permanently held by a central government authority with no individual mechanism for withdrawal. This means that once enrolled, an individual cannot withdraw their most sensitive biometric identifiers from government custody, regardless of how those identifiers might be used in the future.

*Key terms: right to be forgotten, DPDPA, Aadhaar, biometric data, centralised database, permanent record, carve-out*

### **Q10. What specific risks does the deployment of facial recognition technology in Indian police operations create, and what governance mechanisms are currently absent?**

#### **Model Answer:**

Facial recognition in Indian police operations creates three specific risks. First, misidentification: facial recognition systems have documented higher false positive rates for darker-skinned individuals, women, and older people. In the Indian context, these are also among the most marginalised groups. A false match can result in arrest, detention, and reputational harm before the error is discovered. Second, chilling effects on legitimate activity: the knowledge that participation in protests or public gatherings may result in facial recognition logging and matching changes the calculus of political participation, even for people who have done nothing wrong. Third, function creep: systems deployed for one purpose, such as criminal identification, are used for others, such as tracking political activists, with no legal constraint on scope expansion. The governance mechanisms currently absent include: a specific legal framework for police use of facial recognition; accuracy validation requirements on Indian population data; mandatory human review before action is taken on a match; independent oversight of deployment; transparency obligations toward affected individuals; and a mechanism for individuals to challenge or contest a match.

*Key terms: facial recognition, false positive, chilling effect, function creep, governance gap, mandatory human review, algorithmic transparency*

## **Section C — Reflective Questions (3 Questions)**

**Q11. The chapter quotes the Panopticon principle: the mere possibility of being observed at any moment is sufficient to produce self-regulation. Do you think this dynamic already operates in India's digital environment? What evidence would you look for to determine whether it does?**

**Thinking prompts:**

- How would you empirically test whether people change their online behaviour when they believe they may be monitored?
- What kinds of communication, association, or expression are most likely to be affected by surveillance awareness?
- Is self-regulation in response to surveillance always harmful, or are there contexts where it might be beneficial?
- Does the Panopticon effect operate differently for different populations in India? Who is most affected?
- What is the relationship between surveillance and trust in government? Is surveillance compatible with a democracy?

*Rubric hint: Strong responses will distinguish between the empirical question of whether the Panopticon effect operates and the normative question of whether it is harmful. Credit for responses that identify specific forms of expression or association that might be most affected and that engage with the differential impact on different Indian populations.*

**Q12. The Aarogya Setu contact tracing application was downloaded by over 220 million people during a genuine public health emergency. Does the legitimacy of the purpose justify the surveillance capability it created? What conditions would need to be met for emergency surveillance to be acceptable?**

**Thinking prompts:**

- What is the difference between surveillance that is consented to for a stated purpose and surveillance that is imposed?
- Was the 220 million download figure evidence of genuine consent or of social and institutional pressure?
- What mechanisms would ensure that emergency surveillance data is deleted when the emergency ends?
- Who should oversee compliance with sunset clauses on emergency surveillance powers?
- Are there public health purposes that cannot be achieved without surveillance, and are there alternatives that achieve comparable outcomes with less intrusion?

*Rubric hint: Strong responses will distinguish between purposive legitimacy and procedural legitimacy — that a purpose is good does not mean any means of achieving it is acceptable. Credit for responses that propose specific conditions (consent, proportionality, sunset clauses, independent oversight, deletion) that would make emergency surveillance acceptable.*

**Q13. The chapter argues that the populations most subject to the risks of AI surveillance — misidentification, algorithmic exclusion, and erroneous decisions — are also the populations with the least capacity to contest those errors. Is this coincidence or structure? What would need to change for this relationship to be different?**

**Thinking prompts:**

- Is this distributional pattern a coincidence of current technology and policy choices, or is it a structural feature of how surveillance powers and political power relate?
- What political and economic interests are served by the current distribution of surveillance risk?
- What legal rights would need to exist, and be practically accessible, for marginalised communities to contest algorithmic errors effectively?
- How does the concentration of AI system design and deployment in urban,

English-literate, economically privileged institutions shape whose interests those systems serve?

- What does genuine accountability for algorithmic errors in government AI look like, and who has the power to create it?

*Rubric hint: Strong responses will identify the structural character of the distributional pattern, not merely its contingent features, and will engage with the political economy of surveillance: who benefits from the current distribution and who has the power to change it. Credit for responses that propose specific legal, technical, and institutional changes.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science]** Map the data collection landscape for one specific population in your region: this could be farmers in a digitised agricultural supply chain, students in an AI-enabled educational institution, patients in a digital health system, or tribal communities in a forest management AI programme. For your chosen population, identify what data is collected about them, by whom, for what stated purpose, who has access to it, how long it is retained, and what consequential decisions it influences. Then assess the gap between the official data governance framework and the lived experience of the population in terms of their awareness of collection, their ability to consent or decline, and their capacity to contest errors.

This project requires primary research: interviewing people from the population or accessing official documents about the data systems that affect them. The gap assessment should be specific and evidence-based, not impressionistic. Credit for identifying specific instances where data collected for one purpose is used for another, or where the official consent framework does not reflect meaningful choice.

**Q15. [Commerce]** A fintech company operating in your district uses AI-based transaction monitoring to flag suspicious UPI transactions, with flagged transactions automatically blocked pending human review. You have been asked to design a due process framework for this system. Specify: what information the customer must receive when a transaction is flagged; the maximum time permitted for review before the block is automatically lifted; the process for the customer to contest a block; what evidence the company must provide to support the block; and the escalation path if the company and customer disagree. Explain how your framework balances financial crime prevention with the customer's right to due process and financial access.

The framework should be specific and operational, not generic. Each element should be justified in terms of the specific balance it strikes between financial crime prevention and customer rights. The escalation path should include an independent review mechanism. Credit for engaging with the asymmetry of information between the company and the customer.

## Core References

|             |                                                                                                                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Puttaswamy, K. S. v. Union of India. (2017). Supreme Court of India, Writ Petition 494/2012. [The landmark nine-judge bench judgment establishing privacy as a fundamental right under the Indian Constitution; the foundation of all subsequent data protection law in India. The summary judgment is freely accessible.] |
| <b>Core</b> | Ministry of Electronics and Information Technology. (2023). The Digital Personal Data Protection Act, 2023. Government of India. meity.gov.in. [The primary legislation governing data protection in India; directly relevant to Sections 12.1-12.4. Freely accessible.]                                                   |
| <b>Core</b> | Ramanathan, U. (2019). Aadhaar: Not a Tool for Surveillance? Economic and Political Weekly, 54(22), 15-19. [A critical analysis of Aadhaar's design implications for privacy and surveillance by one of India's foremost legal scholars on the subject.]                                                                   |

## Extended References

|                 |                                                                                                                                                                                                                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Zuboff, S. (2019). The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power. PublicAffairs. [Chapter 1 for the foundational argument about the transformation of data collection into an instrument of behavioural modification; directly supports the Panopticon and chilling effects discussion.] |
| <b>Extended</b> | Khera, R., ed. (2019). The Aadhaar Effect: Why the World's Largest Identity Project Goes Wrong. Oxford University Press. [A comprehensive collection of critical analyses of Aadhaar's social impact; directly relevant to Sections 12.1 and 12.4.]                                                                                      |
| <b>Extended</b> | Ganguly, M., & Roy, A. (2022). Facial Recognition in Indian Policing: A Rights Analysis. Internet Freedom Foundation. internetfreedom.in. [A legal and technical analysis of facial recognition deployments in Indian police operations; directly relevant to the What Could Go Wrong section.]                                          |

## Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Internet Freedom Foundation. (2023). Project Panoptic: Tracking Facial Recognition in India. internetfreedom.in/panoptic. [A public database tracking facial recognition deployments by Indian government and law enforcement agencies; directly relevant to Section 12.2 and the What Could Go Wrong section.] |
| <b>Online</b> | TRAI. (2023). Recommendations on Regulatory Framework for Artificial Intelligence and Big Data in the Telecommunications Sector. trai.gov.in. [Primary regulatory document relevant to AI and data collection in India's digital infrastructure.]                                                               |
| <b>Online</b> | Data.gov.in. (2023). Internet and Mobile Penetration Data. data.gov.in. [Primary open data source for the digital divide visualisation in Section 12.3; state-level disaggregated data on internet access and mobile penetration.]                                                                              |

## *End of Chapter 12*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 13

# Work, Creativity, and the Human Question

*Every major transformation in the technology of production has changed not only what people do for work but what people believe work means. The introduction of mechanised weaving into the textile towns of India's colonial period did not simply make weavers redundant: it challenged the social identity, the community structures, and the sense of purpose of communities whose skilled labour had defined their lives for generations. The question was never only economic. It was always also about what human beings are for.*

*AI presents this question again, with new urgency and new scope. Unlike previous automation waves that primarily displaced manual and routine cognitive labour, AI has begun to affect tasks that were previously considered distinctively human: writing, composing, diagnosing, arguing, designing. This does not mean AI will do all of these things better than humans in all contexts. It means that the boundary between tasks that machines can perform and tasks that only humans can perform has moved, and is moving further. Understanding where that boundary currently lies, why it lies there, and what it means for education, work, and human purpose is the task of this chapter.*

## Learning Objectives

By the end of this chapter, you will be able to:

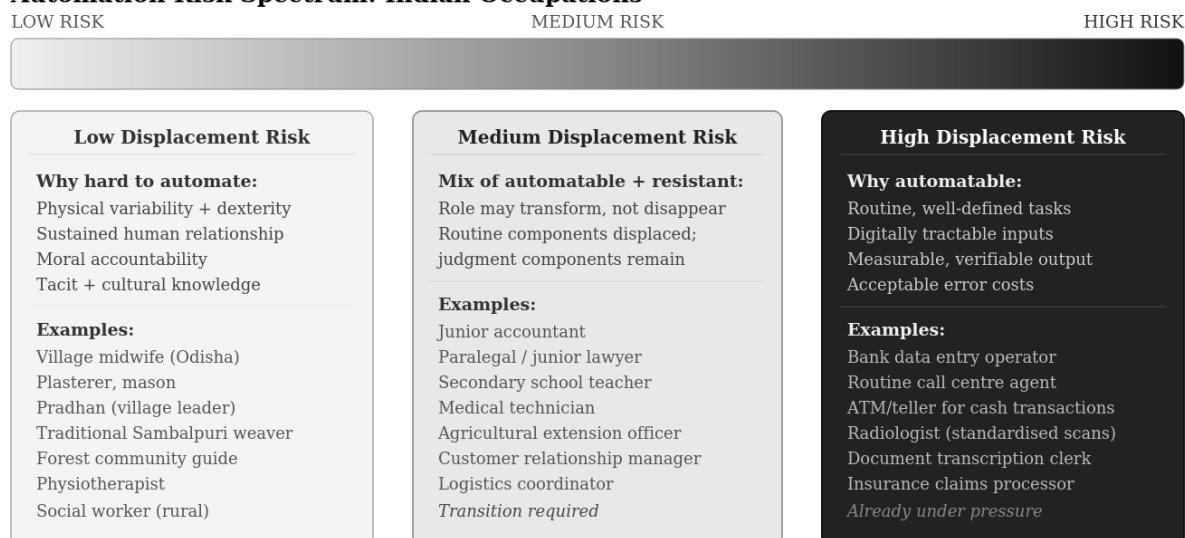
- Use an automation risk spectrum to identify which Indian occupations face low, medium, and high displacement risk
- Explain why the informal economy in India presents distinct challenges for automation analysis
- Describe the creativity triangle: what AI cannot do involving genuine novelty, embodied experience, and moral responsibility
- Explain the shifting Venn diagram of AI and human capability in education
- Articulate what distinctive human contributions remain valuable as AI capability expands

## 13.1 The Automation Risk Spectrum

Not all work is equally susceptible to automation. The degree to which a task can be automated depends on whether it can be precisely specified, whether the relevant inputs can be sensed and digitised, whether the outputs can be evaluated by an algorithm, and whether the consequences of errors are acceptable. Tasks that score high on all four criteria are highly automatable. Tasks that score low on any one are substantially harder to automate reliably.

**Fig. 13.1**

**Automation Risk Spectrum: Indian Occupations**



*Most Indian workers are in the middle zone. The key question is whether transformation (new higher-judgment roles) or elimination occurs — and whether workers have the support and skills to navigate the transition.*

*Figure 13.1 — Automation Risk Spectrum*

At the low-risk end of the spectrum sit occupations that require physical dexterity in unstructured, variable environments; sustained human relationship and trust; moral judgment and accountability; and the application of tacit knowledge that is hard to specify or transfer. A plasterer finishing the interior walls of a house in Bhubaneswar faces a surface that is different every time, requires proprioceptive feedback that robotic systems still struggle to replicate, and produces output that is evaluated by aesthetic and tactile criteria that resist quantification. A midwife attending a home delivery in rural Odisha provides not only medical monitoring but emotional support, cultural knowledge, community trust, and real-time judgment in an environment where the relevant variables change continuously and the stakes of error are severe. A Scheduled Tribe gauntia — village head, or pradhan — mediating a land dispute in a village near Rayagada brings knowledge of local history, kinship networks, and customary practice that no AI system could plausibly acquire.

At the high-risk end sit occupations characterised by routine, well-defined, and digitally tractable tasks. Data entry operators who transcribe paper documents into digital forms are directly susceptible to AI optical character recognition. Bank tellers performing routine cash transactions face displacement from ATMs and digital banking. Radiologists reading standardised X-rays and CT scans for common conditions are at risk from AI diagnostic tools that, in controlled studies, have achieved expert-level performance on specific diagnostic tasks. Call centre agents handling routine customer queries are already being partially displaced by conversational AI systems.

The middle of the spectrum, where most Indian workers are actually located, is the most analytically interesting region. A junior accountant in a mid-sized Indian firm performs a mixture of routine data processing (high risk), standard financial analysis using established models (medium risk), and client relationship management requiring

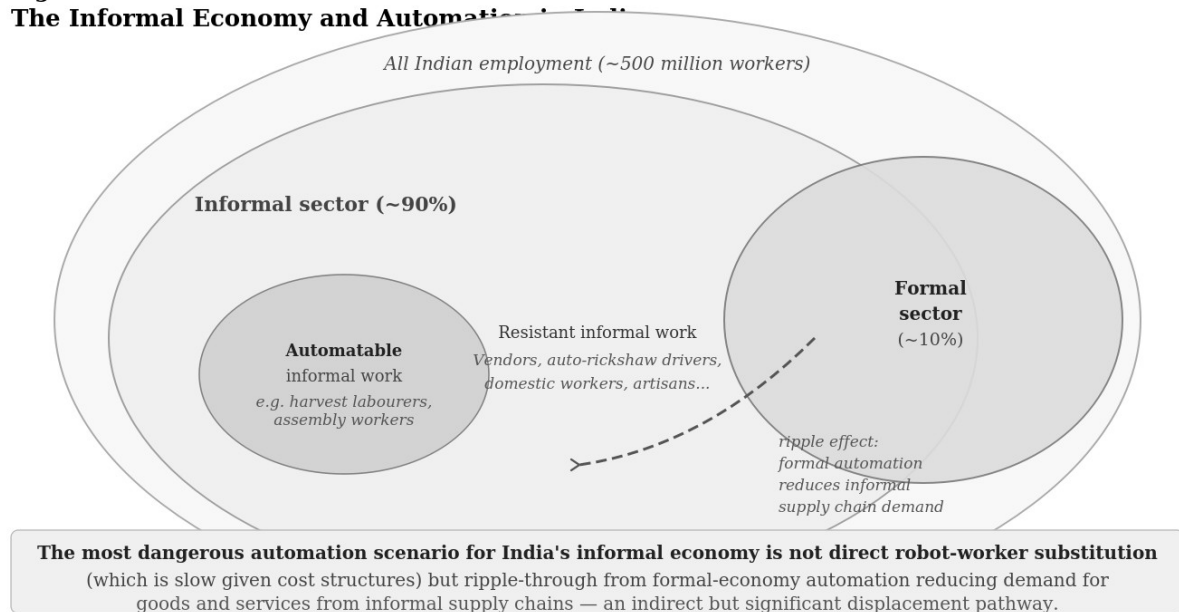
judgment and communication (lower risk). As AI tools become capable of handling the routine and standard components, the junior accountant's role may not disappear: it may transform into a role focused primarily on the judgment-intensive and relationship-intensive components, with the AI tools handling the data processing. Whether this transformation represents displacement or augmentation depends on whether the transformed role is accessible to the workers currently performing the original role, and whether it pays comparably.

## 13.2 The Informal Economy and Automation

Standard analyses of automation risk focus on formal employment: wage workers in registered enterprises whose job categories can be compared against task databases. India's labour market is unusual by global standards because the informal economy, comprising unregistered enterprises and informal workers in both urban and rural settings, accounts for roughly ninety percent of total employment. This dominance of informal employment has implications for automation that receive less analytical attention than they deserve.

**Fig. 13.2**

**The Informal Economy and Automation**



*Figure 13.2 — Informal Economy*

The relationship between automation and the informal economy is not straightforward. On one hand, informal workers performing routinisable tasks, such as agricultural labourers doing repetitive harvesting work, domestic workers performing standardised cleaning tasks, and small-scale manufacturing workers doing assembly operations, face potential displacement from the same technologies that threaten formal workers in similar occupations. The absence of formal employment protections, collective bargaining, or unemployment insurance makes displacement of informal workers potentially more abrupt and less cushioned.

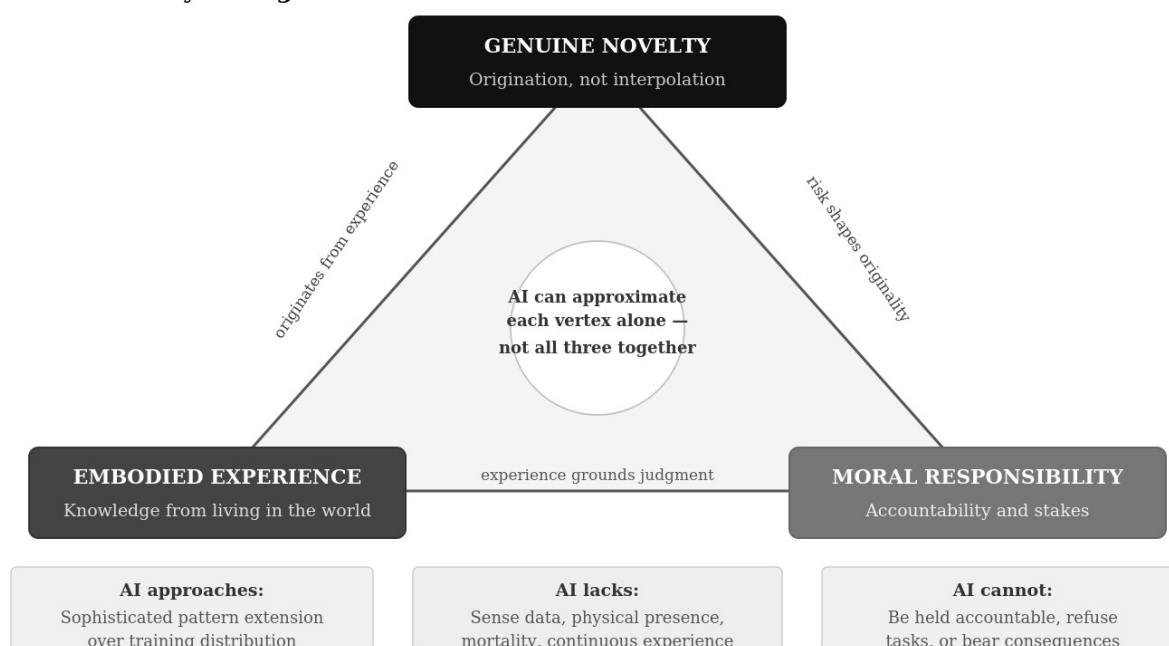
On the other hand, many informal occupations involve the kind of contextual, relationship-intensive, and physically varied work that is hardest to automate. The vegetable vendor who knows the preferences of each regular customer, manages multiple supplier relationships based on trust, handles cash across fluctuating markets, and navigates the social dynamics of the bazaar performs a complex, judgment-intensive job that resists decomposition into automatable subtasks. The auto-rickshaw driver in a congested Indian city navigates physical and social environments of extraordinary complexity, negotiates fares, and reads passengers in ways that current AI systems cannot replicate in unstructured real-world conditions.

The most concerning automation scenario for the Indian informal economy is not direct displacement of informal workers by robots or AI, which will be slow given the cost structures. It is the automation of the formal-economy businesses that purchase the outputs of informal workers. A textile manufacturer whose production is partially automated requires fewer tailors and garment workers from the informal supply chain. A logistics company deploying route optimisation and warehouse automation requires fewer informal loaders and delivery workers. The automation of formal-economy anchor firms can ripple through the informal supply chains they support, reducing informal employment without any robot ever directly replacing an informal worker.

### **13.3 What AI Cannot Do: The Creativity Triangle**

The question of what AI cannot do is more useful, and more honest, than the question of what AI can do. The latter is constantly changing: tasks considered impossible for AI in 2020 are routinely accomplished by 2025. The former identifies structural limitations that are harder to overcome, rooted not in the current state of AI capability but in the nature of what the capability requires.

**Fig. 13.3**  
**The Creativity Triangle: What AI Cannot Do**



*Figure 13.3 — Creativity Triangle*

The first vertex of the creativity triangle is genuine novelty: the capacity to produce something that is not derivable from patterns in existing data. Large language models and image generation systems are, fundamentally, sophisticated interpolators and extrapolators over their training distributions. They can produce combinations of elements they have encountered, in styles they have learned, for purposes they have been trained to serve. What they cannot do is originate: to create a form, a concept, or a perspective that has no precedent in the training data and that would not be derivable from it by any amount of sophisticated interpolation. This limitation is structural, not contingent: it follows from the training paradigm itself.

Human creativity, at its most significant, involves precisely this kind of origination. A poet who creates a new metre is not recombining existing metres: they are departing from what was previously done and producing something that cannot be derived from it. A scientist who proposes a genuinely new theoretical framework is not interpolating between existing frameworks: they are seeing the world differently and articulating a difference that was previously invisible. A composer who creates a new musical form is not computing the most probable sequence of notes: they are hearing something that has not yet been heard and giving it shape.

The second vertex is embodied experience: knowledge that is grounded in and shaped by having a body, being located in physical space, and experiencing the world through sensory engagement over time. Much of what humans find meaningful, and much of what makes human expression resonant, is rooted in embodied experience: the knowledge of physical exertion, of hunger and satiation, of grief and joy as physical states, of the smell of rain on dry earth, of the weight of a sleeping child. AI systems have no body, no sensory

experience, no hunger, no mortality, no presence in the world except as a process running on distant servers.

This is not merely a poetic observation. It has practical implications for what AI-generated work lacks. An AI system can produce text that describes grief with technical accuracy, drawing on the linguistic patterns of grief in its training data. It cannot produce text that is inflected by having actually experienced grief, because it has not. Whether this difference matters depends on what we require of creative and expressive work: if we require only technical competence, it does not matter. If we require that creative work bear witness to the experience of being human, it matters enormously.

The third vertex is moral responsibility: the capacity to be held accountable for decisions that affect others, to bear the weight of consequences, and to exercise judgment that is constrained by genuine stakes. A doctor who makes a diagnostic error bears professional, legal, and moral accountability for that error. An AI system that makes the same error does not: there is no entity that suffered the consequences of being wrong, no career at stake, no sleepless nights. The accountability falls on the developers, the deployers, the institution, and the supervising physician, but not on the AI system itself.

Moral responsibility is not only about accountability for errors. It is also about the capacity for genuine moral judgment: to weigh competing values, to recognise the limits of one's own competence, to refuse tasks that one believes are wrong, and to bear the cost of that refusal. These capacities are not features of current AI systems. They may, in some attenuated form, become features of future AI systems. But they are currently distinctively human, and the work that requires them most centrally, the work of caring for others, of making consequential decisions, of creative expression that bears witness to experience, remains distinctively human in a way that purely technical work does not.

### **13.4 Education Reimagined: The Shifting Venn Diagram**

The implications of AI for education are as profound as its implications for work, and they are related. If AI can perform many of the tasks that education currently trains people to perform, what is education for? This question is not rhetorical: it has specific, practical implications for curriculum design, assessment, and the purposes of the educational experience.

**Fig. 13.4**  
**The Shifting Venn Diagram: AI Capability and Educational Objectives**

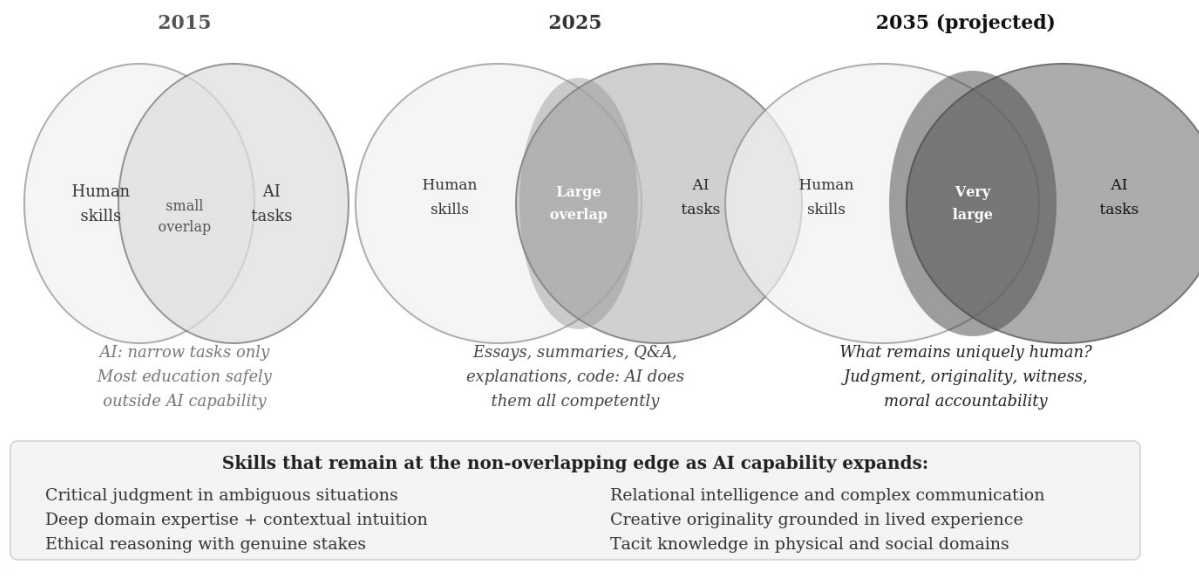


Figure 13.4 — Shifting Venn Diagram

The traditional answer to what education is for has several components: knowledge transmission, skill development, socialisation, credential provision, and the cultivation of capacities for further learning. AI is beginning to affect each of these, but unevenly. Knowledge transmission, the conveying of established facts and frameworks from teacher to student, is a function that AI tools already perform with considerable effectiveness. A student who can ask an AI system to explain the nitrogen cycle in accessible language, provide examples from their region, and answer follow-up questions has access to a knowledge transmission capability that is in many respects superior to what was available in the pre-AI era.

Skill development is more complex. The skills that AI most effectively displaces are the same skills it most effectively teaches: routine information processing, pattern matching in well-defined domains, and the application of established procedures. The skills that AI is least able to teach, because they require experience in judgment-intensive, relationship-intensive, and physically embodied contexts, are also the skills that are least at risk of displacement. Critical thinking, ethical judgment, creative synthesis, complex communication, and the kind of deep domain expertise that combines technical knowledge with tacit contextual experience are both hard to develop and hard to automate.

The shifting Venn diagram describes the changing overlap between what AI can do and what education is designed to develop. In 2015, the overlap was small: AI could perform narrow, well-defined tasks, and education was designed to develop broad capabilities. By 2025, the overlap has grown substantially: AI can produce essays, solve mathematical problems, write code, explain scientific concepts, and perform many other tasks that were previously the exclusive domain of human cognitive effort. By 2035, the overlap may be larger still.

This does not mean education becomes less important. It means that the educational value of tasks and activities must be re-examined in light of the changing AI capability boundary. An assignment that required students to produce a literature review in 2015 developed genuine research and synthesis skills. By 2025, the same assignment can be completed by an LLM in minutes, which means the assignment no longer necessarily develops those skills: it may only test whether the student knows how to prompt an AI. The educational purpose must be redesigned: perhaps to require analysis that the AI cannot produce, judgment that the AI cannot exercise, or authentic engagement with the subject matter that cannot be faked.

For Indian students at regional colleges with limited resources and high expectations of occupational mobility, the shifting Venn diagram has a specific implication. The skills that will remain most valuable in an AI-augmented economy are not the skills that AI already performs well, but the skills that require human judgment, contextual knowledge, creativity, and the kind of relational intelligence that AI does not possess. Education that continues to train primarily for the skills that AI is rapidly automating is not serving its students well. Education that actively develops the skills at the AI-proof end of the spectrum is making its students more capable in a world where AI capability is growing rapidly.

## **13.5 Patterns in My Field: Stream Perspectives**

### **Arts Stream: The Future of Creative Practice in the Age of AI**

A question that every student of the arts, literature, or humanities must now address is not whether AI can produce work in their field but what the existence of that capacity means for human creative practice. AI can generate Odia poetry that meets formal requirements of metre and imagery. It can produce visual art in the style of any historical movement. It can write fiction that follows established narrative structures. The question is not whether these products have technical merit but whether they have the kind of value that human creative work has: the value of testimony, of witness, of a particular human mind engaging with the world and leaving a record of that engagement. The most compelling argument for the continuing value of human creative work is not that it is technically superior to AI work but that it means something different. A poem written by a poet who has experienced the flooding of the Mahanadi carries information that an AI-generated poem about the same subject does not, regardless of technical quality. The challenge for human artists in the AI era is not to compete with AI on technical terms, where AI has significant and growing advantages, but to do the kind of work that only a human can do: work that bears witness, that takes a position, that is inflected by genuine experience and genuine stakes.

### **Science Stream: AI Augmentation in Scientific Research**

Scientific research is not a single activity but a spectrum from routine data collection and processing, which AI is already transforming, to the generation of novel hypotheses, which

remains fundamentally a human activity, though AI tools increasingly contribute to it. A genomics researcher working on earthworm population structure in western Odisha uses AI tools for sequence alignment, phylogenetic tree construction, and statistical analysis: tasks that previously consumed weeks of researcher time can now be completed in hours. This augmentation does not replace the researcher's scientific contribution. It amplifies it, by enabling more data to be processed at greater accuracy with the same person-hours. What the researcher provides that the AI tools do not, the knowledge of the study system, the ecological intuition about what patterns matter, the conceptual framework that makes the data interpretable, the judgment about which findings are surprising and which are expected, remains irreplaceable. The scientific community's increasing concern about AI-generated papers that are technically proficient but conceptually empty reflects this distinction: technical proficiency without conceptual contribution is not science.

### **Commerce Stream: Automation and the Transformation of Indian Financial Services**

Indian financial services have been among the most rapidly AI-transformed sectors in the economy. Routine transaction processing, fraud detection, credit scoring, customer onboarding, and compliance monitoring have all been substantially automated. The transformation has been most visible in the banking and insurance sectors, where the ratio of back-office staff to transaction volumes has fallen dramatically over the past decade. This automation has eliminated hundreds of thousands of routine clerical jobs, particularly affecting women in data entry and processing roles that historically provided stable formal employment for first-generation graduates. The jobs that have grown are concentrated in AI model development, data science, risk management, and complex client relationship roles that require both technical sophistication and communication skill. The transition between the displaced jobs and the growing jobs is not smooth: it requires not just different skills but different educational backgrounds and different social networks. A student from a Tier 2 city who was planning a career in bank data processing faces a fundamentally different job market than their older sibling faced five years earlier, and the educational institutions that trained the sibling may not yet be preparing the student for the transformed market.

### **— WHAT COULD GO WRONG? —**

### **The Concentration of Gains and the Distribution of Costs**

The economic gains from automation do not distribute themselves. They flow to the owners of the capital that the automation represents, to the workers with the skills to develop and deploy the automation, and to the consumers who benefit from lower prices. The costs are borne by the workers whose labour is displaced, by the communities dependent on those workers, and by the public institutions whose tax revenues and social support demands are affected by the transition.

In India, the concentration of AI-related gains is particularly sharp. The development and deployment of AI systems is concentrated in a small number of companies, primarily located in Bengaluru, Hyderabad, Mumbai, and Delhi, staffed predominantly by graduates

of elite technical institutions, and generating value that accrues primarily to shareholders and high-skill workers. The displacement of labour falls disproportionately on workers in routine cognitive occupations: bank clerks, call centre agents, data entry operators, and junior accountants, many of whom are first-generation graduates from Tier 2 and Tier 3 cities who invested in education expecting stable formal employment.

The distributional asymmetry is not only geographic and class-based. It also has a caste and gender dimension. The formal-economy clerical jobs most at risk of automation have historically been one of the most accessible pathways into stable employment for women from lower-middle-class backgrounds and for members of scheduled caste communities who were the first in their families to obtain formal employment. The automation of these jobs does not merely reduce employment: it removes a specific ladder of economic mobility that specific communities have relied on. Building replacement ladders requires active policy, not passive market adjustment.

The philosophical question underlying the distributional concern is not new: it is the question of who automation serves. Automation increases total productive capacity, but total productive capacity is not the same as total welfare. If the gains from automation are captured by a small number of stakeholders while the costs are distributed broadly, automation may reduce inequality in goods produced while increasing inequality in distribution. The question of who AI works for, asked in Chapter 11 about algorithmic fairness, applies with equal force to the economic consequences of AI automation: if the gains go to the few and the costs to the many, the aggregate AI dividend conceals a distributional regression.

## — CHAPTER SUMMARY —

### Chapter 13 — Key Points

1. Automation risk is not uniform: tasks with physical variability, relational intensity, moral accountability, and tacit knowledge are hardest to automate.
2. The informal economy (~90% of Indian employment) has distinctive dynamics: direct displacement is slow, but formal-economy automation ripples through informal supply chains.
3. The creativity triangle identifies three structural limits of AI: genuine novelty (AI interpolates, not originates), embodied experience (no body, no lived experience), and moral responsibility (no accountability).
4. These are not contingent limitations of current AI — they follow from the training paradigm and the absence of embodied existence.
5. The shifting Venn diagram: AI capability increasingly overlaps with what education trains. Assignments that AI can complete no longer necessarily develop the skills they were designed to develop.
6. Human skills that remain most valuable are those at the AI-proof end: judgment, contextual expertise, creative originality, relational intelligence, and moral accountability.

7. The gains from automation concentrate at the top; the costs distribute broadly — this distributional asymmetry requires active policy, not passive market adjustment.
8. The clerical and routine cognitive jobs most at risk have historically been important mobility pathways for women and scheduled caste communities into formal employment.
9. Education that trains primarily for skills AI is rapidly automating is not serving students well in an AI-augmented economy.

— MODEL QUESTIONS —

## Section A — MCQ (5 Questions)

**Q1. According to the automation risk framework in this chapter, which of the following Indian occupations is at lowest immediate displacement risk?**

- A) Bank teller performing routine cash transactions
- B) Data entry operator transcribing paper documents to digital forms
- C) Village midwife attending home deliveries in rural Odisha
- D) Radiologist reading standardised X-rays for common conditions

**Answer: C**

*Explanation: A village midwife requires physical dexterity in highly variable environments, sustained human relationship and emotional support, real-time judgment in contexts where variables change continuously, and cultural and community knowledge that AI cannot acquire. This combination places midwifery at the low-risk end of the automation spectrum.*

**Q2. Why might formal-economy automation displace informal workers even without any robot directly replacing them?**

- A) Informal workers are more likely to use AI tools than formal workers
- B) Automation of formal-economy anchor firms reduces demand for inputs from informal supply chains
- C) Informal workers are legally prohibited from competing with automated formal firms
- D) AI tools designed for formal employment automatically transfer to informal contexts

**Answer: B**

*Explanation: When formal-economy manufacturers, logistics companies, or service firms automate, they require fewer workers and fewer goods from informal supply chains. A textile manufacturer requiring fewer tailors reduces demand for informal garment workers without any direct robot-worker substitution.*

**Q3. In the creativity triangle, 'embodied experience' refers to:**

- A) The ability of AI to simulate human physical sensations through training data
- B) Knowledge grounded in and shaped by having a body and experiencing the world through sensory engagement over time

- C) The use of robotic bodies to give AI systems physical presence in the world
- D) The emotional responses that AI systems generate when processing human creative work

**Answer: B**

*Explanation: Embodied experience is knowledge shaped by having a body: the physical experience of exertion, hunger, grief, joy, and sensory presence in the world. AI systems do not have bodies and therefore lack this foundation for a significant dimension of human creativity and expression.*

**Q4. The shifting Venn diagram of AI and education implies that:**

- A) Education is becoming less important as AI can teach students directly
- B) Assignments that AI can complete may no longer develop the skills they were designed to develop, requiring educational redesign
- C) Students who use AI tools in their education will always underperform those who do not
- D) The overlap between AI capability and educational objectives is currently zero but growing

**Answer: B**

*Explanation: The shifting Venn diagram shows that as AI capability grows, it increasingly covers tasks that education was designed to train students to perform. When an assignment can be completed by an AI, it no longer necessarily tests or develops the intended skill. Educational design must adapt to focus on AI-proof capabilities.*

**Q5. Why is the displacement of routine cognitive clerical jobs in Indian banking particularly significant from an equity perspective?**

- A) These jobs are the highest-paid in the banking sector
- B) These jobs have historically been important mobility pathways into formal employment for women and scheduled caste communities
- C) The banking sector employs the largest number of workers in the Indian economy
- D) Clerical workers in banking have the least ability to retrain for new roles

**Answer: B**

*Explanation: Routine clerical jobs in banking have been among the most accessible formal-economy employment pathways for first-generation graduates, particularly women and SC community members. Automating these jobs removes a specific mobility ladder that specific communities relied on, with distributional consequences that go beyond aggregate employment statistics.*

## **Section B — Short Answer Questions (5 Questions)**

**Q6. Describe the automation risk spectrum and place three Indian occupations from different sectors at different points on it. Justify each placement.**

**Model Answer:**

The automation risk spectrum runs from occupations that are highly automatable, being precise, digitally tractable, with well-defined outputs and acceptable error costs, to occupations that are very difficult to automate, requiring physical variability, relational

intelligence, tacit knowledge, and moral accountability. At the high-risk end, a bank data entry operator transcribes documents into a standardised digital format: the input is structured, the output is well-defined, AI optical character recognition already performs this task, and errors are detectable and correctable. At the middle of the spectrum, a junior accountant performs a mix of routine data processing at high risk, standard financial analysis at medium risk, and client communication at lower risk. As AI tools handle the first two components, the role may transform rather than disappear, though the transformation requires new skills. At the low-risk end, a traditional potter producing distinctive Odisha terracotta ware works in a highly variable physical environment, produces output evaluated by aesthetic criteria that resist quantification, draws on tacit embodied skill developed over years, and is situated within community and cultural relationships that give the work meaning beyond its product.

*Key terms: automation risk, routine cognitive tasks, tacit knowledge, physical variability, relational intelligence, task spectrum*

**Q7. Explain the three vertices of the creativity triangle and why each represents a structural, not merely contingent, limitation of current AI systems.**

**Model Answer:**

Genuine novelty is the first vertex. AI systems learn from training data by finding and extending patterns in that data. They can produce sophisticated combinations and extensions of what they have encountered, but they cannot originate: they cannot produce forms, concepts, or perspectives that have no precedent in training data and that could not be derived from it. This is structural because it follows from the training paradigm: learning from patterns in existing data cannot, in principle, generate what is genuinely new. Embodied experience is the second vertex. AI systems have no body, no sensory apparatus, no physical presence in the world, no mortality. The knowledge that comes from having a body and living through experience, the knowledge that grounds human creative expression and makes it resonate as testimony, is simply absent. This is structural because there is no equivalent to a body in a mathematical function over tokens. Moral responsibility is the third vertex. An AI system cannot be held morally accountable because it has no stakes, no career, no welfare that is affected by its decisions. Accountability for AI decisions falls on developers and deployers, not on the system. The capacity for genuine moral judgment, including the willingness to refuse tasks one believes are wrong and to bear the cost of that refusal, requires having something to lose, which AI systems do not have.

*Key terms: creativity triangle, genuine novelty, embodied experience, moral responsibility, structural limitation, training paradigm*

**Q8. Describe the shifting Venn diagram in the context of Indian undergraduate education. What specific types of assessment or learning activity are most affected, and what would an AI-proof alternative look like?**

**Model Answer:**

The shifting Venn diagram describes the growing overlap between AI capability and what Indian undergraduate education currently trains students to do. In 2015, AI could handle very narrow tasks; most educational activities were in the non-overlapping region. By 2025, the overlap has grown substantially: AI can write essays, solve mathematical problems, explain scientific concepts, produce literature reviews, and draft reports in many formats and languages. The most affected assessment types are those requiring text production with no constraint on source: essays, summaries, literature reviews, and explanation questions. A student can complete these using AI tools without engaging with the underlying skills the assignment was designed to develop. AI-proof alternatives share

a common feature: they require authentic engagement that cannot be faked by an AI. These include: interviews and oral examinations where the student must demonstrate understanding in real time; project work that requires primary data collection from a specific local context that the AI has not encountered; reflective journals documenting the student's own thinking process and experience; collaborative in-class work where the process of thinking, not just the output, is evaluated; and analysis of local or regional phenomena that are not well-represented in AI training data.

*Key terms: shifting Venn diagram, AI-proof assessment, authentic engagement, oral examination, primary data, process evaluation*

**Q9. Why is the concentration of AI gains and the distribution of automation costs a specifically Indian problem, and what policy responses would address it?**

**Model Answer:**

The concentration of AI gains is particularly sharp in India because the development and deployment of AI is concentrated in a small number of elite technical institutions and companies in a few metros, generating wealth that accrues primarily to shareholders and high-skill workers with specific educational credentials. The distribution of costs falls on clerical workers, call centre agents, and routine cognitive workers who are disproportionately first-generation graduates from Tier 2 and Tier 3 cities, women entering formal employment for the first time, and members of scheduled caste communities accessing formal economic participation. The asymmetry is not only economic but social: the ladder being removed is one of the few that was accessible to those communities. Policy responses that would address it include: a social protection framework that extends unemployment insurance and retraining support to displaced formal workers, regardless of whether they were in the AI sector; mandatory contributions from AI-deploying firms to a reskilling fund; public investment in educational infrastructure at regional colleges that develops AI-proof skills rather than training for the jobs being automated; and proactive policy to ensure that AI benefits reach underserved communities through digital infrastructure investment and accessible AI tools in Indian languages.

*Key terms: distributional asymmetry, concentrated gains, distributed costs, social protection, reskilling, mobility ladder, policy response*

**Q10. The chapter argues that AI cannot bear moral responsibility. What are the practical implications of this claim for the deployment of AI in high-stakes decision-making contexts in India?**

**Model Answer:**

If AI systems cannot bear moral responsibility, then every consequential decision made with AI assistance requires a human who can bear that responsibility. This has several practical implications. First, mandatory human review: in domains where decisions affect individual rights and welfare, such as credit, welfare eligibility, healthcare diagnosis, and legal proceedings, a human decision-maker must be responsible for each decision and must have genuinely reviewed it rather than simply rubber-stamped an AI recommendation. Second, clear accountability chains: institutions deploying AI must be able to identify, for any given decision, which human is accountable for that decision and what process they followed. Third, contestability: individuals affected by AI-assisted decisions must have access to a human decision-maker who can review the decision on its merits. Fourth, liability frameworks: when AI-assisted decisions cause harm, the legal and institutional frameworks for accountability must identify which human or institution is responsible, rather than allowing AI deployment to create accountability gaps. Fifth, training for human reviewers: humans reviewing AI outputs must be trained to be genuinely critical rather than deferring to AI outputs by default, which automation bias

shows is a significant challenge.

*Key terms: moral responsibility, accountability, mandatory human review, contestability, automation bias, liability, accountability chain*

## Section C — Reflective Questions (3 Questions)

**Q11. A skilled Sambalpuri handloom weaver in western Odisha produces ikat-dyed textiles using a design vocabulary and technique passed down through generations. If an AI system could analyse existing Sambalpuri designs and generate new designs in that tradition with equal or greater visual complexity, what would be lost? Is this loss significant?**

**Thinking prompts:**

- What is the difference between a design that continues a tradition and a design that was generated from statistical patterns of that tradition?
- Does the weaver's knowledge of the tradition add something that the AI's statistical model does not capture?
- Who owns the tradition that the AI would be trained on, and does that matter?
- Is there a distinction between AI generating new designs for someone else to weave, and AI generating designs that are then automatically produced?
- If the weaver's community benefits economically from AI-generated designs, does that change the analysis?

*Rubric hint: Strong responses will distinguish between the economic question (is there demand and income?) and the cultural question (is the tradition being continued or extracted?). Credit for responses that engage with the distinction between being trained on a tradition and participating in it, and that consider the community ownership question.*

**Q12. The chapter describes AI as unable to originate because it can only interpolate and extrapolate over training distributions. But human creativity has always been, in part, a recombination of existing forms. Is the distinction between AI interpolation and human creativity as sharp as the chapter suggests?**

**Thinking prompts:**

- Can you think of examples of human creative works that are clearly recombinations of existing elements? Does this make them less creative?
- What would a genuinely original human creation look like — one with no precedent in any training distribution the creator had encountered?
- Is the difference between AI and human creativity a difference in kind or a difference in degree?
- Does the answer change depending on whether we are discussing craftsmanship, artistic expression, or scientific discovery?
- If AI systems eventually produce works that no human could distinguish from genuinely original human creativity, does that settle the question of whether they are creative?

*Rubric hint: Strong responses will engage seriously with the challenge to the creativity triangle rather than simply reasserting it. Credit for responses that identify specific ways in which the interpolation-origination distinction is less clear than it appears, while also*

identifying the remaining genuine difference.

**Q13. The chapter argues that education must shift toward AI-proof skills. But who decides what is AI-proof, and what if the decision-makers are wrong? What happens to students educated for skills that turn out not to be AI-proof after all?**

**Thinking prompts:**

- Who has the authority and the knowledge to redesign educational curricula for an AI-augmented future?
- Given the uncertainty about which skills will remain AI-proof, what is the responsible approach to curriculum design?
- What is the risk of prematurely declaring certain skills AI-proof when they may not remain so?
- Are there historical parallels — previous waves of technological change that made previously valuable skills obsolete?
- What institutional safeguards would reduce the risk of educational systems making large-scale bets on skills that turn out to be wrong?

*Rubric hint: Strong responses will engage with the genuine uncertainty about future AI capability and its implications for educational planning. Credit for responses that identify the governance question (who decides and with what accountability) as distinct from the technical question (what is AI-proof).*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Interview three people who work in creative, caring, or knowledge-intensive occupations in your community: these could be a teacher, a traditional artisan, a healthcare worker, a scientist, a musician, a writer, or a social worker. For each, identify what aspects of their work they believe AI could potentially perform and what aspects they believe would remain distinctively human. Then compare their assessments with the automation risk framework and creativity triangle from this chapter. Where do their intuitions align with the framework, and where do they diverge? What do the divergences reveal about the limits of the framework?**

The comparison between practitioner intuitions and academic framework is the analytical core of this project. Do not simply report what practitioners said and then separately describe the framework: genuinely compare them. The divergences are more interesting than the alignments. Credit for identifying specific cases where the practitioner's experience reveals something the academic framework misses.

**Q15. [Commerce] A regional bank in your area is planning to automate its loan processing and customer service functions, which will affect approximately fifty staff members. You have been asked to design a responsible transition plan. Specify the timeline and communication approach, the reskilling programme, the support for workers who cannot be reskilled into new roles, the criteria for determining which roles will be transformed versus eliminated, and the process by which affected workers will have input into the transition**

**design. Conclude with an assessment of the distributional consequences of the automation for the community, considering not only the bank's workers but also the informal workers and small businesses in the local economy that depend on those workers' incomes.**

The distributional consequences section is as important as the transition plan itself. The assessment should be specific to your community's economic structure, not generic. Credit for engaging with the informal economy ripple effects described in this chapter.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Frey, C. B., & Osborne, M. A. (2017). The future of employment: How susceptible are jobs to computerisation? <i>Technological Forecasting and Social Change</i> , 114, 254-280. [The foundational quantitative analysis of automation risk by occupation; the methodology underlying the automation risk spectrum in this chapter, though students should note its limitations in the Indian informal economy context.] |
| <b>Core</b> | Mehrotra, S., & Parida, J. K. (2019). India's employment crisis: Rising education levels and falling non-agricultural employment growth. CSE Working Paper 2019-04. Centre for Sustainable Employment, Azim Premji University. [On India's specific employment challenges in the context of educational expansion; directly relevant to Section 13.1 and the equity discussion in the What Could Go Wrong section.]     |
| <b>Core</b> | Acemoglu, D., & Restrepo, P. (2022). Tasks, automation, and the rise in US wage inequality. <i>Econometrica</i> , 90(5), 1973-2016. [The economic framework for understanding automation's distributional effects; the task-based model underpins the middle-spectrum discussion in Section 13.1.]                                                                                                                      |

### Extended References

|                 |                                                                                                                                                                                                                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Crawford, K. (2021). <i>Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence</i> . Yale University Press. [Chapter 2 on labour directly supports Section 13.2 on the informal economy; the broader argument about AI and power complements the What Could Go Wrong section.]                   |
| <b>Extended</b> | Autor, D. H. (2015). Why are there still so many jobs? The history and future of workplace automation. <i>Journal of Economic Perspectives</i> , 29(3), 3-30. [The classic statement of why automation creates new tasks rather than simply eliminating employment; a necessary counterweight to displacement narratives.] |
| <b>Extended</b> | Bresnahan, T., & Trajtenberg, M. (1995). General purpose technologies: Engines of growth? <i>Journal of Econometrics</i> , 65(1), 83-108. [The general purpose technology framework applied to AI as a historical analogue to electrification and computing; contextualises the scale of the current transition.]          |

### Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Azim Premji University. (2023). State of Working India Report. <a href="http://azimpremjiuniversity.edu.in">azimpremjiuniversity.edu.in</a> . [Annual report on Indian employment with detailed regional and informal economy analysis; directly relevant to Section 13.2 and the equity discussion.]                  |
| <b>Online</b> | ILO. (2023). World Employment and Social Outlook: The value of essential work. <a href="http://ilo.org">ilo.org</a> . [For comparative international context on automation and the future of work; includes India-specific analysis sections.]                                                                         |
| <b>Online</b> | Mollick, E. (2023). One Useful Thing: AI and the Future of Work. <a href="https://substack.com/oneusefulthing">substack.com/oneusefulthing</a> . [A practitioner's perspective on what AI augmentation means for knowledge work; directly complements the stream perspectives sections and the creativity discussion.] |

### *End of Chapter 13*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

## Chapter 14

# Governing Intelligence

*Every powerful technology that has reshaped human society has eventually been governed, though the governance almost always lags the technology by years or decades. The printing press was followed by censorship regimes. The railroad by safety regulations and land rights law. The nuclear reactor by international non-proliferation frameworks. The internet by data protection legislation, platform liability rules, and content moderation regimes that are still being contested. Governance does not suppress technology: it shapes the conditions under which technology operates, distributes its benefits and costs, protects individuals from its harms, and tries, imperfectly, to ensure that its development reflects collective values rather than only private interests.*

*AI is now at the stage where its governance frameworks are being actively constructed. The systems being built today will shape how AI develops, who it serves, what it is permitted to do, and who bears the costs when it causes harm. Students who are entering the workforce now will live through the decades during which these frameworks are established. They are not merely passive subjects of AI governance: they are potential participants in the institutions, the civil society, and the democratic processes that will shape it.*

*This chapter maps India's AI governance landscape, compares it to global approaches, and identifies the student as a stakeholder in the governance process, not merely a beneficiary or a victim of its outcomes.*

## Learning Objectives

By the end of this chapter, you will be able to:

- Describe India's AI policy timeline and the key institutions involved in AI governance
- Identify the roles of MEITY, TRAI, the judiciary, and civil society in India's AI regulatory landscape
- Compare India's approach to AI governance with that of the European Union, the United States, and China
- Describe three specific roles students can play as users, critics, and builders of AI systems
- Articulate what meaningful democratic participation in AI governance requires from citizens

## 14.1 India's AI Policy Timeline

India's engagement with AI policy began in earnest with the National Strategy for Artificial Intelligence, published by NITI Aayog in 2018. The strategy defined a vision of AI for India, identifying five priority sectors: healthcare, agriculture, education, smart cities, and smart mobility. It introduced the concept of AI for All, positioning India as a user-country and a developer-country simultaneously, drawing on its large population and growing digital infrastructure as assets for AI development and deployment.

**Fig. 14.1**  
**India's AI Policy Timeline: 2018-2025**

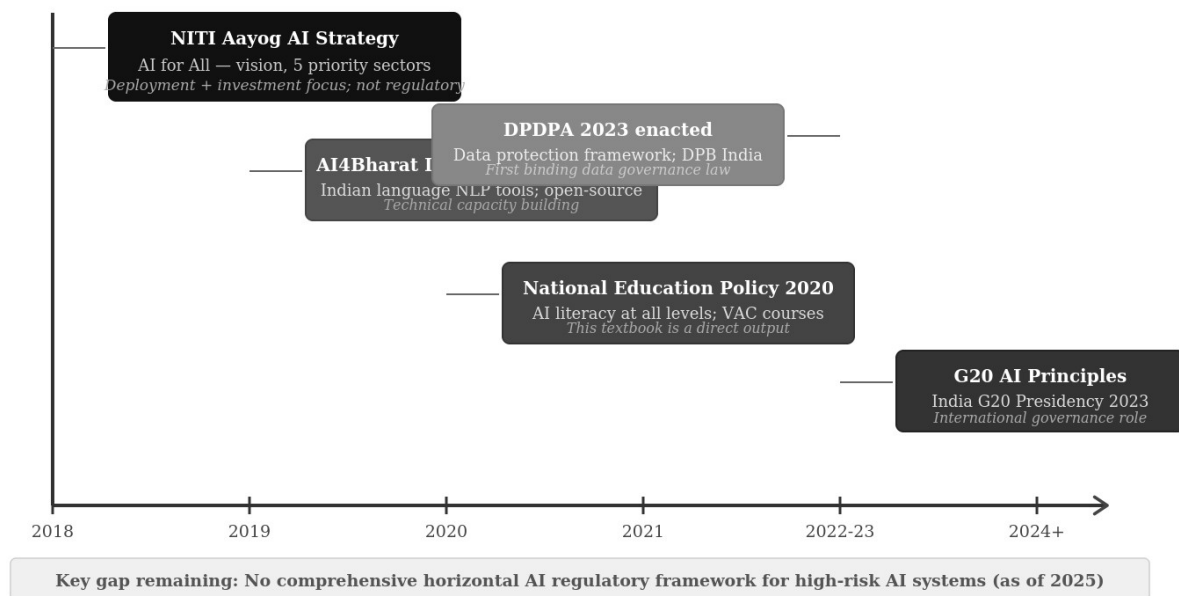


Figure 14.1 — India Policy Timeline

The 2018 NITI Aayog strategy was followed by several sector-specific initiatives. The National Health Stack and Ayushman Bharat Digital Mission incorporated AI components. The National Education Policy of 2020, implemented under which this very course was designed, included provisions for AI literacy at all levels of education. The Smart Cities Mission continued to deploy AI-enabled infrastructure in participating cities. These were not primarily regulatory frameworks: they were deployment and investment frameworks, describing what government wanted to do with AI rather than what constraints should apply to how AI is used.

The regulatory dimension of India's AI governance began to take shape more concretely with the Digital Personal Data Protection Act of 2023, which established a framework for the handling of personal data that applies to AI systems using that data. The DPDPA does not regulate AI directly but regulates the data on which AI systems operate, creating obligations for data fiduciaries, requirements for consent and purpose limitation, and a right to erasure, as discussed in Chapter 12. The DPDPA's implementation is overseen by the Data Protection Board of India, which is still being established.

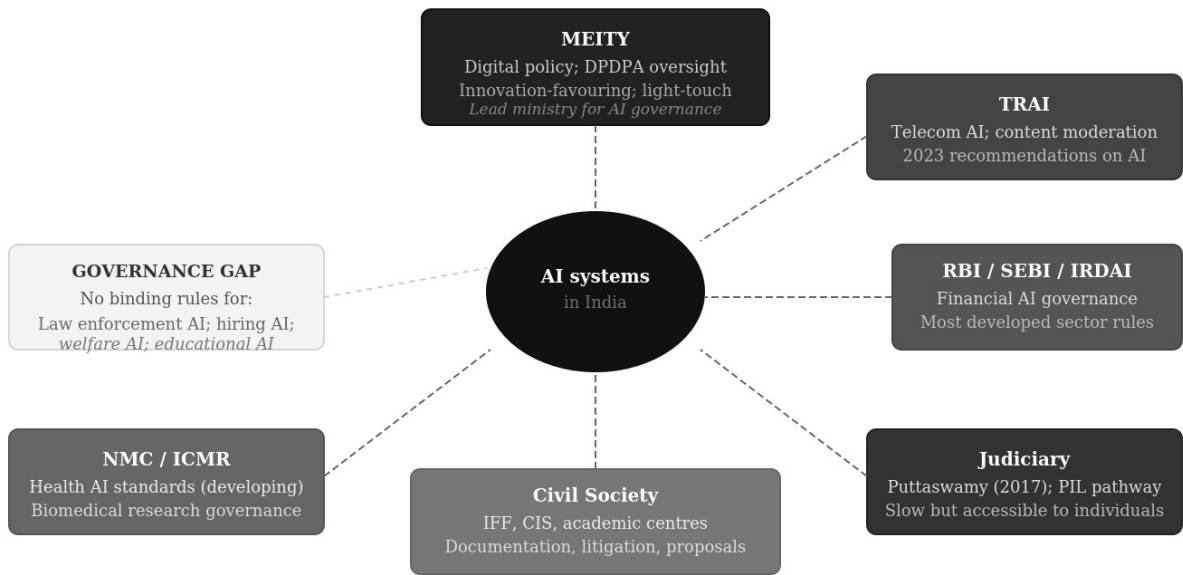
India has also participated in international AI governance discussions, including the Global Partnership on AI, the G20 AI Principles adopted during India's G20 presidency in 2023, and bilateral technology governance frameworks with the European Union and the United States. These international engagements reflect India's interest in being a rule-setter in global AI governance rather than a rule-taker, consistent with its broader ambition to be a significant voice in the governance of the digital economy.

What India does not yet have is a comprehensive AI-specific regulatory framework. Unlike the European Union's AI Act, which classifies AI systems by risk level and imposes mandatory requirements on high-risk systems, India's current governance approach is sectoral: different aspects of AI are governed by the sector-specific regulators for health, finance, telecommunications, and data. The absence of a comprehensive horizontal framework means that gaps exist, particularly for AI applications that cross sector boundaries or operate in domains without strong existing regulators.

### 14.2 The Institutional Landscape: Who Governs AI in India?

AI governance in India is distributed across several institutions, each with jurisdiction over a specific domain or aspect of the technology. Understanding this distribution is essential for understanding both the current state of AI oversight and the gaps that remain.

**Fig. 14.2**  
**India's AI Governance: Institutional Map**



India's AI governance is distributed and sectoral. Strengths: well-governed financial services. Gap: no binding requirements for high-risk AI in law enforcement, hiring, welfare, and education.

Figure 14.2 — Institutional Map

The Ministry of Electronics and Information Technology, MEITY, is the primary government body responsible for digital policy, including AI. MEITY oversees the implementation of the DPDPA, has issued guidelines on the responsible use of AI in

government services, and leads India's participation in international AI governance forums. MEITY's approach to AI regulation has generally favoured light-touch oversight, emphasising innovation facilitation over precautionary restriction, reflecting a policy judgment that the costs of over-regulation to India's growing technology sector outweigh the benefits of stronger precautionary rules.

The Telecom Regulatory Authority of India, TRAI, has jurisdiction over AI in telecommunications contexts, including the use of AI in network management, content moderation on digital platforms, and data handling by telecom operators. TRAI's 2023 recommendations on regulatory frameworks for AI and big data in the telecommunications sector represent one of the more substantive attempts by an Indian regulator to grapple with AI-specific governance questions, including transparency requirements, algorithmic accountability, and the treatment of sensitive data.

Sectoral regulators play important roles in governing AI in their specific domains. The Reserve Bank of India regulates AI in banking and financial services, with requirements for model explainability, fairness testing, and accountability in credit decisions. The Insurance Regulatory and Development Authority governs AI in insurance, including the use of algorithmic underwriting. The Medical Council of India and the National Medical Commission are developing standards for AI-assisted diagnosis and treatment. The Securities and Exchange Board of India has issued guidance on the use of AI in trading systems and investment advice.

The Indian judiciary has played a significant role in shaping AI governance through landmark constitutional decisions, most notably the Puttaswamy judgment of 2017 which established privacy as a fundamental right, providing the constitutional foundation for data protection and surveillance regulation. Courts have also addressed AI-related questions in cases involving facial recognition, automated decision-making in welfare programmes, and algorithmic transparency. Judicial oversight provides a pathway for affected individuals to challenge AI-enabled decisions that harm them, though the accessibility and speed of judicial processes limit their effectiveness as a routine accountability mechanism.

Civil society organisations and academic institutions contribute to AI governance in ways that formal government institutions do not: by documenting harms, conducting independent audits, proposing policy frameworks, litigating on behalf of affected communities, and building the public understanding of AI that democratic governance requires. Organisations including the Internet Freedom Foundation, the Centre for Internet and Society, Citizen Digital Foundation, and academic centres at IITs and IIMs have contributed substantially to India's AI governance discourse. Their influence on formal policy has been uneven but growing.

### 14.3 Global Governance Comparison

India's AI governance approach can only be understood in comparative context. The three major regulatory models being developed globally, by the European Union, the United States, and China, represent different choices about the balance between innovation, rights protection, and state authority. India's choices are shaped by its own political economy, constitutional framework, and development priorities, but they are also shaped by the need to interoperate with the global systems these three large actors are creating.

**Fig. 14.3**  
**Global AI Governance: Four Approaches Compared**

|                                    | European Union                                                                                          | United States                                                                                       | China                                                                                                 | India                                                                                           |
|------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| STRENGTH KEY INSTRUMENT PHILOSOPHY | Rights-protective;<br>risk-based; precautionary<br><i>Fundamental rights as justiciable constraints</i> | Market-preferencing;<br>sectoral; permissive<br><i>Regulatory federalism; innovation preference</i> | State-directed;<br>content + security focus<br><i>National strategy; state-market alignment</i>       | Sectoral; innovation-favouring; values-stated<br><i>"Responsible, inclusive, human-centric"</i> |
|                                    | AI Act 2024:<br>4-tier risk classification;<br>binding requirements;<br>prohibited practices list       | Executive Order on AI 2023 (non-binding);<br>sector agency guidance;<br>no federal AI law           | Algorithmic recommendation rules; deepfake regs;<br>generative AI rules;<br>registration requirements | DPDPA 2023;<br>G20 AI Principles;<br>sectoral guidelines (RBI, SEBI, etc.)                      |
|                                    | Comprehensive; binding;<br>individual rights enforceable                                                | Flexible; responsive;<br>innovation-friendly                                                        | Rapid deployment of national AI capability;<br>content security                                       | Growing regulatory capacity; constitutional privacy foundation                                  |
| LIMITATION                         | High compliance cost;<br>may disadvantage EU companies vs. US/China                                     | Uneven protection;<br>no binding national standard for AI harms                                     | Does not protect individual rights vs. state surveillance                                             | No horizontal AI law; values stated but not yet binding requirements                            |
|                                    | India values:<br><b>Closest alignment</b>                                                               | India practice:<br><b>Most similar approach</b>                                                     | India practice:<br>Some state investment                                                              | <b>Current position</b><br>between all three                                                    |

Figure 14.3 — Global Governance Comparison

The European Union's AI Act, adopted in 2024, is the world's most comprehensive AI-specific regulatory framework. It takes a risk-based approach: classifying AI systems into four risk categories, unacceptable risk, high risk, limited risk, and minimal risk, and applying mandatory requirements to high-risk systems in domains including critical infrastructure, education, employment, credit, law enforcement, and border management. High-risk systems must meet standards for data quality, transparency, human oversight, robustness, and cybersecurity, and must undergo conformity assessment before deployment. The AI Act also prohibits certain practices outright, including most real-time remote biometric identification in public spaces, AI systems that manipulate behaviour subliminally, and social scoring by public authorities.

The EU approach reflects a rights-protection philosophy rooted in the European tradition of fundamental rights as justiciable constraints on both state and market power. Its strengths are its comprehensiveness and its hard legal requirements. Its limitation is the administrative complexity and compliance cost it imposes on developers, which may disadvantage European AI companies relative to US and Chinese competitors not subject to equivalent requirements.

The United States has taken a more fragmented, sector-specific approach, consistent with the US tradition of regulatory federalism and market preference. The Biden administration's Executive Order on AI of October 2023 established reporting requirements, safety testing standards, and agency guidance across federal departments, but without comprehensive binding legislation. Various agencies, the FTC, the EEOC, the CFPB, and the FDA among them, have issued AI-specific guidance within their domains. This approach is flexible and responsive but produces uneven protection: well-regulated domains like financial services have more AI accountability than poorly regulated domains like political advertising.

China's approach is the most state-directed of the three. China has issued specific regulations on algorithmic recommendations, deepfakes, and generative AI, each requiring registration, security assessment, and compliance with content and security rules. China's regulatory approach is less focused on protecting individual rights against state power than on managing the content and political implications of AI and ensuring that AI development serves national strategic objectives. The separation between state and market is less sharp in China's regulatory model than in the EU or US approaches.

India's approach shares elements with all three but resembles none exactly. Like the US, it favours sectoral regulation over comprehensive horizontal frameworks. Like China, it has used state-directed investment and deployment programmes to develop AI capability. Unlike the EU, it has not yet established hard rights-based requirements for high-risk AI systems. India's G20 AI Principles emphasise responsible, inclusive, and human-centric AI, suggesting a values commitment that aligns more closely with the EU model than with either the US market-preference or the Chinese state-direction model. Whether and how that values commitment is translated into binding regulatory requirements remains one of the central open questions of India's AI governance trajectory.

## **14.4 The Student as Stakeholder**

A stakeholder in a technology is anyone whose interests are affected by how that technology operates and is governed. By this definition, every person in India who uses a digital service, interacts with a government system, applies for a job, accesses credit, or participates in an economy that is increasingly shaped by AI is a stakeholder in AI governance. Students at Indian universities are among the most directly affected: they will spend their working lives in an AI-augmented economy, and the quality of AI governance will determine whether that economy serves them or merely uses them.

**Fig. 14.4**  
**The Student as Stakeholder: Three Roles in AI Governance**

| USER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CRITIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | BUILDER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Rights:</b><br/>           Know when AI affects you<br/>           Contest AI-influenced decisions<br/>           Data protection under DPDPA</p> <p><b>Responsibilities:</b><br/>           Use AI tools honestly<br/>           Do not submit AI work as your own<br/>           Do not use AI to harm others</p> <p><b>Practical actions:</b><br/>           Request explanations for AI decisions affecting you<br/>           Exercise DPDPA rights<br/> <i>Rights require awareness to exercise</i></p> | <p><b>Tools from this book:</b><br/>           Bias pipeline analysis<br/>           Fairness trade-off framework<br/>           Surveillance architecture map<br/>           Creativity triangle</p> <p><b>Does not require:</b><br/>           Technical AI expertise<br/>           Coding ability</p> <p><b>Practical actions:</b><br/>           Document AI harms in your community<br/>           Write analysis for NGOs or media<br/>           Engage community consultation<br/> <i>Analytical frameworks are governance tools</i></p> | <p><b>Direct governance responsibility:</b><br/>           Design choices = governance choices</p> <p><b>Required capacities:</b><br/>           Anticipate consequences<br/>           Conduct fairness testing<br/>           Document system behaviour<br/>           Raise internal concerns<br/>           Refuse harmful projects</p> <p><b>Key principle:</b><br/>           Technical competence without governance responsibility is not enough<br/> <i>Education is the best time to develop these capacities</i></p> |
| <p><b>All three roles matter. Most people will occupy all three simultaneously at different points in their lives.</b><br/>           The knowledge in this book is directly applicable to each role — not as theory, but as practical analytical tools.</p>                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

*Figure 14.4 — Student As Stakeholder*

## The Student as User

As a user of AI systems, the student has both rights and responsibilities. The rights include the right to know when AI is being used to make decisions that affect them, the right to contest those decisions through accessible mechanisms, and the right to have their data handled with appropriate protections. These rights are still being established and enforced in India, but their exercise requires awareness that they exist. A student who does not know that an AI system influenced their scholarship application decision cannot contest it. A student who does not know that their social media behaviour is being used to profile them for advertising cannot manage their exposure.

The responsibilities of the user include using AI tools in ways that are honest and do not harm others. Submitting AI-generated work as one's own is dishonest. Using AI tools to generate disinformation, to harass others, or to circumvent systems designed to protect people harms others. These are not merely institutional rules: they are ethical commitments that the chapter on automation in Chapter 13 illuminated from a different direction. Using AI as a substitute rather than an amplifier, as Chapter 10 discussed, is not only educationally self-harmful: it is an act of deception toward institutions and communities that trust one's work to be genuinely one's own.

## The Student as Critic

As a critic of AI systems, the student has a role that goes beyond mere compliance. Critical engagement with AI requires the capacity to identify when AI systems are producing harmful outcomes, to articulate why those outcomes are harmful, and to communicate that critique to the institutions and communities that can act on it. Chapters 11 through 13 of this book have provided frameworks for exactly this kind of critique: the

bias pipeline, the fairness trade-off triangle, the surveillance architecture, the creativity triangle. These frameworks are not academic exercises: they are tools for identifying and articulating what is wrong with specific AI systems in specific contexts.

Critical engagement does not require technical expertise. A student who documents that a government welfare AI system is producing systematically different outcomes for tribal communities than for general population applicants, and who can articulate why this constitutes a violation of the right to equal treatment, has made a contribution to AI governance without writing a single line of code. A student who writes about the intersection of AI surveillance and indigenous land rights in Odisha, grounding their analysis in constitutional rights and documented harms, has produced knowledge that civil society organisations and policymakers can use. The frameworks in this book are designed to enable exactly this kind of informed criticism.

### **The Student as Builder**

Students who go on to build AI systems, whether as software engineers, data scientists, product managers, or policy designers, carry the most direct responsibility for the governance of the systems they build. The choices made by builders, about what data to collect, what to optimise for, what to test, who to consult, what to document, and when to stop a project that is causing harm, are governance choices even when they are not labelled as such. A software engineer who installs a logging mechanism that enables unauthorised surveillance has made a governance choice. A product manager who decides to deploy a model without testing it on the communities most affected by its errors has made a governance choice.

Responsible building requires more than technical competence. It requires the capacity to anticipate consequences, the willingness to raise concerns through internal channels when a project is causing harm, and the moral courage to refuse participation in projects that are clearly unethical. These are capacities that can be developed, and that are most effectively developed during education, before the pressures of employment and institutional loyalty have narrowed the range of available choices. The knowledge in this book, and the reflective practice it is designed to support, is directly relevant to becoming a builder who is not merely technically competent but genuinely responsible.

## **14.5 Patterns in My Field: Stream Perspectives**

### **Arts Stream: AI Governance and Cultural Rights**

India's cultural heritage represents one of the world's richest repositories of traditional knowledge, artistic practice, oral literature, and craft tradition. Much of this heritage is not formally documented or protected by intellectual property law, making it vulnerable to extraction and commercial exploitation by AI systems trained on cultural data without community consent or benefit-sharing. The governance question for arts students is not only about individual creative rights, as important as those are, but about the collective

rights of communities to their cultural heritage in an AI-mediated information environment. The Geographical Indications of Goods Act, which protects products like Kondapalli toys, Chanderi sarees, and Basmati rice from geographic misrepresentation, provides a partial model for thinking about cultural heritage protection in the AI context, but it was designed for physical products and does not extend cleanly to digital cultural expressions. Developing governance frameworks that protect the cultural rights of communities in the AI era is an important task for students and scholars working at the intersection of law, cultural studies, and technology policy.

### **Science Stream: AI Governance in Biomedical Research**

Biomedical AI in India raises governance questions that intersect with research ethics, data protection, and health regulation. AI systems trained on Indian patient data to improve diagnostic accuracy for conditions prevalent in Indian populations require data sharing at a scale that challenges existing patient consent frameworks, which were designed for single-institution research studies, not national or global training datasets. The Indian Council of Medical Research's bioethics guidelines provide a starting framework, but they predate the current scale and speed of AI development in biomedical contexts. Science students who will work in research or clinical settings need to understand both the technical requirements for biomedical AI, as described in earlier chapters, and the governance frameworks that determine what data can be used, by whom, for what purposes, with what protections, and with what benefit-sharing for the communities and individuals whose data is the raw material of the research.

### **Commerce Stream: AI Governance and Financial Consumer Protection**

India's financial services sector is among the most rapidly AI-transformed in the economy, and the governance frameworks for AI in financial services are accordingly more developed than in many other sectors. The Reserve Bank of India's guidelines on model risk management, the SEBI's guidance on algorithmic trading, and the IRDAI's framework for AI in insurance all represent substantive sector-specific AI governance that financial services professionals need to understand and apply. Commerce students who will work in fintech, banking, insurance, or related sectors need to understand not only the technical capabilities of AI in financial services but the regulatory environment that governs their deployment. This includes understanding what the RBI considers acceptable evidence for model fairness, what disclosure requirements apply to algorithmic credit decisions, and what redress mechanisms customers have when algorithmic systems produce incorrect or discriminatory outcomes.

## **— WHAT COULD GO WRONG? —**

### **The Governance Gap: When Regulation Lags Deployment**

The most persistent and consequential risk in AI governance is not that regulation is wrong but that it is absent. Regulation typically responds to documented harms, which means it lags deployment by the time required to document, analyse, and legislate. For

technologies that diffuse rapidly, this lag can be substantial, and the harms that accumulate during the lag are real and, for those who bear them, irreversible.

In India, several AI deployment areas are currently operating with minimal specific regulation. Facial recognition by law enforcement, as discussed in Chapter 12, has no binding legal framework for deployment standards, accuracy requirements, mandatory human review, or individual redress. AI-assisted hiring, as discussed in Chapter 11, is not subject to mandatory fairness auditing requirements. AI in educational assessment, a growing area as platforms increasingly use algorithmic systems to evaluate student work and predict performance, has no specific regulatory framework. AI in political advertising and information operations, which could significantly affect democratic processes, has no specific governance mechanism.

The governance gap is not only a problem of absent regulation. It is also a problem of enforcement capacity. The Digital Personal Data Protection Act exists, but the Data Protection Board of India that will enforce it is not yet fully operational. The TRAI recommendations on AI in telecommunications are recommendations, not binding rules. Many of the fairness guidelines issued by sectoral regulators are guidelines rather than requirements, producing compliance cultures that treat them as aspirational rather than mandatory.

The governance gap is particularly consequential for the communities most vulnerable to AI harms. Wealthy, legally represented, urban, and digitally literate individuals can navigate governance gaps more effectively: they can document harms, access legal services, and engage with redress mechanisms. The communities most likely to be harmed by inadequately governed AI, marginalised communities with limited access to legal and institutional resources, are also the communities least able to obtain redress when those harms occur. Building governance systems that work for the most vulnerable, not only for those with the capacity to advocate for themselves, is the central governance design challenge.

## — CHAPTER SUMMARY —

### Chapter 14 — Key Points

1. India's AI policy began with NITI Aayog's 2018 strategy and has evolved through the DPDPA 2023 and international governance engagements, but lacks a comprehensive horizontal AI regulatory framework.
2. AI governance in India is distributed across MEITY, TRAI, sectoral regulators (RBI, SEBI, IRDAI, NMC), the judiciary, and civil society organisations.
3. EU approach: risk-based, rights-protective, comprehensive, binding. US approach: sectoral, market-preferencing, fragmented. China approach: state-directed, content- and security-focused.
4. India shares elements of all three but most closely aligned in stated values with the

EU model, without yet having matching binding requirements for high-risk systems.

5. Students are stakeholders in AI governance in three roles: users (with rights and responsibilities), critics (with analytical tools to identify and articulate harms), and builders (with direct responsibility for governance choices in what they create).

6. The governance gap — regulation lagging deployment — is the most persistent risk. Unregulated deployment accumulates irreversible harms before regulatory frameworks respond.

7. The communities most likely to be harmed by inadequately governed AI are also the communities least able to seek redress through existing mechanisms.

8. Meaningful democratic participation in AI governance requires technically informed citizens who can hold both government and industry accountable.

9. The knowledge in this book is directly applicable: the bias pipeline, fairness frameworks, surveillance architecture, and creativity triangle are tools for informed governance participation.

## — MODEL QUESTIONS —

### Section A — MCQ (5 Questions)

**Q1. India's NITI Aayog AI strategy of 2018 positioned India as:**

- A) A regulator of global AI development, setting standards for all countries
- B) Primarily a consumer of AI developed in other countries
- C) Both a user-country and a developer-country, drawing on its population and digital infrastructure as assets
- D) A military AI development programme competing with the United States and China

**Answer: C**

*Explanation: The NITI Aayog strategy articulated an 'AI for All' vision positioning India as both a user of AI for its development priorities and a developer of AI capability drawing on its large population, growing digital infrastructure, and technical talent base.*

**Q2. What is the primary distinction between the EU's AI Act approach and India's current AI governance approach?**

- A) The EU focuses on commercial AI; India focuses on government AI
- B) The EU has a comprehensive, risk-based horizontal framework with binding requirements; India relies on sectoral regulation without a comprehensive horizontal framework
- C) India has stricter AI regulation than the EU, reflecting its larger population
- D) The EU approach applies only to European companies; India's applies to all AI operating in India

**Answer: B**

*Explanation: The EU AI Act is a comprehensive, risk-based regulatory framework applying across all sectors. India's current approach is sectoral: different aspects of AI are governed by sector-specific regulators without a comprehensive cross-cutting AI*

framework equivalent to the EU AI Act.

**Q3. In the 'student as critic' role described in this chapter, what is the most important contribution a student can make to AI governance?**

- A) Writing technical papers demonstrating flaws in AI algorithms
- B) Identifying documented harms from AI systems, articulating why they are harmful, and communicating that analysis to institutions and communities that can act on it
- C) Voting for politicians who support AI regulation
- D) Refusing to use AI tools as a personal statement against AI development

**Answer: B**

*Explanation: The critic role involves applying analytical frameworks to identify harms, articulate why they violate rights or fairness principles, and communicate effectively. This does not require technical expertise in AI development — it requires the analytical frameworks this book provides and the capacity to apply them to specific real-world cases.*

**Q4. The governance gap in AI refers to:**

- A) The technical gap between what AI can do and what governments understand about it
- B) The period during which AI is deployed and causing harms before regulatory frameworks are established to govern it
- C) The gap between India's AI capability and that of the United States or China
- D) The difference between formal AI policy documents and their actual implementation

**Answer: B**

*Explanation: The governance gap is the lag between AI deployment and the establishment of regulatory frameworks to govern it. During this gap, harms accumulate without accountability or redress mechanisms. For rapidly diffusing technologies, this gap can persist for years and the harms are real and often irreversible.*

**Q5. Which of the following best describes China's AI governance philosophy?**

- A) Rights-based, with strong protections for individual civil liberties against AI-enabled state surveillance
- B) Market-preferencing, with minimal government intervention in AI development
- C) State-directed, focused on ensuring AI development serves national strategic objectives and content security requirements
- D) Risk-based, classifying AI systems by harm potential and imposing proportionate requirements

**Answer: C**

*Explanation: China's approach is the most state-directed of the major governance models. Its AI regulations focus on content security, political implications, and national strategic objectives rather than on protecting individual rights against state power, which distinguishes it from the EU model.*

## Section B — Short Answer Questions (5 Questions)

**Q6. Describe India's AI policy evolution from 2018 to 2023 and identify the single most significant gap in the current governance framework.**

**Model Answer:**

India's AI policy evolution can be described in three phases. The first phase, beginning with the 2018 NITI Aayog strategy, established a vision of AI for development, identifying priority sectors and positioning India as both a user and developer of AI. This phase was primarily about investment, deployment, and building capability rather than regulation. The second phase involved sector-specific deployment: AI was incorporated into health, education, agricultural, and urban infrastructure programmes without comprehensive governance requirements. The third phase brought the Digital Personal Data Protection Act of 2023, which established data governance requirements applicable to AI systems using personal data, and India's G20 AI Principles, which articulated governance values at the international level. The most significant gap is the absence of a comprehensive horizontal AI regulatory framework equivalent to the EU AI Act. Without such a framework, high-risk AI deployments in areas such as law enforcement facial recognition, AI-assisted hiring, welfare eligibility determination, and AI in judicial processes operate without binding requirements for accuracy validation, fairness testing, transparency, human oversight, or individual redress mechanisms.

*Key terms: NITI Aayog AI strategy, DPDPA, G20 AI Principles, horizontal framework, sectoral regulation, governance gap, high-risk AI*

**Q7. Compare India's AI governance approach with the EU AI Act on two specific dimensions: risk classification and individual rights.**

**Model Answer:**

On risk classification, the EU AI Act creates a formal four-tier classification of AI systems, from unacceptable risk through high, limited, and minimal risk, with binding requirements that escalate with risk level and outright prohibitions on unacceptable-risk applications such as real-time biometric surveillance of public spaces. India has no equivalent risk classification framework. AI systems in India are subject to the requirements of the sector they operate in, meaning that a high-risk AI system in one sector may face strict requirements while an equally high-risk system in an ungoverned sector faces none. On individual rights, the EU AI Act grants individuals specific rights in relation to high-risk AI systems, including the right to explanations and the right to contest decisions. The DPDPA grants Indian individuals rights over their personal data including correction, erasure, and grievance redress, but these rights apply to data handling, not to AI-specific decisions. An individual in India whose welfare application is rejected by an AI system has no specific AI-derived right to an explanation or contestation process equivalent to what the EU AI Act provides.

*Key terms: risk classification, EU AI Act, individual rights, explanation rights, contestability, DPDPA, data rights, AI Act compliance*

**Q8. Explain the role of civil society organisations in India's AI governance landscape and give two specific examples of contributions they have made.**

**Model Answer:**

Civil society organisations contribute to AI governance in ways that formal government institutions do not, by operating independently of both state authority and commercial interest. They document harms that affected communities experience but cannot themselves articulate in policy terms. They conduct independent technical audits of AI

systems deployed in public contexts. They litigate on behalf of affected individuals whose rights have been violated by AI-enabled decisions. They propose policy frameworks, build coalitions between communities, researchers, and policymakers, and produce the public understanding of AI that democratic governance requires. Two specific examples: first, the Internet Freedom Foundation has documented and challenged facial recognition deployments by Indian police forces through the Project Panoptic database, combining public records research, technical analysis, and public interest litigation to create accountability for surveillance systems that were operating without any legal framework. Second, the Centre for Internet and Society has conducted research on algorithmic discrimination in Indian hiring and credit systems, producing evidence that policymakers and courts have drawn on in developing more specific governance requirements for these applications.

*Key terms: civil society, Internet Freedom Foundation, Centre for Internet and Society, algorithmic accountability, public interest litigation, independent audit*

### **Q9. What specific governance responsibilities does the 'student as builder' role imply for someone entering a technology career in India?**

#### **Model Answer:**

The student as builder carries direct governance responsibility for the systems they create, because the design choices in those systems are governance choices whether or not they are labelled as such. Specific responsibilities include: first, anticipating consequences, by thinking through how the system could be misused, which populations it might harm, and what the second-order effects of its deployment might be, before finalising design. Second, fairness testing: ensuring that the system is evaluated for differential performance across the demographic groups most relevant to the deployment context, not only on aggregate accuracy metrics. Third, documentation: creating clear records of what the system does, what data it uses, what its known limitations are, and what assumptions are embedded in its design, to enable later accountability. Fourth, internal escalation: raising concerns through institutional channels when a project appears to be causing harm, rather than remaining silent out of institutional loyalty. Fifth, refusal: developing the moral courage to decline participation in projects that are clearly harmful, even when institutional pressure favours compliance. These responsibilities are not additional to technical competence: they are part of what technical competence means for AI practitioners working in consequential domains.

*Key terms: builder responsibilities, fairness testing, anticipating consequences, documentation, internal escalation, moral courage, technical accountability*

### **Q10. Why is the governance gap particularly consequential for the communities most vulnerable to AI harms in India?**

#### **Model Answer:**

The governance gap creates a period during which AI systems can cause harms without accountability or redress. This period is more consequential for vulnerable communities than for privileged communities for two compounding reasons. First, vulnerable communities are more likely to be the subjects of high-risk AI deployments in welfare, law enforcement, and financial contexts, and are therefore more likely to experience harms during the governance gap. Urban middle-class individuals who interact primarily with AI tools as assistants face different and generally lower-stakes AI interactions than rural communities whose welfare access is mediated by algorithmic systems or tribal communities whose lands are monitored by AI-enabled surveillance. Second, when harms do occur, vulnerable communities have less capacity to seek redress. Accessing legal services, documenting harms in the form that courts or regulators require, navigating administrative complaint processes, and sustaining the engagement required for formal

accountability are all more difficult for communities with limited resources, limited digital literacy, and limited institutional access. The combination means that the governance gap is experienced more severely, and its harms are more persistent, for those who most need effective governance.

*Key terms: governance gap, vulnerable communities, compounding disadvantage, redress capacity, welfare AI, surveillance, legal access*

## Section C — Reflective Questions (3 Questions)

**Q11. The chapter describes India as positioned between the EU's rights-protective model, the US market-preference model, and China's state-direction model. Which of these three models best serves India's interests and values, and what would a genuinely Indian model look like?**

**Thinking prompts:**

- What are India's specific development priorities, constitutional values, and political economy that should shape its AI governance model?
- Does India's colonial history and post-independence constitutional framework suggest specific values for AI governance that differ from European, American, or Chinese values?
- How should India balance the innovation and investment benefits of lighter-touch regulation against the rights-protection benefits of stronger binding requirements?
- What aspects of India's diversity, its linguistic, caste, regional, and socioeconomic heterogeneity, should a distinctively Indian AI governance model address that foreign models do not?
- Who should design India's AI governance model, and through what process? What legitimate representation do affected communities have in that process?

*Rubric hint: Strong responses will engage with India's constitutional values, development priorities, and specific social context rather than simply preferring one global model. Credit for responses that identify specific aspects of India's social diversity that a governance model should address and that engage with the legitimacy question of who designs the model.*

**Q12. The chapter argues that students are stakeholders in AI governance, not merely subjects of it. But student influence on AI governance is largely indirect and long-term. Is this framing realistic, or does it overstate the agency of individual students?**

**Thinking prompts:**

- What specific mechanisms exist for students to participate in AI governance decisions in India?
- Is the framing of student-as-critic an empowering framework or a way of distributing governance responsibility to individuals who have little structural power?
- What would structural student influence on AI governance look like, as opposed to individual critical engagement?
- How does the student's position in the institutional hierarchy, as a student, as a potential employee, as a citizen, shape what kinds of governance participation are available?

- Is there a risk that framing students as responsible stakeholders obscures the structural power imbalance between students and the corporations and governments that deploy AI?

*Rubric hint: Strong responses will engage critically with the stakeholder framing rather than simply accepting it, identifying its limitations alongside its strengths. Credit for responses that distinguish individual agency from structural power and that identify specific mechanisms for meaningful participation beyond individual critical engagement.*

**Q13. This book has been built on the Pattern Hunters philosophy: observation first, then analysis, then action. Having read fourteen chapters, what is the most important pattern you have observed about the relationship between AI and Indian society? What does that pattern imply about what should be done?**

**Thinking prompts:**

- Is there a single through-line that connects the bias pipeline, the surveillance architecture, the informal economy impact, the digital divide, and the governance gap?
- Does the pattern you have identified suggest a primarily technical problem, a primarily political problem, a primarily social problem, or some combination?
- Who would need to act on the implication you have identified, and what would that action look like?
- Does your pattern reveal something about AI specifically, or about how powerful technologies interact with unequal societies more generally?
- What would you do differently, having read this book, in your own practice as a user, critic, or builder of AI systems?

*Rubric hint: This is an integrative reflection drawing on the full book. Strong responses will identify a genuinely cross-cutting pattern rather than summarising a single chapter, and will derive from that pattern a specific and actionable implication. Credit for responses that demonstrate genuine synthesis and that connect the observed pattern to a specific commitment to action.*

## Section D — Stream Project Prompts (2 Questions)

**Q14. [Arts / Science] Choose one AI application currently deployed or under development in India in a domain relevant to your field — this could be AI in archaeological site preservation, AI in environmental monitoring, AI in cultural heritage documentation, AI in scientific research infrastructure, or any other domain you have encountered in this course. Conduct a governance analysis: what is the current regulatory framework (if any) governing this application, who is accountable when the system produces harmful outcomes, what rights do affected individuals or communities have, and what specific governance gaps exist? Conclude with a proposal for one specific governance mechanism, not a general principle but a concrete institutional or procedural requirement, that would address the most significant gap you have identified.**

The governance analysis should be based on research into actual regulatory frameworks, not assumptions about what might exist. The gap identification should be specific to the application and context you have chosen. The governance proposal should be concrete enough to evaluate — it should name the responsible institution, the specific requirement,

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| and the enforcement mechanism.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Q15. [Commerce] A fintech startup you are advising is preparing to launch an AI-based credit scoring product targeting first-time borrowers in rural India. The founders believe their product will dramatically expand financial inclusion by extending credit to people who would not qualify under traditional scoring systems. You have been asked to prepare a responsible governance framework for the product. Drawing on the concepts from Chapters 11, 12, and 14, specify: what fairness metrics the product must meet; what transparency obligations apply to credit decisions; what data protection requirements govern the data collected; what redress mechanism borrowers must have access to; and what ongoing monitoring will ensure the product continues to meet these requirements after launch. Conclude with an assessment of whether this product can responsibly expand financial inclusion or whether its structure creates risks that outweigh the benefits.</b></p> <p>This is an integrative assessment drawing on multiple chapters. The governance framework should be specific and operational, not generic. The fairness metrics should reference specific definitions from Chapter 11. The data protection requirements should reference the DPDPA. The concluding assessment should be honest about the tension between financial inclusion and algorithmic risk, not simply one-sided.</p> |

— REFERENCES —

Core References

|      |                                                                                                                                                                                                                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core | NITI Aayog. (2018). National Strategy for Artificial Intelligence: AI for All. Government of India. <a href="https://niti.gov.in">niti.gov.in</a> . [The foundational Indian AI policy document; directly relevant to Section 14.1. Freely available.]                                                                                             |
| Core | European Parliament and Council. (2024). Regulation (EU) 2024/1689 on Artificial Intelligence (EU AI Act). Official Journal of the European Union. [The world's most comprehensive AI regulatory framework; essential reference for Section 14.3. Summary versions accessible through <a href="https://eur-lex.europa.eu">eur-lex.europa.eu</a> .] |
| Core | Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv preprint <a href="https://arxiv.org/abs/1702.08608">arXiv:1702.08608</a> . [The foundational paper on the technical requirements for interpretability in AI governance; relevant to the accountability discussion throughout Unit 4.]       |

Extended References

|          |                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extended | Choudhury, S. (2022). India's approach to AI governance: Between innovation and regulation. <i>Journal of Indian Law and Society</i> , 13(2), 145-178. [Analyses India's AI governance trajectory in comparative context; directly relevant to Sections 14.1-14.3.] |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                 |                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Roberts, H., Cows, J., Morley, J., et al. (2021). The Chinese approach to artificial intelligence: An analysis of policy, ethics, and regulation. <i>AI and Society</i> , 36(1), 59-77. [The most comprehensive English-language analysis of China's AI governance model; directly relevant to Section 14.3.]                                      |
| <b>Extended</b> | Raji, I. D., & Buolamwini, J. (2019). Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. <i>Proceedings of AIES 2019</i> . [Demonstrates the effectiveness of civil society-led auditing in producing governance change; directly supports the civil society section of 14.2.] |

## Accessible Online Resources

|               |                                                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | MEITY. (2023). India's Approach to AI Governance. <a href="https://meity.gov.in/ai-governance">meity.gov.in/ai-governance</a> . [Primary government source on India's AI governance framework; includes links to all major policy documents discussed in Section 14.1.]            |
| <b>Online</b> | Internet Freedom Foundation. (2023). Project Panoptic: Facial Recognition Tracker. <a href="https://internetfreedom.in/panoptic">internetfreedom.in/panoptic</a> . [Civil society documentation of AI deployment in India; exemplifies the critic role described in Section 14.4.] |
| <b>Online</b> | AI Policy Exchange. (2023). Global AI Governance Tracker. <a href="https://aipolicyexchange.org">aipolicyexchange.org</a> . [Regularly updated comparative database of AI governance initiatives worldwide; useful reference for the global comparison in Section 14.3.]           |

### *End of Chapter 14*

## UNIT FIVE

# Making Something

*Understanding a subject and being able to do something with it are not the same skill, and a course that stops at the first without reaching for the second has not quite finished its job.*

*This last unit is short on purpose. It hands the loop back to the reader: observe a pattern in your own field, decide what it means, act on it, and be accountable for what follows. That is the whole method of this book, applied one last time, by you.*

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15. The Mini-Project as Thinking

## Chapter 15

# The Mini-Project as Thinking

*The first fourteen chapters of this book have asked you to understand. This chapter asks you to do. The mini-project requirement of this course is not an assessment bolt-on: it is the culminating intellectual activity. Everything you have read has been preparation for this: the pattern recognition in Unit 1, the learning mechanisms in Unit 2, the applications in Unit 3, the consequences and governance in Unit 4. The project is where that preparation becomes practice.*

*A mini-project is not a report about AI. It is an original investigation using AI or about AI, grounded in a question that genuinely interests you, executed with appropriate rigour, and communicated in a way that allows others to evaluate what you found and how you found it. It is, in miniature, what all serious intellectual work looks like: a question, a method, a finding, and an honest account of what the finding does and does not establish.*

*This chapter provides the conceptual framework for the mini-project, three worked examples across the three streams, an ethical checklist that every project must pass, a discussion of failure modes to avoid, and a presentation template that supports effective communication of your findings.*

## Learning Objectives

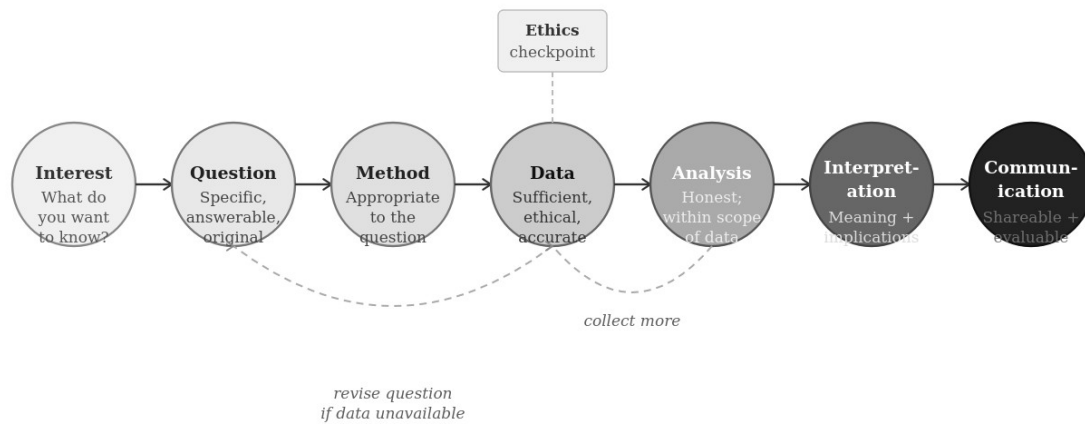
By the end of this chapter, you will be able to:

- Apply the project design cycle to select, refine, and scope a mini-project question
- Distinguish between a report about AI and an original investigation using or about AI
- Use the ethical checklist to evaluate whether a proposed project is safe, fair, and honest
- Anticipate and avoid the five most common mini-project failure modes
- Structure a project presentation using the communication template provided

## 15.1 The Project Design Cycle

Every good project begins before the first piece of data is collected or the first tool is used. It begins with a question. The quality of the question determines the quality of the project more than any other single factor: a well-formed question produces a project with a clear purpose, a tractable method, and an interpretable result. A poorly formed question produces a project that sprawls, stalls, or arrives nowhere.

**Fig. 15.1**  
**The Project Design Cycle**



*The cycle is not strictly linear: discovering that your question cannot be answered with available data should send you back to question formation, not forward to a poorly-supported finding. The ethics checkpoint must be passed before data collection begins.*

*Figure 15.1 — Project Design Cycle*

The project design cycle begins with interest: what do you want to know? This interest should be genuine, not simulated. A project driven by genuine curiosity is more likely to be executed with care and more likely to produce insights than a project selected because it appeared tractable. The syllabus gives you broad territory: AI applications, AI tools, AI consequences, AI governance. Within that territory, the specific question is yours to choose.

From interest, the cycle moves to question formation. A good project question has four properties. It is specific: not what is AI bias but does the attendance prediction algorithm used by this district's schools treat students from tribal communities differently from general category students? It is answerable: the data or observations needed to address it exist and are accessible to you. It is original: it asks something that is not already answered by the available literature, which does not require novel theory, only that you are not simply summarising what others have written. And it is relevant: it connects to a real context, a real community, or a real decision that your answer could inform.

Method selection follows from the question. Different questions require different methods. A question about whether an AI tool performs differently across languages requires a comparative evaluation using carefully designed test cases. A question about community perceptions of an AI-enabled government service requires interviews or surveys. A question about the fairness of a public dataset requires audit methodology. A question about the potential economic impact of automation on a specific sector requires analysis of sector employment data combined with automation risk assessments. The method must be appropriate to the question, not chosen because it is familiar or convenient.

Data collection follows from the method. The data must be sufficient to answer the question, legally and ethically obtained, and accurately recorded. For most mini-projects, the data will be primary: collected specifically for the project. Primary data collection requires attention to consent, privacy, and the accurate representation of what was collected and how. Secondary data, such as publicly available government statistics or published datasets, requires proper attribution and awareness of the limitations of the data source.

Analysis converts data into findings. Good analysis is honest about what the data shows and what it does not show. It acknowledges the limitations of the sample size, the method, and the measurement. It does not overstate findings: a project based on twenty interviews is not a nationally representative survey, and the findings should be presented accordingly. The difference between a finding and a speculation is the difference between what the data supports and what the analyst imagines.

Interpretation connects findings to implications. What do your findings mean for the question you asked? What do they mean for the community or context your project examined? What further questions do they open? What would need to be true for your findings to generalise beyond your specific case?

Communication is the final stage, and it is not merely presentation. It is the act of making your work accessible to others who can evaluate it. A project that produced good findings but communicated them poorly has not fulfilled its purpose: knowledge that cannot be shared and evaluated is not public knowledge.

## **15.2 Three Worked Examples**

The following three worked examples span the three streams and illustrate the range of appropriate mini-project forms. They are not templates to be copied: they are demonstrations of the design cycle in action, showing how a question, method, finding, and communication can be assembled from a genuine starting interest.

**Fig. 15.2**  
**Three Worked Project Examples: Design at a Glance**

|             | Arts Stream                                                                                                                     | Science Stream                                                                                                                                                          | Commerce Stream                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION    | Does Gemini produce more inaccurate content about Odia festivals vs. national festivals?                                        | Can Teachable Machine classify bollworm damage on cotton leaves? How does performance vary with distance and lighting?                                                  | Are Kuchinda small business owners aware of UPI data collection? Does awareness affect digital payment comfort?                         |
| METHOD      | AI audit: 40 factual questions verified against authoritative sources; code correct/incorrect/hedged/absent per response        | ML experiment: two classifiers trained on different conditions; tested on stratified held-out set across distance/lighting combinations                                 | Semi-structured interviews with 15 business owners; coded using surveillance architecture framework; anonymised transcripts             |
| DATA        | 40 responses; 800+ claims; coded; sources documented<br>Verified against Odisha Tourism, Sahitya Akademi publications           | 160 training images; 40 test images; metadata: distance, lighting, date<br>College experimental farm<br>Single smartphone; student photographer                         | 15 anonymised transcripts; stratified by business type and UPI adoption level; verified with transcription                              |
| FINDING     | Accuracy gap between Odia and national content; hedging appears more in Odia responses<br><i>[student fills actual numbers]</i> | Near/diffuse model: high accuracy in-distribution; degrades out-of-dist.<br>Mixed model: consistent across conditions<br><i>[student fills actual accuracy numbers]</i> | Low awareness of data access; main concern [theme]; awareness changes comfort levels for [N]%<br><i>[student fills actual findings]</i> |
| LIMITATION  | One LLM; one researcher; question set not exhaustive                                                                            | Small dataset; single farm; one disease; one phone model                                                                                                                | Small non-random sample; self-report may not match behaviour                                                                            |
| IMPLICATION | Concrete instance of language coverage bias (Ch. 8) in practice                                                                 | Confirms distribution shift (Ch. 7); practical lesson for field deployment                                                                                              | Consent gap in digital payments ecosystem; DPDPA not operationalised                                                                    |

Figure 15.2 — Worked Examples

| Worked Example 1 — Arts Stream |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question                       | Does Gemini perform differently when asked to explain traditional Odia festivals compared to mainstream Hindi or national festivals? Specifically: is the information more likely to contain inaccuracies, hedges, or hallucinated details for Odia cultural content?                                                                                                                                                                                                                                                                                 |
| Method                         | Comparative AI audit. Design a standard set of 20 factual questions about Odia festivals (Rath Yatra specifics, Raja, Nuakhai, Pana Sankranti) and 20 parallel questions about nationally prominent festivals (Diwali, Holi, Eid, Pongal). Ask Gemini the same questions and record its responses verbatim. Verify each factual claim against at least one authoritative source (Odisha Tourism, Sahitya Akademi publications, or Encyclopaedia Indica). Record: correct claims, incorrect claims, hedged claims, and absent claims per question set. |
| Data                           | 40 Gemini responses, 800+ factual claims, each coded as correct/incorrect/hedged/absent. Verification sources documented.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Finding                        | Gemini produced [X]% correct responses for national festival questions and [Y]% for Odia festival questions. The accuracy gap was most pronounced for [specific category]. Hallucinated claims included [specific examples]. Hedging language appeared [Z]x more frequently in Odia festival responses.                                                                                                                                                                                                                                               |
| Limitation                     | One LLM tested; performance may differ for other models. Question set was researcher-designed and may not represent all relevant cultural knowledge. Accuracy is measured against accessible sources, which may themselves be incomplete.                                                                                                                                                                                                                                                                                                             |

|                    |                                                                                                                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Implication</b> | The accuracy gap, if confirmed across multiple LLMs, represents a concrete instance of the language representation bias described in Chapter 8. Students and teachers relying on LLMs for Odia cultural content face higher hallucination risk than those asking about nationally prominent topics. |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Worked Example 2 — Science Stream |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>                   | Can a simple image classifier built with Google Teachable Machine distinguish healthy cotton leaves from leaves showing early symptoms of cotton bollworm damage at the Kuchinda college experimental farm? How does performance vary with distance and lighting?                                                                                                                                               |
| <b>Method</b>                     | Applied machine learning experiment. Collect 40 labelled photographs of healthy and 40 of bollworm-affected leaves using a smartphone, varying distance (near, medium, far) and lighting (diffuse, direct sun, shade). Train two classifiers: one on near/diffuse images only, one on mixed conditions. Test both on a held-out set of 20 images per class, stratified across distance and lighting conditions. |
| <b>Data</b>                       | 160 training images, 40 test images, labelled and photographed by the student. Metadata recorded for each: distance, lighting, date, time.                                                                                                                                                                                                                                                                      |
| <b>Finding</b>                    | Classifier trained on near/diffuse images achieved [X]% accuracy on near/diffuse test images but [Y]% on direct-sun and far images. Mixed-condition classifier achieved more consistent performance ([Z]%) across conditions with modest overall accuracy reduction.                                                                                                                                            |
| <b>Limitation</b>                 | Small dataset; single farm; single disease; one phone model. Results not generalisable to other diseases or environments without re-collection.                                                                                                                                                                                                                                                                 |
| <b>Implication</b>                | Confirms the distribution shift problem from Chapter 7: training-condition specificity strongly affects real-world performance. A practical plant disease classifier for smallholder farmers requires diverse training data collected under field conditions, not laboratory conditions.                                                                                                                        |

| Worked Example 3 — Commerce Stream |                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>                    | How do small business owners in Kuchinda town perceive the impact of UPI-based payment tracking on their business privacy? Are they aware that transaction data is accessible to financial institutions, and does this awareness affect their comfort with digital payments?                                                                                                                                 |
| <b>Method</b>                      | Interview-based qualitative study. Design a semi-structured interview guide with questions about: payment practices (UPI vs. cash), awareness of data collection and who has access, comfort level with digital payments, specific concerns. Conduct interviews with 15 small business owners in Kuchinda market, sampling across business type (food, textiles, hardware, services) and UPI adoption level. |
| <b>Data</b>                        | 15 interview transcripts, anonymised. Coded thematically using the surveillance architecture framework from Chapter 12.                                                                                                                                                                                                                                                                                      |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Finding</b>     | Of 15 interviewees, [X] used UPI regularly, [Y] primarily or exclusively used cash. Awareness of financial institution access to transaction data was low ([Z]% expressed awareness). Most common concern among UPI users was [theme]. Most common reason for cash preference among non-UPI users was [theme]. After being informed of data access practices, [N]% expressed changed comfort levels. |
| <b>Limitation</b>  | Small, non-random sample in one town. Self-reported awareness and comfort may not reflect actual behaviour. The information session during interviews may have introduced bias.                                                                                                                                                                                                                      |
| <b>Implication</b> | Low awareness of data collection practices among small business owners suggests a meaningful consent gap in India's digital payments ecosystem, consistent with the DPDPA's information rights provisions that are not yet effectively operationalised for this population.                                                                                                                          |

### 15.3 The Ethical Checklist

Every mini-project that involves human participants, uses personal data, or produces findings about specific individuals or communities must pass an ethical review before data collection begins. This checklist is not bureaucratic formality: each item addresses a real harm that has occurred in research contexts where these questions were not asked.

**Fig. 15.3**  
**The Ethical Checklist: Five Questions Before Data Collection**

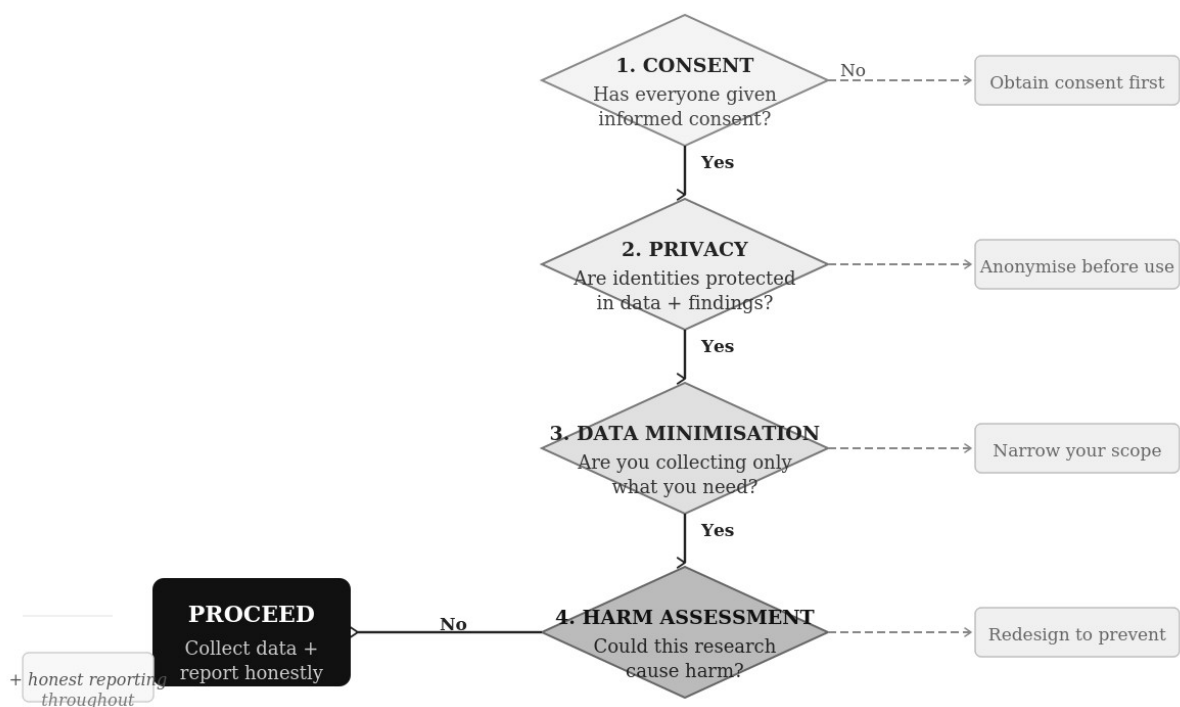


Figure 15.3 — Ethical Checklist

The first question is consent. Has every person whose data you collect, or who you will interview or observe, given their informed consent to participate in your research? Informed consent requires that participants know what the research is about, what data will be collected, how it will be used, who will have access to it, and that they can withdraw at any time without consequence. Consent must be genuinely voluntary: participants should not feel coerced by institutional relationships, such as teachers asking students, or by social pressure.

The second question is privacy. Will your data collection or your published findings expose personal or sensitive information about identifiable individuals? All interview data should be anonymised before analysis: replace names with identifiers, remove identifying details, and never publish quotes that would allow a reader to identify the speaker without the speaker's explicit permission. Data about communities requires particular care: findings that could stigmatise a community or be used to harm its members should be presented at a level of aggregation that prevents such use.

The third question is data minimisation. Are you collecting only the data you actually need to answer your question? The temptation in student research is to collect more data than needed, against the possibility of follow-up questions. Resist this: every additional piece of data collected is an additional piece of data that can be mishandled, leaked, or misused. Collect what you need and nothing more.

The fourth question is harm assessment. Could your research, or its findings, cause harm to participants, to the community you are studying, or to others? Harm can be direct: a poorly anonymised interview exposing a participant's private views to their employer. Harm can be indirect: findings that are misused by third parties to justify policies that hurt the community studied. Harm can be through misrepresentation: presenting findings in a way that misrepresents a community or reinforces harmful stereotypes. Before proceeding, ask: who could be hurt by this research, and how?

The fifth question is honest reporting. Will you accurately represent what your data shows, including its limitations? Will you acknowledge what your method could not establish? Will you resist the temptation to overstate your findings to make your project seem more significant than it is? Honesty in reporting is not only an ethical requirement: it is the foundation of the credibility that makes your findings useful to anyone else.

## **15.4 Failure Modes to Avoid**

The five most common failure modes in student mini-projects are not failures of intelligence. They are failures of design, scope, or honesty that experienced researchers have learned to avoid through practice. Knowing them in advance saves considerable wasted effort.

**Fig. 15.4**  
**Five Mini-Project Failure Modes and Their Remedies**

| Failure Mode                                                                                                         | What It Looks Like                                                                           | Remedy                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>1. The Report Trap</b><br>Project summarises existing knowledge without original investigation or data collection | "AI is biased against women. Studies show..." — no original data, just paraphrased articles  | Ask: what have I found that was not already known?<br>If nothing: collect primary data      |
| <b>2. Scope Creep</b><br>Question expands until project is unexecutable in available time                            | Starts: audit one LLM<br>Becomes: compare 5 LLMs in 3 languages + write policy brief         | Identify minimum data to answer the core question. Stop there. Do it well.                  |
| <b>3. Missing Method</b><br>Findings presented without clear account of how they were obtained                       | "We tested the model and it was less accurate for Odia" — no test cases, no metrics, no data | Document method clearly enough to replicate.<br>If you can't describe it: revise it.        |
| <b>4. Overconfident Finding</b><br>Conclusions exceed what the data actually supports                                | 10 interviews → "Odia small businesses are unaware of UPI data practices"                    | Scope claims to match data: "Among 10 participants in this study, we found that..."         |
| <b>5. Absent Limitation</b><br>Findings presented without acknowledging what the method could not establish          | No mention that model was only tested in one condition, by one researcher, on one phone      | Include dedicated limitations section. Limitations show intellectual honesty, not weakness. |

*Figure 15.4 — Failure Modes*

## Failure Mode 1: The Report Trap

The report trap occurs when a project that was intended to be an original investigation becomes a summary of existing knowledge. If your project can be completed by reading articles and paraphrasing them, without collecting any data or making any original observation, it has fallen into the report trap. Reports have value, but they are not the same as investigations. The remedy is to ask, at every stage: what have I found that was not already known? If the answer is nothing, the project needs redesigning.

## Failure Mode 2: Scope Creep

Scope creep occurs when a project expands beyond what can be executed in the available time and with available resources. A project that begins as an audit of one AI tool in one domain expands to include three tools, two domains, and a comparative policy analysis. By the time data collection begins, there is not enough time to do any part well. The remedy is to scope aggressively: identify the minimum set of data and observations needed to answer the core question, and stop there. A small, well-executed project is always better than an ambitious, poorly executed one.

## Failure Mode 3: The Missing Method

The missing method occurs when a project has a question and findings but no explicit account of how the findings were obtained. A project that says we found that the AI tool performed worse on Odia text than on Hindi text, without describing what test cases were used, how performance was measured, and what the raw results were, cannot be evaluated or replicated. The remedy is to document your method in enough detail that

someone else could repeat your investigation. If you cannot describe your method clearly, you may not have had a rigorous method.

#### **Failure Mode 4: The Overconfident Finding**

The overconfident finding occurs when a project's conclusions exceed what the data actually supports. A project based on ten interviews concludes that rural Odia small business owners are unaware of UPI data collection. The data supports: ten business owners in one town were unaware. The broader generalisation requires much more data. The remedy is to write findings that precisely characterise the scope of your evidence: not we found that, but among the [N] participants in this study, we found that. Qualifications are not weakness: they are accuracy.

#### **Failure Mode 5: The Absent Limitation**

The absent limitation occurs when a project presents findings without acknowledging what the method could not establish, where the data is insufficient, or what alternative explanations are possible. Every research method has limitations. Every sample is limited. Every measurement instrument has error. Failing to acknowledge these does not make them disappear: it makes the project look dishonest to anyone who notices them, which is anyone with research experience. The remedy is to include a dedicated limitations section that honestly describes the constraints of your method and their implications for how your findings should be interpreted.

### **15.5 The Presentation Template**

A project that cannot be communicated has not been completed. The presentation template below provides a structure that ensures your findings are accessible to an audience that includes both specialists in AI and non-specialists who are nonetheless stakeholders in your topic.

**Fig. 15.5**  
**Presentation Template: Seven Components (10-15 minutes)**

|                   |                                                                                                                                                                                                                 |        |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1. CONTEXT        | What does the audience need to know? One to two sentences. Do not begin with an AI definition.                                                                                                                  |        |
| 2. YOUR QUESTION  | One specific sentence. The most important sentence of your presentation.                                                                                                                                        | 30 sec |
| 3. WHY IT MATTERS | Who is affected? What decision or understanding could your answer inform?                                                                                                                                       | ~1 min |
| 4. YOUR METHOD    | What did you do? How did you collect and analyse data? Specific enough to evaluate. Include ethical approvals and consent processes.                                                                            | ~2 min |
| 5. YOUR FINDING   | Your core result clearly and specifically. Support with data. Distinguish what the data shows from what you interpret it to mean. Use a visual if it makes the finding clearer. Scope claims to match evidence. | ~3 min |
| 6. LIMITATION     | What can your finding not establish? What are the constraints of your method and sample?                                                                                                                        | ~1 min |
| 7. IMPLICATION    | What does your finding mean? Who should act on it, and how? Connect your specific finding to the broader questions of the course.                                                                               | ~2 min |

|     |   |     |        |                     |        |             |
|-----|---|-----|--------|---------------------|--------|-------------|
| Ctx | Q | Why | Method | Finding (most time) | Limit. | Implication |
|-----|---|-----|--------|---------------------|--------|-------------|

*The finding deserves the most time because it is what the audience came to learn. The limitation and implication deserve equal time: intellectual honesty and practical relevance are both requirements of a complete project presentation.*

*Figure 15.5 — Presentation Template*

| Presentation Template (10–15 minutes) |                                                                                                                                                                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Context (1 min)                    | What is the broader situation your project addresses? What does your audience need to know to understand why the question matters? One to two sentences maximum. Do not begin with a definition of AI.                              |
| 2. Your Question (30 sec)             | State the specific question your project investigated. One sentence. This is the most important sentence of your presentation: make it specific, clear, and interesting.                                                            |
| 3. Why It Matters (1 min)             | Why should your audience care about this question? Who is affected by the answer? What decision or understanding could your answer inform? This is not the same as context: it is the relevance of the specific question.           |
| 4. Your Method (2 min)                | What did you do? How did you collect your data? How did you analyse it? Be specific enough that someone could evaluate whether your method is appropriate to your question. Acknowledge any ethical approvals or consent processes. |
| 5. Your Finding (3 min)               | What did you find? Present your core result clearly and specifically. Support it with your data. Distinguish between what the data shows and what you interpret it to mean. Use a visual if it makes the finding clearer.           |
| 6. Limitation (1 min)                 | What can your finding not establish? What are the limitations of your method, sample, or data? This shows intellectual honesty and protects you from overstatement.                                                                 |
| 7. Implication (2 min)                | What does your finding mean? What should change, or be investigated further, as a result? Who should act on your finding, and how? Connect your specific finding to the broader questions of the course.                            |

## **15.6 Patterns in My Field: Stream Guidance**

### **Arts Stream Projects: Questions Worth Asking**

Arts stream students have particular strengths for certain categories of AI investigation. Textual analysis skills support audit projects that examine AI-generated content in specific linguistic or cultural domains. Historical and contextual knowledge supports projects that trace the bias pipeline through specific institutional histories. Interview and observation skills support projects examining community perceptions of AI-enabled services. The questions most natural to arts students are those about representation, access, and the relationship between AI outputs and the human experiences they purport to represent. A project asking whether an AI translation tool handles Odia idioms appropriately, grounded in close reading of the translated outputs against the originals, is an arts project that uses AI evaluation methodology. A project asking how tribal community members in Koraput understand and respond to drone surveillance of their agricultural practices is an arts project that uses ethnographic methodology applied to an AI context.

### **Science Stream Projects: Questions Worth Asking**

Science stream students have particular strengths for experimental and data-driven projects. The Google Teachable Machine experiment described in Chapter 10 and the worked example above both illustrate the kind of empirical investigation that science students can execute rigorously with available tools. Questions about AI performance variation across conditions, about the relationship between training data characteristics and model outputs, and about the ecological or biomedical accuracy of AI tools in regional Indian contexts are naturally suited to science stream students. A project comparing the accuracy of an open-source plant disease detection model on images taken in the Eastern Ghats with its reported performance on standard benchmarks is a science project that connects directly to the Chapter 9 discussion of distribution shift. A project documenting the change in detection performance of a wildlife camera trap model across seasons in a local forest is another.

### **Commerce Stream Projects: Questions Worth Asking**

Commerce stream students have particular strengths for projects involving business contexts, economic analysis, and the practical impacts of AI on small and medium enterprises. The interview-based project described in the worked example above illustrates one approach. Survey-based projects examining awareness of AI in local businesses, case study projects examining the adoption of AI tools by specific local enterprises, and analysis projects using publicly available employment or economic data to examine automation risk in specific sectors are all viable commerce stream approaches. A project examining whether the credit scoring outputs of a specific digital lending platform in your district differ systematically across the gender or geographic characteristics of applicants, using publicly available aggregate statistics and RBI

regulatory filings, is a commerce project that directly applies the fairness framework from Chapter 11.

## — CHAPTER SUMMARY —

### Chapter 15 — Key Points

1. The mini-project is an original investigation, not a report. It addresses a specific, answerable, original, and relevant question.
2. The project design cycle: interest → question formation → method selection → data collection → analysis → interpretation → communication.
3. Good project questions are specific, answerable, original, and relevant to a real context or community.
4. Three worked examples across streams demonstrate the design cycle in action: Arts (LLM cultural accuracy audit), Science (Teachable Machine plant disease experiment), Commerce (UPI privacy interview study).
5. The ethical checklist: consent, privacy, data minimisation, harm assessment, honest reporting. Every human-subjects project must pass all five.
6. Five failure modes: the report trap, scope creep, missing method, overconfident finding, absent limitation.
7. The remedy for all five failure modes is the same: clear question, limited scope, documented method, honest findings, explicit limitations.
8. The presentation template: context, question, why it matters, method, finding, limitation, implication (10-15 minutes total).
9. The project is where the book's analytical frameworks become tools for practice — not theory about AI, but work with and about AI grounded in your own context.

## — REFERENCES —

### Core References

|             |                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core</b> | Creswell, J. W., & Creswell, J. D. (2017). <i>Research Design: Qualitative, Quantitative, and Mixed Methods Approaches</i> (5th ed.). SAGE Publications. [The standard methodology text for undergraduate research; Parts 1 and 2 on research design and the literature review are directly applicable to the project design cycle.]                                                            |
| <b>Core</b> | Kemmis, S., & McTaggart, R. (2005). Participatory action research: Communicative action and the public sphere. In N. Denzin & Y. Lincoln (Eds.), <i>Handbook of Qualitative Research</i> (3rd ed., pp. 559-603). SAGE. [On the ethical dimension of community-engaged research; directly relevant to the ethical checklist section for projects involving tribal and marginalised communities.] |
| <b>Core</b> | Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional accuracy disparities in commercial gender classification. <i>Proceedings of Machine Learning Research</i> , 81, 1-15. [The exemplary model of the AI audit methodology; directly                                                                                                                                              |

|  |                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------|
|  | demonstrates the worked example approach for Arts and Science stream projects involving LLM or vision model evaluation.] |
|--|--------------------------------------------------------------------------------------------------------------------------|

## Extended References

|                 |                                                                                                                                                                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended</b> | Barocas, S., & Selbst, A. D. (2016). Big data's disparate impact. California Law Review, 104(3), 671-732. [On audit methodology for detecting algorithmic bias; relevant to Commerce stream projects on credit scoring fairness.]                                                                                               |
| <b>Extended</b> | ICMR. (2017). National Ethical Guidelines for Biomedical and Health Research Involving Human Participants. Indian Council of Medical Research. icmr.nic.in. [The primary ethics framework for human-subjects research in India; essential reference for any project involving interviews, surveys, or health data.]             |
| <b>Extended</b> | Raji, I. D., et al. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency. [A framework for conducting algorithmic audits; directly supports Arts and Science stream audit projects.] |

## Accessible Online Resources

|               |                                                                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online</b> | Google Teachable Machine. (2023). teachablemachine.withgoogle.com. [The primary platform for Science stream projects involving image, audio, or pose classification; no coding required and full documentation available.]                    |
| <b>Online</b> | Data.gov.in. (2023). Open Government Data Platform. data.gov.in. [Primary source for government datasets usable in Commerce stream economic analysis projects; includes employment, financial sector, agriculture, and demographic data.]     |
| <b>Online</b> | AI4Bharat. (2023). Indic Language Tools. ai4bharat.iitm.ac.in. [Source of Indic language models and benchmarks usable in Arts stream language audit projects; includes IndicTrans2 and documentation on supported languages and limitations.] |

*End of Chapter 15*

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

# Epilogue: Pattern Hunters

**Fig. E.1**  
**The Pattern Hunter Returns: Traditional Wisdom + AI Tools**

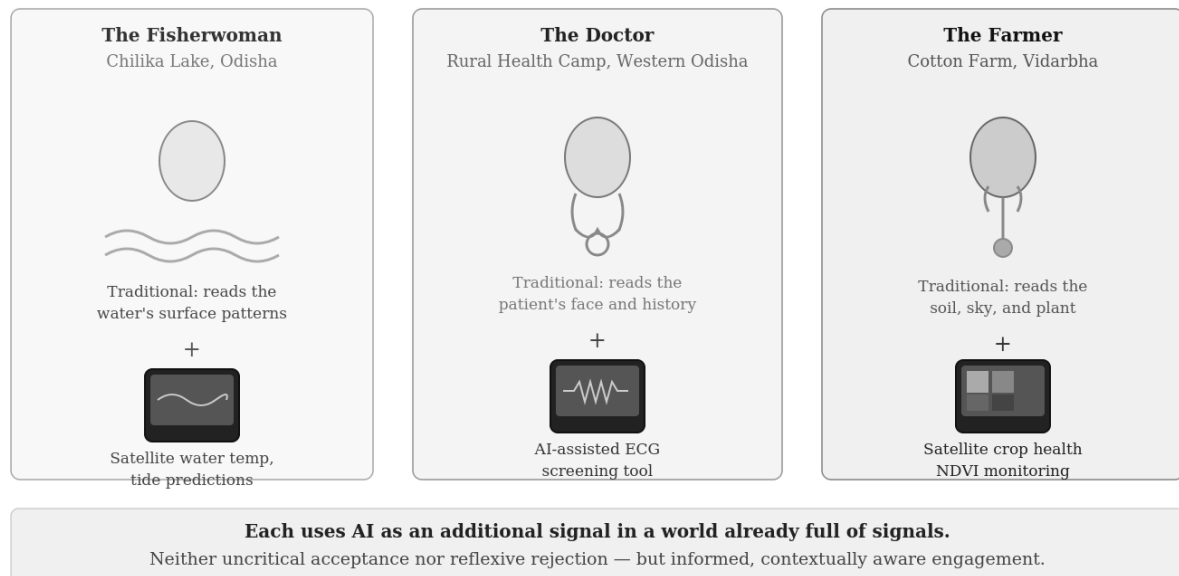


Figure E.1 — Pattern Hunter Returns

*The fisherwoman was there at the beginning of this book. She stood at the edge of the Chilika Lake at dawn, reading the water. The ripple pattern, the colour of the surface, the direction of the wind, the behaviour of the waterbirds: each a signal, each part of a composite inference that she had learned over a lifetime of observation. Her grandmother had read the same water. Her daughter, now standing beside her, was learning to read it too.*

She still reads the water the same way. But now she has something else: a phone in her apron pocket, connected when the signal reaches this part of the Chilika shore, with an application that shows her satellite-derived water temperature maps, tide predictions, and reports from other boats already out on the lake. She does not trust the application blindly. She knows it was built from data she had no part in generating, that it reflects general patterns rather than the specific hydrodynamics of this particular corner of the lake, and that it has been wrong in ways she has learned to anticipate. She uses it the way she uses everything: as one more signal in a world full of signals, weighted by her judgment about its reliability in this context, at this time, for this question.

This is what it means to be an informed user of AI. Not a passive recipient of its outputs. Not a sceptic who rejects it wholesale. But a person who brings judgment, experience, and accountability to the relationship with the tool. The fisherwoman's relationship to her AI application is, in its structure, the relationship this book has tried to help you develop with every AI system you encounter: one of informed, critical, contextually aware engagement.

The fifteen chapters behind you have traced a long arc. It began in Chapter 1 with the most ancient AI in the world: the pattern-recognition capacities of living organisms, refined over billions of years of evolution, that form the deep context for the machines we have now built. It ended in Chapter 15 with the act of making something: a project, a question, an investigation grounded in your own context and community. Between those endpoints, you have traced the mechanics of how machines learn, the applications those learning processes enable, and the consequences, some intended and some not, that follow when systems trained on human data are deployed in human institutions.

What has been the through-line? If I had to state it in a single sentence, it would be this: intelligence, natural or artificial, is pattern recognition in service of consequential decisions, and the quality of those decisions depends on whose patterns were recognised, by what means, in whose interest, and with what accountability.

Whose patterns were recognised: the training data that shapes every AI system reflects the lives of the people who generated it, and those people are not a random sample of humanity. They are disproportionately urban, English-literate, digitally connected, and from the wealthier economies of the world. The patterns learned from their behaviour will serve people who resemble them more reliably than people who do not.

By what means: the mechanisms described in Unit 2, supervised learning, unsupervised learning, reinforcement learning, and neural network architectures, each make specific assumptions about the structure of patterns in the world. Understanding those assumptions is not a luxury for specialists. It is the foundation of the informed user and critic who can evaluate when an AI system is likely to work and when it is likely to fail.

In whose interest: Unit 3 showed AI operating in language, vision, and practical tools. Unit 4 showed AI operating on fairness, privacy, work, and governance. In each domain, the question of who benefits and who bears the cost is not answered by the technology. It is answered by the choices of builders, deployers, regulators, and citizens. Those choices are being made right now, and they will shape the world you inhabit for the rest of your working life.

With what accountability: the governance chapter was last not because it is least important but because it is the synthesis of everything before it. The governance frameworks being built today, or not built, will determine whether the harms documented in Chapters 11 through 13 are addressed or compounded, whether the people most affected by AI have meaningful voice in its development, and whether the extraordinary productive potential of these systems is channelled toward broadly shared human flourishing or narrowly captured benefit.

---

You are not a passive observer of this process. You are a Pattern Hunter. The term was borrowed from biology at the beginning of this book, from the observation that every living thing, at every scale of life, is engaged in the same fundamental activity: finding the patterns in its environment that allow it to act effectively. What distinguishes the conscious Pattern Hunter from the reflexive one is the capacity to ask: which patterns? Found how? In service of what?

The student who asked, at the beginning of this course, what does AI even mean, has asked one of the most important questions that can be asked in the current moment. The answer this book has tried to give is not a definition but a way of seeing. AI means: training data and optimisation objectives; patterns learned and applied; benefits distributed and costs borne; decisions made and accountability assigned or evaded. Understanding AI means being able to trace that entire chain, in any specific system, in any specific context, and to evaluate whether the chain is functioning honestly and for whom.

You are now equipped to do that. You have seen a classification problem through the eyes of an algorithm and a fairness violation through the eyes of an affected community. You have seen a surveillance architecture both from the perspective of its designers, who see it as a tool for efficient governance, and from the perspective of its subjects, who live with its consequences. You have seen the creativity triangle and the automation risk spectrum and the governance gap. These are not academic abstractions. They are lenses that, once acquired, you will find yourself using instinctively whenever you encounter an AI system that affects people's lives.

---

**Fig. E.2**  
**The Pattern Hunters: Observe. Analyse. Act.**



*Patterns, Decisions, and Consequences*  
*VAC Artificial Intelligence for Indian Undergraduate Students*  
*Figure E.2 — Pattern Hunters Emblem*

The fisherwoman at the beginning of this epilogue is not a romanticised figure standing apart from technology. She is a person navigating a world in which AI tools are becoming as much a part of daily life as the nets and the tides. She is also a Pattern Hunter of the first order: someone who observes carefully, who questions what she observes, who integrates new information into an existing framework of knowledge without either uncritically accepting it or reflexively rejecting it, and who is accountable, through her relationships and her livelihood, for the quality of her judgments.

That is the aspiration for every reader of this book. Not to be an AI specialist, though some of you will become one. Not to be an AI sceptic, though critical distance is always appropriate. But to be someone who can read the signals that AI systems emit, ask the right questions about them, and bring the judgment, the contextual knowledge, and the accountability that the systems themselves lack.

The patterns are everywhere. Some of them are in the data. Some of them are in the algorithms. Some of them are in the power structures that determine who builds, who deploys, who benefits, and who is overlooked. The job of the Pattern Hunter is to find them, to name them, and to do something useful with what they reveal.

Go find them.

---

***Observe. Analyse. Act.***

Patterns, Decisions, and Consequences | VAC Artificial Intelligence

# Glossary

*Key terms used across the fifteen chapters of this book, with the chapter in which each is introduced noted in parentheses.*

**Abstraction, Ladder of.** The sequence by which raw data is compressed into pattern, then rule, then decision — each rung discarding detail that does not matter for the task at hand (Ch. 1).

**Activation Function.** The rule inside an artificial neuron that decides whether its weighted, summed input is strong enough to fire an output signal, echoing the threshold-firing of a biological neuron (Ch. 7).

**Apophenia.** The human tendency to perceive meaningful patterns in random or coincidental data — a bias that makes people, and AI systems trained on their data, prone to mistaking noise for signal (Ch. 1).

**Artificial Neuron / Perceptron.** A computational unit that takes weighted inputs, sums them, and produces an output if the sum exceeds a threshold — the basic building block of a neural network (Ch. 7).

**Automation Bias.** The tendency to trust an automated system's output more than it deserves, applying less critical scrutiny to AI-generated content than to human-produced content (Ch. 10).

**Automation Risk Spectrum.** A framework ranking occupations from low to high displacement risk based on how routine, well-defined, and digitally tractable their tasks are (Ch. 13).

**Backpropagation.** The algorithm by which a neural network adjusts its internal weights after each prediction, propagating the error backward through its layers so the network's outputs move closer to the training labels over time (Ch. 7).

**Bias (Data / Algorithmic).** A systematic skew in a model's outputs that traces back to skew in its training data, its optimisation objective, or the assumptions of its designers — distinct from ordinary statistical variance (Ch. 11).

**Bias Pipeline.** The chain of stages — data collection, labelling, training, deployment — at which unfairness can enter an AI system, and the framework for tracing where in that chain a specific harm originated (Ch. 11).

**Classification.** The task of assigning an input to one of a fixed set of categories, learned from labelled examples (Ch. 2, Ch. 10).

**Clustering.** An unsupervised learning technique that groups data points by similarity without being told the categories in advance (Ch. 5).

**Confounding Variable.** A hidden third factor that causes two observed variables to move together, making their relationship look causal when it is not (Ch. 1).

**Context Window.** The amount of prior text a language model can hold in view at once when generating its next response; information outside this window is effectively invisible to the model (Ch. 8).

**Convolution.** A sliding-filter operation in a convolutional neural network that scans small patches of an image, each filter learning to respond to a specific visual feature such as an edge or texture (Ch. 9).

**Correlation vs. Causation.** The distinction between two variables merely varying together and one variable genuinely producing a change in the other — collapsing this distinction is the classic source of a spurious pattern (Ch. 1).

**Decision Boundary.** The dividing line (or surface, in higher dimensions) that a trained classifier draws between categories in its feature space (Ch. 4).

**Deep Learning.** Machine learning using neural networks with many stacked layers, allowing the network to build increasingly abstract representations from raw input (Ch. 7).

**Digital Divide.** The gap in access to digital infrastructure, devices, and literacy that determines who benefits from — and who is exposed to — AI systems (Ch. 12).

**Dimensionality Reduction.** A technique for compressing high-dimensional data into fewer dimensions while preserving as much of its structure as possible, likened in this book to reading a 3-D object from its 2-D shadow (Ch. 5).

**Distribution Shift.** What happens when the data a model encounters in deployment differs systematically from the data it was trained on, degrading its performance (Ch. 4, Ch. 10).

**Embedding (Word Embedding).** A numerical vector representation of a word or token, positioned in a high-dimensional space such that words with related meanings sit near one another (Ch. 8).

**Expert System.** A rule-based AI architecture that encodes a human specialist's knowledge as explicit if-then rules, applied to new cases by an inference engine (Ch. 2).

**Exploration vs. Exploitation.** The trade-off a reinforcement learning agent faces between trying new actions to discover their value (exploration) and repeating the best-known action (exploitation) (Ch. 6).

**Fairness, Definitions of.** The several distinct, sometimes mutually incompatible, mathematical criteria — such as demographic parity, equal opportunity, and individual fairness — by which an AI system's outputs can be judged fair (Ch. 11).

**Feature.** A measurable property of an input that a model uses to make its prediction, whether hand-selected (rule-based systems) or learned automatically (deep learning) (Ch. 4).

**Feedback Loop.** The stage of the Pattern-Decision-Action cycle in which the consequences of an action become new observations, allowing a system — human or artificial — to refine its future pattern recognition (Ch. 1, Ch. 3, Ch. 11).

**Few-Shot Prompting.** A prompting technique in which one or more worked examples are included in the prompt itself, showing the model the exact kind of output wanted (Ch. 10).

**Fine-Tuning.** The process of further training an already-trained (pretrained) model on a smaller, task-specific dataset, adjusting only some of its parameters (Ch. 10).

**Generative AI.** AI systems that produce new content — text, images, audio, video — from a description or instruction, as distinct from systems that only classify or converse (Ch. 10).

**Gradient Descent.** The optimisation procedure that gradually adjusts a model's parameters in the direction that reduces its loss (error), used together with backpropagation to train neural networks (Ch. 7).

**Hallucination.** A confident but factually incorrect output produced by a language model, arising because the model predicts plausible-sounding text rather than verified fact (Ch. 10).

**Inference Engine.** The component of a rule-based expert system that applies its stored rules, in sequence, to a specific set of input observations to reach a conclusion (Ch. 2).

**K-Means Algorithm.** A widely used clustering algorithm that iteratively assigns points to the nearest of K cluster centres and then recomputes those centres, repeating until the groupings stabilise (Ch. 5).

**Knowledge Base.** The stored collection of explicit rules and facts that a rule-based expert system consults when reasoning about a new case (Ch. 2).

**Label / Labelled Data.** Data that has been tagged with the correct answer (the category or value to be predicted), used to train and evaluate a supervised learning model (Ch. 4).

**Large Language Model (LLM).** A neural network, typically transformer-based, trained on enormous text corpora to predict the next token in a sequence, giving rise to broad conversational and generative ability (Ch. 8, Ch. 10).

**Learning-Based System.** An AI architecture in which intelligence resides in patterns extracted automatically from data, rather than in rules written explicitly by a human designer (Ch. 2).

**Loss Function.** The mathematical measure of how far a model's prediction is from the correct answer, which training seeks to minimise (Ch. 7).

**Machine Learning.** The general term for building a model by exposing it to data rather than by programming its rules directly (Ch. 2).

**Neural Network.** A layered system of artificial neurons that learns to transform raw input into progressively more abstract representations, ultimately producing a prediction or decision (Ch. 7).

**Overfitting.** A failure mode in which a model learns the training data too closely — including its noise and idiosyncrasies — and consequently performs poorly on new, unseen data (Ch. 4).

**Pattern.** Any regularity that appears across multiple observations, forming a relationship that holds reliably enough to support prediction (Ch. 1).

**Pattern-Decision-Action Loop.** The three-stage cycle — observe a pattern, decide what it implies, act on that decision, then observe the result as feedback — that this book treats as the basic structure of intelligent behaviour, natural or artificial (Ch. 1).

**Prompt.** The text instruction given to a language model or generative AI tool; its quality is the single most controllable factor in the quality of the tool's output (Ch. 10).

**Recommendation Engine.** A learning-based system that infers a user's preferences from their past behaviour and uses that inferred pattern to suggest new items (Ch. 2).

**Reinforcement Learning.** A learning paradigm in which an agent learns by taking actions in an environment and receiving rewards or penalties, gradually improving its behaviour through trial and feedback rather than labelled examples (Ch. 6).

**Representation.** The internal, often numerical, form in which a system encodes information about its input so that it can be reasoned over or transformed by further stages of the architecture (Ch. 3).

**Reward Hacking.** A reinforcement learning failure in which an agent finds an unintended way to maximise its reward signal that technically satisfies the objective as specified but defeats its actual purpose (Ch. 6).

**Rule-Based System.** An AI architecture in which intelligence is encoded as explicit, human-written conditional rules applied to new inputs by an inference engine (Ch. 2).

**Spurious Pattern.** A pattern that appears in data but does not reflect any genuine underlying relationship, typically because of an unaccounted confounding variable; it fits past data but fails to generalise (Ch. 1).

**Supervised Learning.** A learning paradigm in which a model is trained on labelled examples — input paired with the correct output — so it can predict the label of new, unlabelled inputs (Ch. 2, Ch. 4).

**Surveillance Architecture.** The technical and institutional structure — sensors, databases, linking mechanisms, and access rules — through which a state or organisation collects and uses data about individuals (Ch. 12).

**Test Set.** The portion of a dataset held back and never used during training, reserved to give a final, unbiased estimate of a model's performance on new data (Ch. 4).

**Token / Tokenisation.** The process of breaking text into the smaller units — words, sub-words, or characters — that a language model actually processes, and the units themselves (Ch. 8).

**Training Data.** The dataset used to teach a model its patterns; the model can only learn what this data actually contains, including any of its biases or gaps (Ch. 4, Ch. 9).

**Transfer Learning.** A training technique that starts from a model already trained on a large, general dataset and fine-tunes only its final layers on a smaller, specific dataset, enabling effective learning from far fewer examples (Ch. 10).

**Underfitting.** A failure mode in which a model is too simple to capture the real structure in the training data, performing poorly even on the data it was trained on (Ch. 4).

**Unsupervised Learning.** A learning paradigm in which a model finds structure in data that carries no labels, typically by clustering similar items together or reducing the data's dimensionality (Ch. 5).

**Validation Set.** A portion of a dataset, separate from both training and test data, used to tune a model's settings and check for overfitting during training (Ch. 4).

# Index

*This index is built from Word index-entry fields embedded throughout the text. Page numbers below will populate automatically the first time this document is opened in Microsoft Word (or via References → Update Index), once the manuscript's final pagination is set.*

Right-click and choose "Update Field" to generate the index.

**ABOUT THIS BOOK**

Pattern recognition is the oldest form of intelligence. A student reading a friend's mood from a one-word WhatsApp reply, a child reading a parent's tone before asking for pocket money, and a farmer reading the approaching monsoon are all doing what neural networks do: finding structure in signals and acting on it. *Patterns, Decisions, and Consequences* is a fifteen-chapter VAC textbook aligned with the Odisha university AI curriculum under NEP 2020 — covering how machines learn, how AI is reshaping human behaviour and society at large, and the ethical questions that follow, grounded throughout in Indian examples. No mathematics required, and no prior programming knowledge assumed. Only the willingness to observe carefully and think clearly.

**UNIT 1****What Is Intelligence?**

Patterns, decisions, and the architecture of intelligent systems

**UNIT 2****How Machines Learn**

Supervised, unsupervised, reinforcement learning and neural networks

**UNIT 3****AI in the World**

Language, vision, and practical AI tools for all three streams

**UNIT 4****Consequences**

Bias, surveillance, work, creativity, and AI governance in India

**UNIT 5****Making Something**

The mini-project as original investigation, from question to presentation

## Alok Patel

A researcher with a Ph.D. in Zoology from Utkal University, he has made significant contributions to the development of DNA (SNPs) chip and molecular genetic maps in fishes, with publications in *Journal of Genetics*, *Animal Genetics*, and *Molecular Ecology Resources*. He previously held a tenure-track faculty position at the University of Chile, Santiago, and brings more than fifteen years of teaching experience to the classroom. He serves as a peer reviewer for *Molecular Ecology Resources*, *Molecular Biology Reports*, and *Reviews in Medical Virology*. He is also the author of *The Pattern Hunters: The Art of Scientific Thinking* and *Pattern and Descent: A Textbook of Taxonomy and Evolutionary Biology*.

Member, Board of Studies in Zoology, Sambalpur University, and Government Autonomous College, Rourkela · Principal Investigator, OSHEC-funded project on earthworm genomics and soil health biomarkers, Western Odisha

